

The Machine

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Kuch banane ke sawal

Jab kuch toota ho

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Paise aur career ke sawal

Sirf spine padhna ho toh

Poora naksha

THE MACHINE

BOOK 1 OF TWO

Yeh kitaab aapko batati nahi hai

Zyada kitaabein aapko ek definition deti hain aur ummeed karti hain ki aap maan lo. Aap padhte ho, sar hilate ho, aur ek hafte baad kuch bacha nahi hota.

Aap yeh already jaante ho ki yeh tareeka aap pe kaam nahi karta.

Toh yeh kitaab ulta chalti hai.

Har chapter aapke saamne ek problem rakhta hai. Asli problem, jismein log hote hain. Phir poochta hai ki aap kya karte. Aap sochte ho. Aap kuch nikaalte ho. Aur phir kitaab batati hai ki jo cheez aapne abhi banayi, woh already exist karti hai, aur duniya use is naam se bulati hai.

Aap naam yaad nahi kar rahe. Aap cheezein bana rahe ho, aur naam baad mein utha rahe ho.

Knowledge sirf isi tarah se tikta hai.

Do log

Yeh kitaab do logon ki baat-cheet se chalti hai.

Madhav. Umar saath ke aas-paas. Poori zindagi cheezein banayi hain, pehle machine, phir software. Ab padhata hai. Jawab seedha nahi deta, kyunki uska maanna hai ki jo jawab diya jaaye woh tikta nahi.

Kabir. Umar pachchees. Padhai poori ho chuki hai. Kai cheezein shuru ki hain, koi khatam nahi hui. Roz AI istemaal karta hai aur use bilkul samajh nahi aata ki andar kya hai. Har hafte kisi nayi cheez ka naam sunta hai jo baaki sabko pehle se pata tha. Woh

sust nahi hai, aur woh bewakoof toh bilkul nahi hai. Uske paas bahut saare tukde hain jo aapas mein judte nahi.

Kabir aksar galat hota hai. Uske galat jawab iss kitaab ka sabse kaam ka hissa hain, kyunki woh wahi galtiyan hain jo lagbhag har koi karta hai. Jab woh galat ho, toh ruk kar socho ki aap bhi wahi sochte the ya nahi.

Har chapter do hisson mein hai

SAMVAAD	Madhav aur Kabir baat karte hain. Yahan idea banta hai. Isse padho jaise aap kamre mein baithe ho.
USKE BAAD	Naam, asli duniya ka example, log kya galat samajhte hain, map, aur sawal-jawab. Yeh hissa seedha likha hai taaki baad mein dhoondhne mein aasan ho.

Samvaad mein aap seekhte ho. Uske baad wala hissa isliye hai ki chhe mahine baad jab aapko kuch yaad karna ho, toh poora baat-cheet dobara na padhna pade.

Chaar vaade

Koi shabd aapki zaroorat se pehle nahi aayega. Aap kabhi aisa word nahi milenge jiski kami aapne pehle mehsoos na ki ho. Agar koi word bold mein dikhe, toh matlab do page pehle aapne woh idea bana liya tha, aur ab bas uska label mil raha hai.

Bina wajah kuch exist nahi karta. Technology ki har cheez kisi insaan ne banayi thi, jiske paas ek problem thi. Yeh kitaab hamesha pehle problem deti hai. Agar kabhi kuch bewajah lage, toh woh likhne ki galti hai, aapki nahi.

Ek chapter, ek idea. Chapters jaan-boojh ke chote hain. Do ya teen page. Ek nayi cheez. Agar chapter khatam karke aap us ek idea ko bol sako, toh woh chapter kaam kar gaya.

Baat hamesha aap tak wapas aati hai. Har chapter aise khatam hota hai jo aapne khud dekha hai, aksar isi hafte.

Do kitaabein, aur yeh pehli kyun hai

BOOK 1 THE MACHINE

Switches, languages, networks, data, serving.
Jo 2005 mein sach tha aur 2045 mein bhi sach rahega.
Yeh kitaab kabhi purani nahi hogi.

BOOK 2 THE THINKING MACHINE

Probability, learning, meaning, transformers, agents.
Yeh har saal thodi badalti rahegi.

Book 2 zyada exciting lagti hai, aur log wahin se shuru karna chahte hain. Problem yeh hai ki Book 2 ka har chapter Book 1 ki cheezon pe khada hai. AI ek file hai numbers ki, jo ek machine pe chalti hai, ek network ke through aati hai, aur ek database se judti hai.

Pehle machine, phir thinking machine. Us order mein kuch bhi udhaar nahi lena padta.

Kitni tezi se padhein

58 chapters. Do chapter roz padho toh chaar hafte. Ek roz padho toh do mahine.

Ek baithak mein poora khatam mat karo. Yeh kitaab is tarah likhi gayi hai ki har chapter ke baad rukna pade. Jo minute aap ruk ke sochte ho, asli kitaab wahin hoti hai.

Do tarah ke chapters

[SPINE] Poora map isse banta hai. Yeh sab padho.

[DEPTH] Mechanism andar se. Pehli baar mein chhod sakte ho.

Sirf SPINE padhoge toh 37 chapters hain, aur poora naksha phir bhi mil jaayega. DEPTH wale tab padho jab jaanna ho ki andar se kaam kaise karta hai.

Sawal jo chapter ke aakhir mein aate hain

Har chapter do ya teen sawal ke saath khatam hota hai. Teen tarah ke sawal hain:

SAMAJH CHECK	is chapter ka idea pakda ya nahi
JODNE WALA	yeh chapter pehle wale kis cheez se judta hai
DESIGN SAWAL	ek part mein ek baar. Kuch banane ka sawal.

Yeh test nahi hain. Computer ki zaroorat nahi. Likhne ki bhi nahi, sirf design wale sawal ke liye.

Ek minute baitho har sawal pe, aage badhne se pehle.

Har sawal ka jawab usi ke neeche likha hai. Pehle apna jawab socho, sahi ho ya galat, phir neeche padho. Galat jawab jo aapne socha, woh sahi jawab se behtar hai jo aapko diya gaya.

Design sawalon ke jawab `WORKBOOK.md` mein likhna. Woh iss kitaab ka asli output hai.

Ek cheez jo poori kitaab mein chalti rahegi

Chapter 1.1 mein hum ek chhota product uthayenge: **ek chat app jo aapke saare messages store karta hai aur unmein se dhoondh sakta hai.**

Simple lagta hai. Har part mein hum uspe wapas aayenge, aur har baar pata chalega ki uske andar ek aur layer chhupi thi.

Kitaab ke aakhir mein aap us app ko poora khol chuke honge, upar se neeche tak.

Ek hi instruction

Aage mat kudo.

Har chapter pichhle wale pe khada hai. Iss kitaab mein jargon nahi hai, jiska matlab yeh bhi hai ki beech mein se ghus ke catch up karne ka koi rasta nahi hai. Chain hi poora design hai.

Page palto.

Chapter 0.1 [SPINE]

The village: paisa exist hi kyun karta hai

SAMVAAD

Madhav: Ek gaon socho. Sau log, ek valley mein. Paani hai, zameen hai, patthar hai, lakdi hai. Paisa nahi hai, kyunki abhi tak kisi ne invent hi nahi kiya. Sabko khaana chahiye, paani chahiye, chhat chahiye, bimari mein ilaaj chahiye. Ab batao, woh jiyenge kaise?

Kabir: Har banda apna kaam kar lega. Apna khaana ugayega, apna ghar banayega.

Madhav: Achha. Toh jis din uski haddi tootegi, us din?

Kabir: Woh... khud jod lega?

Madhav: Kisne sikhaya use haddi jodna?

Kabir: Theek hai. Woh kaam nahi karega.

Madhav: Aur bhi kuch nahi chalega. Ek aadmi saat kaam kare toh saaton kharaab karega. Din mein itni roshni hi nahi hai. Gaon zinda toh rahega, lekin jaanwaron ke level pe. Toh doosra tareeka batao.

Kabir: Kaam baant lein. Bees kheti karein, das paani laayein, bees ghar banayein, das ilaaj seekhein.

Madhav: Haan. Ab dhyaan do ki kya hua. Gaon ko nayi zameen mili?

Kabir: Nahi.

Madhav: Naye auzaar? Naye log?

Kabir: Nahi. Kuch bhi naya nahi mila.

Madhav: Phir bhi ab woh pehle se zyada banate hain. Sirf ek arrangement badla, aur output badh gaya. Yeh yaad rakhna, aage kaam aayega. Ab agla sawal. Kisan ke paas anaaj hai, chhat nahi. Mistri ke paas hunar hai, khaane ko kuch nahi. Kya karenge?

Kabir: Aasan hai. Kisan anaaj dega, mistri chhat banayega.

Madhav: Bilkul. Ab vaidya. Usne mochi ka haath theek kiya. Vaidya ko joote chahiye.

Kabir: Toh mochi joote de dega. Hisaab barabar.

Madhav: Aur teen mahine baad vaidya ko phir joote chahiye. Iss baar mochi bilkul theek-thaak hai. Usko vaidya se kuch nahi chahiye.

Kabir: Toh vaidya intezaar karega ki mochi bimar pade.

Madhav: Toh vaidya ki roti iss baat pe tiki hai ki log bimar padein?

Kabir: Nahi. Yeh toh galat hai.

Madhav: Galat hai, aur yeh koi ajeeb case bhi nahi hai. Master ko anaaj chahiye lekin kisan ke bachche bade ho gaye. Pehredaar ko chhat chahiye lekin mistri ka ghar bana hua hai. Seedha exchange sirf tab chalta hai jab do log ek hi waqt pe ek doosre ka kaam chahte hon.

Kabir: Toh likh lo. Ek register banao. Vaidya ne mochi ka kaam kiya, likh do. Baad mein mochi se le lega.

Madhav: Achha jawab. Sach mein achha, aur duniya ne yeh kiya bhi hai. Lekin do sawal. Register rakhega kaun?

Kabir: Koi ek aadmi.

Madhav: Aur jis din usne apna naam thoda bada likh diya?

Kabir: Hmm.

Madhav: Doosra sawal. Sau log the toh register chal gaya. Ab das hazaar log hain, alag gaon hain, log aate jaate hain. Kitne register?

Kabir: Bahut zyada. Aur kisi ko kisi pe bharosa bhi nahi hoga.

Madhav: Toh register ka idea sahi hai, uski shakal galat hai. Ab socho. Aisi kaunsi cheez ho jo mochi le lega, chahe use ilaaj bilkul nahi chahiye?

Kabir: Aisi cheez jo uske kaam ki ho. Anaaj?

Madhav: Kabhi kabhi. Aur jis din uske paas anaaj bhara ho?

Kabir: Toh phir... aisi cheez jo woh kisi aur ko de sake.

Madhav: Ruko. Yeh important hai. Woh use kyun lega?

Kabir: Kyunki use pata hai ki baaki sab bhi lenge.

Madhav: Bas. Woh cheez uske kisi kaam ki nahi hai. Woh use isliye leta hai kyunki use bharosa hai ki aage badha dega. Ab batao, us cheez mein kya kya hona chahiye?

Kabir: Sadni nahi chahiye. Warna jab tak main use karun, khatam.

Madhav: Aur?

Kabir: Uthane mein aasan ho. Bhaari patthar se kaam nahi chalega.

Madhav: Aur?

Kabir: Nakli banana mushkil ho. Warna koi bhi bana lega aur bina kuch diye sab le lega.

Madhav: Aur? Ek chhoti cheez khareedni ho toh?

Kabir: Toh woh tukdon mein bat jaani chahiye.

Madhav: Aur aakhri, sabse zaroori?

Kabir: Sab log use maanein. Ek bhi na maane toh chain toot jaati hai.

Madhav: Tumne abhi paisa invent kar diya.

NAAM

Jo cheez Kabir ne nikaali, uska naam hai **paisa**.

Ab dhyaan do ki woh actually hai kya, kyunki yahin par lagbhag sab log galat samajhte hain.

Paisa daulat nahi hai. Paisa **ek record hai ki aapne kisi ko value di**, jise aap gaon mein kisi ke bhi saamne rakh sakte ho.

Anaaj daulat hai. Chhat daulat hai. Judi hui haddi daulat hai. Tokens doosre logon ke kaam pe ek claim hain.

Aur Kabir ka register wala jawab galat nahi tha. Paisa hi ek register hai, bas woh kisi ek aadmi ke paas nahi rakha. Woh sabke haathon mein tukdon mein baant diya gaya hai. Isiliye use kisi rakhwaale pe bharose ki zaroorat nahi padti.

Ek doosri cheez bhi paisa chupke se karta hai. Jo aadmi leta hai lekin deta kuch nahi, uske paas kuch jama nahi hota. Paise se pehle yeh dikhna mushkil tha.

ASLI DUNIYA SE EK EXAMPLE

1945 mein ek British arthshastri, R. A. Radford, ne apne war-time prisoner camp ke baare mein likha. Har kaidi ko Red Cross se ek jaisa parcel milta tha: cigarettes, jam, butter, chocolate, biscuits.

Kisi ne paisa jaari nahi kiya. Koi sarkar nahi thi. Phir bhi kuch hi hafton mein cigarettes paisa ban gaye.

Aur exactly wahi hua jo samvaad mein nikla. Jo aadmi smoke nahi karta tha woh bhi cigarettes leta tha, kyunki use pata tha ki baaki sab lenge. Log cigarettes mein daam bolne lage: ek shirt itni cigarettes, ek chocolate utni. Jab parcel late aate the toh cigarettes kam ho jaati thin aur har cheez sasti ho jaati thi. Jab zyada aate the toh daam chadh jaate the.

Kisi ne yeh design nahi kiya. Kuch sau log, ek jagah band, aur exchange ki problem ne apne aap paisa bana diya. Aur usne wahi paanch shartein poori kin jo Kabir ne ginayi thin.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

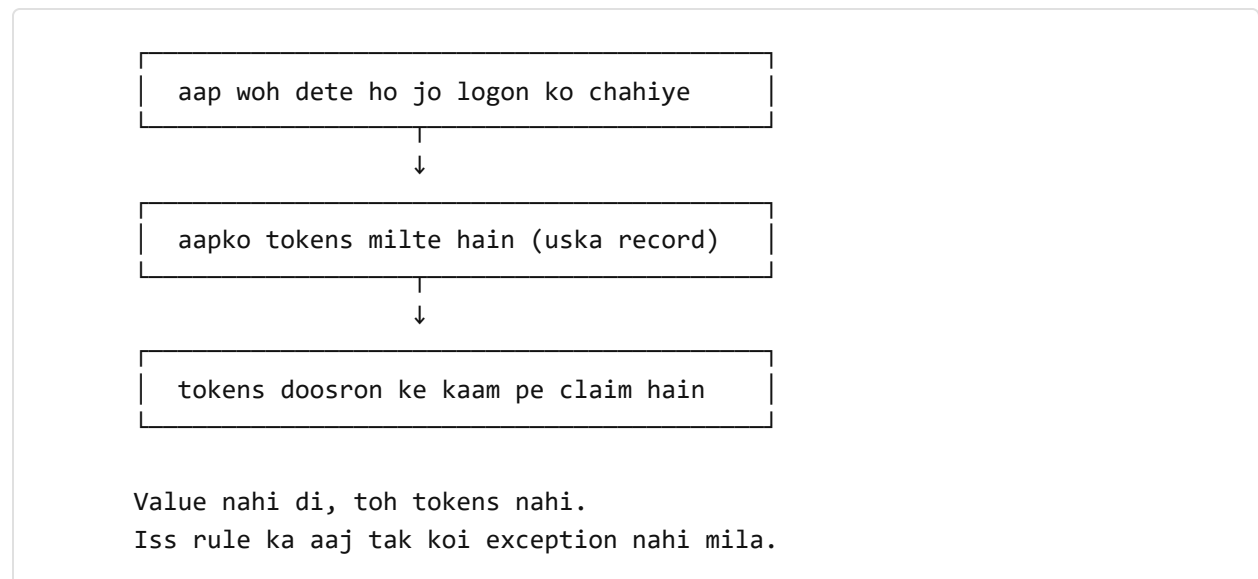
Sabse aam galti: **paise ko hi daulat samajh lena.**

Isse ek khatarnak nateeja nikalta hai. Log sochte hain ki agar sabke paas zyada paisa aa jaaye toh sab ameer ho jaayenge. Sarkarein bhi kabhi kabhi yehi sochti hain.

Lekin paisa ek claim hai, cheez nahi. Zyada claim banane se cheezein nahi banti. Sirf har claim chhota ho jaata hai.

Yeh galti tempting isliye hai kyunki ek aadmi ke level pe woh sach lagti hai. Agar sirf aapke paas zyada paisa aa jaaye, toh aap sach mein zyada cheezein khareed sakte ho. Dimaag ise sab pe laagu kar deta hai. Lekin ek aadmi ka zyada paisa doosron ke claims ka hissa cheen kar aata hai. Sabka zyada paisa kisi se kuch nahi cheenta, aur isliye kuch deta bhi nahi.

MAP PE



SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Raat-o-raat gaon ke har aadmi ke paas das guna zyada tokens aa jaate hain. Anaaj utna hi hai, ghar utne hi hain, joote utne hi hain. Cheezon ka daam kya hoga? Aapka jawab tokens ke baare mein kya batata hai?

Jawab: Daam lagbhag das guna ho jaayenge. Kisi ka kuch nahi badla.

Wajah: tokens ka koi apna kaam nahi hai. Woh sirf iss baat ke record hain ki gaon ki cheezon pe aapka kitna hissa hai. Cheezein utni hi hain, toh hissa bhi utna hi hai. Sirf ginti ke number bade ho gaye.

Isse ek aur baat nikalti hai: **paisa gaon ki daulat nahi hai. Woh us daulat ko baantne ka tareeka hai.** Camp wale example mein bilkul yehi dikha, jahan zyada cigarettes aate hi har cheez mehngi ho gayi.

2. (samajh check) Ek aadmi ko tokens ka bada dher viraasat mein milta hai aur woh zindagi bhar kisi ko koi value nahi deta. Woh khaata hai, ghar mein rehta hai, bimari mein ilaaj paata hai. Kya use gaon se asli value mil rahi hai? Agar haan, toh woh value banayi kisne, aur gaon ko badle mein kya mila?

Jawab: Haan, use bilkul asli value mil rahi hai. Woh anaaj kisi kisan ne ugaya, woh ghar kisi mistri ne banaya.

Value kisi aur ne banayi, aur gaon ko iss aadmi se badle mein kuch nahi mila.

Lekin dhyaan do: value kisi waqt di zaroor gayi thi, sirf iss aadmi ne nahi di. Jisne di, usne woh record iss aadmi ko de diya. Tokens ki khaas baat yahi hai, ki woh haath badal sakte hain. Woh yeh nahi batate ki kaam kisne kiya, sirf yeh ki record ab kiske paas hai.

Yeh paise ki kamzori hai, bug nahi. Agar tokens haath na badal sakein toh mochi wala deadlock wapas aa jaayega.

3. (samajh check) Paise ne mochi wala deadlock saaf kar diya. Ek aisi problem batao jo gaon ke paas paise se pehle nahi thi, aur ab hai.

Jawab: Kai sahi jawab hain. Kuch mazboot wale:

Jama karna. Anaaj sad jaata tha, isliye koi use jama nahi kar sakta tha. Tokens sadte nahi, isliye ab ek aadmi bahut saara hissa rok ke rakh sakta hai jabki baaki ke paas kam ho.

Kaam se kat jaana. Pehle dikhta tha ki kisne kya diya. Ab sirf ek number reh gaya. Tokens yaad nahi rakhte ki woh kaise aaye. Chori aur imaandaari ke tokens ek jaise dikhte hain.

Nakli banana. Ab shoonya se claim banane ka matlab hai gaon ki daulat ka hissa bina kuch diye le lena.

Paise ke liye kaam. Pehle aap us cheez ke liye kaam karte the jo kisi ko chahiye thi. Ab aap tokens ke liye kaam kar sakte ho, chahe jo bana rahe ho woh kisi ke kaam ka ho ya na ho.

Chapter 0.2 [SPINE]

Same hours, different pay

SAMVAAD

Madhav: Usi gaon mein do log hain. Ravi paani dhota hai, sooraj se pehle nikalta hai, andhere mein lautta hai, bees saal se. Ek din chhutti nahi li. Uska ghar chhota hai. Meera haddi jodti hai. Kuch hafte chaar ghante kaam karti hai, kuch hafte bilkul nahi. Uska ghar gaon mein sabse bada hai.

Kabir: Toh gaon Ravi ke saath galat kar raha hai.

Madhav: Kaun kar raha hai galat? Naam batao.

Kabir: Panchayat.

Madhav: Koi panchayat nahi hai. Kisi ne rate tay nahi kiye. Har aadmi apni marzi se de raha hai.

Kabir: Toh phir aisa hua kaise?

Madhav: Yehi sawal hai. Teen tareeke se dekhte hain. Pehla: Ravi aaj paani na laaye toh tumhara kya jaayega?

Kabir: Mujhe khud nadi tak jaana padega. Ek ghanta.

Madhav: Aur Meera aaj tumhari tooti haddi na jode toh?

Kabir: Haddi tedhi jud jaayegi.

Madhav: Kitne din ke liye?

Kabir: Hamesha ke liye. Main zindagi bhar langdaunga.

Madhav: Toh ek ghanta, aur chaalis saal. Dono ne mehnat ki. Lekin dono ne jo cheez roki, woh barabar hai?

Kabir: Nahi. Bilkul nahi.

Madhav: Toh pehli baat: value mehnat se nahi aati. Woh us problem ke size se aati hai jo tum hatate ho. Ab doosra sawal. Paani gaon mein aur kitne log dho sakte hain?

Kabir: Sab. Matka uthana koi hunar nahi hai.

Madhav: Aur haddi kitne jod sakte hain?

Kabir: Sirf Meera.

Madhav: Ravi kal chhod de toh?

Kabir: Dopahar tak koi aur dho raha hoga.

Madhav: Meera kal chhod de toh?

Kabir: Das saal koi nahi hoga.

Madhav: Toh gaon ko Meera ko rokna padega. Ravi ko rokna zaroori nahi hai. Yeh zulm hai ya hisaab?

Kabir: Hisaab. Lekin bura lagta hai.

Madhav: Bura lagna aur galat hona alag cheezein hain. Ab teesra aadmi. Anil naam ka banda Ravi ko roz aate-jaate dekhta hai. Do saal lagakar woh nadi se gaon tak ek naali khodta hai. Do saal tak woh aisa kuch nahi banata jo koi pee sake. Phir naali पूरी ho jaati hai aur paani roz subah khud aata hai. Anil ghar chala jaata hai. Ab ginti karo. Ravi ne kitne saal kaam kiya?

Kabir: Bees.

Madhav: Uske bees saal ka kaam aaj kahan hai?

Kabir: Khatam. Jo paani dhoya woh pee liya gaya.

Madhav: Anil ne kitne saal kaam kiya?

Kabir: Do.

Madhav: Aur uske do saal ka kaam aaj kahan hai?

Kabir: Naali mein. Woh abhi bhi chal rahi hai.

Madhav: Anil ne Ravi se zyada mehnat ki?

Kabir: Nahi. Bahut kam ki.

Madhav: Toh farak kya tha?

Kabir: Anil ki mehnat ek cheez ke andar chali gayi, aur woh cheez uske bina bhi chalti rahi.

Madhav: Ab teeno ko saath rakho. Kisi ko kitna milega, yeh kya tay karta hai?

Kabir: Problem ka size. Kitne log woh kaam kar sakte hain. Aur kaam tikta hai ya khatam ho jaata hai.

Madhav: Teeno jodkar, ya guna karke?

Kabir: Guna. Kyunki agar ek bhi zero ho toh kuch nahi bachta. Anil ki naali se agar koi paani na peeta, toh woh kitni bhi durlabh hoti, uska kuch nahi milta.

Madhav: Sahi. Ab formula dobara bolo, aur mujhe batao ki usmein kya nahi hai.

Kabir: Size, guna kitne kam log, guna kitni baar wapas aata hai. Ghante usmein nahi hain.

Madhav: Ek baar bhi nahi. Aur yeh us gaon mein sach tha jahan na paisa tha, na malik, na koi economy. Yeh kisi ka banaya hua rule nahi hai. Yeh exchange se apne aap nikalta hai.

Kabir: Toh phir sab log yeh kyun maante hain ki ghanton ka paisa milta hai?

Madhav: Kyunki jab teeno dials neeche hon, tab ghante hi ek matra lever bachta hai. Aur jo ek matra lever bachta hai, woh mechanism lagne lagta hai.

NAAM

Teen cheezein, guna hokar:

```
jo problem aap solve karte ho, uska size
x kitne kam log woh solve kar sakte hain
x aapka kaam kitni baar wapas aata rehta hai
-----
= aapko kitna milta hai
```

Doosri cheez ka naam hai **scarcity**. Teesri ka naam hai **leverage**.

Ghante formula mein nahi hain. Na guna, na jode, kahin nahi.

Lagbhag har koi maanta hai ki use ghanton ka paisa milta hai. Woh mechanism nahi hai. Woh woh hai jo bachta hai jab mechanism aapki pahunch se bahar ho.

ASLI DUNIYA SE EK EXAMPLE

2014 mein Facebook ne WhatsApp ko lagbhag 19 billion dollar mein khareeda.

Us waqt WhatsApp mein **55 log** kaam karte the, jinmein sirf 32 engineers the. Aur unke app ko lagbhag 45 crore log har mahine istemaal karte the.

Ek engineer, lagbhag ek crore chalees lakh users.

Formula pe chalao. Size: logon ko ek doosre se baat karni hai, jo seedhi zaroorat hai. Scarcity: us waqt bahut kam log itne bade paimane pe chalne wala messaging system bana sakte the. Leverage: poora upar. Code ek baar likha gaya aur woh raat-din chalta raha, chahe likhne wala so raha ho.

Ab isi ke saath ek postman ki tulna karo, jo bhi sandesh pahunchane ka hi kaam karta hai. Wahi zaroorat. Postman roz chalta hai aur uska kaam har shaam khatam ho jaata hai.

Ravi aur Anil, doosri shakal mein. Field nahi badla. Position badli.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh maan lena ki sahi field chun lena hi asli faisla hai.**

Log saal-do-saal iss sawal pe atak jaate hain: doctor banun ya engineer, coding karun ya business. Jaise ek darwaza sahi ho aur baaki galat.

Lekin har field mein dono taraf ke log hain. Padhane mein bhi crore kamane wale hain aur mushkil se guzara karne wale bhi. Ilaaj mein bhi. Khaane mein bhi. Ek hi field ke andar ka farak, do alag fieldon ke farak se kahin bada hai.

PADHANA

das students wala tutor ↔ jiski kitaab ek crore log padhte hain

KHAANA

ek kitchen mein cook ↔ jiske do sau kitchen hain

ILAAJ

roz tees mareez dekhne wala karte hain ↔ jisne woh ilaaj banaya jo sab istemaal

SANGEET

shaadi mein bajane wala ↔ jiski recording sote waqt bajti rehti hai

Yeh galti tempting isliye hai kyunki field ka faisla **dikhta hai**. Uska ek naam hai, ek degree hai, log poochte hain. Position ka faisla dikhta nahi. Uska koi naam nahi hota aur woh ek din mein nahi hota, isliye lagta hai ki woh faisla hai hi nahi.

MAP PE

KAM

size chhoti problem, bina bhi chal jaaye
scarce bahut log kar sakte hain
leverage kaam karte hi khatam hai

ZYADA

badi problem, bina nahi chalti
↔ bahut kam log kar sakte hain
↔ kaam aapke bina bhi aata rehta

Teen dials. Jo milta hai woh teeno ka guna hai, isiliye ek dial ko neeche se upar karna sab kuch badal sakta hai, aur kisi dial ko chhue bina sirf zyada der kaam karne se lagbhag kuch nahi badalta.

SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Ravi Anil ki naali dekhta hai aur sochta hai ki use aur mehnat karni chahiye. Woh bees ki jagah chalees matke roz dhone lagta hai aur uski kamai lagbhag dugni ho jaati hai. Meera aur Anil ki nahi badalti. Iss tareeke se uski kamai zyada se zyada kahan tak ja sakti hai, aur wahan ruk kyun jaati hai?

Jawab: Ek din mein jitne matke ek insaan ka sharir dho sakta hai, bas wahan tak. Shayad saath, shayad assi. Uske baad kuch nahi.

Ceiling ki wajah yeh hai ki Ravi apna **waqt** bech raha hai, aur waqt roz utna hi milta hai. Uska leverage dial zero pe atka hai, aur zero se guna karo toh mehnat kitni bhi ho, jawab chhota hi rehta hai.

Yeh important isliye hai ki Ravi ka faisla galat nahi hai. Zyada mehnat se sach mein zyada milta hai. Bas woh raasta ek deewar pe khatam hota hai, aur woh deewar uske sharir mein hai. Anil ke raaste mein aisi koi deewar nahi thi.

2. (samajh check) Teen dials mein se kaunsa ek aadmi sabse jaldi badal sakta hai, aur kaunsa sabse dheere? Har ek ke liye actually kya chahiye, woh socho.

Jawab: Sabse tez **problem ka size** hai. Aap kal se hi badi problem pe kaam karna shuru kar sakte ho. Kisi ki ijaazat nahi chahiye, sirf yeh chunna hai ki kis problem pe waqt lagana hai. Ravi agar kal se naali khodna shuru kare toh usne yeh dial usi din ghuma diya.

Sabse dheema **scarcity** hai. Woh cheez seekhne mein saal lagte hain jo bahut kam log kar sakte hain. Meera ke saalon ka koi shortcut nahi tha.

Leverage beech mein hai, aur uski shakal alag hai. Woh dheere-dheere nahi badhta. Woh ek din chalu hota hai. Anil ki naali do saal tak zero thi, phir ek din poori. Isliye leverage sabse zyada chhoda jaata hai: uska bada hissa aisa dikhta hai jaise kuch ho hi nahi raha.

3. (jodne wala) Pichhle chapter mein paisa “value dene ka record” tha. Anil ko do saal baad bhi paani ka hissa milta rehta hai. Woh record kis cheez ka hai, aur woh value di kab gayi thi?

Jawab: Woh record us naali ka hai, na ki aaj ke kisi kaam ka. Value do saal pehle di gayi thi, jab woh khod raha tha.

Ravi roz value deta hai aur roz record kamata hai. Anil ne do saal ek saath value di aur aisa record kamaya jo baar baar kaam aata hai.

Toh gaon Anil ko “kuch na karne” ke paise nahi de raha. Woh us kaam ke paise de raha hai jo poora ho chuka hai aur ab bhi kaam kar raha hai. Yeh Chapter 0.1 wale rule ka seedha nateeja hai: value di gayi thi, isliye tokens aa rahe hain.

Yahin par leverage ki asli definition milti hai: **aisi value dena jo ek baar di jaaye aur baar baar mile.**

Chapter 0.3 [SPINE]

Society ke chaar levels

SAMVAAD

Madhav: Gaon badh gaya. Sau log ab ek crore hain. Valley sheher ban gayi. Ek cheez toot gayi jiske baare mein kisi ne socha hi nahi tha. Batao kya?

Kabir: Bheed ho gayi?

Madhav: Usse gehri baat. Gaon mein tumhe pata tha ki Meera haddi jodti hai. Uske ghar tak chal ke ja sakte the. Ab ek crore logon mein tumhe kitne log pata hain?

Kabir: Do sau, teen sau.

Madhav: Toh aaj subah tumne jo chawal khaya, woh kisne ugaya?

Kabir: Pata nahi.

Madhav: Paani kis pipe se aaya, kisne daali?

Kabir: Pata nahi.

Madhav: Wahi kaam abhi bhi ho raha hai. Roz. Lekin purana arrangement ise organise nahi kar sakta, kyunki purana arrangement ek doosre ko jaanne pe khada tha. Toh ab yeh chalta kaise hai? Sabse neech se shuru karo. Ek insaan ko chahiye kya?

Kabir: Khaana, paani, ghar, ilaaj, suraksha, padhai. Aur kisi ka saath.

Madhav: Aur? Ek cheez chhoot rahi hai.

Kabir: Kuch aisa jisse din bitane laayak lagein.

Madhav: Haan. Ab batao, yeh list gaon wale ke liye alag thi?

Kabir: Nahi. Bilkul yehi thi.

Madhav: Paanch hazaar saal pehle?

Kabir: Wahi.

Madhav: Toh yeh list hilti nahi. Sab kuch iske upar khada hai. Ab agla kadam. Meera sau logon ko sambhal leti thi. Ek crore ko khaana kaun khilayega?

Kabir: Ek aadmi toh nahi kar sakta.

Madhav: Toh?

Kabir: Log mil kar karenge. Ek group.

Madhav: Aur woh group kis kis ko khilayega?

Kabir: Un logon ko jinhe woh jaanta bhi nahi.

Madhav: Bas, yahi scale ka jawab hai. Ek aadmi unhe serve karta hai jinhe woh jaanta hai. Ek group unhe serve karta hai jinhe usmein se koi nahi jaanta. Ab teesra kadam. Har group ek hi sawal poochta hai. Kya?

Kabir: Zyada paisa kaise kamayein?

Madhav: Usse pehle wala sawal. Woh zyada logon tak kaise pahunchein, bina zyada aadmi lagaye?

Kabir: Koi machine. Koi auzaar.

Madhav: Hal. Pahiya. Likha hua hisaab. Telephone. Ab dhyaan se batao: hal kitne logon ko khaana khilata hai?

Kabir: Sabko. Usise toh kheti hoti hai.

Madhav: Nahi. Hal ne aaj tak ek bhi aadmi ko khaana nahi diya. Woh kisi ke muh mein kuch nahi daalta.

Kabir: Toh phir woh karta kya hai?

Madhav: Kisan ko guna karta hai. Aur kisan logon ko khilata hai.

Kabir: Haan. Auzaar seedha insaan ko serve nahi karta. Woh us cheez ko badata hai jo insaan ko serve kar rahi hai.

Madhav: Ise yaad rakhna, agle chapter ka poora chapter isi pe hai. Ab aakhri kadam. Ab tak har auzaar ek hi kaam karta tha. Hal hal chalata hai. Use mana kar hisaab-kitaab nahi karwa sakte. Sahi?

Kabir: Sahi.

Madhav: Aur aaj?

Kabir: Aaj aisi cheez hai jo hisaab bhi likhti hai, message bhi, scan bhi padhti hai, raasta bhi banati hai.

Madhav: Toh woh ek kaam ka auzaar hai?

Kabir: Nahi. Woh... auzaar banane ka auzaar hai.

Madhav: Ab poora dhaancha bolo. Neeche se upar.

Kabir: Insaan ki zaroorat. Uske upar group jo use scale pe poora karta hai. Uske upar auzaar jo group ko guna karta hai. Aur uske upar woh cheez jo auzaaron ko behtar karti hai.

Madhav: Chaar levels. Aur aakhri sawal: paisa in mein kahan ghusta hai?

Kabir: Sabse upar? AI companies ke paas toh bahut paisa hai.

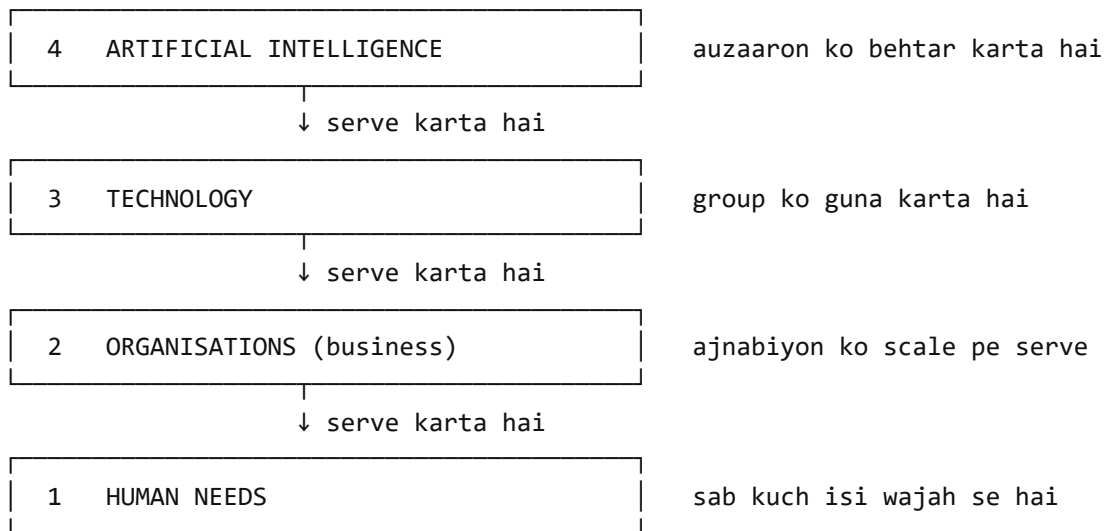
Madhav: Ghusta kahan hai, rukta kahan hai, yeh alag sawal hai. Jab tum khaana khareedte ho, tumhare haath se paisa kis level pe nikalta hai?

Kabir: Level ek pe. Meri zaroorat pe.

Madhav: Bas. Paisa hamesha sabse neeche se ghusta hai, jab ek insaan apni zaroorat ke liye deta hai. Uske baad woh upar chadhta hai. Upar wale levels paisa banate nahi, woh neeche se aaya hua paisa aapas mein baantte hain.

NAAM

Chaar levels. Har ek apne neeche wale ko serve karne ke liye hai.



Service NEECHE ki taraf behti hai.

Paisa UPAR ki taraf, aur woh sirf level 1 pe hi ghusta hai,
jab ek insaan apni zaroorat ke liye paise deta hai.

Teen cheezein check karne laayak hain, kyunki saaf-suthra diagram inhe chhupa deta hai.

Har zaroorat business se poori nahi hoti. Maa apne bachche ko khila rahi hai, yeh level 1 seedha serve ho raha hai, na koi organisation, na paisa kahin. Insaan ko ek doosre se jo chahiye, uska bahut bada hissa kabhi transaction tha hi nahi aur kabhi hoga bhi nahi.

Har business ko technology se madad nahi milti. Kuch ko mushkil se chhoti hai. Kuch jaan-boojh ke mana karte hain, aur wahi mana karna hi unka product hai.

Har technology ko AI behtar nahi karta. Paani ki pipe technology hai. Usmein sochne wali machine ke liye karne ko kuch hai hi nahi.

Yeh stack ek jhukav batata hai, kanoon nahi. Jahan levels gayab hain, aksar wahi jagah sabse interesting hoti hai.

ASLI MEIN

Koi bhi zaroorat uthao, uske upar chaaron levels abhi baithe milenge.

ZAROORAT khaana	→	LEVEL 2 restaurant, farm	→	LEVEL 3 fridge, tractor	→	LEVEL 4 demand ka andaaza
sehat	→	hospital	→	scanner, dawaiyan	→	scan padhna
gyaan	→	school, university	→	chhapai, video	→	har sawal ka jawab dene wala tutor
suraksha	→	police, insurance	→	radio, camera	→	pattern pakadna
judav	→	post, telecom	→	phone, internet	→	kisi bhi do bhaashaon ke beech tarjuma

Har column bhara hua hai. Us table ka har level 4 wala entry aapse chhota hai.

ASLI DUNIYA SE EK EXAMPLE

Aaj raat aap phone se khaana order karte ho. Chaaron levels ek saath chal rahe hain, aur aap sirf ek button dekhte ho.

- LEVEL 1 Bhookh. Bas. Yeh cheez badalti hi nahi.
- LEVEL 2 Ek company jo aapko un rasoion se jodti hai jinse aap kabhi mile nahi. Restaurant khaana banata hai, delivery wala laata hai, dono aapke liye ajnabi hain. Yeh woh "group" hai jo ajnabiyon ko scale pe serve karta hai.
- LEVEL 3 Phone, internet, GPS, map, payment system. Inmein se koi bhi aapko khaana nahi khilata. Yeh sab us company ko guna karte hain. GPS ke bina ek delivery wala ek din mein 4 order kar paata. GPS ke saath 25.
- LEVEL 4 Kitne order kahan se aayenge iska andaaza. Kaunsa raasta chuna jaaye. Kaunsa delivery wala kaunse order ke liye. Yeh level 3 ke auzaaron ko behtar karta hai.

Ab neeche se upar ginti karo. Level 1 hazaaron saal purana hai. Level 2 ki shakal purani hai, dhaabe hamesha the. Level 3 ke tukde 20 se 40 saal purane hain. Level 4 lagbhag 10 saal purana hai.

Ek order, chaar levels, aur unki umron mein hazaaron saal ka farak.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh maan lena ki upar wale level pe hone se apne aap zyada paisa milta hai.**

"Al mein chala jaaunga" sunne mein aisa lagta hai jaise woh khud ek jawab hai. Woh nahi hai. Woh sirf batata hai ki aap kaunsi manzil pe khade honge, na ki us manzil pe aapka kya haal hoga.

Level 4 pe hazaaron log naukri karte hain, ghante bechte hain, aur unka leverage dial bilkul zero pe hai. Woh bas Ravi hain, ek nayi manzil pe. Udhar Level 1 pe koi kisan hai jiske paas apni zameen hai aur woh Ravi se kai guna kama raha hai.

Yeh galti tempting isliye hai kyunki level dikhta hai aur position nahi. Level ka naam hota hai jo aap kisi ko bata sakte ho. Position ka koi naam nahi hota.

Do maps ko saath rakhna hi asli kaam hai: **level batata hai ki aap kis kaam ke aas-paas ho, dials batate hain ki us kaam se aapko kya milega.**

MAP PE

Ab aapke paas do map hain jo aapas mein fit hote hain.

Pehla map, pichhle chapter se, batata hai ki **kisi ko kitna milega yeh kya tay karta hai**: value, scarcity aur leverage.

Yeh map batata hai ki **society mein saara kaam hota kahan hai**: chaar levels, har ek apne neeche wale ko serve karta hua.

Dono ko saath rakho aur ek cheez dikhti hai: har level pe aise log hain jo teeno dials pe neeche hain, aur aise bhi jo teeno pe upar hain. Koi level apne aap ameer nahi hai. Sirf levels ke andar ki positions hoti hain.

SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Pichhle hafte aapne kisi cheez ke paise diye. Use neeche ki taraf trace karo: kaunsi organisation ne di, woh organisation kis technology pe depend karti hai, aur woh aakhir mein us list mein se kaunsi zaroorat पूरी karti hai? Agar aap list tak pahunch hi nahi paate, toh woh us cheez ke baare mein kya batata hai jo aapne khareedi?

Jawab: Zyadatar cheezein 2 ya 3 kadmon mein list tak pahunch jaati hain. Chai → dukaan → supply chain → khaana. Recharge → telecom company → tower aur fibre → judav.

Agar aap list tak nahi pahunch paate, toh do mein se ek baat hai.

Ek: woh cheez kisi aur ki zaroorat पूरी karti hai, aapki nahi. Business ke liye khareedi gayi cheezein aksar aisi hoti hain. Woh list tak pahunchti hain, bas kisi aur ke raaste se.

Do: woh zaroorat ki nahi, chaah ki cheez hai. Yeh galat nahi hai, insaan hamesha aisi cheezein khareedte aaye hain. Lekin dhyaan rakho: zaroorat ki cheezein mandi mein tikti hain aur chaah ki cheezein sabse pehle katti hain.

2. (jodne wala) Chapter 0.2 ke teen dials yaad karo. Kya level 4 pe hone se apne aap leverage mil jaata hai? Ek aisa aadmi socho jo level 4 pe kaam karta hai aur jiska leverage phir bhi kam hai.

Jawab: Nahi, bilkul nahi milta.

Ek aadmi jo kisi company mein AI ka kaam karta hai aur ghante ke hisaab se tankhwah leta hai, uska leverage utna hi hai jitna Ravi ka. Woh chhutti pe jaaye toh uski kamai rukti hai. Kaam woh kar raha hai, lekin jo cheez baar baar chalti rahegi uska maalik koi aur hai.

Leverage level se nahi aata. Woh iss baat se aata hai ki jo cheez aapke bina chalti rahegi, uspe aapka haq hai ya nahi.

Isi tarah, level 1 pe bhi bahut leverage ho sakta hai. Jis kisan ke paas apni zameen hai, uske paas ek aisi cheez hai jo har saal fasal deti hai. Zameen uski naali hai.

3. (design sawal, iss part ka) Ek zaroorat chuno jo aapko lagta hai theek se poori nahi hoti. Uske chaaron levels likho jaisi woh aaj hain. Phir woh jagah nishaan lagao jahan koi level gayab hai ya kamzor hai. Yeh `WORKBOOK.md` mein likho.

Jawab: Iska ek sahi jawab nahi hai, kyunki yeh aapka apna kaam hai. Lekin dekhne ke liye do cheezein batata hoon.

Gayab level 3: ek business chal raha hai lekin usko guna karne wala koi auzaar nahi hai. Log haath se register bhar rahe hain, phone pe hisaab kar rahe hain. Yeh sabse aam khaali jagah hai aur sabse aasan bharne wali.

Gayab level 2: zaroorat asli hai lekin use scale pe serve karne wala koi group hi nahi hai. Yeh dhoondhna mushkil hai aur milne pe bada hota hai.

Ek chetavni: khaali jagah dikhne ka matlab hamesha mauka nahi hota. Kabhi kabhi level isliye khaali hai kyunki wahan paisa nahi hai, ya kisi ne koshish karke chhod diya. Chapter 8.5 mein hum isi pe wapas aayenge.

Chapter 0.4 [SPINE]

Technology level dikhta kyun nahi

SAMVAAD

Madhav: Aaj subah nashte se pehle tumne kitni technologies istemaal ki?

Kabir: Do teen. Phone, light.

Madhav: Ginti karo. Switch dabaya, roshni aayi. Nal khola, saaf paani aaya. Phone uthaya, kisi se baat ki jo sheher se bahar tha. Message bheja jo second se kam mein pahuncha. Fridge se doodh nikaala jo garmi mein bhi kharaab nahi tha.

Kabir: Theek hai, das ke aas-paas.

Madhav: In dus mein se kitni tum kisi ko samjha sakte ho?

Kabir: Sach kahun toh ek bhi nahi.

Madhav: Aur tumhe iss baat ki sharam aati hai.

Kabir: Haan. Lagta hai baaki sabko pata hai.

Madhav: Toh chalo dekhte hain kiski galti hai. Ek nal ka kaam kya hai?

Kabir: Paani dena.

Madhav: Bas itna?

Kabir: Haan.

Madhav: Ab socho. Agar nal aisa hota ki har baar tumhe yaad rakhna padta ki paani kis bandh se aa raha hai, kaunsa pump use upar bhej raha hai, kaunsi pipe kis gali se guzarti hai, kaunse kemikal se woh saaf hua. Woh nal kaisa hota?

Kabir: Bahut bura. Kaam hi nahi karta.

Madhav: Toh achha nal woh hai jo tumhe yeh sab bhoolne deta hai.

Kabir: Haan.

Madhav: Ab isko poore technology pe lagao. Gaadi tumse kya chhupa rahi hai?

Kabir: Engine. Gearbox. Tel kahan se aaya.

Madhav: Switch?

Kabir: Power plant. Grid. Taar.

Madhav: Search box?

Kabir: Poori duniya ke computer.

Madhav: Toh ek technology ka kaam kya hai?

Kabir: Apne neech wali cheez ko chhupa dena.

Madhav: Ab yeh bolo, aur dhyaan se bolo. Ek technology jitni achhi hoti hai, tum usse utna...

Kabir: Utna zyada samajhte ho.

Madhav: Ulta.

Kabir: Utna kam samajhte ho?

Madhav: Bolo phir se, poora.

Kabir: Technology jitni achhi hogi, main usse utna kam samjhunga. Kyunki achhi hone ka matlab hi hai ki woh gayab ho jaaye.

Madhav: Ab batao, tumhe grid ke baare mein kab pata chalta hai?

Kabir: Jab light chali jaati hai.

Madhav: Internet ke baare mein?

Kabir: Jab band ho jaata hai.

Madhav: Toh tum ek aisi duniya mein rehte ho jahan sab kuch theek chal raha hai, aur isiliye kuch samajh nahi aata. Ab wapas apni sharam pe aao. Kiski galti hai?

Kabir: Kisi ki nahi. Cheezein banai hi aisi gayi hain.

Madhav: Sahi. Lekin ab main tumse ek aur cheez poochunga, aur iska jawab aasan nahi hai. Jo cheez chhupi hui hai, uske saath tum kya kar sakte ho?

Kabir: Use istemaal kar sakta hoon.

Madhav: Aur kya nahi kar sakte?

Kabir: Main... use chun nahi sakta. Do cheezon mein se behtar nahi chun sakta, kyunki mujhe farak hi nahi pata.

Madhav: Aur?

Kabir: Main uske upar kuch bana nahi sakta.

Madhav: Aur?

Kabir: Main yeh nahi keh sakta ki uske saath kya mumkin hai aur kya nahi.

Madhav: Bas. Toh galti tumhari nahi hai, aur keemat phir bhi tum hi chukate ho. Yeh do baatein ek saath sach hain, aur zyadatar log inmein se ek chun lete hain aur doosri bhool jaate hain.

NAAM

Neeche wali cheez ko chhupa dene ke iss tareeke ka naam hai **abstraction**.

Yeh technology ka sabse zaroori aur sabse kam samjha jaane wala idea hai. Poori computer ki duniya iske alaawa lagbhag kuch bhi nahi hai: parton ka dher, jahan har parat neeche wali ko chhupa deti hai, taaki uske upar wala usse soche bina kaam kar sake.

aap ek app istemaal karte ho
program ek language istemaal
language ek OS istemaal karti
OS chip istemaal karta hai

taaki program ke baare mein na sochna pade
taaki 1 aur 0 ke baare mein na sochna pade
taaki chip ke baare mein na sochna pade
taaki electron ke baare mein na sochna pade

Koi bhi saari parton ko gehrai se nahi jaanta. Sab kisi chhupi hui parat pe khade hain. Yeh aapke gyaan ka gaddha nahi hai. Yeh poora design hai.

ASLI MEIN

Jab bijli pehli baar sheheron mein aayi, log ise samajhte the. Ghar ke malik apna fuse badalte the aur unhe pata hota tha ki circuit kaise banta hai. Woh isliye nahi ki woh aaj ke logon se hoshiyaar the. Woh isliye ki technology abhi itni achhi nahi thi ki gayab ho sake.

Ab bijli aapke liye ek switch hai. Behtar technology, kam samajh. Yeh exchange har ek cheez ke saath hua hai.

Aur wahi abhi AI ke saath ho raha hai, sirf bahut tezi se. Aap ek box mein type karte ho aur jawab aa jaata hai. Beech ka sab kuch pehle hi gayab ho chuka hai, jabki cheez bani hue paanch saal bhi theek se nahi hue.

ASLI DUNIYA SE EK EXAMPLE

1958 mein Leonard Read ne ek chhota lekh likha jiska naam tha “I, Pencil”. Usmein ek pencil apni kahani khud sunati hai.

Uski baat ek hi thi: **duniya mein ek bhi aisa insaan nahi hai jo ek pencil bana sake.**

Lakdi kisi ne kaati, aari kisi aur ne banayi, us aari ka steel kisi teesre ne, us steel ka loha kisi khadaan se. Graphite Sri Lanka se, mitti Mississippi se, mom, rang, rubber, us rubber ko jodne wala kemikal. Us pencil ke peeche hazaaron log hain, alag deshon mein, jo ek doosre ko jaante tak nahi, aur jinmein se koi bhi akela pencil nahi bana sakta.

Ab yeh yaad rakho aur socho ki ek phone kya cheez hai. Ya ek AI model.

Aur ab dhyaan do ki iss baat ne aapko chaunkaya. **Pencil sabse simple cheez hai jo aap soch sakte ho.** Woh bhi chhupi hui hai. Yeh soch ki “mujhe technology samajh nahi aati” us pencil se hi shuru ho jaati hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki technology samajh na aana meri kami hai.**

Yeh baat andar hi andar chalti rehti hai. Log sochte hain ki baaki sabko yeh sab pata hai aur woh akele peeche reh gaye. Isse sharm aati hai, aur sharm ki wajah se log sawal poochna band kar dete hain, jisse gaddha aur bada hota hai.

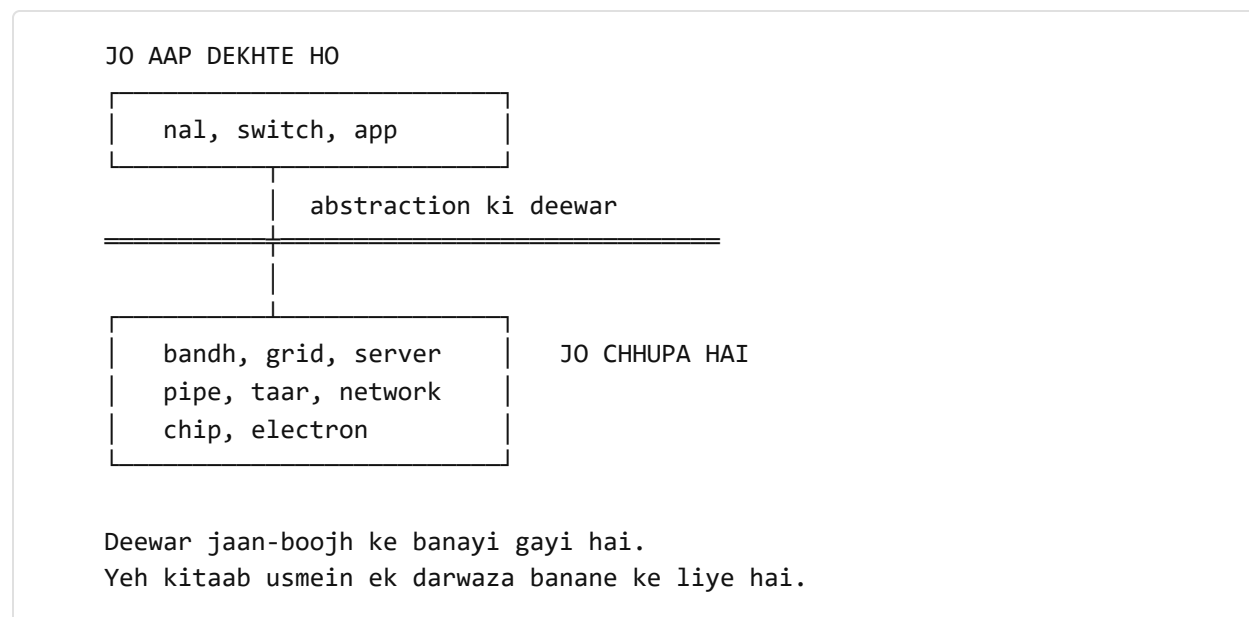
Lekin jo abhi aapne dekha, uske hisaab se yeh galat hai. Cheezein banai hi is tarah gayi hain ki samjhe bina chal jaayein. **Aapko woh nahi dikh raha jo dikhane ke liye banaya hi nahi gaya tha.**

Ab yahan par ek baat saaf kar deta hoon, kyunki iska ulta bhi galat hai.

Yeh iss baat ka bahana nahi hai ki samajhne ki koshish hi na ki jaaye. Pichhla section abhi bata chuka hai ki chhupi hui cheez ko aap istemaal toh kar sakte ho, lekin uspe kuch bana nahi sakte. Toh kami aapki nahi hai, aur keemat phir bhi aapko hi chukani padti hai.

Sahi baat yeh hai: **peeche hona aapki galti nahi hai, lekin usse nikalna aapka hi kaam hai.**

MAP PE



SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Aaj aapne jo cheezein istemaal ki hain, unmein se ek chuno, aur us sabki list banao jo woh aapse chhupa rahi hai. Kahan tak neeche ja sakte ho, jab tak aapko sach mein pata na ho ki neeche kya hai?

Jawab: Zyadatar log 2 ya 3 kadam neeche jaakar ruk jaate hain. Nal se pipe tak, pipe se tanki tak, aur uske baad andhera.

Woh rukne ki jagah hi kaam ki cheez hai. Woh batati hai ki aapki abstraction ki deewar kahan khadi hai.

Aur dhyaan do ki har cheez ke liye woh deewar alag jagah hai. Jo cheez aapne khud thodi bahut kholi hai, uske liye aap gehre ja paate ho. Yeh saabit karta hai ki deewar aapke dimaag ki seema nahi hai. Woh sirf yeh hai ki aapne ab tak kya khola aur kya nahi.

2. (samajh check) Chapter kehta hai ki achhi technology gayab ho jaati hai. Toh phir ek achhi technology aur ek aisi cheez mein kya farak hai jo hai hi nahi? Aap doosron ko kaise samjhaoge ki koi chhupi hui cheez wahan maujood hai?

Jawab: Farak tab dikhta hai jab woh **ruk jaaye**.

Jo cheez hai hi nahi, woh ruk nahi sakti. Achhi technology jab rukti hai toh poora ka poora andar ka jaal ek saath saamne aa jaata hai. Light jaane ke din aapko pata chalta hai ki bijli ka grid maujood tha.

Isiliye zyadatar log technology ke baare mein sirf uske toot-ne se seekhte hain. Aur isiliye woh seekh tukdon mein rehti hai: aapko sirf woh hissa dikhta hai jo us din toota tha.

Ek doosra tareeka bhi hai: keemat dekho. Har chhupi hui cheez ko koi na koi chala raha hai aur uska kharcha uth raha hai. Jahan paisa ja raha hai, wahan kuch maujood hai. Part 7 mein hum yehi karenge.

3. (jodne wala) Chapter 0.2 kehta tha ki pay value, scarcity aur leverage se aata hai. Agar abstraction ka matlab hai ki bahut kam log neeche wali parat samajhte hain, toh us parat ko samajhne wale ke liye scarcity ka dial kahan baithta hai? Aur woh dial waqt ke saath kis taraf khisak raha hai?

Jawab: Jitni gehri parat, utna upar scarcity ka dial. Yeh seedha nikal aata hai: samajhne wale kam hain, isliye jo samajhta hai woh durlabh hai.

Dooosra hissa zyada important hai, aur uska jawab do tarah se sach hai.

Ek taraf: har naya abstraction ek purani parat ko scarce se common bana deta hai. Pehle server chalane ke liye machine ka gehra gyaan chahiye tha. Cloud aane ke baad kam log ko uski zaroorat rahi. Us parat ka dial neeche khisak gaya.

Doosri taraf: wahi abstraction ek nayi, aur upar wali parat bana deta hai, jahan dial ekdum upar hai, kyunki cheez nayi hai aur abhi kisi ko samajh nahi aayi.

Toh scarcity gayab nahi hoti. Woh upar khisak jaati hai.

Yeh iss poori kitaab ka sabse kaam ka nateeja hai, aur Book 2 isi jagah pe khadi hai. AI abhi wahi nayi parat hai.

Chapter 0.5 [SPINE]

Toh ab hume kya kholna padega

SAMVAAD

Madhav: Ab tak tumhare paas do cheezein hain. Ek naksha, jo batata hai ki society ka kaam kahan hota hai. Aur ek formula, jo batata hai ki kisi ko kitna milega. Ab in dono ko saath rakho aur mujhe batao ki tumhe seekhna kya chahiye.

Kabir: Level chaar. AI.

Madhav: Kyun?

Kabir: Kyunki wahi sabse upar hai.

Madhav: Sabse upar hone se kya milta hai?

Kabir: Zyada... paisa?

Madhav: Pichhle chapter mein hi humne dekha ki level se kuch nahi milta. Dobara socho. Formula ke teen dials lo, ek-ek karke. Pehla, problem ka size. Woh kis level pe tay hota hai?

Kabir: Level ek pe. Insaan ki zaroorat pe.

Madhav: Aur woh list badalti hai?

Kabir: Nahi. Paanch hazaar saal se wahi hai.

Madhav: Toh us dial ko samajhne ke liye tumhe kya seekhna padega?

Kabir: Kuch bhi nahi. Main khud insaan hoon, mujhe pata hai logon ko kya chahiye.

Madhav: Bas. Pehla dial khula hua hai. Ab doosra, scarcity. Woh kis cheez pe khada hai?

Kabir: Iss baat pe ki kitne log woh kaam kar sakte hain.

Madhav: Aur kaunsa level aisa hai jise sabse kam log samajhte hain?

Kabir: Level teen. Jo chhupa hua hai.

Madhav: Toh scarcity ka dial apne aap kahan sabse upar hoga?

Kabir: Wahin, jahan chhupa hua hai.

Madhav: Ab teesra. Leverage. Anil ki naali. Aaj ke zamane mein woh “naali” kya hai?

Kabir: Software. Ya koi system. Koi machine.

Madhav: Woh kis level pe baithte hain?

Kabir: Teen pe. Phir se teen pe.

Madhav: Toh teeno dials ek saath kahan ishaara kar rahe hain?

Kabir: Ek dial khula hai, aur baaki do ek hi chhupi hui parat ke peeche hain.

Madhav: Bas. Yeh poori kitaab ka tark hai, ek line mein. Ab ek chetavni, kyunki tum abhi ek galat nateeja nikalne wale ho. Kya main yeh keh raha hoon ki level teen pe jaane se tum ameer ho jaoge?

Kabir: Nahi... kyunki har level pe neeche wale log bhi hain.

Madhav: Sahi. Main sirf itna keh raha hoon ki jo do dials tumhare haath mein hain, dono ek hi darwaze ke peeche hain. Darwaza kholna zaroori hai. Kaafi nahi hai. Ab aakhri sawal, aur yeh sabse practical hai. Us parat ko samajhne ke do tareeke hain. Batao.

Kabir: Ek toh yeh ki jo cheez saamne aaye woh padh lo. Aaj database, kal cloud.

Madhav: Tum abhi tak yehi kar rahe the.

Kabir: Haan.

Madhav: Aur nateeja?

Kabir: Har hafte kuch naya sunta hoon jo baaki sabko pehle se pata tha.

Madhav: Kyun? Tum sust toh nahi ho.

Kabir: Kyunki har cheez alag se aati hai. Uske liye mere paas koi jagah nahi hoti.

Madhav: Aur jab jagah na ho toh har nayi cheez kya lagti hai?

Kabir: Khabar. Aur khabar hamesha der se aati hai.

Madhav: Toh doosra tareeka kya hoga?

Kabir: Neeche se poora dhaancha banana. Pehle jagah banao, phir cheezein apne aap baith jaayengi.

Madhav: Woh dheema lagta hai.

Kabir: Shuru mein.

Madhav: Aur baad mein?

Kabir: Baad mein... har agli cheez sasti ho jaati hai. Kyunki mujhe yaad nahi karni padti, sirf jagah dhoondhni padti hai.

Madhav: Toh hum kahan se shuru karenge?

Kabir: Sabse neeche se. Jo meri zindagi se sabse door cheez lagti hai.

Madhav: Ek switch se. Chaar hafte lagenge. Uske baad AI tumhe ek alag duniya nahi, ek aur chapter lagega.

NAAM

Iss kitaab ka baaki hissa woh doosra tareeka hai, poora ka poora.

Aur order ulta lagega, toh yeh saaf keh deta hoon. Hum sabse neeche se shuru karenge, ek switch se, jo aapki rozmarra ki zindagi se sabse door cheez lagti hai.

Woh isliye ki har parat sirf tabhi samajh aati hai jab uske neeche wali parat pehle se khadi ho. Beech se shuru karoge toh udhaar lena padega, aur udhaar liya hua gyaan hi woh gyaan hai jo tikta nahi.

Chaar hafte lagenge. Uske baad Book 2 mein jo AI hai, woh khud ba khud ek chapter jaisa lagega, na ki ek alag duniya.

ASLI DUNIYA SE EK EXAMPLE

Jab koi nayi programming language ya naya framework aata hai, ek cheez baar baar dikhti hai.

Naya seekhne wala usmein mahine lagata hai. Purana kaam karne wala usmein ek hafta lagata hai.

Wajah yeh nahi hai ki woh tez hai. Wajah yeh hai ki uske paas dhaancha pehle se hai. Jab woh nayi cheez padhta hai, toh use naye ideas nahi mil rahe. Use yeh mil raha hai ki jo ideas woh pehle se jaanta hai, unhe iss baar kis naam se bulaya gaya hai aur kaunse do-teen faisle alag liye gaye hain.

Woh yaad nahi kar raha. Woh sirf farak note kar raha hai.

Yehi cheez har layer pe kaam karti hai. Ek baar aap samajh gaye ki data store kaise hota hai, toh har naya database ek naya subject nahi rehta. Woh ek jaana-pehchana subject ban jaata hai jismein kuch alag chunav kiye gaye hain.

Yeh iss kitaab ka poora vaada hai. Aap 112 cheezein yaad nahi karoge. Aap ek dhaancha banaoge, aur uske baad nayi cheezein yaad karne ki zaroorat hi nahi padegi.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki neech se shuru karna dheema raasta hai.**

Aisa lagta hai ki seedha kaam ki cheez pe jaana tez hoga. AI seekhna hai toh AI padho, switch se kya matlab.

Yeh pehle do hafte tak sach lagta hai, aur uske baad ulta ho jaata hai. Kyunki jo aadmi tukdon mein seekh raha hai, use har nayi cheez zero se seekhni padti hai. Jo aadmi dhaancha bana raha hai, uski har agli cheez pichhli se sasti hoti jaati hai.

Ek saal baad tukde wale ke paas sau alag cheezein hoti hain, jinmein se aadhi bhool chuki hoti hain. Dhaanche wale ke paas ek cheez hoti hai jismein sau jagah hain.

Yeh galti tempting isliye hai ki tukdon mein seekhne mein **shuruaat mein zyada tez lagta hai**. Aapko roz kuch naya pata chalta hai. Dhaancha banane mein shuruaat mein kuch bhi kaam ka nahi milta. Ravi aur Anil, phir se, ab seekhne mein.

EK CHEEZ JO SAATH CHALEGI

Iss kitaab mein hum ek chhote se product ko saath lekar challenge:

Ek chat app, jo aapke saare messages sambhal ke rakhta hai, aur unmein se kuch bhi dhoondh sakta hai.

Bas itna. Aisa lagta hai ki ek aadmi ek hafte mein bana le.

Har part ke aakhir mein hum us app pe wapas aayenge. Aur har baar pata chalega ki uske andar ek aur parat chhupi thi jo pichhli baar dikhi hi nahi thi.

Kitaab ke aakhir tak aap us app ko poora khol chuke honge, upar se neeche tak, aur us waqt aapko yeh bhi pata hoga ki use banane mein kitna waqt lagta.

MAP PE

Part 0 khatam. Naksha abhi aisa dikhta hai:

4	AI	[Book 2]	
3	TECHNOLOGY		
	■■■■■ sab chhupa hua ■■■■■		← yeh Book 1 hai
2	ORGANISATIONS	✓ samajh gaye	
1	HUMAN NEEDS	✓ samajh gaye	

Agle chapter se hum us kaale hisse mein ghusenge, sabse neeche wale kone se.

PART 0 KA GATE

Aage badhne se pehle in teen sawalon ka jawab do. Agar nahi aata, toh bataye gaye chapter pe wapas jao.

1. Ek aadmi hafte mein saath ghante kaam karta hai aur bahut kam kamata hai. Uski kamai badhane ke teen alag raaste batao, aur har raaste ke liye batao ki kaunsa dial ghum raha hai. *(nahi aaya toh 0.2)*

Jawab: Teen raaste, teen dials.

Badi problem pe jaana. Wahi hunar, lekin aisi jagah lagana jahan na hone ka nuksaan bada ho. Ek hi kaam karne wala aadmi kisi aur jagah kai guna kama sakta hai, sirf isliye ki wahan uska rukna zyada mehnga padta hai. Dial: size.

Aisa kuch seekhna jo kam log kar sakte hain. Dheema, saalon ka kaam. Dial: scarcity.

Aisi cheez banana jo uske bina chale. Ek auzaar, ek system, ek likhi hui cheez, ek business jismein doosre log kaam karein. Dial: leverage.

Ek chauthi cheez bhi log aksar kehte hain: aur zyada ghante. Woh dial hai hi nahi, aur uski ek deewar hai. Chapter 0.2 ka pehla sawal wahi tha.

2. Level 3 ki koi cheez uthao, aur teen level 2 wale kaam batao jinhe woh guna karti hai. Phir batao ki woh level 1 ki kaunsi zaroorat tak pahunchti hai. *(nahi aaya toh 0.3)*

Jawab: Ek udaharan: GPS.

Woh delivery ko guna karta hai (ek banda 4 ki jagah 25 order kar leta hai), taxi ko guna karta hai (bina raasta poochhe kahin bhi), aur samaan bhejne wali company ko guna karta hai (poora truck kahan hai, hamesha pata).

Level 1 tak: khaana, judav, aur samaan ka pahunchna.

Sahi jawab kai hain. Check yeh hai ki aapne teeno kaamon ke liye wahi ek auzaar rakha, aur teeno alag business the. Jo cheez sirf ek hi business ko guna karti hai, woh aksar level 3 nahi, us business ka ek hissa hoti hai.

3. Ek achhi technology aur ek buri technology mein, aapko kaunsi zyada samajh aati hai, aur aisa kyun hai? *(nahi aaya toh 0.4)*

Jawab: Buri wali.

Buri technology aapko andar ka hissa dikhati hai kyunki woh use chhupa nahi paati. Woh atakti hai, error deti hai, aur aapko andar jaakar theek karna padta hai. Us process mein aapko pata chal jaata hai ki andar kya hai.

Achhi technology kuch nahi dikhati. Woh bas chal jaati hai.

Isiliye purane zamane ke log apni cheezein zyada samajhte the. Woh hoshiyaar nahi the. Unki cheezein zyada kharaab thin.

Part 0 aur paise ka rishta: yeh poora part hi paise ke baare mein tha. Aage ke parts machine ke baare mein hain. Woh dhaaga Part 7 ke aakhir mein wapas judega, jab hum poochenge ki iss sab ko chalane ka kharcha kaun uthata hai.

Chapter 1.1 [SPINE]

Ek switch dabane se kya hota hai

SAMVAAD

Madhav: Deewar pe jo switch hai, use dabao.

Kabir: Dabaya. Light jal gayi.

Madhav: Ab batao, tumne kiya kya?

Kabir: Switch dabaya.

Madhav: Switch ne kya kiya?

Kabir: Light chalu ki.

Madhav: Nahi. Switch ne light ko chhua bhi nahi. Woh deewar mein hai, light chhat pe hai. Switch ne kya kiya?

Kabir: Bijli jaane di.

Madhav: Kaise?

Kabir: Taar jud gaya. Pehle raasta toota hua tha, ab jud gaya.

Madhav: Bas. Ek switch sirf ek kaam karta hai: raasta jodta hai ya todta hai. Aur uski kitni haalatein hain?

Kabir: Do. Juda hua, ya toota hua.

Madhav: Sirf do? Aadha juda hua nahi ho sakta?

Kabir: Nahi. Ya toh bijli jaa rahi hai ya nahi jaa rahi.

Madhav: Ise likhne ka ek tareeka batao. Sabse chhota tareeka.

Kabir: Haan ya na. Chalu ya band.

Madhav: Aur agar sirf do nishaan istemaal karne hon?

Kabir: 1 aur 0.

Madhav: Theek hai. Ab ek sawal poochta hoon jo ajeeb lagega. Tumhare ghar mein kitne switch hain?

Kabir: Pandrah, bees.

Madhav: Aur tumhare phone ke andar?

Kabir: Phone mein switch kahan hote hain?

Madhav: Poocha maine. Andaza lagao.

Kabir: Kuch nahi honge. Woh toh chip hai.

Madhav: Ek aam phone ki chip mein aaj kuch **arab** switch hote hain. Itne chhote ki ek baal ki mothai mein hazaaron aa jaayein. Lekin kaam wahi karte hain jo deewar wala karta hai. Raasta jodo ya todo.

Kabir: Arab switch se kya hoga? Woh toh sirf light jalate hain.

Madhav: Yehi iss poori kitaab ka sawal hai. Ek switch se kuch nahi hota. Do switch se thoda kuch hota hai. Arab switch se woh sab hota hai jo tumne aaj subah se ab tak kiya hai. Har photo, har message, har video, har jawab jo tumhe AI ne diya.

Kabir: Yeh maanna mushkil hai.

Madhav: Maanna mat. Hum ise banayenge. Agle kuch chapter mein tum khud dekh loge ki ek switch se do tak, do se ginti tak, ginti se faisle tak, aur faisle se har cheez tak kaise pahunchte hain. Us din tumhe maanne ki zaroorat nahi padegi.

Kabir: Aur beech mein kuch aur bhi hai? Ya bas switch hi switch?

Madhav: Bas switch. Poori kitaab mein hum ek bhi nayi cheez neeche nahi daalenge. Jo bhi aayega, wahi arab switch alag tareeke se jude honge.

NAAM

Woh chhota switch jo chip ke andar hota hai, use **transistor** kehte hain.

Woh deewar wale switch se sirf teen tarah se alag hai:

DEEWAR KA SWITCH		TRANSISTOR
haath se dabana padta hai	→	bijli se hi dabta hai
ek second mein ek baar	→	ek second mein arbon baar
ek ungli jitna bada	→	itna chhota ki dikhta hi nahi

Teesri baat sabse zaroori hai, aur woh yeh hai: **ek transistor ko doosra transistor daba sakta hai.**

Iska matlab hai ki switch ek doosre ko control kar sakte hain, bina kisi insaan ke. Aur wahi cheez sab kuch mumkin banati hai. Aage do chapter isi ek baat pe hain.

ASLI DUNIYA SE EK EXAMPLE

Pehla transistor 1947 mein bana, Bell Labs mein. Woh haath mein pakadne laayak tha.

1971 mein Intel ne ek chip banayi jismein 2,300 transistor the.

Aaj ek aam phone ki chip mein lagbhag 15 se 20 arab transistor hote hain.

Ginti karo: 2,300 se 20,00,00,00,000 tak, pachas saal mein. Aur unmein se har ek wahi ek kaam karta hai jo deewar wala switch karta hai.

Poori computer ki duniya mein jo kuch bhi hua hai, woh iss ek cheez ke chhota aur sasta hone se hua hai. Koi naya idea nahi aaya. Wahi switch, bas zyada.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki computer ke andar kuch aur bhi hai.**

Log maante hain ki kahin koi jaadu hai. Koi aisi cheez jo sirf engineers ko pata hai. Kuch aisa jo samajh se bahar hai.

Andar kuch nahi hai. Sirf switch hain, aur woh ek doosre ko dabate hain.

Yeh galti tempting isliye hai kyunki jo cheez computer karta hai woh switchon se bahut door lagti hai. Ek gaana, ek chehra, ek jawab. In cheezon aur “chalu ya band” ke beech ki doori itni badi hai ki dimaag maanne se mana kar deta hai.

Woh doori asli hai. Lekin usmein koi jaadu nahi hai, sirf parat pe parat hai. Aur hum har parat ko ek-ek karke rakhenge.

EK CHEEZ JO POORI KITAAB MEIN CHALEGI

Iss kitaab mein hum ek chhote product ko saath lekar challenge:

Ek chat app. Aapke saare messages sambhal ke rakhta hai, aur unmein se kuch bhi dhoondh sakta hai.

Bas itna. Sunne mein lagta hai ek aadmi ek hafte mein bana le.

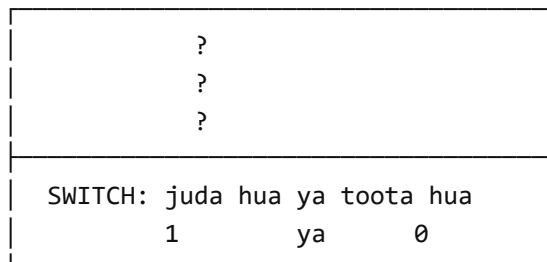
Har part ke aakhir mein hum iss app pe wapas aayenge, aur har baar uske andar ek nayi parat khulegi. Abhi ke liye sirf ek sawal apne dimaag mein rakh lo:

Jab aap ek message type karte ho, woh message jaata kahan hai?

Kitaab ke aakhir tak aapko iska poora jawab pata hoga, upar se neeche tak, aur yeh bhi ki aisa app banane mein kitna waqt lagta.

MAP PE

Naksha yahan se shuru hota hai.



sab kuch abhi khaali hai

← sirf yeh rakha hai

Iske neeche kuch nahi jayega.
Poori kitaab isi ke upar banegi.

SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Ek switch ki do haalatein hain. Do switch ko saath rakho. Ab kitni alag haalatein ban sakti hain? Teen switch se?

Jawab: Do switch se chaar. Teen se aath.

Ginti aise karo: pehla switch do haal mein ho sakta hai, aur uski har haalat mein doosra bhi do haal mein ho sakta hai. Toh $2 \times 2 = 4$. Teesra aaya toh phir se dugna, 8.

Yeh chhota sa hisaab poori kitaab ka sabse taakatwar hisaab hai. Har naye switch pe **dugna**. Das switch se 1,024. Bees se dus lakh se zyada.

Isiliye arab switch itni badi cheez hai. Aur isiliye ek transistor ka chhota hona itna mayne rakhta hai.

2. (samajh check) Madhav ne kaha ki ek transistor ko doosra transistor daba sakta hai, aur yeh sabse zaroori baat hai. Deewar wale switch ke saath aisa nahi ho sakta. Kyun mayne rakhta hai ki ek switch doosre ko dabaye?

Jawab: Kyunki tab insaan ki zaroorat khatam ho jaati hai.

Deewar wale switch ko har baar koi ungli chahiye. Yaani woh utni hi baar chal sakta hai jitni baar koi aadmi use dabaye.

Jab ek switch doosre ko dabata hai, toh ek switch dabane se ek poori shrinkhla chal padti hai, aur uske liye kisi ki zaroorat nahi. Aap ek baar tay kar dete ho ki kaun kise dabayega, aur uske baad machine khud chalti hai.

Wahi “khud chalna” hi poori technology hai. Chapter 0.2 ki bhasha mein: yahi Anil ki naali hai. Ek baar bana do, phir woh aapke bina chalti rahe.

3. (jodne wala) Chapter 0.4 kehta tha ki har technology apne neeche wali cheez ko chhupa deti hai. Aapne pandrah saal phone istemaal kiya aur aaj tak transistor ke baare mein nahi socha. Kya yeh phone ki kami hai ya uski kaamyaabi?

Jawab: Uski kaamyaabi. Poori tarah se.

Agar phone istemaal karne ke liye aapko transistor samajhna padta, toh woh ek bekaar phone hota.

Lekin ab woh doosri baat bhi yaad karo jo Kabir ne 0.4 mein khud nikaali thi: chhupi hui cheez ko aap istemaal kar sakte ho, uske upar kuch bana nahi sakte.

Isliye phone theek hai. Sirf aapka kaam ab yeh hai ki uss deewar mein ek darwaza banao.

Chapter 1.2 [SPINE]

Yeh maan kar chal raha hoon

SAMVAAD

Kabir: Ek baat poochun? Iske liye mujhe maths aani chahiye?

Madhav: Kitni aati hai?

Kabir: School wali. Uske baad chhoot gayi.

Madhav: Do aur do?

Kabir: Chaar.

Madhav: Aadha kitna hota hai?

Kabir: Sau ka pachas.

Madhav: Bas. Poori kitaab ke liye itni kaafi hai.

Kabir: Sach mein? Computer toh poora maths hi hai.

Madhav: Computer ke andar maths hai. Use samajhne ke liye maths nahi chahiye. Yeh do alag baatein hain, aur log inhe hamesha mila dete hain.

Kabir: Farak samjhao.

Madhav: Tumhare pet ke andar kya ho raha hai abhi?

Kabir: Khaana pach raha hai.

Madhav: Usmein kaunse kemikal lag rahe hain, bata sakte ho?

Kabir: Nahi.

Madhav: Phir bhi tum yeh samajhte ho ki khaana kaise pachta hai. Khaya, tuta, sharir ne le liya, bacha hua nikal gaya. Yeh samajh galat hai?

Kabir: Nahi. Bas gehri nahi hai.

Madhav: Woh galat nahi hai. Woh **ek parat ki** samajh hai. Usse aap khaane ke faisle le sakte ho, doctor se baat kar sakte ho, aur yeh samajh sakte ho ki bimari kyun hui. Kemikal ki zaroorat sirf tab padti hai jab tum nayi dawai banao.

Kabir: Toh yeh kitaab mujhe kis parat pe le jaayegi?

Madhav: Utni gehri jitni faisle lene ke liye chahiye. Tum yeh samajh jaoge ki har cheez kaam kaise karti hai aur kyun waisi hai. Tum yeh nahi seekhoge ki use banayein kaise.

Kabir: Toh main code likhna nahi seekhunga.

Madhav: Nahi. Aur yeh main pehle din bata raha hoon, taaki teesre hafte mein tumhe dhokha na lage.

Kabir: Toh phir faayda kya hai?

Madhav: Do aadmi socho. Ek ko code likhna aata hai lekin naksha nahi pata. Doosre ko naksha pata hai lekin code nahi aata. Kaunsa aadmi yeh tay kar sakta hai ki kya banaya jaaye?

Kabir: Doosra.

Madhav: Aur kaunsa ek hafte mein code seekh sakta hai, kyunki uske paas har cheez ke liye jagah pehle se hai?

Kabir: Doosra hi.

Madhav: Ulta karna mushkil hai. Code se naksha nahi banta. Log saalon code likhte hain aur naksha phir bhi nahi hota.

Kabir: Aur mujhe kya kya pehle se aana chahiye?

Madhav: Teen cheezein. Padhna aata ho. Ginti aati ho. Aur tumne kabhi phone ya computer istemaal kiya ho.

Kabir: Bas?

Madhav: Bas. Aur ek cheez jo maine maan li hai aur woh hunar nahi hai: har chapter ke baad tum ek minute ruk kar sochoge. Agar woh nahi karoge toh yeh kitaab kaam nahi karegi, chahe tumhe kitni bhi maths aati ho.

NAAM

Yeh chapter iss kitaab ka **contract** hai. Dono taraf se.

Jo maan kar chal raha hoon:

aap padh sakte ho
aap gin sakte ho, aur aadha-dugna samajhte ho
aapne kabhi phone ya computer istemaal kiya hai
aap har chapter ke baad ek minute rukoge

Jo nahi maan raha:

aapko maths aati hai
aapne pehle kabhi coding ki hai
aap kisi technical shabd ka matlab jaante ho
aapko yaad hai school mein kya padha tha

Jo yeh kitaab degi:

har parat kaam kaise karti hai, aur kyun waisi hai
kisi bhi nayi cheez ko naksh e pe rakhne ki taakat
ek product ko kaagaz pe design karne ki samajh
AI se woh nikaalna jo expert nikaalte hain

Jo yeh kitaab nahi degi:

code likhne ki kabiliyat
kisi language mein maharat
haath se kaam karne ka tajurba
dikhane laayak koi banaya hua saaman

Samajh pehle aati hai aur use chhoda nahi ja sakta. Kabiliyat alag cheez hai, woh karne se aati hai, aur woh iske baad aati hai.

ASLI DUNIYA SE EK EXAMPLE

Ek building ka architect aur ek mistri.

Architect ne shayad zindagi mein ek deewar nahi banayi. Woh cement nahi ghol sakta, sariya nahi bandh sakta. Us hisaab se woh mistri se kam kaabil hai.

Lekin architect yeh tay karta hai ki building kaisi hogi, kitni manzil sahi hai, kahan wazan padega, kahan roshni aayegi, aur kya cheez giregi.

Mistri ko yeh sab nahi aata, aur usne poori zindagi deewarein banayi hain.

Dono kaam ke hain. Dono ko paisa milta hai. Lekin building ki shakal architect tay karta hai.

Yeh kitaab aapko architect banati hai. Mistri banna alag padhai hai, aur bahut aasan ho jaati hai jab naksha pehle se dimaag mein ho.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki samajhne se pehle karna zaroori hai.**

”Pehle coding seekh lo, phir sab samajh aa jaayega.” Yeh salaah har jagah milti hai.

Woh kabhi kabhi kaam karti hai, aur bahut mehngi hai. Kyunki bina naksha ke code seekhne ka matlab hai hazaaron chhote tukde yaad karna bina yeh jaane ki woh kis cheez ka hissa hain. Log saal-do-saal lagate hain aur unhe abhi bhi nahi pata ki jo cheez woh likh rahe hain woh kahan chal rahi hai.

Yeh galti tempting isliye hai ki karne se **turant kuch dikhta hai**. Pehle din kuch chal jaata hai screen pe. Samajhne mein pehle hafte kuch nahi dikhta.

Aur ek aur wajah: karne ki salaah dene wale zyadatar log woh hain jinke paas naksha **already hai**. Unhe yaad nahi ki woh unhe kab aur kaise mila, isliye woh maan lete hain ki woh karne se hi aa gaya tha.

MAP PE

KABILİYAT	code likhna, banana	← alag padhai
SAMAJH	naksha, kya kahan hai	← yeh kitaab
SWITCH	1 ya 0	← kal rakha tha

Samajh ke bina kabiliyat = tukde
Kabiliyat ke bina samajh = design
Dono = builder

SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Madhav ne kaha ki khaana pachne ki “ek parat ki samajh” galat nahi hai, bas gehri nahi hai. Aisi teen cheezein batao jo aap roz istemaal karte ho aur jinki aapko ek parat ki samajh hai. Kya kabhi us kami se aapka kaam ruka?

Jawab: Aam jawab: gaadi, bijli, bank, dawai, internet.

Aur zyadatar logon ke liye kaam kabhi nahi rukta, kyunki woh un cheezon ko sirf **istemaal** karte hain.

Kaam tab rukta hai jab aap kuch chunna chahte ho ya banana chahte ho. Do dawaiyon mein se behtar chunni ho. Do phone mein se behtar chunna ho. Kisi ko batana ho ki kya banaya jaaye.

Isse ek saaf rule nikalta hai: **jab tak aap sirf istemaal kar rahe ho, ek parat kaafi hai. Jis din chunna ya banana ho, us din nahi.**

2. (samajh check) Kitaab kehti hai ki woh coding nahi sikhayegi. Kabir ne poocha “toh faayda kya hai.” Ab aap uska jawab do, apne shabdon mein, ek line mein.

Jawab: Kai theek jawab hain. Sabse mazboot yeh hai:

Code likhna aaj sabse sasti cheez hai. Yeh jaanna ki kya likhna chahiye, sabse mehngi.

Aur Chapter 0.2 ki bhasha mein: code likhna ek hunar hai jo lakhon logon ke paas hai, toh uska scarcity dial neeche hai. Yeh tay karna ki kya banaya jaaye, woh bahut kam log kar paate hain.

Ek aur seedha jawab: is kitaab ke baad koi bhi nayi technology aayegi toh aapko uske liye jagah dhoondhni nahi padegi. Woh khabar nahi lagegi.

3. (jodne wala) Chapter 0.5 mein Kabir ne kaha tha ki tukdon mein seekhne se har cheez “khabar” lagti hai. Iss chapter ka architect wala example uss baat se kaise judta hai?

Jawab: Architect ke paas ek dhaancha hai: wazan, roshni, jagah, kharcha. Jab woh koi nayi building dekhta hai, toh use nayi cheez nahi dikhti. Use dikhta hai ki wahi jaani-pehchani cheezein iss baar kaise jodi gayi hain.

Mistri jo sirf deewarein banata aaya hai, use har nayi design ek nayi cheez lagti hai.

Wahi farak Kabir ka hai. Jab dhaancha nahi hota toh har cheez alag se aati hai aur alag se bhoolni padti hai. Jab dhaancha hota hai toh nayi cheez sirf ek farak hoti hai, poori cheez nahi.

Chapter 1.3 [SPINE]

Computer actually hai kya

SAMVAAD

Madhav: Ek switch ki do haalatein hain. Ab do switch ek taar mein ek ke baad ek lagao. Bulb jalega kab?

Kabir: Jab dono chalu hon. Ek bhi band hua toh raasta toota.

Madhav: Aur agar dono switch alag-alag raaston pe hon, aur dono raaste bulb tak jaate hon?

Kabir: Toh ek bhi chalu ho toh bulb jal jaayega.

Madhav: Ruko. Tumne abhi jo bola, use dhyaan se suno. Pehle case mein tumne kya kaha?

Kabir: Dono chalu hon tabhi.

Madhav: Aur doosre mein?

Kabir: Ek bhi chalu ho toh kaafi.

Madhav: Tumne abhi do faisle banaye. Sirf taar jodne ke tareeke se. “Dono” aur “koi ek.” Yeh sirf bijli nahi hai. Yeh ek shart hai.

Kabir: Theek hai, lekin yeh toh bahut chhoti baat hai.

Madhav: Ek teesra tareeka bhi hai. Aisa switch socho jo ulta chalta hai. Bijli aaye toh raasta tode, na aaye toh jode.

Kabir: Woh kis kaam ka?

Madhav: Woh “nahi” hai. Ab tumhare paas teen cheezein hain: **dono**, **koi ek**, aur **nahi**. Ab main tumse ek badi baat kahunga aur tum use abhi maanoge nahi. In teen cheezon se, kaafi saari lagakar, tum har woh sawal hal kar sakte ho jiska jawab haan ya na mein

ho.

Kabir: Har sawal? Jaise?

Madhav: Jaise: yeh do numbers barabar hain? Yeh number us number se bada hai? Yeh tareekh guzar chuki hai? Yeh password sahi hai?

Kabir: Yeh sab sirf “dono, koi ek, nahi” se?

Madhav: Sirf isse. Ek udaharan lo. Do switch, A aur B. Main jaanna chahta hoon ki dono alag hain ya nahi. Batao kaise pata karoge.

Kabir: Hmm. Agar A chalu hai aur B band, toh alag hain. Ya A band aur B chalu, tab bhi alag hain.

Madhav: Toh usko teen cheezon mein likho.

Kabir: “A aur B-nahi” wala case. Ya “A-nahi aur B” wala case. Koi ek ho jaaye toh alag hain.

Madhav: Tumne abhi taaron se tulna banayi. Kisi ne tumhe formula nahi diya. Ab yeh socho: agar do cheezein barabar hain ya nahi, yeh switchon se pata chal sakta hai, toh kya cheez nahi chal sakti?

Kabir: Pata nahi. Shayad kuch bhi nahi.

Madhav: Ab teesri baat, aur yeh sabse zaroori hai. Pichhle chapter mein maine kaha tha ki ek transistor doosre ko daba sakta hai. Iska matlab kya hua?

Kabir: Ki ek shart ka jawab doosri shart mein daala ja sakta hai.

Madhav: Aur uska jawab teesri mein. Aur aage. Kitni door tak?

Kabir: Jitne switch hon.

Madhav: Aur tumhare phone mein kitne hain?

Kabir: Bees arab.

Madhav: Toh tumhare paas bees arab shartein hain, ek doosre mein daali hui, aur woh ek second mein arbon baar chal sakti hain. Ab batao, computer kya hai?

Kabir: Switch ka dhher, jo ek doosre ko dabate hain, aur milkar shartein banate hain.

Madhav: Aur uske alaawa?

Kabir: Kuch nahi?

Madhav: Kuch nahi. Ab poori kitaab mein hum yehi karenge: dekhte jaayenge ki iske upar parat pe parat kaise chadhi.

NAAM

Woh teen cheezein jo Kabir ne banayi, unke naam hain **AND**, **OR** aur **NOT**.

AND	dono chalu hon tabhi	(ek ke baad ek taar)
OR	koi ek chalu ho toh kaafi	(alag alag raaste)
NOT	ultra kar do	(chalu ho toh band)

Inhe **logic gates** kehte hain. Gate isliye ki woh bijli ko rok bhi sakte hain aur jaane bhi de sakte hain, shart ke hisaab se.

Aur jo baat Kabir ne khud nikaali, uska naam bhi hai. Yeh saabit ho chuka hai ki **in teen se** har woh cheez banayi ja sakti hai jo koi bhi computer karta hai. Aur jodna, ghatana, tulna, yaad rakhna, sab.

Toh ek computer ki poori definition itni hai:

Bahut saare switch, aise jude hue ki woh AND, OR aur NOT banayein, aur ek ka jawab doosre mein jaaye.

Iske alaawa computer mein kuch nahi hai. Baaki sab jo aap dekhte ho, wo iske upar chadhi hui parat hai.

ASLI DUNIYA SE EK EXAMPLE

1937 mein ek 21 saal ka student, Claude Shannon, ne apni master's thesis likhi. Usne dikhaya ki bijli ke switchon ke jaal aur tark ke niyam bilkul ek jaisi cheez hain.

Us waqt switch phone ke exchange mein lagte the, call jodne ke liye. Kisi ne unhe sochne wali cheez nahi maana tha.

Shannon ne dikhaya ki agar aap switch ko tark ki tarah likh sakte ho, toh aap tark ko switch se bana bhi sakte ho.

Woh thesis lagbhag har digital cheez ki neev hai jo uske baad bani. Aur usmein koi nayi machine nahi thi, sirf yeh dekhna tha ki do purani cheezein ek hi cheez hain.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki computer “sochta” hai.**

Log kehte hain computer ne decide kiya, computer ko laga, computer samajh gaya. Yeh sirf bolne ka tareeka nahi hai, log sach mein aisa mehsoos karte hain.

Computer koi faisla nahi leta. Usmein bees arab shartein pehle se lagi hui hain, aur bijli un shartों mein se guzar jaati hai. Jo bahar aata hai, woh jawab hai.

Ek chaabi wale taale se socho. Sahi chaabi ghumti hai, galat nahi ghumti. Taale ne kuch soch kar faisla nahi kiya. Uski shakal aisi hai. Computer ek taala hai jismein bees arab pins hain aur unhe har second badla ja sakta hai.

Yeh galti tempting isliye hai ki bahar se dono ek jaise dikhte hain. Aap sawal poochte ho, jawab aata hai. Andar kya hua, woh dikhta nahi.

Yeh baat Book 2 mein bahut kaam aayegi, jab hum AI kholenge. Wahan yeh galti sabse zyada hoti hai.

MAP PE

SHARTEIN	AND, OR, NOT ek ka jawab doosre mein	← naya
SWITCH	1 ya 0	

SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Ek ghar mein aisa taala lagana hai jo tabhi khule jab do log ek saath apni chaabi lagayein. Kaunsa gate hai yeh? Ab aisa taala jo kisi ek chaabi se bhi khul jaaye?

Jawab: Pehla AND hai. Doosra OR.

Bank ki tijoriyon mein sach mein AND lagta hai: do alag logon ki chaabi, ek saath. Ghar ke darwaze pe OR hota hai: aapki chaabi ya ghar ke kisi aur ki, dono chalti hain.

Dhyaan do ki yeh koi computer ki cheez nahi hai. Yeh sirf shart hai. Computer ne shart invent nahi ki, usne use bijli mein daal diya.

2. (samajh check) Madhav ne kaha ki AND, OR aur NOT se har haan-na wala sawal hal ho sakta hai. Ek aisa sawal sochne ki koshish karo jo iss tareeke se hal na ho.

Jawab: Agar sawal ka jawab pakka haan ya na hai, aur jo cheezein us jawab ko tay karti hain woh sab pata hain, toh woh hal ho jaayega. Iska koi apwaad nahi hai. Yeh saabit ho chuka hai.

Jo sawal hal nahi honge, woh do tarah ke hain:

Jinka jawab haan-na mein hai hi nahi. “Yeh gaana achha hai?” Yahan pehle yeh tay karna padega ki achhe ka matlab kya hai. Woh tay ho jaaye toh sawal hal ho jaayega, lekin jawab aapki di hui paribhasha ka hoga, sachai ka nahi.

Jinka jawab tay karne wali cheezein pata hi nahi hain. “Kal baarish hogi?” Yeh switch ki kami nahi hai, yeh jaankari ki kami hai.

Yeh farak yaad rakhna. Book 2 mein AI ki lagbhag saari seemayein inhi do mein se ek hain.

3. (jodne wala) Chapter 1.1 mein aapne dekha ki har naye switch pe haalatein dugni ho jaati hain. Ab jodo: agar bees arab switch hain, toh kya sach mein har mumkin haalat kabhi istemaal hoti hai?

Jawab: Nahi, bilkul nahi. Aur farak kalpana se bahar hai.

Bees arab switch se jitni haalatein banti hain, woh brahmand ke saare paramanuon se bhi behisaab zyada hain. Unmein se ek bahut hi chhota hissa hi kabhi kaam mein aata hai.

Toh phir itne switch kyun? Kyunki aapko pehle se nahi pata ki kaunsi haalat chahiye hogi. Machine ko har mumkin cheez ke liye taiyaar rehna padta hai, isliye nahi ki sab hongi, balki isliye ki koi bhi ho sakti hai.

Yeh soch aage kai jagah dikhegi: **jagah isliye nahi banayi jaati ki woh bhar jaayegi, balki isliye ki pehle se pata nahi ki kya aayega.**

Chapter 1.4 [SPINE]

Switches numbers kaise rakhte hain

SAMVAAD

Madhav: Ek switch se kitne numbers dikha sakte ho?

Kabir: Do. 0 aur 1.

Madhav: Do switch se?

Kabir: Chaar haalatein hain. Toh 0, 1, 2, 3.

Madhav: Likh ke dikhao. Kaunsi haalat kaunsa number hai?

Kabir: Dono band ko 0 kehta hoon. Phir 01 ko 1. Phir 10 ko 2. Phir 11 ko 3.

Madhav: Tumne yeh order kyun chuna?

Kabir: Bas... ginti jaisi lagi.

Madhav: Woh ginti jaisi hai. Ab teen switch se ginti karo, 0 se shuru.

Kabir: 000, 001, 010, 011, 100, 101, 110, 111. Aath haalatein. 0 se 7.

Madhav: Aur tumne ginti kaise ki? Kya niyam laga rahe the?

Kabir: Dayen se badhata gaya. Jab 1 ke aage badhna tha toh use 0 kar ke agle ko 1 kar diya.

Madhav: Bilkul wahi jo tum aam ginti mein karte ho. 9 ke baad 0 likh kar aage 1 badhate ho. Sirf yahan 9 ki jagah 1 pe hi bharav ho jaata hai, kyunki sirf do nishaan hain.

Kabir: Toh yeh koi nayi ginti nahi hai. Bas kam nishaan hain.

Madhav: Bas itna hi. Aam ginti mein das nishaan hain, 0 se 9. Yahan do hain. Aur kuch nahi badla.

Kabir: Lekin fayda kya hua? 7 likhne ke liye teen switch lag gaye. 7 toh ek nishaan hai.

Madhav: Sahi sawal. Ab dhyaan se dekho. Teen switch se 8 haalatein. Chaar se?

Kabir: Solah.

Madhav: Aath switch se?

Kabir: 2 ko aath baar guna... 256.

Madhav: Solah switch se?

Kabir: 65,000 ke aas-paas.

Madhav: Tees switch se?

Kabir: Bahut zyada. Karodon.

Madhav: Ek arab se upar. Ab yeh dekho: switch tees hain, lekin numbers ek arab. Har naya switch **dugna** karta hai. Isiliye tumhara sawal ulta ho gaya. Sawal yeh nahi hai ki 7 ke liye teen switch kyun lage. Sawal yeh hai ki tees switch se ek arab kaise aa gaye.

Kabir: Theek hai. Toh numbers ban gaye. Lekin mujhe toh naam bhi likhne hain, gaane bhi.

Madhav: Ek kaam karo. Ek kaagaz pe likho: A ke liye 1, B ke liye 2, C ke liye 3. Aage bhi.

Kabir: Kar liya.

Madhav: Ab "CAB" likho, sirf numbers mein.

Kabir: 3, 1, 2.

Madhav: Aur maine tumhe woh kaagaz diya hota, toh main 3-1-2 padh kar kya samajhta?

Kabir: CAB.

Madhav: Toh akshar kya hain?

Kabir: Numbers. Bas hum unhe alag dikhte hain.

Madhav: Rang?

Kabir: Rang bhi number ho sakta hai. Laal ka ek number, neela ka doosra.

Madhav: Ek photo?

Kabir: Photo... bahut saare chhote dabbe, har dabbe ka ek rang, har rang ek number.

Madhav: Aur awaaz?

Kabir: Awaaz toh lehar hai. Uski unchai bhi number ho sakti hai. Har pal ka ek number.

Madhav: Toh ek switch se 0 aur 1 mile. Kaafi switch se koi bhi number mila. Aur ek tay-shuda list se, koi bhi number kisi bhi cheez ka matlab ban gaya. Ab batao, aisa kya hai jo computer mein nahi daala ja sakta?

Kabir: Aisi cheez jise number mein badalne ka koi tareeka hi na ho.

Madhav: Bilkul sahi. Aur yeh yaad rakhna, kyunki agla pura hissa isi pe khada hai.

NAAM

Ek switch ki haalat ko **bit** kehte hain. Bit ka matlab hai: ek haan-na, 0 ya 1.

Sirf do nishaanon wali ginti ko **binary** kehte hain.

Aur woh tay-shuda list jismein likha ho ki kaunsa number kaunsi cheez ka matlab hai, use **encoding** kehte hain.

Yeh teesri cheez sabse zaroori hai aur sabse kam samjhi jaati hai. Number khud kuch nahi kehta.

number 3	→	agar akshar ki list dekhein	→	C
number 3	→	agar rang ki list dekhein	→	koi rang
number 3	→	agar seedha number ho	→	teen
number 3	→	agar niyam ki list dekhein	→	"jodo"

Ek hi 3, chaar alag matlab. Farak sirf iss baat ka hai ki aap kaunsi list dekh rahe ho.

Computer ke andar sirf numbers hain. Matlab kabhi bhi andar nahi hota. Matlab is baat mein hai ki unhe padha kis tarah jaa raha hai.

Yeh baat aage bahut kaam aayegi. Jab file kharaab ho jaati hai, jab akshar ki jagah ajeeb nishaan aate hain, jab ek program doosre ki file nahi khol paata, tab lagbhag hamesha wajah yehi hoti hai: numbers sahi hain, list galat lagayi gayi.

ASLI DUNIYA SE EK EXAMPLE

1963 mein America mein ek list tay ki gayi jiska naam ASCII tha. Usmein 128 nishaanon ke liye numbers tay kiye gaye: angrezi ke akshar, ginti ke ank, aur kuch chinh.

128 numbers ke liye saat bit kaafi hain. Us zamane mein har bit mehnga tha, isliye chhoti list rakhi gayi.

Problem yeh thi ki usmein duniya ki baaki bhaashaayein thi hi nahi. Hindi nahi, Chinese nahi, Arabic nahi. Har desh ne apni alag list banayi, aur ab ek desh ki file doosre desh mein kholo toh bakwaas dikhta tha. Numbers theek pahunchte the, list galat lagti thi.

Iska hal 1990 ke baad aaya, jab **Unicode** bani: ek hi list, duniya ki lagbhag har likhavat ke liye, aur usmein aaj lagbhag 1,50,000 se zyada nishaan hain.

Aaj aapke phone mein hindi, angrezi aur emoji ek hi message mein saath chalte hain. Woh isliye ki poori duniya ne ek list pe sehmati bana li. Naya hardware nahi laga. Sirf ek list.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki computer ke andar akshar, photo aur gaane rakhe hote hain.**

Log kehte hain “file mein meri photo hai.” Photo file mein nahi hai. File mein numbers hain, aur ek tay-shuda tareeka hai unhe rang mein badalne ka.

Yeh sirf shabdon ka khel nahi hai. Iska seedha nateeja hai. Jab aapki file kharaab ho jaati hai aur ajeeb nishaan dikhte hain, tab log sochte hain ki file “toot gayi.” Aksar file bilkul theek hoti hai. Sirf padhne wali list galat lag rahi hai.

Yeh galti tempting isliye hai ki aap photo ko photo ki tarah hi dekhte ho. Number kabhi dikhte hi nahi. Screen pe woh parat pehle hi hat chuki hoti hai.

Ek jumla yaad rakh lo, aage bees jagah kaam aayega:

Andar sirf numbers hain. Matlab hamesha bahar se aata hai.

MAP PE

ENCODING	number ka matlab kya hai	← naya
BINARY	kaafi switch = koi number	← naya
SHARTEIN	AND, OR, NOT	
SWITCH	1 ya 0	

SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Aath bit se 256 alag haalatein banti hain. Agar aapko duniya ki har bhaasha ke akshar rakhne hain, toh 256 kaafi hai? Agar nahi, toh kya karoge?

Jawab: Bilkul kaafi nahi hai. Akele Chinese mein hazaaron nishaan hain.

Hal seedha hai: **zyada bit lo**. Solah bit se 65,000 se zyada. Ikkis bit se bees lakh se zyada, jo duniya ki har likhvat ke liye kaafi hai.

Aur Unicode ne yehi kiya, ek hoshiyaari ke saath: aam angrezi akshar aaj bhi aath bit lete hain, aur jo nishaan kam aate hain woh zyada bit lete hain. Isse purani ASCII files bhi chalti rahin aur nayi bhaashaayein bhi aa gayin.

Yeh soch aage bar bar dikhegi: **jo cheez zyada aati hai use sasta rakho, jo kam aati hai use mehnga hone do.**

2. (samajh check) Aapke paas ek file hai jismein numbers hain: 72, 101, 108, 108, 111. Aapko nahi pata ki yeh photo hai, gaana hai, ya likhawat. Kya aap sirf numbers dekh kar bata sakte ho?

Jawab: Nahi. Aur yeh iss chapter ki sabse zaroori baat hai.

Woh numbers ASCII list se “Hello” hain. Rang ki list se paanch alag halke rang hain. Awaaz ki list se ek chhoti si lehar.

Numbers khud nahi batate ki woh kya hain.

Isiliye file ke naam ke aage `.jpg` ya `.mp3` lagta hai. Woh file ka hissa nahi hai. Woh ek chit hai jo batati hai ki kaunsi list lagani hai. Chit badal do, aur wahi numbers kuch aur ban jaate hain.

3. (jodne wala) Chapter 0.4 kehta tha ki har parat neeche wali ko chhupa deti hai. Iss chapter mein kaunsi parat kis cheez ko chhupa rahi hai?

Jawab: Encoding, binary ko chhupa rahi hai.

Jab aap “A” likhte ho, toh aap 65 nahi soch rahe. Aur jab machine 65 rakhti hai, toh woh 01000001 nahi soch rahi. Aur jab woh 01000001 rakhti hai, toh woh aath switchon ke baare mein nahi soch rahi.

Teen parat, ek hi cheez pe khadi.

Yeh iss kitaab ki pehli jagah hai jahan aapne parat banti hui dekhi hai, bani hui nahi. Aage har part mein yeh dobara hoga.

Chapter 1.5 [DEPTH]

Switches jodte kaise hain

Yeh DEPTH chapter hai. Ismein aap ek machine ko andar se banta hua dekhoge. Pehli baar mein chhod sakte ho, lekin agar ek baar bhi dekh liya ki shartein kaam kaise karti hain, toh “computer jaadu hai” wali soch hamesha ke liye khatam ho jaati hai.

SAMVAAD

Madhav: Ek switch faisla karta hai. Kaafi switch numbers rakhte hain. Ab dono ko jodo: kya switch **jod** sakte hain?

Kabir: Pata nahi.

Madhav: Chalo banate hain. Sabse chhota jod. Ek bit aur ek bit. Chaar mumkin case hain, batao.

Kabir: 0 aur 0. 0 aur 1. 1 aur 0. 1 aur 1.

Madhav: Jawab kya hain?

Kabir: 0, 1, 1, aur... 2.

Madhav: Ruko. Ek bit mein 2 aa sakta hai?

Kabir: Nahi. Ek bit mein sirf 0 ya 1.

Madhav: Toh kya karoge?

Kabir: Do jagah chahiye. Jaise 9 aur 9 jodte hain toh 18 hota hai, do ank.

Madhav: Bilkul. Toh jawab ke do hisse honge. Ek jo wahin dikhta hai, aur ek jo aage jaata hai. Chaaron case dobara likho, ab do hisson mein.

Kabir:


```
0 + 0 → wahin 0, aage 0
0 + 1 → wahin 1, aage 0
1 + 0 → wahin 1, aage 0
1 + 1 → wahin 0, aage 1
```

Madhav: Ab pehle “aage” wale khaane ko dekho. Woh kab 1 hota hai?

Kabir: Sirf jab dono 1 hon.

Madhav: Aur “dono 1 hon” ka gate?

Kabir: AND!

Madhav: Toh aage jaane wala hissa ek AND gate hai. Bas. Ab “wahin” wale khaane ko dekho. Woh kab 1 hota hai?

Kabir: Jab ek 1 ho aur doosra 0. Yaani jab dono alag hon.

Madhav: Aur “dono alag hain” ko humne pichhle chapter mein banaya tha. Yaad hai?

Kabir: Haan. “A aur B-nahi” ya “A-nahi aur B.” AND, NOT aur OR se.

Madhav: Toh tumne abhi kya bana diya?

Kabir: Jodne wali machine? Sirf gates se?

Madhav: Ek bit ki. Ab badi karte hain. Aath bit ke do numbers jodne hain. Kya karoge?

Kabir: Wahi cheez aath baar lagaunga. Har jagah pe do bit jodunga.

Madhav: Aur jo “aage” wala hissa nikla tha?

Kabir: Woh agli jagah mein daal dunga. Jaise haath se jodte waqt hasiya aage le jaate hain.

Madhav: Toh ab tumhe har jagah pe teen cheezein jodni hain: do numbers ke bit, aur pichhli jagah se aaya hua.

Kabir: Toh do jodne wali machine ek ke baad ek laga dunga. Pehle do jodo, phir uss jawab mein teesra jodo.

Madhav: Bas. Aur ab aath jagah, ek line mein, har ek agli ko apna hasiya de rahi hai. Tumne abhi ek asli chip ka hissa bana diya, sirf AND, OR aur NOT se.

Kabir: Lekin sirf jodna hi toh hai.

Madhav: Ghatana kya hai?

Kabir: Ulta jodna.

Madhav: Guna?

Kabir: Baar baar jodna.

Madhav: Bhaag?

Kabir: Baar baar ghatana.

Madhav: Toh ek jodne wali machine se saara ganit nikal aaya. Aur woh machine sirf switch hai. Ab aakhri sawal: ek chip mein aisi kitni machinein hoti hain?

Kabir: Kai hazaar?

Madhav: Karodon. Aur woh ek second mein arbon baar chalti hain. Isiliye tumhara phone ek video chala pata hai, jo har second mein karodon numbers ka hisaab hai.

NAAM

Jo aapne abhi banaya, uska naam hai **adder**.

Ek bit wale hisse ko **full adder** kehte hain, kyunki woh teen cheezein jodta hai: do bit aur pichhla hasiya.

Aur jo teen hisse mile:

"wahin dikhne wala" hissa	→ XOR gate ("dono alag hain")
"aage jaane wala" hissa	→ AND gate
inhe jodne wala	→ OR gate

Aath ya solah ya chausath aise hisse ek line mein lagao, aur aapke paas ek poora jodne wala circuit hai.

Aur ab ek baat jo pehli baar ajeeb lagti hai. **Chip ko yeh nahi pata ki woh jod rahi hai.**

Usmein bijli aati hai, shartein us bijli ko rokti aur jaane deti hain, aur doosri taraf se kuch nikal aata hai. Jodna sirf iss baat ka naam hai ki humne jo bahar aaya use kaise padha.

Chapter 1.4 ka jumla, wapas:

| Andar sirf numbers hain. Matlab hamesha bahar se aata hai.

ASLI DUNIYA SE EK EXAMPLE

1642 mein Blaise Pascal ne, 19 saal ki umar mein, ek jodne wali machine banayi. Usmein bijli nahi thi, sirf gear the. Har gear pe das dant, aur jab ek gear poora ghoom jaata tha toh woh agle gear ko ek dant aage kar deta tha.

Wahi hasiya. Sirf loha.

Uske teen sau saal baad wahi cheez switchon se bani, aur ek gear ki jagah ek transistor aaya. Idea nahi badla. Sirf hasiya le jaane wali cheez dhaatu se bijli ban gayi, aur uske saath speed lakhon guna ho gayi.

Yeh dekhna kaam ka hai: computer ka koi bhi idea naya nahi hai. Naya sirf yeh hai ki woh kitna chhota aur kitna tez ho gaya.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki chip ke andar ganit ke niyam likhe hain.**

Log sochte hain ki kahin ek jagah “ $2 + 2 = 4$ ” jaisa kuch bhara hua hai, ya koi table hai jise dekh kar machine jawab batati hai.

Aisa kuch nahi hai. Koi jawab kahin rakha hua nahi hai. Bijli sharton mein se guzarti hai aur jawab **ban jaata hai**, har baar naye sire se.

Yeh galti tempting isliye hai ki jab hum jodte hain toh hum yaad se karte hain. Humne bachpan mein table ratay the. Toh lagta hai machine bhi yaad se karti hogi.

Machine yaad nahi karti. Woh har baar naye sire se banati hai, aur usse itni tezi se banati hai ki yaad karna dheema padta.

MAP PE

GANIT	jod, ghatao, guna, bhaag	← naya
ENCODING	number ka matlab kya hai	
BINARY	kaafi switch = koi number	
SHARTEIN	AND, OR, NOT	
SWITCH	1 ya 0	

SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Aath bit se sabse bada number 255 hota hai. Ab 255 aur 1 jodo. Machine kya karegi?

Jawab: Har jagah pe hasiya aage jaayega, aur aakhri hasiya ke paas jaane ki jagah hi nahi bachegi. Jawab 0 aa jaayega.

Machine ise galti nahi maanegi. Uske paas aath jagah hain, usne aath jagah bhar di.

Iska naam **overflow** hai, aur yeh asli duniya mein cheezein toda karta hai. Purane games mein score ek had ke baad zero ho jaata tha. Kuch systems mein tareekh ek din ke baad wapas peeche chali jaati hai.

Seekhne wali baat: **machine ke paas jagah tay hoti hai, aur jagah khatam hone pe woh rukti nahi, woh chup-chaap galat jawab de deti hai.**

2. (samajh check) Adder mein har jagah apna hasiya agli jagah ko deti hai. Toh chausath bit ke numbers jodne mein, aakhri jagah ko kitna intezaar karna padega?

Jawab: Use pichhli teiyasath jagahon ka intezaar karna padega, kyunki hasiya ek-ek karke aage aata hai.

Isiliye zyada bit wala jod thoda dheema hota hai. Har jagah ka intezaar jud jaata hai.

Asli chip mein iske liye alag design lagaye jaate hain jo hasiya pehle se andaz laga lete hain, taaki intezaar kam ho. Woh zyada transistor lete hain aur tez chalte hain.

Yeh iss kitaab ka pehla **trade-off** hai: **zyada jagah kharch karo aur tez ho jao, ya kam jagah lo aur dheere chalo.** Yeh chunav aage har parat pe milega.

3. (jodne wala) Chapter 1.3 mein Kabir ne “dono alag hain” wali shart AND, OR aur NOT se banayi thi. Aaj wahi shart adder ka aadha hissa nikli. Isse aap kya seekhte ho?

Jawab: Ki chhoti shartein bade tukdon ke purze ban jaati hain, aur wahi tukde aur bade tukdon ke.

Aapne kabhi bhi teen se zyada cheezein istemaal nahi kin. AND, OR, NOT. Unse “alag hain” bana. Usse adder bana. Adder se ganit bana.

Yehi poori kitaab ka dhaancha hai. Har parat neeche wali ke tukdon ko jodkar ek naya tukda banati hai, aur uske baad us naye tukde ko koi neeche ki taraf nahi dekhta.

Isiliye jab aage koi badi cheez aaye, toh yeh mat poochna ki woh kaise mumkin hai. Poochna ki **woh kis cheez se bani hai.**

Chapter 1.6 [SPINE]

Ek machine sab kuch kyun kar sakti hai

SAMVAAD

Madhav: Ek sawal jo shayad tumne kabhi socha hi nahi. Tumhare paas ek chaaku hai, ek chammach hai, ek taala hai. Kya ek chaaku taala ban sakta hai?

Kabir: Nahi.

Madhav: Aur tumhara phone? Woh camera hai, ghadi hai, radio hai, kitaab hai, calculator hai, naksha hai, aur chitthi bhejne ki machine bhi.

Kabir: Haan.

Madhav: Toh phone ke andar itni saari alag machinein hain?

Kabir: Nahi. Ek hi chip hai.

Madhav: Toh ek hi machine itni saari cheezein kaise ban jaati hai? Aur chaaku kyun nahi ban paata?

Kabir: Kyunki phone mein alag-alag app hain.

Madhav: App hai kya?

Kabir: Ek program.

Madhav: Aur program hai kya?

Kabir: Nirdeshon ki list. Kya karna hai, kis order mein.

Madhav: Aur woh list rakhi kahan hai?

Kabir: Phone ke andar.

Madhav: Kis shakal mein?

Kabir: Numbers mein. Kyunki andar sirf numbers hi hote hain.

Madhav: Ab ruko. Yeh baat abhi tumhare haath se nikal gayi, isliye dobara bolo. Program kis shakal mein rakha hai?

Kabir: Numbers ki tarah.

Madhav: Aur photo kis shakal mein rakhi hai?

Kabir: Numbers ki tarah.

Madhav: Toh machine ke liye program aur photo mein kya farak hai?

Kabir: Koi... koi farak nahi hai?

Madhav: Koi farak nahi hai. Dono numbers hain. Farak sirf itna hai ki machine ek ko **padhti** hai aur doosre ko **karti** hai.

Kabir: Yeh ajeeb hai.

Madhav: Yeh ajeeb hai aur yehi sab kuch hai. Ab socho. Agar nirdesh bhi sirf numbers hain, aur numbers badle ja sakte hain, toh machine kya kar sakti hai?

Kabir: Alag numbers daal do toh woh alag kaam karegi.

Madhav: Machine badalni padegi?

Kabir: Nahi. Sirf numbers badalne padenge.

Madhav: Toh ab batao, chaaku taala kyun nahi ban sakta?

Kabir: Kyunki chaaku ki shakal hi uska kaam hai. Use badalne ke liye chaaku ko hi badalna padega.

Madhav: Aur computer?

Kabir: Computer ki shakal uska kaam nahi hai. Uska kaam uske andar rakhe numbers hain. Woh badal do, kaam badal gaya.

Madhav: Bas. Isiliye ek hi machine camera bhi hai aur ghadi bhi. Machine kabhi nahi badalti. Sirf list badalti hai.

Kabir: Toh jab main ek naya app install karta hoon...

Madhav: Tum machine mein kuch nahi jodte. Tum sirf numbers ka ek naya set daalte ho.

Kabir: Aur woh number kahan se aate hain?

Madhav: Koi insaan unhe likhta hai. Seedhe numbers mein nahi, kyunki woh kaafi takleef deh hai. Woh lagbhag angrezi jaisi shakal mein likhta hai aur ek doosra program use numbers mein badal deta hai.

Kabir: Ek program doosre program ko banata hai?

Madhav: Haan. Aur woh banane wala program bhi khud numbers hi hai. Yeh Part 3 hai, abhi mat kholo.

Madhav: Aakhri sawal. Agar ek machine har woh kaam kar sakti hai jiske liye nirdesh likhe ja sakein, toh kaunsa kaam woh nahi kar sakti?

Kabir: Jiske liye nirdesh likhe hi na ja sakein.

Madhav: Ise yaad rakhna. Poori kitaab mein aur poori Book 2 mein, har seema isi ek line se nikalti hai.

NAAM

Woh numbers jo nirdesh hain, unhe **program** kehte hain.

Aur woh khoj ki nirdesh bhi data ki tarah hi rakhe ja sakte hain, use **stored program** kehte hain. Yeh 1940 ke aas-paas ki baat hai aur yeh computer ka sabse bada ek idea hai.

Farak samajhne ke liye:

CHAAKU jaisi machine

- |— shakal hi kaam hai
- |— naya kaam chahiye toh nayi machine
- |— ek machine, ek kaam

COMPUTER jaisi machine

- |— shakal khaali hai, kaam andar ke numbers mein hai
- |— naya kaam chahiye toh naye numbers
- |— ek machine, anginat kaam

Iska ek naam aur hai: aisi machine ko **general purpose** kehte hain. Poori duniya mein aisi machinein bahut kam hain. Lagbhag har cheez jo insaan ne banayi, woh ek kaam ki hai. Computer un ginti ki cheezon mein se hai jiska kaam pehle se tay nahi hota.

Aur yahi wajah hai ki Chapter 0.3 mein level 4 alag nikla tha. Kabir ne wahan kaha tha ki AI “auzaar banane ka auzaar” hai. Woh isliye mumkin hua kyunki uske neeche jo machine hai, woh khud pehle se hi kisi bhi kaam ke liye khaali thi.

ASLI DUNIYA SE EK EXAMPLE

Dusre vishwa yudh ke waqt ENIAC naam ki ek badi machine banayi gayi thi, hisaab ke liye. Woh ek kamre jitni thi.

Use naya kaam dene ke liye log usmein **taar badalte the**. Switch haath se ghumate the, plug nikaal kar doosri jagah lagate the. Ek naya program lagane mein kai din lag jaate the, aur woh kaam zyadatar auraton ki ek team karti thi jo us machine ki asli programmer thin.

Kuch hi saal baad wali machinein alag thin. Unmein program bhi wahin rakha jaata tha jahan data rakha jaata tha. Naya kaam dene ke liye taar nahi badalte the, sirf naye numbers bhar dete the.

Woh farak, taar badalne se numbers badalne tak, computer ka asli janm hai. Uske pehle woh ek machine thi. Uske baad woh anginat machinein ban gayi.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **software ko koi alag jaadui cheez samajhna**.

Log “hardware” aur “software” ko do alag duniya maante hain. Ek cheez jo hai, aur ek cheez jo... pata nahi kya hai. Kuch aisa jo hawa mein hai.

Software hawa mein nahi hai. Software woh numbers hain jo machine ke andar rakhe hain, aur woh utne hi asli hain jitni ek photo.

Iss galti ka seedha nateeja hai: log yeh nahi samajh paate ki software ko banane mein itna waqt aur paisa kyun lagta hai, jab “usme kuch hai hi nahi.” Aur woh yeh bhi nahi samajh paate ki software gayab kyun ho sakta hai, kharaab kyun ho sakta hai, ya use jagah kyun chahiye.

Yeh galti tempting isliye hai ki software ko aap chhoo nahi sakte. Lekin bijli ko bhi nahi chhoo sakte, aur woh bilkul asli hai.

Sahi tasveer yeh hai:

Hardware woh machine hai jo nirdesh maanti hai. Software woh nirdesh hain. Dono numbers hain. Ek switchon mein baitha hai, doosra unke beech se guzarta hai.

MAP PE

PROGRAM	nirdesh, bhi numbers hain numbers badlo, kaam badla	← naya
GANIT	jod, ghatao, guna, bhaag	
ENCODING	number ka matlab	
BINARY	kaafi switch = koi number	
SHARTEIN	AND, OR, NOT	
SWITCH	1 ya 0	

Part 1 khatam. Ek switch se ek general machine tak.

PART 1 KA GATE

Aage badhne se pehle in teen sawalon ka jawab do.

1. Ek dost kehta hai “computer bahut hoshiyaar hai.” Aap Part 1 ke aadhaar pe kya kahoge? (*nahi aaya toh 1.3*)

Jawab: Computer mein hoshiyaari kahin nahi hai. Usmein switch hain jo shartein banate hain, aur bijli un sharton se guzarti hai.

Hoshiyaari, agar kahin hai, toh un nirdeshon mein hai jo kisi insaan ne likhe. Machine unhe samajh kar nahi chal rahi, woh unhe chala rahi hai.

2. Aapke phone mein camera app aur calculator app dono hain. Andar kya alag hai? (*nahi aaya toh 1.6*)

Jawab: Sirf numbers alag hain. Machine bilkul wahi hai.

Yeh sunne mein chhota lagta hai lekin yehi wajah hai ki ek chhoti si cheez aapki jeb mein sau alag machinon ka kaam kar leti hai.

3. Ek file mein numbers hain: 200, 15, 99. Yeh photo hai, gaana hai, ya nirdesh? (*nahi aaya toh 1.4*)

Jawab: Sirf numbers dekh kar bataya hi nahi ja sakta. Woh teeno ho sakte hain.

Matlab andar nahi hota. Matlab iss baat mein hota hai ki unhe kaunsi list se padha jaa raha hai, aur machine unhe padhne wali hai ya karne wali.

CHAT APP PE WAPAS

Yaad karo, hum ek chat app saath lekar chal rahe hain: aapke messages sambhalne aur unhe dhoondhne wala.

Part 1 ke baad aapko itna pata hai:

Aapka message akshar nahi hai. Woh numbers hai.
Woh numbers switchon mein rakhe hain, 0 aur 1 ki shakal mein.
App khud bhi numbers hai, sirf woh numbers jo "kiye" jaate hain.
Aur jo machine app chala rahi hai, woh app ke liye banayi hi nahi gayi.
Woh khaali thi. App ne use apni shakal di.

Ek sawal jo abhi khula hai, aur agla part usi pe hai:

Aapke ek message ko kitni jagah chahiye? Aur ek photo ko usse kitni zyada?

Chapter 2.1 [SPINE]

Ek cheez kitni jagah leti hai

SAMVAAD

Madhav: Ek akshar rakhne ke liye kitne switch chahiye?

Kabir: Angrezi ke akshar 128 ki list mein aate hain, toh saat.

Madhav: Log aath istemaal karte hain. Ek fazool bacha lete hain kyunki aath ka hisaab aasan hai. Ab ek shabd, paanch akshar ka?

Kabir: Chalis switch.

Madhav: Ek page, do hazaar akshar?

Kabir: Solah hazaar.

Madhav: Ek kitaab, teen sau page?

Kabir: Lagbhag paanch karod.

Madhav: Ab dhyaan do. Paanch karod switch. Ek switch ek baal se bahut chhota hai. Toh ek poori kitaab jagah leti hai...

Kabir: Bahut kam.

Madhav: Ab ek photo. Ek photo mein chhote dabbe hote hain, har dabbe ka apna rang. Ek aam phone ki photo mein lagbhag ek karod dabbe hote hain. Aur ek rang rakhne ke liye teen numbers chahiye: laal kitna, hara kitna, neela kitna. Har number aath switch. Toh ek photo?

Kabir: Ek karod guna teen guna aath. Chobees karod switch.

Madhav: Aur kitaab paanch karod thi. Toh ek photo ek kitaab se kitni badi hai?

Kabir: Paanch guna. Ek photo, paanch kitaabein.

Madhav: Ab video. Video kya hai?

Kabir: Bahut saari photos, ek ke baad ek.

Madhav: Ek second mein kitni?

Kabir: Tees ke aas-paas?

Madhav: Toh ek second ka video kitni kitaabein hua?

Kabir: Tees guna paanch. Ek sau pachas kitaabein. Ek second mein?

Madhav: Ek second mein. Ab ek ghante ka video ginti karo.

Kabir: Ek ghante mein 3,600 second. Yaani paanch lakh se zyada kitaabein.

Madhav: Ab ek sawal. Tumhare phone mein ghanton ke video hain. Toh phone mein karodon kitaabon jitni jagah hai?

Kabir: Nahi. Ho hi nahi sakti.

Madhav: Toh video usmein aata kaise hai?

Kabir: Kuch toh chhoda ja raha hoga.

Madhav: Kya chhoda ja sakta hai? Socho. Do lagatar photos mein, video ke andar, kitna farak hota hai?

Kabir: Bahut kam. Aadmi thoda hila, baaki sab wahi hai.

Madhav: Toh agar main pehli photo poori rakhun, aur uske baad sirf **farak** rakhun?

Kabir: Toh bahut kam jagah lagegi. Kyunki farak chhota hai.

Madhav: Aur ek hi photo ke andar? Aasman ka bada hissa ek jaisa neela hai. Kya karoge?

Kabir: Har dabbe ka rang alag se nahi rakhunga. Likh dunga: yeh rang, itne dabbon tak.

Madhav: Tumne abhi do sabse badi hoshiyaariyan khud nikaal li. Ek: jo cheez dohrayi ja rahi hai use ek baar likho. Do: jo cheez pehle jaisi hai use dobara mat likho, sirf farak likho.

NAAM

Aath bit ko ek **byte** kehte hain. Yeh sabse aam naap hai, kyunki ek angrezi akshar aksar ek byte leta hai.

Uske baad har kadam pe lagbhag ek hazaar guna:

1 byte	ek akshar	
1 KB (kilobyte)	hazaar akshar	ek chhota paragraph
1 MB (megabyte)	das lakh akshar	ek moti kitaab
1 GB (gigabyte)	sau karod akshar	lagbhag ek hazaar kitaabein
1 TB (terabyte)	das lakh MB	ek chhoti library

Aur jo Kabir ne nikaala, us poore idea ka naam hai **compression**.

Uske do tareeke wahi hain jo usne bataye:

DOHRAV HATANA	"neela, neela, neela x 500" → "neela x 500"
FARAK RAKHNA	poori photo mat rakho, pichhli se farak rakho

Aur ek teesra bhi hai, jo sirf photo, awaaz aur video mein chalta hai:

JO DIKHTA HI NAHI USE PHENK DO

Insaan ki aankh kuch rangon ka halka farak pakad hi nahi paati. Kaan kuch awaazein sun hi nahi sakta. Toh unhe rakhna hi kyun.

Yahan se do tarah ki compression bantee hai:

LOSSLESS	kuch nahi khota. Wapas kholo toh bilkul wahi milta hai. Likhavat, program, hisaab. Yahan ek bit bhi kho nahi sakta.
LOSSY	kuch hamesha ke liye chala jaata hai. Photo, gaana, video. Jo aankh-kaan pakad nahi paate, woh gaya.

Isiliye ek program ki file ko lossy compression nahi karte. Ek bhi number badal jaaye toh program chalega hi nahi. Lekin ek photo mein hazaron numbers badal do, aankh ko pata bhi nahi chalega.

ASLI DUNIYA SE EK EXAMPLE

Ek aam phone ki photo agar kuch chhoda hi na jaaye toh lagbhag 30 MB ki hoti hai. Jab woh aapke phone mein JPEG banke aati hai, toh woh 2 se 4 MB ki hoti hai.

Yaani lagbhag 90 pratishat phenk diya gaya, aur aapko dekh kar farak mushkil se pata chalta hai.

Video mein yeh aur zyada hai. Ek ghante ka bina kuch chhode video kai hazaar GB ka hota. Netflix aapko wahi ghanta lagbhag 3 GB mein bhej deta hai. Yaani hazaar guna se zyada nichod diya gaya.

Aur yeh sirf jagah bachane ki baat nahi hai. **Compression ke bina video internet pe chal hi nahi sakta tha.** Taar utna data le hi nahi ja sakte. YouTube, Netflix, video call, ye sab ek hisaab ki wajah se mumkin hain, kisi naye taar ki wajah se nahi.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki compression se kuch khota nahi.**

Log ek photo ko baar baar WhatsApp pe bhejte hain aur poochte hain ki woh dhundhli kyun ho gayi. Har baar bhejne pe compression dobara lagti hai, aur har baar thoda aur phenka jaata hai.

Isi tarah, ek lossy file ko wapas “poora” nahi kiya ja sakta. Jo phenk diya gaya woh gaya. Koi app use wapas nahi la sakta, chahe woh kuch bhi daawa kare. Woh sirf andaza laga kar naya bana sakta hai.

Yeh galti tempting isliye hai ki photo dekhne mein waisi hi lagti hai. Aankh ko farak nahi dikhta, toh dimaag maan leta hai ki farak hai hi nahi.

MAP PE

COMPRESSION	dohrav hatao, farak rakho	← naya
SIZE	byte, KB, MB, GB	← naya
PROGRAM	nirdesh bhi numbers hain	
ENCODING	number ka matlab	
BINARY, SHARTEIN, SWITCH		

SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Do photos hain. Ek mein saaf neela aasman hai. Doosri mein ek ped hai jismein hazaaron patte hain. Dono ka size ek jaisa hai bina compression ke. Compression ke baad kaunsi chhoti hogi?

Jawab: Aasman wali, aur bahut zyada chhoti.

Aasman mein dohrav bahut hai. Lakhon dabbe lagbhag ek hi rang ke hain, aur unhe ek line mein likha ja sakta hai.

Patton mein har dabba apne padosi se alag hai. Dohrav hai hi nahi, toh hatane ko kuch nahi hai.

Isse ek gehri baat nikalti hai: **compression dohrav pe chalti hai. Jitna zyada dohrav, utni zyada bachat.** Aur jis cheez mein koi dohrav na ho, use compress kiya hi nahi ja sakta.

2. (samajh check) Aap ek zip file banate ho, phir usi zip file ko dobara zip karte ho. Doosri baar mein kitni jagah bachegi?

Jawab: Lagbhag kuch nahi. Aksar file thodi **badi** ho jaati hai.

Kyunki pehli zip ne saara dohrav pehle hi nikaal diya. Jo bacha hai usmein koi pattern nahi hai.

Aur zip file mein thodi si jaankari bhi rakhni padti hai ki kholna kaise hai. Woh jud jaati hai.

Yeh saabit karta hai ki compression jaadu nahi hai. Woh dohrav ka faayda uthati hai. Dohrav khatam, faayda khatam.

3. (jodne wala) Chapter 1.4 kehta tha ki matlab andar nahi hota, bahar se aata hai. Compression usse kaise judti hai?

Jawab: Compressed file ke numbers dekh kar aapko kuch samajh nahi aayega. Woh asli photo ke numbers se bilkul alag hain.

Unka matlab tabhi banta hai jab aap kholne ka sahi tareeka lagayein. Galat tareeka lagao, bakwaas milega.

Toh compression ek aur parat hai, usi dher pe. Photo ke numbers ke upar, ek aur encoding.

Isiliye `.jpg` aur `.png` alag hote hain. Andar dono photo hain, lekin nichodne ka tareeka alag hai, toh kholne ka tareeka bhi alag hai.

Chapter 2.2 [SPINE]

Bade numbers ka matlab

SAMVAAD

Madhav: Ek khabar padhta hoon. “Nayi chip mein 20 arab transistor hain.” Iska tumhe kya matlab samajh aaya?

Kabir: Bahut zyada hain.

Madhav: Aur “purani chip mein 15 arab the.” Ab?

Kabir: Woh bhi bahut zyada the.

Madhav: Toh dono ka matlab tumhare liye ek hi hai: bahut zyada. Number ne tumhe kuch bataya hi nahi.

Kabir: Haan.

Madhav: Yeh sabse badi wajah hai ki tech ki khabar padh kar kuch samajh nahi aata. Number toh hote hain, lekin unka koi paimana nahi hota. Toh ab ek paimana banate hain. Ek second kitna hota hai?

Kabir: Ek second.

Madhav: Hazaar second?

Kabir: Sochne do... lagbhag pandrah minute.

Madhav: Das lakh second?

Kabir: Pata nahi. Kuch din?

Madhav: Gyarah din se thoda zyada. Ab ek arab second?

Kabir: Kuch mahine?

Madhav: Battis saal.

Kabir: Kya?

Madhav: Phir se dekho. Hazaar second: pandrah minute. Das lakh second: gyarah din. Ek arab second: battis saal.

Kabir: Yeh toh bilkul andaza hi nahi tha.

Madhav: Kyunki dimaag mein “hazaar,” “lakh,” “arab” sab ek hi dabbe mein rakhe hain, jiska naam hai “bahut.” Lekin unke beech ka farak hi asli baat hai. Ek arab, das lakh se **hazaar guna** bada hai. Gyarah din aur battis saal.

Kabir: Toh mujhe kya karna chahiye?

Madhav: Jab bhi bada number dikhe, use kisi aisi cheez se joddo jise tum mehsoos kar sakte ho. Chalo abhi karte hain. Ek chip ek second mein lagbhag teen arab baar chalti hai. Iska matlab kya hai?

Kabir: Bahut tez.

Madhav: Ulta karo. Ek kaam mein kitna waqt lagta hai?

Kabir: Ek second ka teen arabva hissa.

Madhav: Ab paimana lagao. Us waqt mein roshni kitni door jaati hai?

Kabir: Roshni ek second mein tin lakh kilometre jaati hai. Toh teen arabve hisse mein... das centimeter?

Madhav: Lagbhag. Toh chip ek kaam karti hai, aur us waqt mein roshni ek haath ki lambai bhi nahi jaati.

Kabir: Yeh mehsoos hota hai. “Teen arab” nahi hota tha.

Madhav: Ab doosri taraf. Chapter 2.1 mein humne dekha ki ek kitaab lagbhag 1 MB ki hai. Aaj ek aam phone mein 256 GB hote hain. Kitni kitaabein?

Kabir: GB matlab hazaar MB. Toh 256 guna hazaar. Do lakh se zyada kitaabein.

Madhav: Tum ek zindagi mein kitni kitaabein padhoge?

Kabir: Do hazaar? Shayad usse bhi kam.

Madhav: Toh tumhari jeb mein sau zindagiyan ka padhna rakha hai.

Kabir: Aur main usmein video bhar deta hoon.

Madhav: Bilkul. Ab aakhri. Ek AI model ke baare mein padhte ho: “70 arab parameters.” Har parameter lagbhag do byte leta hai. Kitni jagah?

Kabir: 140 arab byte. Yaani 140 GB.

Madhav: Aur tumhare phone mein 256 GB hain. Toh?

Kabir: Woh toh phone mein aa sakta hai. Aadhe se thoda zyada.

Madhav: Jagah mein aa jaayega. Chalane mein nahi. Aur kyun nahi, woh Chapter 2.4 hai. Lekin dekho, ab woh number tumhare liye khaali nahi raha. Tumne use apne phone se naap liya.

NAAM

Iss aadat ka naam hai **orders of magnitude** dekhna. Yaani number ko us baat se naapna ki usmein kitne shoonya hain, na ki uske asli ank se.

Sabse kaam ka paimana:

har baar hazaar guna, aur har baar duniya badal jaati hai

1	ek	
1,000	hazaar	
10,00,000	das lakh	(hazaar ka hazaar)
1,00,00,00,000	ek arab	(das lakh ka hazaar)

Aur yaad rakhne wale sahaare:

WAQT

1,000 second = 15 minute

10 lakh second = 11 din

1 arab second = 32 saal

JAGAH (1 kitaab ≈ 1 MB)

1 GB = 1,000 kitaabein

256 GB (phone) = 2.5 lakh kitaabein

1 TB = 10 lakh kitaabein

SPEED

chip ka ek kadam = itne waqt mein roshni ek haath bhi nahi jaati

Ek aur cheez jo bahut kaam aati hai: **jab tak ek number dus guna na badle, aksar kuch nahi badla.** 15 arab transistor se 20 arab jaana ek behtari hai. 2,300 se 20 arab jaana ek nayi duniya hai.

Isliye khabar padhte waqt sawal yeh nahi hai ki “number bada hai kya.” Sawal yeh hai ki **kitne shoonya jude.**

ASLI DUNIYA SE EK EXAMPLE

1965 mein Gordon Moore ne ek chhota lekh likha jismein usne dekha ki ek chip mein lagne wale transistor har do saal mein lagbhag dugne ho jaate hain.

Dugna hona dhoka deta hai. Sunne mein chhota lagta hai. Lekin dugna, baar baar, lagbhag panch dashak tak chalta raha.

Do saal mein dugna, matlab bees saal mein lagbhag hazaar guna, aur chalees saal mein das lakh guna.

Isiliye 1971 ki chip mein 2,300 transistor the aur aaj ki mein 20 arab. Koi ek badi khoj nahi hui. Bas ek chhota dugna hona, itni baar chala ki nateeja kalpana se bahar chala gaya.

Aur isiliye woh cheezein jo 1990 mein “kabhi mumkin nahi” thin, aaj phone mein chal rahi hain. Idea 1990 mein bhi tha. Machine chhoti thi.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **bade numbers ko ek hi dabbe mein rakhna.**

”Lakh,” ”crore,” ”arab,” ”kharab” sab ”bahut” ban jaate hain. Aur jis din sab bahut hain, us din koi bhi khabar aapko kuch nahi bata sakti.

Iska seedha nateeja: log AI ki khabar padh kar tay nahi kar paate ki kuch bada hua hai ya nahi. Har mahine koi bada number aata hai. Bina paimane ke, sab ek jaise lagte hain, aur aadmi ya toh har cheez pe uchhal jaata hai ya kisi pe nahi.

Yeh galti tempting isliye hai ki dimaag ko bade numbers ka koi tajurba hi nahi hai. Hum sau tak dekh sakte hain. Hazaar tak kalpana kar sakte hain. Uske aage sab ek jaisa mehsoos hota hai, chahe farak hazaar guna ho.

Iska ilaaj yaad karna nahi hai. Ilaaj **jodna** hai. Har bade number ko kisi aisi cheez se naapo jo aapne apni aankh se dekhi ho.

MAP PE

PAIMANA	shoonya ginno, ank nahi	← naya
COMPRESSION	dohrav hatao	
SIZE	byte, KB, MB, GB	
PROGRAM, ENCODING, BINARY, SWITCH		

SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Do khabarein. Pehli: ”naye model mein 10 pratishat behtari.” Doosri: ”naya model 50 guna sasta.” Kaunsi zyada mayne rakhti hai?

Jawab: Doosri, aur bahut zyada.

10 pratishat ek hi paimane ke andar ka farak hai. Jo cheez pehle mumkin thi woh thodi behtar ho gayi.

50 guna sasta hone ka matlab hai ki woh sab kuch jo pehle mehnga hone ki wajah se banaya hi nahi jaa sakta tha, ab ban sakta hai. Yeh paimana badal deta hai.

Rule: jo cheez pratishat mein badalti hai woh khabar hai. Jo cheez guna mein badalti hai woh badlav hai.

2. (samajh check) Aapke phone mein 256 GB hain. Aap chahte ho ki usmein har woh baat rakhi jaaye jo aap zindagi bhar bolenge, sirf likhawat ki tarah. Aa jaayegi?

Jawab: Aaram se aa jaayegi, aur bahut jagah bachegi.

Ek aadmi din mein lagbhag 15,000 shabd bolta hai. 80 saal mein lagbhag 45 crore shabd. Har shabd lagbhag 6 byte. Yaani lagbhag 2.7 GB.

Poori zindagi ki baatein, 256 GB mein se 3 GB.

Aur usi phone mein kuch ghante ka video usse zyada jagah le leta hai. **Likhawat bahut sasti hai. Dekhne aur sunne wali cheezein bahut mehngi.** Yeh farak aage bar bar dikhega, khaaskar jab hum AI ki keemat dekhenge.

3. (jodne wala) Chapter 2.1 mein compression tha. Ab Moore ka dugna hona dekha. In dono ne milkar video ko mumkin banaya. Kaise?

Jawab: Dono taraf se doori kam hui.

Compression ne video ka size hazaar guna neeche laaya.

Aur chip ke dugna hone ne machine ki taakat lakhon guna badhayi, jisse woh compression ka hisaab itni tezi se kar payi ki video chalte-chalte khul sake.

Ek akele se kaam nahi hota. Bina compression ke taar kam padte. Bina tez chip ke compression khulti hi nahi.

Yeh aage bar bar milega: **koi bhi nayi cheez aksar ek khoj se nahi aati. Woh do ya teen purani cheezon ke ek saath kaafi ho jaane se aati hai.**

Chapter 2.3 [DEPTH]

Ek instruction chalti kaise hai

DEPTH chapter. Yahan aap dekhoge ki “program chal raha hai” ka asli matlab kya hai. Chhod sakte ho, lekin iske baad “computer dheema ho gaya” jaisi baat pehli baar samajh mein aati hai.

SAMVAAD

Madhav: Chapter 1.6 mein tumne kaha tha ki program bhi numbers hain, aur machine unhe “karti” hai. Ab batao, karti kaise hai?

Kabir: Pata nahi. Padhti hogi aur kar deti hogi.

Madhav: Chalo ek insaan se samjhte hain. Tum ek rasoie ho. Tumhe ek parchi milti hai jispe likha hai: “pyaaz kaato.” Tum kya karte ho, kadam ke hisaab se?

Kabir: Parchi uthata hoon. Padhta hoon. Phir kaatta hoon.

Madhav: Aur agli parchi?

Kabir: Woh uthata hoon.

Madhav: Toh teen kadam hain, baar baar. **Uthao, samjho, karo.** Machine bilkul yehi karti hai. Ab thoda gehra chalte hain. Parchiyan rakhi kahan hain?

Kabir: Ek dher mein.

Madhav: Aur tumhe kaise pata ki agli kaunsi hai?

Kabir: Main upar wali uthata hoon. Ya... mujhe yaad rehta hai ki main kahan tak pahuncha.

Madhav: Machine ke paas yaaddasht nahi hoti. Uske paas ek chhota khaana hota hai jismein likha hota hai: “abhi main nirdesh number 47 pe hoon.” Kaam khatam, toh usmein 48 likh deti hai.

Kabir: Toh woh khaana ek ginti karne wala hai.

Madhav: Bas. Ab ek aur cheez. Tum pyaaz kaat rahe ho. Pyaaz kahan hai?

Kabir: Mere haath mein. Ya saamne ke plate mein.

Madhav: Aur baaki saara saaman?

Kabir: Almari mein, peeche.

Madhav: Kyun saamne nahi rakhte sab?

Kabir: Jagah nahi hai. Aur zaroorat bhi nahi.

Madhav: Machine mein bhi wahi hai. Uske paas kuch bahut chhote khaane hote hain, ekdam paas mein, jismein sirf woh numbers rehte hain jinpe abhi kaam ho raha hai. Baaki sab door rakha hota hai.

Kabir: Kitne chhote khaane?

Madhav: Aam taur pe solah se kuch dozen. Har ek mein ek number.

Kabir: Sirf itne? Bees arab transistor hain aur khaane sirf solah?

Madhav: Yehi baat samajhne laayak hai. Bees arab transistor kaam **karne** mein lagte hain, rakhne mein nahi. Rakhna alag jagah hota hai, aur woh door hoti hai.

Kabir: Door matlab kitni door?

Madhav: Chip ke andar hi, lekin chip ke hisaab se door. Aur ab sabse zaroori baat: **door se laane mein waqt lagta hai.** Ek nirdesh chalane mein ek kadam lagta hai. Door se ek number uthane mein sau kadam lag sakte hain.

Kabir: Sau guna?

Madhav: Aur agar woh number aur bhi door ho, disk pe, toh das lakh kadam.

Kabir: Toh machine ka zyadatar waqt...

Madhav: Intezaar mein jaata hai. Chip khaali baithi rehti hai, number ke aane ka intezaar karti hui.

Kabir: Yeh toh bekaar hai. Kuch kiya nahi jaa sakta?

Madhav: Bahut kuch kiya jaata hai, aur poora Chapter 2.4 usi pe hai. Abhi bas yeh pakad lo: machine ki asli seema uski speed nahi hai. Uski seema yeh hai ki number use time pe milte hain ya nahi.

NAAM

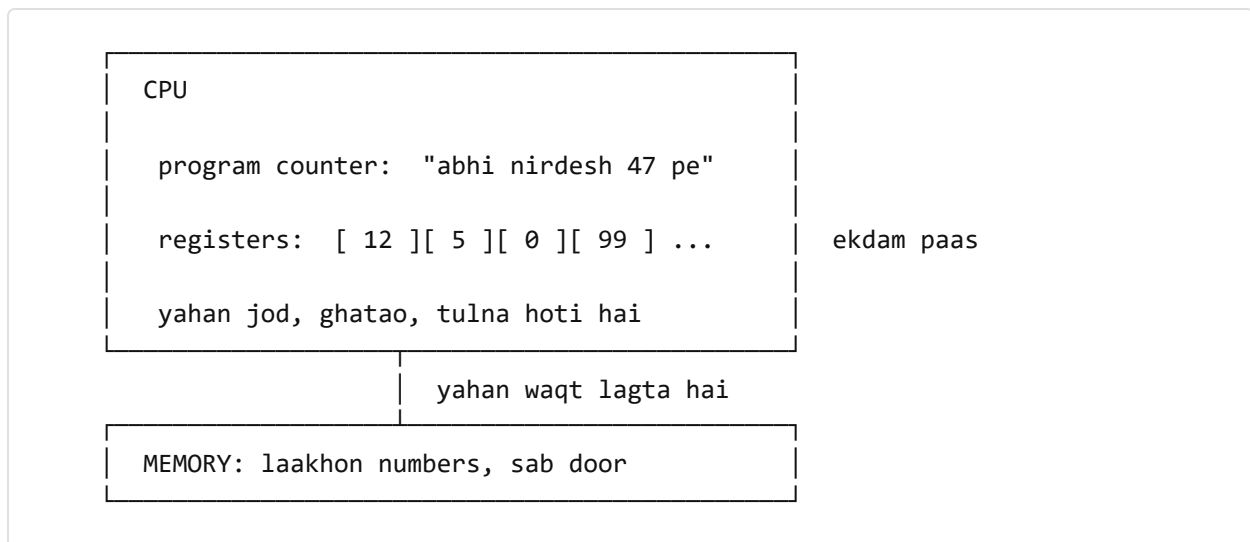
Woh teen kadam, baar baar, unka naam hai **fetch, decode, execute**. Uthao, samjho, karo.

Woh ginti wala khaana jo batata hai ki abhi kaunsa nirdesh chal raha hai, use **program counter** kehte hain.

Woh chhote khaane jo ekdam paas hain, unhe **registers** kehte hain.

Aur woh poora hissa jo yeh sab karta hai, use **CPU** kehte hain, yaani woh jagah jahan nirdesh chalte hain.

Ab poori tasveer:



Aur sabse kaam ki baat, jise log aakhir tak nahi samajh paate:

ek nirdesh chalana	→ 1 kadam
paas wale khaane se number	→ 1 kadam
memory se number	→ lagbhag 200 kadam
disk se number	→ lagbhag 10 lakh kadam

CPU tez hai. Number laana dheema hai. **Zyadatar programs isliye dheere nahi hain ki hisaab bahut hai. Woh isliye dheere hain ki data door hai.**

Yeh baat aage kai jagah dohrayi jaayegi: Part 6 mein jab database aayenge, Part 7 mein jab server aayenge, aur Book 2 mein jab AI ka kharcha samjhenge.

ASLI DUNIYA SE EK EXAMPLE

1980 ke aas-paas CPU aur memory lagbhag ek hi raftaar se chalte the. Number maangna aur number milna, dono ek jaisa waqt lete the.

Uske baad CPU har saal lagbhag 50 pratishat tez hote gaye, aur memory sirf 10 pratishat ke aas-paas.

Chalees saal baad woh farak sau guna se zyada ho chuka hai. Iss badhti hui khaai ka apna naam pad gaya: **memory wall**.

Aaj ki chips ka bahut bada hissa hisaab karne mein nahi, balki iss khaai ko chhupane mein lagta hai. Numbers pehle se mangwa lena, paas mein copy rakh lena, aur jab tak ek number aa raha ho tab tak koi doosra kaam kar lena.

Yeh dhyaan dene laayak hai: chip ka aadhe se zyada design us problem ka hal hai jo chip ke andar hai hi nahi. Woh doori ka hal hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki tez processor matlab tez computer.**

Log GHz dekh kar phone khareedte hain, jaise woh ek hi number sab kuch bata deta ho.

Lekin agar CPU zyadatar waqt intezaar mein baithi hai, toh use aur tez karne se kuch nahi hoga. Woh bas aur jaldi intezaar karegi.

Isiliye aksar purane computer mein SSD daalne se woh naye jaisa lagne lagta hai, jabki CPU wahi purani hai. Aapne hisaab tez nahi kiya. Aapne **doori kam ki**.

Yeh galti tempting isiliye hai ki speed ka ek hi number dikhaya jaata hai, aur “tez” ek hi shabd hai. Lekin computer mein kam se kam do alag “tez” hote hain: hisaab kitni jaldi hota hai, aur data kitni jaldi pahunchta hai. Doosra wala aksar zyada mayne rakhta hai.

MAP PE

CPU	uthao, samjho, karo	← naya
DOORI	paas tez, door dheema	← naya
PAIMANA, COMPRESSION, SIZE		
PROGRAM, ENCODING, BINARY, SWITCH		

SOCHNE KE LIYE

Har sawal pe ek minute socho, phir neeche jawab padho.

1. (samajh check) Ek program ko das lakh numbers jodne hain. Doosre ko sirf sau numbers jodne hain, lekin har number disk se laana padta hai. Kaunsa tez hoga?

Jawab: Pehla, aur bahut aage se.

Das lakh jod = das lakh kadam ke aas-paas.

Sau numbers disk se = sau guna das lakh = dus karod kadam.

Doosra program das hazaar guna kam hisaab kar raha hai aur phir bhi sau guna dheema hai.

Isse ek rule nikalta hai jo poori kitaab mein kaam aayega: **hisaab sasta hai, data laana mehnga hai**. Jab bhi kuch dheema ho, pehle yeh poocho ki data kahan se aa raha hai, na ki yeh ki hisaab kitna hai.

2. (samajh check) Rasoiye wale example mein, aap saamne ki plate mein kya rakhoge aur almari mein kya?

Jawab: Saamne woh jo abhi lag raha hai, ya abhi lagne wala hai. Almari mein baaki sab.

Aur agar आपको पता हो कि अगला काम क्या है, तो आप उसे पहले से निकाal कर रख लेंगे, ताकि उस वाqt चलना न पड़े.

Machine bilkul yehi karti hai, aur uske teen naam hain jo aage milenge: paas mein copy rakhna (cache), पहले से mangwa lena (prefetch), aur intezaar ke waqt koi doosra kaam kar lena.

Teeno ek hi soch hain: **doori ko chhupa do**.

3. (jodne wala) Chapter 1.6 mein tay hua tha ki program aur data dono numbers hain, machine ek ko padhti hai aur doosre ko karti hai. Ab yeh chapter kehta hai ki dono laane mein waqt lagta hai. Isse kya nikalta hai?

Jawab: Ki nirdesh laana bhi utna hi mehnga hai jitna data laana.

Machine ko har kadam pe kam se kam do baar door jaana padta hai: ek baar agla nirdesh laane, ek baar us nirdesh ka data laane.

Isiliye asli chips mein nirdeshon ke liye alag paas wali jagah hoti hai, data ki paas wali jagah se alag.

Aur isiliye chhota program aksar tez chalta hai. Isiliye nahi ki usmein kam kaam hai, balki isiliye ki woh poora paas wali jagah mein aa jaata hai aur use baar baar door jaana nahi padta.

Chapter 2.4 [SPINE]

Yaad rakhna aur bhool jaana

SAMVAAD

Madhav: Pichhle chapter mein tumne dekha ki paas wali jagah tez hai aur door wali dheemi. Ab sawal: sab kuch paas hi kyun nahi rakh lete?

Kabir: Jagah nahi hogi.

Madhav: Aur?

Kabir: Mehngi hogi?

Madhav: Dono sahi hain, aur ek teesri baat bhi hai jo tum abhi khud nikaaloge. Ek switch se ek bit banta hai. Us bit ko yaad rakhne ke liye kya chahiye?

Kabir: Switch ko usi haalat mein rakhna hoga.

Madhav: Aur switch usi haalat mein rehta kaise hai?

Kabir: Bijli aati rahegi toh rahega.

Madhav: Aur bijli chali gayi?

Kabir: Toh switch band ho jaayega. Sab kuch 0 ho jaayega.

Madhav: Bas. Tumne abhi computer ki sabse badi seema nikaal li. Jo cheez bijli se yaad rakhi jaati hai, woh bijli jaate hi gayab ho jaati hai.

Kabir: Toh phir meri photos kaise bachti hain?

Madhav: Kyunki woh switchon mein nahi rakhi hain. Unhe kisi aisi cheez mein rakha jaata hai jo bina bijli ke bhi apni haalat nahi badalti.

Kabir: Jaise?

Madhav: Jaise ek chhota sa dabba jismein kuch electron band kar diye jaate hain. Bijli chali jaaye toh bhi woh wahin fanse rehte hain. Unhe nikaalne ke liye khaas mehnat karni padti hai.

Kabir: Toh yeh behtar hai. Sab kuch usmein kyun nahi rakhte?

Madhav: Kyunki woh dheema hai. Electron ko band karne aur nikaalne mein waqt lagta hai, switch dabane se bahut zyada.

Kabir: Kitna zyada?

Madhav: Hazaar guna ke aas-paas. Aur agar ghoomne wali purani disk ho toh das lakh guna.

Kabir: Toh ek tez hai lekin bhool jaata hai, doosra yaad rakhta hai lekin dheema hai.

Madhav: Bas. Aur ab batao, agar tumhe machine banani ho, toh kya karoge?

Kabir: Dono rakhunga. Jispe abhi kaam chal raha hai woh tez wali mein, aur baaki sab dheemi wali mein.

Madhav: Aur jab program shuru hota hai?

Kabir: Toh dheemi se tez mein le aana padega.

Madhav: Isiliye phone chalu hone mein waqt lagta hai. Aur isiliye app pehli baar kholne mein dheema hota hai aur doosri baar tez.

Kabir: Aur isiliye bijli jaane pe jo maine save nahi kiya woh chala gaya.

Madhav: Save karne ka matlab kya hai, ab bolo.

Kabir: Tez wali jagah se dheemi wali jagah mein copy karna. Taaki bijli jaane pe bache.

Madhav: Bas yehi. "Save" ek jaadui shabd nahi hai. Woh ek jagah se doosri jagah copy karna hai.

NAAM

Do tarah ki jagah, aur inke naam har jagah milenge:


```

RAM    (tez, bhool jaane wali)
├─ switchon jaisi cheezon mein rakha
├─ bahut tez
├─ bijli gayi, sab gaya
├─ mehngi, isliye kam hoti hai
└─ phone mein 6 se 16 GB

DISK / SSD    (dheemi, yaad rakhne wali)
├─ electron fanse hue, ya chumbak
├─ bahut dheemi
├─ bijli gayi toh bhi bacha rehta hai
├─ sasti, isliye zyada hoti hai
└─ phone mein 128 se 512 GB

```

Aur ab poora dher, sabse tez se sabse dheema:

registers	ekdam CPU ke andar	1 kadam	kuch dozen numbers
cache	CPU ke bilkul paas	3 se 40 kadam	kuch MB
RAM	door, alag chip	~200 kadam	kuch GB
SSD	aur door	~10 lakh kadam	kuch sau GB

Har kadam pe wahi sauda: **tez, chhota, mehnga** ya **dheema, bada, sasta**.

Yeh dher poori computer ki duniya mein dohraya jaata hai. Aage aap ise phir dekhoge: database mein, network mein, server mein, aur AI mein. Har jagah wahi shakal.

ASLI DUNIYA SE EK EXAMPLE

2007 tak lagbhag har computer mein ghoomne wali disk hoti thi: ek dhaatu ki plate jo minute mein hazaaron baar ghoomti thi, aur ek sui uspar se padhti thi.

Us sui ko sahi jagah pahunchne mein lagbhag 10 millisecond lagte the. Chip ke hisaab se woh lagbhag **teen karod kadam** ka intezaar hai.

SSD mein koi cheez ghoomti nahi. Sirf electron hain. Wahi kaam lagbhag 0.1 millisecond mein hota hai, yaani sau guna tez.

Isiliye purane laptop mein SSD daalte hi woh naya lagne lagta hai. CPU wahi hai. RAM wahi hai. Sirf intezaar sau guna kam ho gaya.

Aur ab wapas Chapter 2.3 ki baat pe: **speed hisaab se nahi, doori se aati hai.**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **RAM aur storage ko ek hi cheez samajhna.**

Log kehte hain “mere phone mein 128 GB memory hai.” Woh storage hai. RAM alag hai, aur woh 8 GB hogi.

Yeh sirf shabdon ka farak nahi hai. Iska seedha nateeja hai:

Jab phone kehta hai “jagah nahi hai,” toh woh storage ki baat kar raha hai. Photos delete karni padegi.

Jab phone dheema ho jaata hai aur app band ho jaate hain, toh woh RAM ki baat hai. Photos delete karne se kuch nahi hoga. App band karne se hoga.

Do bilkul alag problem, do bilkul alag hal, aur log aksar galat wala lagate hain.

Yeh galti templing isliye hai ki dono ko “memory” kehte hain, aur dono GB mein naapte hain. Lekin ek yaad rakhti hai aur doosri bhool jaati hai, aur wahi asli farak hai.

MAP PE

DO TARAH KI JAGAH tez + bhool jaane wali (RAM) dheemi + yaad rakhne wali (disk)	← naya
CPU, DOORI	
PAIMANA, COMPRESSION, SIZE	
PROGRAM, ENCODING, BINARY, SWITCH	

Part 2 khatam.

PART 2 KA GATE

1. Aapka phone dheema chal raha hai aur app baar baar band ho rahe hain. Photos delete karne se faayda hoga? *(nahi aaya toh 2.4)*

Jawab: Nahi. Yeh RAM ki problem hai, storage ki nahi.

App band karo, ya phone restart karo. Photos delete karne se sirf storage khaali hoti hai, jiska iss problem se koi lena dena nahi.

2. "Naya AI model 400 GB ka hai." Kya woh aapke phone mein aa jaayega, aur kya woh chalega? *(nahi aaya toh 2.2, 2.4)*

Jawab: Rakhne mein nahi aayega, kyunki aam phone mein 128 se 512 GB storage hoti hai aur usmein pehle se sab bhara hai.

Aur agar aa bhi jaata, toh chalta nahi, kyunki chalane ke liye use RAM mein aana padta, aur phone mein 8 se 16 GB RAM hai.

Rakhna aur chalana do alag sawal hain. Log inhe hamesha mila dete hain.

3. Ek program dheema hai. Do wajahein ho sakti hain. Batao. *(nahi aaya toh 2.3)*

Jawab: Ek: hisaab bahut zyada hai. Do: data door se aa raha hai.

Aur zyadatar mamlon mein doosri wali hi hoti hai.

CHAT APP PE WAPAS

Part 2 ke baad, aapke chat app ke baare mein yeh saaf ho gaya:

Ek text message lagbhag 100 byte ka hai. Bahut sasta.

Ek photo lagbhag 30 lakh byte ki. Tees hazaar guna zyada.

Ek minute ka video ek lakh message ke barabar.

Message likhte waqt woh RAM mein hai. Bijli gayi toh gaya.

"Bhej diya" ka matlab hai woh kisi aisi jagah pahunch gaya jo bijli jaane pe bhi use yaad rakhegi.

Aur jab aap purana message dhoondhte ho, toh woh disk se aana padta hai, jo RAM se das lakh guna dheemi hai.

Agla sawal, jo Part 3 kholta hai:

Woh nirdeshon ki list, jo app hai, likhi kisne? Aur insaan seedha numbers mein toh likhta nahi hoga?

Chapter 3.1 [SPINE]

Translation problem

SAMVAAD

Madhav: Machine sirf numbers samajhti hai. Aur nirdesh bhi numbers hain. Toh program likhne ka matlab kya hua?

Kabir: Sahi numbers sahi jagah likhna.

Madhav: Chalo likho. Do numbers jodne ka nirdesh likho.

Kabir: Mujhe kaise pata hoga ki kaunsa number “jodo” hai?

Madhav: Kisi kitaab mein likha hoga. Har chip ki apni list hoti hai. Maan lo “jodo” ka number 137 hai.

Kabir: Toh main 137 likh dunga.

Madhav: Aur kis cheez mein kya jodna hai?

Kabir: Woh bhi likhna padega. Kaunse khaane se, kaunse khaane mein.

Madhav: Toh ek chhote se kaam ke liye teen-chaar numbers. Ab ek poora app socho. Kitne nirdesh?

Kabir: Lakhon.

Madhav: Lakhon numbers, haath se, sahi order mein, aur ek bhi galat hua toh?

Kabir: Toh sab kuch galat.

Madhav: Aur agar tumhe beech mein ek nirdesh jodna ho?

Kabir: Uske aage ke saare... aage khisak jaayenge. Poora dobara likhna padega.

Madhav: Ab batao, tum kya chahoge?

Kabir: Main chahunga ki main aam bhasha mein likhun. "Ise usme jodo." Aur koi use numbers mein badal de.

Madhav: Kaun badlega?

Kabir: Koi aadmi.

Madhav: Aadmi thak jaayega aur galti karega.

Kabir: Toh koi machine.

Madhav: Machine ko kaise pata chalega ki kya karna hai?

Kabir: Uske andar bhi ek program hoga.

Madhav: Aur woh program kis shakal mein hoga?

Kabir: Numbers mein. Kyunki har program numbers mein hi hota hai.

Madhav: Toh tum ek aisa program chahte ho, jo numbers mein likha ho, aur jiska kaam ho doosre programs ko numbers mein badalna.

Kabir: Haan. Yeh ajeeb lag raha hai.

Madhav: Ajeeb hai, lekin ek baar bana lo toh phir kabhi nahi banana padta. Kisi ne ek baar woh takleef uthayi, aur uske baad kisi ko nahi uthani padi.

Kabir: Yeh toh Anil ki naali hai.

Madhav: Bilkul wahi hai. Do saal ki takleef, phir hamesha ke liye paani.

Kabir: Lekin sabse pehla wala? Jab koi translator tha hi nahi?

Madhav: Woh haath se likha gaya, numbers mein, ek-ek karke. Aur uske baad kisi ko dobara nahi likhna pada.

NAAM

Woh problem jo Kabir ne mehsoos ki, uska naam nahi hai, lekin uske hal ka hai.

Woh aam-bhasha jaisi shakal jismein insaan likhta hai, use **programming language** kehte hain.

Woh program jo use numbers mein badalta hai, use **compiler** ya **interpreter** kehte hain. Farak agle chapter mein.

Aur woh seedhe numbers wali shakal, jismein pehla aadmi likhta tha, use **machine code** kehte hain.

Beech mein ek aur shakal hoti hai, jahan har nirdesh ko ek chhota naam de diya jaata hai, jaise `ADD` ya `MOV`. Use **assembly** kehte hain. Woh insaan ke liye machine code se behtar hai, lekin sirf thoda.

insaan likhta hai	→	total = price + tax
		↓ compiler
assembly	→	MOV R1, price ADD R1, tax MOV total, R1
		↓
machine code	→	10110001 00000101 00000001 00000110 10001001 00000111
		↓
switch	→	bijli aayi ya nahi

Chaar parat. Har parat neeche wali ko chhupa rahi hai. Yeh wahi cheez hai jo Chapter 0.4 mein Kabir ne khud nikaali thi, ab kaam karti hui dikh rahi hai.

Aur ek baat jo pehli baar ajeeb lagti hai: **compiler khud ek program hai, aur use bhi kisi ne likha tha**. Aaj ke compiler aksar usi language mein likhe hote hain jise woh translate karte hain. Yeh mumkin hai kyunki pehla wala haath se likha gaya tha, aur uske baad har naye compiler ko purane wale ne bana diya.

ASLI DUNIYA SE EK EXAMPLE

1950 ke dashak tak sab log seedhe machine ke numbers mein hi likhte the. Yeh maan liya gaya tha ki koi aur tareeka ho hi nahi sakta.

Grace Hopper naam ki ek naval officer aur mathematician ne kaha ki program aam angrezi jaise shabdon mein likha jaana chahiye, aur ek program use machine ke numbers mein badal dega.

Us waqt logon ne kaha ki yeh bewakoofi hai. Do wajahein deen: ki machine kabhi itni hoshiyaar nahi ho sakti, aur ki jo translator banayega woh insaan se ghatiya code banayega.

Doosri baat kai saal tak sach thi. Shuru ke compiler sach mein aise programs banate the jo haath se likhe programs se dheeme the.

Lekin pehli baat galat thi, aur woh zyada zaroori thi. Aaj lagbhag koi bhi machine code haath se nahi likhta. Aur compiler ab itne achhe ho chuke hain ki aksar woh insaan se behtar code banate hain, kyunki woh lakhon possibilities ek saath dekh sakte hain.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki programming language machine ki bhasha hai.**

Log kehte hain “computer Python samajhta hai.” Computer Python nahi samajhta. Computer sirf apni chip ke numbers samajhta hai.

Python ek insaan ki suvidha hai. Machine ne usmein kabhi koi ruchi nahi li.

Iska seedha nateeja hai: yeh samajhna ki ek hi program alag machinon pe kyun nahi chalta, aur kyun use har jagah ke liye dobara translate karna padta hai. Aur yeh bhi ki “language” chunna asal mein machine ka nahi, insaan ka faisla hai.

Yeh galti templing isliye hai ki aap code likhte ho aur woh chal jaata hai. Beech ka translation dikhta hi nahi. Ek aur parat, pehle se chhupi hui.

MAP PE

LANGUAGE	insaan ki shakal	← naya
TRANSLATOR	use numbers mein badalta	← naya
PROGRAM	nirdesh bhi numbers hain	
CPU, DOORI, RAM/DISK		
BINARY, SHARTEIN, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Compiler khud ek program hai. Toh sabse pehla compiler kisne banaya?

Jawab: Kisi insaan ne, haath se, machine ke numbers mein. Ek baar.

Uske baad us pehle compiler se ek behtar compiler banaya gaya. Aur us behtar wale se ek aur behtar.

Aaj ke compiler lagbhag hamesha usi language mein likhe hote hain jise woh translate karte hain. Yeh ghoomta hua lagta hai lekin nahi hai, kyunki har naya wala kisi purane wale se bana tha, aur woh chain sabse pehle haath wale kaam pe jaakar rukti hai.

Yeh technology ki ek aam shakal hai: **ek baar takleef uthao, phir us cheez se agli cheez banao.**

2. (samajh check) Ek aadmi kehta hai “main Python jaanta hoon, isliye main kisi bhi computer pe kaam kar sakta hoon.” Kya yeh sahi hai?

Jawab: Lagbhag sahi hai, aur wajah dilchasp hai.

Woh isliye sahi nahi hai ki computer Python jaanta hai. Woh isliye sahi hai ki lagbhag har computer ke liye koi Python translator maujood hai.

Yaani woh aadmi language pe nahi, un logon pe bharosa kar raha hai jinhone har machine ke liye translator likh diya.

Yeh baat kaam ki hai: **jab kuch “har jagah chalta hai,” toh iska matlab yeh nahi ki woh sarvavyapi hai. Iska matlab hai ki kisi ne har jagah ke liye alag se mehnat ki hai.**

3. (jodne wala) Chapter 0.2 ki teen dials yaad karo. Compiler banana kis dial ka udaharan hai?

Jawab: Leverage ka, aur shayad kitaab ka sabse saaf udaharan hai.

Ek baar likha gaya. Uske baad har programmer, har din, uska faayda uthata hai, bina us likhne wale ke kuch kiye.

Scarcity bhi upar thi: bahut kam log compiler likh sakte hain.

Aur size bhi bada tha: bina uske programming ek chhoti si duniya rehti.

Teeno dials upar. Isiliye woh chand log jinhone yeh cheezein banayi, unka asar us paise se kahin zyada hai jo unhe mila.

Chapter 3.2 [SPINE]

Translator ke do tareeke

SAMVAAD

Madhav: Tumhe ek angrezi kitaab hindi mein chahiye. Do tareeke hain. Batao.

Kabir: Ek toh poori kitaab pehle translate karwa lo. Phir padho.

Madhav: Doosra?

Kabir: Ek aadmi paas baith jaaye aur padhte padhte bolta jaaye.

Madhav: Ab dono ke faayde nuksaan batao. Pehla tareeka.

Kabir: Shuru mein waqt lagega, poori kitaab karni padegi. Lekin uske baad padhna tez hoga. Aur baar baar padho toh har baar tez.

Madhav: Aur agar kitaab mein ek galti hai, page 300 pe?

Kabir: Woh translate karte waqt hi pakdi jaayegi. Padhna shuru karne se pehle.

Madhav: Achha. Ab doosra tareeka.

Kabir: Turant shuru ho jaata hai. Lekin padhna dheema hoga, kyunki har line par tarjuma ho raha hai.

Madhav: Aur page 300 wali galti?

Kabir: Woh tab pata chalegi jab hum page 300 pe pahunchenge.

Madhav: Aur agar hum kabhi page 300 tak pahunche hi nahi?

Kabir: Toh kabhi pata nahi chalegi.

Madhav: Ab ek aur baat. Doosre tareeke mein, agar main beech mein kahun “yeh line badal do, dobara padho”?

Kabir: Woh ho sakta hai. Aadmi wahin baitha hai.

Madhav: Aur pehle tareeke mein?

Kabir: Poori kitaab dobara translate karwani padegi.

Madhav: Toh ab batao, kaunsa tareeka behtar hai?

Kabir: Depends. Agar kitaab baar baar padhni hai toh pehla. Agar experiment karna hai toh doosra.

Madhav: Bas. Aur programming mein bilkul yahi do tareeke hain, unhi faaydon aur nuksaan ke saath.

NAAM

COMPILER poora program pehle numbers mein badal do,
phir chalao

INTERPRETER chalate chalate translate karo, line by line

Aur seedha nateeja:

	COMPILER	INTERPRETER
shuru hone mein	der lagti hai	turant
chalne mein	tez	dheema
galti kab pakdi	chalne se pehle	wahan pahunchne pe
badalna	dobara compile	turant
dena kis shakal mein	numbers	asli code + translator

Aur isse ek baat nikalti hai jo log aksar nahi jodte:

Compiled program dene ke liye aapko apna code nahi dena padta. Aap numbers de dete ho. Interpreted program dene ke liye aapko asli code dena padta hai, ya kam se kam kuch aisa jo uske bahut kareeb hai.

Isiliye bikne wale software aksar compiled hote hain, aur khule software aksar interpreted.

Aaj ki asli duniya beech mein hai. Java aur C# pehle ek beech ki shakal mein compile hote hain, aur woh shakal chalte waqt translate hoti hai. JavaScript aur Python interpreted hain, lekin unke andar ek hissa chalte-chalte compile bhi karta hai, un jagahon ke liye jo baar baar chalti hain.

Yeh khud ek kaam ka soch hai: **jo cheez baar baar chalti hai use pehle se taiyaar kar lo, jo ek baar chalti hai use waqt pe hi kar lo.** Yeh soch aage cache mein, database mein, aur AI mein phir milegi.

ASLI DUNIYA SE EK EXAMPLE

C, jo 1972 mein bani, compiled hai. Operating system, database aur game engine aaj bhi usmein ya usi jaisi languages mein likhe jaate hain, kyunki wahan har kadam ki speed mayne rakhti hai.

Python, jo 1991 mein aayi, interpreted hai. Woh C se lagbhag 10 se 100 guna dheemi hai.

Ab dhyaan dene wali baat: **aaj lagbhag saara AI ka kaam Python mein likha jaata hai**, jo dheemi language hai.

Yeh ulta lagta hai. Lekin wajah saaf hai: Python khud kaam nahi karti. Woh sirf order deti hai. Asli hisaab C ya usse bhi neeche likhe hisson mein hota hai. Python bas yeh tay karti hai ki kaunsa bada kaam kab shuru ho.

Yaani insaan wahan baitha hai jahan sochna hai, aur tez cheez wahan hai jahan mehnat hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki dheemi language matlab dheema program.**

Log yeh sun kar ghabra jaate hain ki Python C se sau guna dheemi hai, aur maan lete hain ki unka program bhi sau guna dheema hoga.

Lekin agar program ka 99 pratishat waqt kisi tez hisse mein ja raha hai, aur Python sirf order de rahi hai, toh farak lagbhag nahi padta.

Yeh Chapter 2.3 wali baat hai, doosri shakal mein: **speed wahan mayne rakhti hai jahan waqt lagta hai, har jagah nahi.**

Yeh galti tempting isliye hai ki speed ka ek hi number bata diya jaata hai, aur woh number kabhi nahi batata ki program apna waqt kahan bitata hai.

MAP PE

DO TAREEKE	pehle poora, ya chalte chalte	← naya
LANGUAGE,	TRANSLATOR	
CPU,	DOORI,	RAM/DISK
PROGRAM,	BINARY,	SWITCH

SOCHNE KE LIYE

1. (samajh check) Ek game bana rahe ho jo har second saath baar screen banata hai. Compiler ya interpreter?

Jawab: Compiler, aur bina soche.

Kyunki wahan har kadam ki speed sach mein mayne rakhti hai. Har frame ke liye lakhon hisaab hain, aur woh hisaab aapka apna code kar raha hai, kisi aur ka nahi.

Yeh tab badalta hai jab aapka code sirf order de raha ho. Tab language ki speed lagbhag bemaani ho jaati hai.

2. (samajh check) Ek program compile hone ke baad chalta hai lekin ek khaas haalat mein crash ho jaata hai. Compiler ne use kyun nahi pakda?

Jawab: Kyunki compiler sirf **shakal** dekh sakta hai, **nateeja** nahi.

Woh pakad sakta hai ki aapne galat likha, ya do cheezein jodne ki koshish ki jo jud hi nahi sakti.

Woh yeh nahi pakad sakta ki chalte waqt koi number zero ho jaayega, ya user kuch aisa likhega jiska aapne socha hi nahi tha.

Yeh ek gehri seema hai: **kuch galtiyan sirf chalne pe hi dikhti hain, kyunki woh code mein nahi, data mein hoti hain.** Isiliye testing ek alag cheez hai, aur woh Part 4 hai.

3. (jodne wala) Chapter 1.6 kehta tha ki program aur data dono numbers hain. Interpreter uss baat ka kaunsa udaharan hai?

Jawab: Ek saaf udaharan.

Interpreter ke liye, aapka program **data** hai. Woh use padhta hai, jaise koi program kisi photo ko padhta hai.

Aur us data ko padhkar woh tay karta hai ki kya karna hai.

Toh ek hi cheez, aapka code, do nazariyon se do alag cheezein hai: aapke liye woh nirdesh hai, interpreter ke liye woh data hai.

Yeh baat Book 2 mein bahut kaam aayegi. Wahan aap dekhoge ki AI ke liye bhi aapke shabd data hain, nirdesh nahi, aur usse bahut kuch samajh mein aata hai.

Chapter 3.3 [SPINE]

Itni saari languages kyun hain

SAMVAAD

Kabir: Agar sab languages ka kaam ek hi hai, machine ko nirdesh dena, toh itni saari kyun hain? Sau se zyada.

Madhav: Tumhare ghar mein kitne chaaku hain?

Kabir: Chaar paanch.

Madhav: Sabka kaam kaatna hi hai. Phir chaar kyun?

Kabir: Roti ka alag, sabzi ka alag, bada wala alag.

Madhav: Aur agar sirf ek rakhna ho?

Kabir: Rakh sakte hain, lekin har kaam thoda mushkil ho jaayega.

Madhav: Bas. Ab languages ka bhi wahi hai. Lekin unke chunav thode alag hote hain. Ek sawal se shuru karte hain. Tumne likha: $x = 5$. Machine ko yeh kaise pata chalega ki 5 ek number hai, akshar nahi?

Kabir: Kyunki 5 number hai.

Madhav: Chapter 1.4 yaad karo. Andar sirf numbers hain. Matlab bahar se aata hai. Toh machine ko kaise pata?

Kabir: Kisi ko batana padega.

Madhav: Kaun bataye? Do tareeke hain. Ek: likhne wala pehle se likh de, “yeh khaana numbers ke liye hai.” Do: koi na kuch likhe, aur chalte waqt machine dekhe ki usmein kya pada hai.

Kabir: Pehla zyada kaam hai likhne mein.

Madhav: Aur uska faayda?

Kabir: Agar main galti se usmein akshar daal doon toh pehle hi pakda jaayega.

Madhav: Aur doosre mein?

Kabir: Chalte waqt pata chalega. Ya shayad chalta rahega aur galat jawab dega.

Madhav: Toh ek tareeka likhne mein dheema aur bharosemand hai, doosra likhne mein tez aur risky hai. Kaunsa behtar hai?

Kabir: Chhote kaam ke liye doosra. Bade ke liye pehla.

Madhav: Yeh sirf ek chunav hai. Aise dozen chunav hote hain, aur har language unhe alag tarah se chunti hai. Ek aur lo. Jab tumhe RAM mein jagah chahiye, kaun de?

Kabir: Machine de degi.

Madhav: Aur jab kaam khatam, wapas kaun le?

Kabir: Machine hi le legi?

Madhav: Kaise pata chalega ki kaam khatam ho gaya?

Kabir: Hmm. Ya toh main bataun, ya woh khud dekhti rahe ki ab yeh kisi ke kaam ka nahi hai.

Madhav: Dono hote hain. Agar tum batao toh tez chalta hai lekin bhool gaye toh jagah bharti jaayegi. Agar machine khud dekhe toh surakshit hai lekin woh dekhne mein bhi waqt lagta hai, aur woh waqt kabhi bhi aa sakta hai, beech mein.

Kabir: Toh game mein woh beech mein aa gaya toh atak jaayega.

Madhav: Bilkul. Isiliye game aur operating system aksar aisi languages mein likhe jaate hain jahan aap khud batate ho. Aur website aksar aisi mein jahan machine khud dekhti hai.

Kabir: Toh koi language sabse achhi nahi hai.

Madhav: Koi bhi nahi. Har ek ne kuch chunav kiye, aur har chunav mein kuch mila aur kuch gaya. Ab aakhri sawal, aur yeh sabse kaam ka hai. Tumne C seekh li. Ab Python seekhne mein kitna lagega?

Kabir: Pata nahi. Shuru se?

Madhav: Shuru se kya kya seekhna padega? Shartein?

Kabir: Nahi, woh toh wahi hain.

Madhav: Ginti, dohrav, numbers rakhna, tulna?

Kabir: Woh sab bhi wahi hain.

Madhav: Toh kya naya hai?

Kabir: Sirf likhne ka tareeka. Aur woh chunav jo unhone alag kiye.

Madhav: Toh kitna waqt lagega?

Kabir: Hafta, do hafte.

Madhav: Aur pehli language seekhne mein?

Kabir: Mahine.

Madhav: Kyunki pehli baar tum language nahi seekh rahe the. Tum sochne ka tareeka seekh rahe the. Woh ek baar seekhna padta hai.

NAAM

Woh chunav jinke aadhaar pe languages alag hoti hain, unmein sabse bade do yeh hain:

```

TYPES: pehle batana ki khaane mein kya aayega, ya nahi
|— pehle batao (C, Java, Rust, TypeScript)
|   likhne mein dheema, galtiyan pehle pakdi jaati hain
|— mat batao (Python, JavaScript, Ruby)
|   likhne mein tez, galtiyan chalte waqt

```

```

MEMORY: RAM ki jagah wapas kaun leta hai
|— tum khud (C, C++)
|   tez aur poora control, aur bhoolne pe program toot ta hai
|— machine khud dekhti rehti hai (Java, Python, Go, JavaScript)
|   surakshit, lekin beech mein ruk sakta hai
|— compiler pehle hi tay kar deta hai (Rust)
|   tez aur surakshit dono, lekin likhna mushkil

```

Aur teesri baat, jo aksar sabse zyada mayne rakhti hai aur kitaabon mein sabse kam likhi jaati hai: **pehle se bana hua saaman**.

Ek language isliye nahi chuni jaati ki woh sundar hai. Woh isliye chuni jaati hai ki uske aas-paas pehle se hazaaron cheezein bani hui hain jo aapko dobara nahi banani padengi.

Python AI mein isliye nahi hai ki woh AI ke liye achhi hai. Woh isliye hai ki AI ka saara saaman Python ke liye pehle se bana hua hai. Aur woh saaman isliye bana kyunki shuru ke kuch log Python istemaal kar rahe the. Ek baar shuru ho jaaye toh khud ko badhata rehta hai.

Yeh baat Chapter 8.5 mein dobara aayegi, jahan hum poochenge ki ek technology jeetati kyun hai. Aksar jawab “behtar thi” nahi hota.

ASLI DUNIYA SE EK EXAMPLE

JavaScript 1995 mein lagbhag **das din** mein banayi gayi thi, ek browser mein chhote-mote kaam karne ke liye. Kisi ne nahi socha tha ki woh badi cheezein banane ke kaam aayegi.

Uska design jaldbaazi mein hua aur usmein aisi cheezein reh gayin jinhe aaj tak galtiyan maana jaata hai.

Aaj woh duniya ki sabse zyada istemaal hone wali languages mein se ek hai.

Wajah unke design mein nahi hai. Wajah yeh hai ki woh browser ke andar thi, aur browser har machine pe tha. Jo language wahan pehle se maujood thi, wahi jeet gayi.

Aur jab log uski kamiyon se pareshan hue, toh unhone use hataya nahi. Unhone uske upar ek nayi parat banayi, TypeScript, jo uski shakal sudhaar deti hai aur phir use wapas JavaScript mein badal deti hai.

Ek aur parat, phir se. Purani cheez hatayi nahi gayi, chhupa di gayi.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki “kaunsi language seekhun” ek bada faisla hai.**

Log mahine iss sawal pe atak jaate hain. Kaunsi sabse achhi hai, kaunsi ka future hai, kaunsi mein naukri hai.

Lekin pehli language seekhne ka lagbhag saara waqt language pe nahi jaata. Woh sochne ke tareeke pe jaata hai: cheezein rakhna, shartein lagana, dohrav, tukdon mein todna. Yeh sab har language mein ek hi hai.

Ek baar woh aa gaya toh doosri language ek hafte ka kaam hai.

Yeh galti tempting isliye hai ki bahar se languages bilkul alag dikhti hain. Alag shabd, alag nishaan. Andar se woh lagbhag ek jaisi hain.

MAP PE

CHUNAV	types, memory, saaman	← naya
DO TAREEKE	compiler, interpreter	
LANGUAGE, TRANSLATOR		
CPU, RAM/DISK, PROGRAM, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Ek hospital ki machine ka software likhna hai jo mareez ko dawai deti hai. Kaunse chunav karoge, aur kyun?

Jawab: Types pehle batao. Memory ka control khud lo, ya aisi language lo jo compile ke waqt hi tay kar de.

Wajah: yahan galti ki keemat jaan hai. Aap chahte ho ki jitni bhi galtiyan pakdi ja sakti hain, woh chalne se pehle pakdi jaayein.

Aur aap nahi chahte ki machine beech mein apne aap ruk jaaye jagah saaf karne ke liye, kyunki woh ruk-ruk kar chalna ek dawai dene wali machine mein khatarnak hai.

Dhyaan do: yeh chunav “achhi language” ka nahi hai. Yeh **is kaam ke liye sahi chunav** ka hai.

2. (samajh check) Aapka dost kehta hai “Python dheemi hai isliye woh ek ghatiya language hai.” Kya jawab doge?

Jawab: Ki speed ek chunav ka nateeja hai, kami nahi.

Python ne likhne ki aasani chuni aur chalne ki speed di. C ne ulta kiya.

Sawal yeh nahi hai ki kaunsi tez hai. Sawal yeh hai ki aapke kaam mein mehnga kya hai: machine ka waqt, ya insaan ka waqt.

Zyadatar kaamon mein insaan ka waqt mehnga hota hai. Isiliye dheemi languages itni zyada istemaal hoti hain.

3. (jodne wala) Chapter 0.4 kehta tha ki har parat neeche wali ko chhupa deti hai. TypeScript ka udaharan uss baat se kaise judta hai?

Jawab: Seedha judta hai, aur ek nayi baat bhi batata hai.

JavaScript mein kamiyan thin. Unhe theek karne ka matlab hota duniya ke har browser ko badalna, jo mumkin hi nahi tha.

Toh logon ne kuch nahi hataya. Unhone upar ek parat rakh di.

Nayi baat yeh hai: **jo cheez ek baar phail jaati hai, use hataya nahi ja sakta, sirf dhaka ja sakta hai.** Isiliye technology mein purani galtiyen hamesha zinda rehti hain, bas unke upar naye kapde aa jaate hain.

Yeh yaad rakhna. Jab bhi aapko lage ki koi cheez bewajah ulajhi hui hai, aksar wajah yeh hoti hai ki neech kuch purana hai jise koi hata nahi sakta.

Chapter 3.4 [SPINE]

Woh manager jo machine baantta hai

SAMVAAD

Madhav: Tumhare phone mein abhi kitne app khule hain?

Kabir: Chaar paanch. Aur peeche kuch aur bhi chal rahe honge.

Madhav: Aur CPU kitni hai?

Kabir: Ek chip.

Madhav: Chapter 2.3 mein humne dekha ki CPU ek waqt mein ek nirdesh chalati hai. Toh paanch app ek saath kaise chal rahe hain?

Kabir: Ho sakta hai chip mein kai hisse hon.

Madhav: Aath hisse hote hain aam taur pe. Aur app peeche milakar tees-chalees. Ab?

Kabir: Toh sach mein saath nahi chal rahe honge.

Madhav: Toh kya ho raha hai?

Kabir: Baari baari?

Madhav: Kitni jaldi baari badalti hogi ki tumhe pata hi na chale?

Kabir: Bahut jaldi. Second ke kuch hisson mein.

Madhav: Har app ko lagbhag das millisecond milte hain, phir agla. Tumhari aankh us farak ko pakad hi nahi sakti, isliye sab ek saath lagta hai.

Kabir: Toh baari kaun tay karta hai?

Madhav: Yehi sawal hai. Socho, kya app khud tay kar sakte hain?

Kabir: Nahi. Har app apni hi baari lena chahega.

Madhav: Aur agar ek app kabhi na chhode?

Kabir: Toh baaki sab ruk jaayenge. Phone hang ho jaayega.

Madhav: Toh?

Kabir: Koi upar wala hona chahiye. Koi jo sabse zyada taakat rakhta ho aur baari tay kare.

Madhav: Aur woh khud kya hai?

Kabir: Woh bhi ek program hi hoga.

Madhav: Aur agar woh bhi ek program hai, toh use apni taakat kahan se milti hai? App use hata kyun nahi sakte?

Kabir: Kyunki... woh pehle chalu hota hai?

Madhav: Ek wajah yeh hai. Doosri wajah chip ke andar hai. Chip ke do modes hote hain. Ek mein program sab kuch kar sakta hai. Doosre mein kuch cheezein rok di jaati hain. App doosre mode mein chalte hain.

Kabir: Toh app chahkar bhi kuch nahi kar sakta?

Madhav: Nahi. Agar woh koshish kare toh chip use rok deti hai aur manager ko bula leti hai.

Kabir: Achha. Toh manager sirf baari nahi baantta.

Madhav: Aur kya baantega, socho. Do app ek saath RAM maangte hain.

Kabir: Toh koi tay kare ki kisko kaunsa hissa mile. Aur ek app doosre ka hissa na padh paaye.

Madhav: Aur agar padh le?

Kabir: Toh mera bank app mere doosre app ka data padh lega.

Madhav: Ab tum samajh gaye ki manager kyun hai. Aur bhi? Do app ek saath file kholna chahein?

Kabir: Wahi baat. Koi tay kare kaun kholega, aur kis order mein.

Madhav: Camera? Ek waqt mein kaun istemaal karega?

Kabir: Wahi.

Madhav: Ab ek line mein bolo ki manager karta kya hai.

Kabir: Jo cheezein ek hain aur maangne wale kai, unhe baantta hai. Aur sabko ek doosre se alag rakhta hai.

NAAM

Us manager ka naam hai **operating system**. Windows, Android, iOS, Linux, macOS.

Uske chaar bade kaam:

BAARI BAANTNA	kaunsa program ab chalega, aur kitni der (har program ko lagta hai woh akela chal raha hai)
JAGAH BAANTNA	RAM ka kaunsa hissa kise mile (aur koi doosre ka hissa na dekh sake)
SAAMAN BAANTNA	screen, camera, speaker, network, ek hain aur maangne wale kai
ALAG RAKHNA	ek program doosre ko na chhoo sake (yeh sabse zaroori hai, aur kam dikhta hai)

Aur woh sabse andar wala hissa jiske paas poori taakat hoti hai, use **kernel** kehte hain.

Ab woh cheez jo Kabir ne nikaali, use naam dena zaroori hai. Har program ko lagta hai ki poori machine sirf uski hai. Use nahi pata ki koi aur bhi chal raha hai. Use nahi pata ki uski baari beech mein rok kar kisi aur ko de di jaati hai.

Yeh jhooth jaan-boojh ke bola jaata hai, aur woh hi poore operating system ka asli product hai.

Aur ab Chapter 0.4 ki baat wapas dekho. Yeh abstraction ka sabse saaf udaharan hai. OS neeche wali sachai, ki ek chip hai aur woh baari baari chal rahi hai, poori tarah chhupa deta hai.

ASLI DUNIYA SE EK EXAMPLE

Purane phone aur computer mein yeh alag rakhna itna sakht nahi tha. Ek app doosre ka data padh sakta tha. Ek app crash hota tha toh poori machine baith jaati thi.

Aaj ke phone mein har app apne dabbe mein band hota hai. Woh doosre app ki files nahi dekh sakta. Woh camera ya location tab tak nahi le sakta jab tak aap ijaazat na do.

Isiliye jab aap koi app install karte ho toh woh poochta hai. Woh shishtachar nahi hai. Woh iss baat ka nateeja hai ki OS ne use koi taakat di hi nahi, aur woh sirf manager se maang sakta hai.

Aur isiliye ek kharaab app aapka phone ab utni aasani se barbaad nahi kar pata jitna pehle karta tha. Woh apne dabbe se bahar nahi ja sakta.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki operating system woh cheez hai jo screen pe dikhti hai.**

Log Windows ka matlab samajhte hain woh neela background, woh menu, woh icons. Woh saara hissa upar ki ek parat hai. Use hataya ja sakta hai aur OS phir bhi chalega.

Asli OS woh hissa hai jo dikhta hi nahi: kaun kab chalega, kiski jagah kahan hai, kaun kya chhoo sakta hai.

Iska seedha nateeja: log samajh nahi paate ki server pe OS bina kisi screen ke kaise chalta hai. Aur woh yeh bhi nahi samajh paate ki phone dheema kyun hai, kyunki woh sirf woh dekh rahe hain jo dikh raha hai.

Yeh galti tempting isliye hai ki OS ka jo hissa aapse milta hai, wahi ek hissa hai jo dikhne ke liye banaya gaya hai. Baaki sab apne aap ko chhupane ki koshish kar raha hai.

MAP PE

OS	baari, jagah, saaman aur sabko alag rakhna	← naya
LANGUAGE, TRANSLATOR, CHUNAV		
CPU, RAM/DISK, DOORI		
PROGRAM, BINARY, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Ek app hang ho jaata hai lekin aap use band kar sakte ho aur baaki phone chalta rehta hai. Kaun band kar raha hai use, aur woh kaise kar pa raha hai?

Jawab: OS kar raha hai, aur woh isliye kar pa raha hai kyunki app ke paas kabhi asli taakat thi hi nahi.

App ki baari OS deta hai. Toh OS bas uski baari dena band kar deta hai aur uski jagah wapas le leta hai.

App yeh rok nahi sakta. Woh chip ke us mode mein chal raha hai jahan usse yeh kar hi nahi sakta.

Isiliye purane systems mein ek app poori machine baitha deta tha, aur ab nahi. Farak taakat ka hai, speed ka nahi.

2. (samajh check) Aapka phone kehta hai “yeh app aapka location dekhna chahta hai.” Kaun pooch raha hai, aur agar aap na kaho toh kya hota hai?

Jawab: OS pooch raha hai, app nahi.

App ne OS se location maanga. OS ne aapse poocha. Aap mana karo toh OS app ko keh deta hai ki nahi mila.

App iske aas-paas se nikal nahi sakta, kyunki uske paas location tak jaane ka koi seedha raasta hai hi nahi. Har raasta OS se hokar jaata hai.

Yeh soch aage phir milegi. Book 2 mein aap dekhoge ki AI agent ke paas bhi apna koi haath nahi hota. Har cheez kisi ke through jaati hai, aur wahi jagah hai jahan rok lagti hai.

3. (jodne wala) Chapter 0.3 mein Kabir ne kaha tha ki ek group ajnabiyon ko scale pe serve karta hai. OS us baat ka kaunsa roop hai?

Jawab: OS ek manager hai jo un programs ke beech baantta hai jo ek doosre ko jaante hi nahi.

Jaise sheher mein log ek doosre ko jaane bina ek hi sadak istemaal karte hain, aur koi niyam tay karta hai ki kaun kab chalega.

Aur wahi wajah bhi ek hai: gaon mein sau log the toh aapas mein tay ho jaata tha. Programs bhi jab do teen the tab tay ho jaata tha. Jab tees-chalees ho gaye, tab manager zaroori ho gaya.

Scale hi manager ko paida karta hai. Yeh baat Part 7 mein poore zor se wapas aayegi.

Chapter 3.5 [DEPTH]

”Running” ka matlab kya hai

DEPTH chapter. Iske baad “app khula hai,” “app band hai,” aur “app crash ho gaya” ka asli matlab pata chalega.

SAMVAAD

Madhav: Ek app tumhare phone mein hai lekin khula nahi hai. Woh kahan hai?

Kabir: Storage mein. Disk pe.

Madhav: Aur kis shakal mein?

Kabir: Numbers ki ek file.

Madhav: Woh file kuch kar rahi hai?

Kabir: Nahi. Bas padi hai.

Madhav: Ab tum icon dabate ho. Kya hota hai?

Kabir: App khul jaata hai.

Madhav: Kadam ke hisaab se batao. Pehla kadam kya?

Kabir: Woh file... RAM mein aani padegi. Kyunki CPU disk se seedha nahi chala sakti.

Madhav: Achha. Ab woh RAM mein hai. Aur?

Kabir: CPU ko batana padega ki shuru kahan se karna hai.

Madhav: Woh program counter, Chapter 2.3 wala. Usmein pehla pata daal do. Aur?

Kabir: Bas, chalna shuru?

Madhav: Ek cheez aur chahiye. App ko apne kaam ke numbers rakhne ke liye jagah chahiye. Woh jagah kaun degi?

Kabir: OS.

Madhav: Toh ab ginti karo. App chalne ke liye kya kya chahiye?

Kabir: RAM mein nirdesh. Ek jagah apne data ke liye. Aur ek nishaan ki abhi kahan hoon.

Madhav: Aur ek cheez aur, jo tum bhool rahe ho. Agar OS beech mein iski baari le le, aur baad mein wapas de, toh use kaise pata chalega ki yeh kahan tha?

Kabir: Uske sab khaane sambhal kar rakhne padenge. Registers.

Madhav: Bas. Ab in sab ka ek dher bana lo. Yeh dher hi “chal raha app” hai.

Kabir: Toh file aur chalta hua app do alag cheezein hain.

Madhav: Bilkul alag. File ek hai. Ab batao, kya ek hi file se do app ek saath chal sakte hain?

Kabir: Do browser window? Haan, chal sakte hain.

Madhav: Toh file ek hai aur chalte hue kitne?

Kabir: Do. Alag alag data ke saath.

Madhav: Toh file ek nuskha hai, aur chalta hua app us nuskhe se banaya gaya khaana. Ek nuskhe se kitne bhi khaane ban sakte hain, aur har khaana alag hota hai.

Kabir: Aur jab main app band karta hoon?

Madhav: Kya hota hai, socho.

Kabir: RAM wapas le li jaati hai. Jo usmein tha, gaya.

Madhav: Aur file?

Kabir: File wahi ki wahi hai. Disk pe.

Madhav: Isiliye band karke dobara kholne se app naye jaisa ho jaata hai. Tumne app ko theek nahi kiya. Tumne purana khaana phenk kar naya bana liya.

Kabir: Toh “restart karo” ka matlab yehi hai.

Madhav: Har baar yehi. Aur isiliye woh itni baar kaam kar jaata hai.

NAAM

Woh file jo disk pe padi hai, use **program** ya executable kehte hain.

Woh chalta hua dher, use **process** kehte hain.

Ek process mein kya kya hota hai:

PROCESS	
nirdesh	(file se copy kiye hue)
data ki jagah	(OS ne di hui)
program counter	(abhi kahan hoon)
registers	(abhi kya haath mein hai)
khuli hui files, network ke raaste	
kis user ka hai, kya kar sakta hai	

Ek program se kai process ban sakte hain. Har process ka apna dabba hota hai, aur woh ek doosre ko dekh nahi sakte. Yeh Chapter 3.4 wala “alag rakhna” hai, ab shakal ke saath.

Chalne ki haalatein

Ek process hamesha chal nahi rahi hoti. Uski teen haalatein hoti hain:

CHAL RAHI	CPU abhi isi ke nirdesh chala rahi hai
TAIYAAR	chal sakti hai, bas baari ka intezaar hai
RUKI HUI	kisi cheez ka intezaar hai: disk, network, ya user

Aur ab ek baat jo bahut kuch samjha deti hai: **zyadatar process zyadatar waqt ruki hui hoti hain.**

Aapka chat app 99 pratishat waqt kuch nahi kar raha. Woh intezaar mein hai. Isiliye ek phone tees-chalees app “khule” rakh leta hai, jabki chip mein sirf aath hisse hain. Woh sab chal nahi rahe. Woh sab so rahe hain.

Chapter 2.3 ki baat, phir se: **CPU tez hai, intezaar lamba hai.** Poora system isi ek sachai ke aas-paas bana hua hai.

ASLI DUNIYA SE EK EXAMPLE

Jab aap phone pe ek app se doosre pe jaate ho, toh pehla app aksar band nahi hota. Woh “ruki hui” haalat mein chala jaata hai, aur uska data RAM mein pada rehta hai.

Agar RAM bhar jaaye, toh OS chup-chaap kisi purane app ka poora dher phenk deta hai.

Isiliye kabhi kabhi aap ek purane app pe wapas jaate ho aur woh shuru se khulta hai, jaise aapne use kabhi khola hi na ho. Woh crash nahi hua. OS ne use maar diya, kyunki jagah chahiye thi.

Aur isiliye zyada RAM wale phone “smooth” lagte hain. Woh tez nahi hain. Unmein app zinda rehte hain.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki “app band karne” se phone tez hota hai.**

Log baar baar saare app swipe karke band karte hain, yeh sochkar ki phone halka ho jaayega.

Zyadatar mamlon mein ulta hota hai. Ruki hui process CPU nahi le rahi thi. Woh sirf RAM mein padi thi, aur RAM ko khaali rakhne ka koi faayda nahi hai. Ab jab aap woh app dobara khologe, toh use disk se poora naye sire se load karna padega, jo mehnga hai.

OS pehle se hi jaanta hai ki kise phenkna hai aur kab. Woh aapse behtar jaanta hai.

Yeh galti tempting isliye hai ki “khula hua” ka matlab lagta hai “chal raha hai aur mehnat kar raha hai.” Woh nahi hai. Zyadatar khule app sirf so rahe hain.

MAP PE

PROCESS	file ≠ chalta hua app chal rahi / taiyaar / ruki	← naya
OS	baari, jagah, alag rakhna	
LANGUAGE, TRANSLATOR, CHUNAV		
CPU, RAM/DISK, PROGRAM, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Aap ek document likh rahe ho, save nahi kiya, aur bijli chali jaati hai. Kya gaya aur kyun?

Jawab: Jo aapne likha woh process ki RAM wali jagah mein tha. Bijli gayi toh RAM khaali ho gayi.

File disk pe abhi bhi hai, us haalat mein jaisi aapne aakhri baar save ki thi.

”Save” ka matlab hai process ki jagah se disk pe copy karna, jo Chapter 2.4 mein aaya tha.

Aur isiliye aaj ke app har kuch second mein khud save karte rehte hain. Unhone problem hal nahi ki, unhone bas woh khidki chhoti kar di jismein nuksaan ho sakta hai.

2. (samajh check) Ek hi program se do process chal rahe hain. Ek crash ho jaata hai. Doosre pe kya asar?

Jawab: Kuch nahi. Bilkul kuch nahi.

Har process ka apna alag dher hai: apni jagah, apne registers, apna sab kuch. Woh ek hi file se bane hain, lekin ek doosre se jude nahi hain.

Isiliye browser mein har tab aksar apna alag process hota hai. Ek page kharaab ho toh sirf woh tab marta hai, poora browser nahi.

Yeh ek design ka faisla hai: alag rakhne mein zyada RAM lagti hai, lekin ek hissa toot ne pe baaki bach jaate hain. **Suraksha ki keemat jagah hai.**

3. (jodne wala) Chapter 1.6 mein “machine badalti nahi, sirf numbers badalte hain” tha. Process usi baat ka kaunsa hissa hai?

Jawab: Process woh numbers hain, chalte hue.

Machine wahi ek chip hai. Aap ek app kholte ho toh chip nahi badalti. RAM mein numbers ka ek naya dher aata hai, aur chip ab un numbers se guzarti hai.

Band karo toh dher hat jaata hai. Chip phir bhi wahi hai.

Isiliye ek chhoti si chip ek hi din mein hazaron alag “machinein” ban jaati hai. Har process usi khaali machine ka ek naya roop hai.

Chapter 3.6 [SPINE]

Free cheezein exist kyun karti hain

SAMVAAD

Madhav: Chapter 0.1 ka niyam bolo.

Kabir: Value nahi di toh tokens nahi.

Madhav: Ab ek problem hai. Linux ek operating system hai. Duniya ke lagbhag saare servers uspe chalte hain, aur har Android phone ke andar bhi wahi hai. Uski keemat zero hai. Koi bhi utha kar istemaal kar sakta hai, badal sakta hai, aur bech bhi sakta hai.

Kabir: Toh usne banane wale ko kya diya?

Madhav: Yehi sawal hai. Aur Linux akela nahi hai. Python free hai. Woh database jo aadhi duniya chalata hai, free hai. Woh saara AI ka saaman jo abhi tum sunte ho, bahut sa free hai.

Kabir: Toh log ise banate kyun hain?

Madhav: Pehle ek aur sawal. Ek company ko ek database chahiye. Do raaste hain: khud banaye, ya khareede. Ab teesra raasta socho.

Kabir: Kisi aur ke banaye hue ko istemaal kare, free mein.

Madhav: Aur agar usmein koi kami ho?

Kabir: Toh woh use theek kar le.

Madhav: Ab woh theek kiya hua hissa uske paas hai. Do raaste: apne paas rakhe, ya wapas de de. Rakhne mein kya hoga?

Kabir: Jab agla version aayega, toh use apna hissa dobara jodna padega. Har baar.

Madhav: Hamesha ke liye. Aur de de toh?

Kabir: Toh woh agle version mein khud aa jaayega. Uska kaam khatam.

Madhav: Toh company apna kaam free mein kyun de rahi hai?

Kabir: Kyunki rakhne mein use zyada mehnga pad raha hai.

Madhav: Bas. Yeh daan nahi hai. Yeh hisaab hai.

Kabir: Lekin sabse pehla banane wala? Jab koi company thi hi nahi?

Madhav: Woh log alag wajah se banate hain. Kabhi unhe khud woh cheez chahiye hoti hai aur milti nahi. Kabhi woh naam kamana chahte hain, jo aage kaam deta hai. Kabhi unhe woh problem hi dilchasp lagti hai.

Kabir: Aur woh naam se kya kamate hain?

Madhav: Chapter 0.2 ke dials lagao. Jisne woh cheez banayi jo hazaaron log roz istemaal karte hain, uski scarcity kahan hai?

Kabir: Bahut upar. Bahut kam log woh bana sakte hain, aur ab woh saabit bhi hai.

Madhav: Toh use tokens mile ya nahi?

Kabir: Seedhe nahi mile. Lekin uske teeno dials upar chale gaye.

Madhav: Ab ek aur wajah, jo sabse badi hai. Ek company kyun apni bani hui cheez free mein de degi, jaan-boojh ke, shuru se?

Kabir: Samajh nahi aaya.

Madhav: Socho. Tumhare paas ek cheez hai jo log istemaal karte hain, aur ek cheez hai jo uske upar chalti hai aur bikti hai. Kaunsi free karoge?

Kabir: Neeche wali. Taaki zyada log upar wali tak aayein.

Madhav: Aur agar woh neeche wali sab jagah pahunch jaaye, toh baaki sabko kya karna padega?

Kabir: Usi ke hisaab se chalna padega.

Madhav: Toh free dene se kya mila?

Kabir: Woh cheez sabki ban gayi, aur uske aas-paas ki har cheez uske hisaab se banegi.

Madhav: Bas. Free hona ek chaal bhi ho sakti hai, daan bhi, aur hisaab bhi. Teeno ek saath chalte rehte hain.

NAAM

Aisa software jiska asli code sabke liye khula ho, use **open source** kehte hain.

Char alag wajahein, aur teeno mein se koi bhi “achhai” nahi hai:

1. APNA KHARCHA KAM KARNA
theek kiya hua hissa wapas do, taaki har baar dobara na jodna pade
2. NAAM AUR SAABITI
scarcity ka dial upar, dikhaane laayak kaam
(aksar naukri aur mauke isi se aate hain)
3. NEECHE WALI PARAT PE KABZA
apni parat free karo taaki upar wali parat bik sake
4. KHUD KO ZAROORAT THI
koi banata hai kyunki use chahiye tha aur mil nahi raha tha

Aur ab ek zaroori farak, jo log hamesha mila dete hain:

FREE ka matlab "muft"	→ paisa nahi lagta
FREE ka matlab "khula"	→ code dikhta hai, badal sakte ho

Aksar dono saath hote hain, lekin hone zaroori nahi hain. Kuch software muft hai lekin uska code bilkul band hai. Aap use istemaal kar sakte ho aur andar dekh nahi sakte.

Jab bhi koi cheez muft mile, ek hi sawal poochna kaafi hai: **iska kharcha kaun utha raha hai, aur use badle mein kya mil raha hai?** Iska jawab hamesha hota hai. Kabhi woh chanda hai, kabhi company ka hisaab, kabhi aapka data, aur kabhi upar wali parat jo bikti hai.

ASLI DUNIYA SE EK EXAMPLE

1991 mein ek 21 saal ka Finnish student, Linus Torvalds, ne shauk mein ek operating system ka core likhna shuru kiya. Usne likha ki yeh “kuch bada nahi hoga.”

Aaj woh Linux hai. Woh duniya ke lagbhag saare web servers pe chalta hai, har Android phone ke andar hai, aur lagbhag har badi cloud company usi pe khadi hai.

Ab dilchasp hissa. Aaj Linux mein sabse zyada code aise programmers se aata hai jinhe badi companies **tankhwah** deti hain yeh kaam karne ke liye. Woh companies ise daan nahi maan rahin. Unke liye yeh sasta hai: sabko ek hi neev chahiye, aur usko akele sambhalna kisi ek ke liye bahut mehnga hai.

Toh sabse bada muft software duniya ke sabse bade karobaar chala raha hai, aur unhi ke paise se chal raha hai.

Yeh Chapter 0.1 ke नियम ko todta nahi hai. Value di ja rahi hai aur mil bhi rahi hai. Bas woh tokens ki shakal mein nahi guzarti.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki free software ghatiya hota hai.**

Log maante hain ki jiska paisa nahi liya gaya woh kamzor hoga, ya usmein koi chaal hogi.

Sachai lagbhag ulti hai. Jo cheezein sabse zyada bhrose ki hain, woh aksar khuli hain: Linux, aur woh cryptography jo aapke bank ki suraksha karti hai, aur woh database jo aspatal chalte hain.

Wajah seedhi hai: jab code sabke saamne khula ho, toh hazaaron log usmein kami dhoondh sakte hain. Band code mein sirf usi company ke log dekh sakte hain.

Yeh galti templing isliye hai ki baaki har cheez mein daam aur quality ka rishta hota hai. Software mein woh rishta toota hua hai, kyunki software ki nakal banane ka kharcha lagbhag zero hai. Ek hi cheez ek aadmi ke paas ho ya das karod ke paas, banane wale ka kharcha wahi rehta hai.

Aur yeh ek gehri baat hai: **jis cheez ki nakal muft hai, uska daam uski quality nahi batata.** Yeh baat aage software ke poore karobaar ko samjhati hai, aur Book 2 mein AI models pe bhi lagegi.

MAP PE

OPEN SOURCE khula code nakal ka kharcha $\approx$ zero	← naya
OS, PROCESS	
LANGUAGE, TRANSLATOR, CHUNAV	
CPU, RAM/DISK, PROGRAM, SWITCH	

Part 3 khatam.

PART 3 KA GATE

1. Ek dost kehta hai “computer Python samajhta hai.” Aap kya kahoge? (*nahi aaya toh 3.1*)

Jawab: Computer sirf apni chip ke numbers samajhta hai. Python ek insaani suvidha hai, jise ek doosra program numbers mein badalta hai.

2. Aapke phone mein tees app “khule” hain lekin chip mein aath hisse hain. Yeh kaise chal raha hai? (*nahi aaya toh 3.4, 3.5*)

Jawab: Woh chal nahi rahe. Zyadatar ruki hui haalat mein hain, RAM mein pade hue. OS baari baantta hai, itni jaldi ki aapko pata na chale.

3. Ek company apna bana hua software free mein de deti hai. Chapter 0.1 ka niyam toota? (*nahi aaya toh 3.6*)

Jawab: Nahi. Value di bhi ja rahi hai aur mil bhi rahi hai, bas tokens ki shakal mein nahi. Kabhi apna kharcha bachta hai, kabhi scarcity ka dial upar jaata hai, kabhi upar wali parat bikne lagti hai.

CHAT APP PE WAPAS

Aapka chat app kisi ne likha, aam-bhasha jaisi language mein.
Ek translator ne use numbers mein badla.
Woh numbers disk pe ek file mein pade hain.

Jab aap icon dabate ho, OS unhe RAM mein laata hai, jagah deta hai,
aur ek process bana deta hai. App ko lagta hai poora phone uska hai.
Woh jhooth OS bol raha hai.

Aur us app ka bahut bada hissa, shayad aadhe se zyada, kisi ne
free mein likha tha, aur app banane wale ne use bas utha liya.

Ek baat abhi tak nahi chhui gayi. Aapka message aapke phone se nikal kar kisi doosre phone tak pahunchta hai.

Woh jaata kaise hai? Yeh Part 4 hai.

Chapter 4.1 [SPINE]

Ek machine kaafi nahi hai

SAMVAAD

Madhav: Ab tak hum ek machine ke andar the. Switch, numbers, program, OS. Ab bahar nikalte hain. Tumhare phone mein internet band kar do. Kya kya chalta rahega?

Kabir: Calculator. Camera. Purani photos. Notes.

Madhav: Aur kya band ho jaayega?

Kabir: Message. Search. Map. Bank. Video. Lagbhag sab kuch jo main sach mein istemaal karta hoon.

Madhav: Toh machine ki apni taakat toh wahi hai. Bees arab transistor abhi bhi wahin hain. Phir kya gaya?

Kabir: Doosri machinein chali gayin.

Madhav: Yeh baat dhyaan se dekho. Tumhare phone ki taakat mein zero farak pada, aur uski upyogita lagbhag khatam ho gayi. Iska matlab kya nikla?

Kabir: Ki uski asli value uski apni taakat mein thi hi nahi.

Madhav: Toh kahan thi?

Kabir: Doosron se jude hone mein.

Madhav: Ab ek sawal jo iski gehrai batayega. Ek phone duniya mein akela ho. Uski keemat kitni?

Kabir: Ek calculator jitni.

Madhav: Do phone hon, jude hue?

Kabir: Ab ek baat kar sakta hai. Ek rishta bana.

Madhav: Teen?

Kabir: Teen rishte. A-B, B-C, A-C.

Madhav: Chaar?

Kabir: Chhe.

Madhav: Paanch?

Kabir: Das.

Madhav: Dhyaan do. Phone ek-ek karke badh rahe hain, aur rishte kis raftaar se badh rahe hain?

Kabir: Bahut tez. Har naye phone pe utne naye rishte jitne pehle se phone the.

Madhav: Toh sau phone pe?

Kabir: Lagbhag paanch hazaar rishte.

Madhav: Aur ek hazaar phone pe lagbhag paanch lakh. Ab batao, jab ek nayi machine judti hai, toh value kiski badhti hai?

Kabir: Sirf uski nahi. Sabki.

Madhav: Isiliye ek naya phone khareedne se tumhare purane dost ka phone bhi thoda zyada kaam ka ho jaata hai, chahe usne kuch bhi na kiya ho.

Kabir: Toh sabse pehla phone bekaar tha.

Madhav: Bilkul bekaar. Aur yehi har network wali cheez ki sabse badi mushkil hai. Shuruaat mein uski value zero hoti hai. Kisi ko woh chahiye hi nahi jab tak baaki sab uspe na hon, aur baaki sab tab tak nahi aayenge jab tak use koi istemaal na kare.

Kabir: Toh phir shuru kaise hota hai?

Madhav: Woh ek behtareen sawal hai aur uska jawab Chapter 8.5 mein hai. Abhi bas itna pakdo: **network ki value machine mein nahi hoti, judav mein hoti hai.** Aur judav ek machine ke andar se nahi banta.

NAAM

Jab kai machinein aapas mein jud jaati hain, use **network** kehte hain.

Aur woh baat jo Kabir ne ginti se nikaali, uska naam hai **network effect**: har naya judne wala pehle se jude hue sabki value badha deta hai.

1 machine	→	0 rishte	→	akeli, bekaar
2 machines	→	1 rishta		
5 machines	→	10 rishte		
100	→	~5,000		
1,000	→	~5,00,000		
10,000	→	~5 karod		

machinein jodke badhti hain
rishte lagbhag guna hokar

Isse do baatein nikalti hain, aur dono aage kaam aayengi.

Ek: shuruaat sabse mushkil hissa hai. Jab tak kaafi log na hon, cheez bekaar hai. Isiliye nayi network wali cheezein aksar mar jaati hain, chahe woh technically behtar hon.

Do: jo pehle bhar gaya, use hatana lagbhag namumkin hai. Behtar cheez aane par bhi log nahi hilte, kyunki hilne ka matlab hai apne saare rishte chhod dena.

Aur ab Chapter 0.2 ke dials pe wapas jao. Ek network wali cheez ka leverage kya hai?

Ek baar bana do, aur har naya judne wala use bina aapki mehnat ke aur mazboot kar deta hai. Anil ki naali, lekin aisi naali jo khud lambi hoti jaati hai.

ASLI DUNIYA SE EK EXAMPLE

1876 mein pehla telephone bana. Us waqt uski value theek zero thi, kyunki doosri taraf koi telephone tha hi nahi. Do banane padte the, aur unke beech taar daalna padta tha.

Uske baad har naya telephone pehle se lage sabhi telephones ki value badhata gaya.

Aaj wahi cheez har jagah dikhti hai. WhatsApp aapke liye isliye kaam ka hai ki aapke sab log usi pe hain, na ki isliye ki woh sabse achha bana hua app hai. Ek behtar messaging app aaye toh bhi aap nahi hiloge, kyunki aap akele hil jaoge aur aapke rishte peeche reh jaayenge.

Yeh Chapter 3.3 wali JavaScript ki baat hai, doosri shakal mein: **jo cheez pehle sab jagah pahunch gayi, woh behtar hone ki wajah se nahi jeeti. Woh pehle pahunchne ki wajah se jeeti.**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh maan lena ki behtar cheez jeet jaayegi.**

Log ek nayi cheez dekhte hain jo har tarah se behtar hai aur sochte hain ki purani ab khatam. Aksar aisa nahi hota.

Kyunki network wali cheezon mein aap sirf cheez nahi chun rahe. Aap yeh chun rahe ho ki kaun aapke saath hoga. Aur akele behtar jagah pe khade hone se koi faayda nahi hota.

Yeh galti tempting isliye hai ki technology ki tulna aasan hai. Feature ginna aasan hai. Rishte ginna kisi ko yaad nahi rehta.

Iska seedha nateeja: agar aap kuch banane ki soch rahe ho, toh yeh poochna zaroori hai ki **kya iski value akele mein hai, ya sirf tab jab aur log bhi aayein?** Doosri wali cheezein bahut zyada mushkil hoti hain, aur bahut zyada badi bhi.

MAP PE

NETWORK	value judav mein hai, machine mein nahi	← naya
OS, PROCESS, LANGUAGE		
CPU, RAM/DISK, PROGRAM, SWITCH		

Ab tak sab kuch ek machine ke andar tha.
Yahan se hum bahar hain.

SOCHNE KE LIYE

1. (samajh check) Ek nayi messaging app aati hai jo har tarah se WhatsApp se behtar hai. Kya log hil jaayenge?

Jawab: Zyadatar nahi.

Kyunki aap akele hilte ho aur aapke rishte peeche reh jaate hain. Behtar app pe akele baithne se bura app pe sabke saath baithna behtar hai.

Log tab hilte hain jab woh **group mein** hil sakein: koi ek badi wajah jo sabko ek saath le jaaye, ya koi aisa hissa jo dono jagah kaam kare.

Isiliye nayi cheezein aksar kisi chhote group se shuru karti hain jahan poora group ek saath aa sakta hai, na ki sabko ek-ek karke todne ki koshish karti hain.

2. (samajh check) Aapka phone offline hai. Calculator chalta hai, map nahi. Dono app phone ke andar hi hain. Farak kya hai?

Jawab: Calculator ko kisi doosri machine ki zaroorat nahi hai. Uske paas apna saara hisaab hai.

Map ko har waqt naye data ki zaroorat hai: kaunsa raasta hai, traffic kahan hai, aap kahan ho.

Toh app “phone ke andar” hone se yeh tay nahi hota ki woh akela chal sakta hai ya nahi. Sawal yeh hai ki uska kaam sirf hisaab hai, ya usmein kisi aur ki jaankari bhi chahiye.

Yeh farak aage bahut kaam aayega, khaaskar Book 2 mein jab hum poochenge ki AI aapke phone mein chal sakta hai ya nahi.

3. (jodne wala) Chapter 0.2 ke teen dials lagao ek network wali cheez pe. Kaunsa dial sabse alag vyavhaar karta hai?

Jawab: Leverage, aur woh yahan apni sabse teekhi shakal leta hai.

Aam leverage mein: ek baar banao, phir woh chalta rehta hai. Naali roz paani deti hai, lekin naali khud badi nahi hoti.

Network mein: ek baar banao, aur phir **har naya istemaal karne wala use aur mazboot kar deta hai**, bina aapki kisi mehnat ke.

Isiliye network wali cheezein duniya ki sabse badi cheezein banti hain. Aur isiliye woh shuruaat mein sabse zyada marti hain, kyunki jab tak woh chakkar chalu na ho, unke paas kuch bhi nahi hota.

Chapter 4.2 [SPINE]

Har machine ko ek address

SAMVAAD

Madhav: Tum ek chitthi bhejna chahte ho. Kya kya chahiye?

Kabir: Chitthi, aur pata.

Madhav: Pate mein kya hona chahiye?

Kabir: Aisa kuch jo duniya mein sirf ek jagah bataye.

Madhav: Aur agar do gharon ka ek hi pata ho?

Kabir: Toh chitthi galat jagah ja sakti hai.

Madhav: Toh pehla नियम: **ek pata, ek jagah**. Ab machines ke liye pata banao. Kaise banaoge?

Kabir: Har machine ko ek number de do.

Madhav: Kitne bade number?

Kabir: Jitni machinein hain.

Madhav: 1970 ke dashak mein jab yeh tay hua, tab duniya mein kuch sau computer the. Unhone chaar byte rakhe. Chaar byte se kitne pate?

Kabir: Chaar byte matlab battis bit. Do ko battis baar guna... lagbhag chaar arab.

Madhav: Us waqt chaar arab kaisa laga hoga?

Kabir: Bahut zyada. Kabhi khatam na hone wala.

Madhav: Aaj duniya mein kitne phone hain?

Kabir: Chaar arab se zyada. Aur laptop, server, TV, ghadi bhi.

Madhav: Toh?

Kabir: Pate khatam ho gaye.

Madhav: 2011 ke aas-paas hi lagbhag khatam ho gaye the. Ab batao kya karoge.

Kabir: Bade pate bana do.

Madhav: Bana diye gaye, 1998 mein hi. Solah byte wale. Unse itne pate bante hain ki zameen ke har ret ke kan ko bhi arbon pate mil jaayein.

Kabir: Toh problem hal ho gayi.

Madhav: Nahi hui. Aaj bhi duniya ka bada hissa purane pate hi istemaal karta hai.

Kabir: Kyun? Naye toh behtar hain.

Madhav: Yaad karo. Pichhle chapter mein tumne kya seekha tha ki behtar cheez ke saath kya hota hai?

Kabir: Jo pehle sab jagah pahunch gaya use hatana namumkin hai. Sabko ek saath badalna padega.

Madhav: Toh phir kaam kaise chal raha hai? Chaar arab pate hain aur bees arab cheezein judi hui hain.

Kabir: Kuch batwara ho raha hoga.

Madhav: Socho. Tumhare ghar mein kitni cheezein internet se judi hain?

Kabir: Do phone, ek laptop, TV.

Madhav: Aur ghar ka ek hi connection hai. Toh chaaron ke paas alag public pate hain ya ek?

Kabir: Shayad ek.

Madhav: Ek hi hai. Ghar ke andar sabke apne chhote pate hain, jo sirf ghar ke andar chalte hain. Bahar jaane wali har cheez ek hi pate se jaati hai, aur ek machine yaad rakhti hai ki kaunsa jawab kis andar wale ka hai.

Kabir: Toh yeh building ke flat jaisa hai. Building ka ek pata, andar flat number.

Madhav: Bilkul. Aur postman ko flat number nahi pata hota. Woh building tak laata hai, aur andar koi baant deta hai.

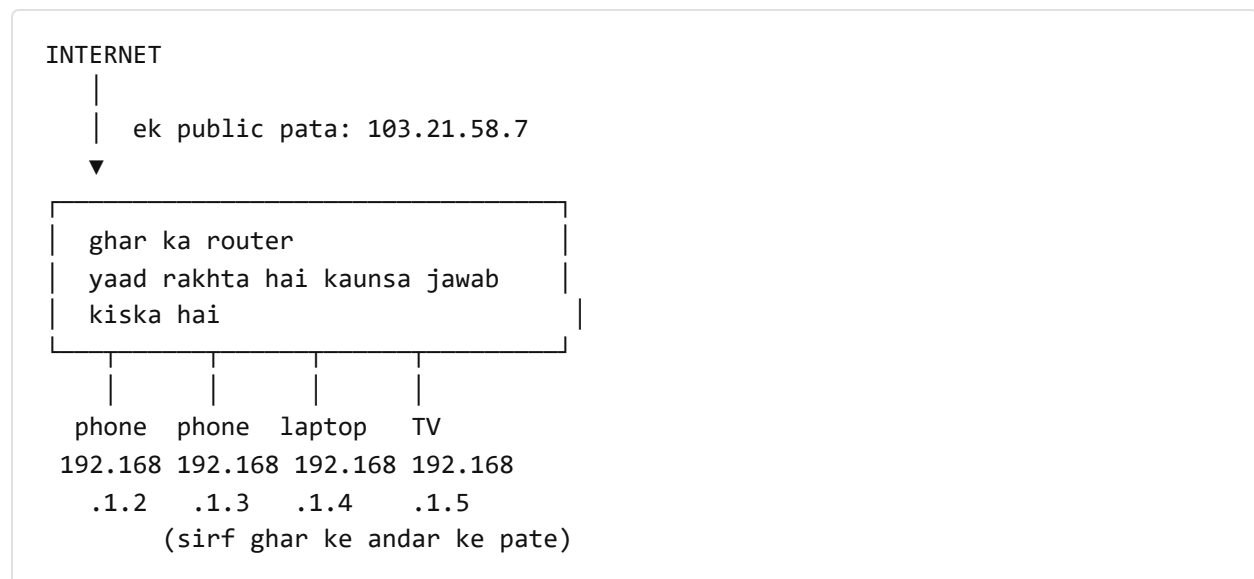
NAAM

Machine ke us pate ko **IP address** kehte hain.

Purane wale, chaar byte ke, **IPv4** hain. Woh aise dikhte hain: `142.250.183.14`. Chaar hisse, har ek 0 se 255 tak, kyunki har hissa ek byte hai.

Naye wale, solah byte ke, **IPv6** hain.

Aur woh ghar wali cheez, jo ek public pate ke peeche kai andar wale pate chhupa deti hai, use **NAT** kehte hain.



Ab ek baat jo iska seedha nateeja hai aur bahut kuch samjhati hai:

Aapka phone bahar se pahuncha nahi ja sakta. Woh bahar baat kar sakta hai, aur jawab wapas aa sakta hai. Lekin koi ajnabi aapke phone pe seedha chitthi nahi bhej sakta, kyunki uske paas jo pata hai woh building ka hai, flat ka nahi.

Isiliye har cheez ko ek beech wale server se hokar jaana padta hai. Aapka message doosre phone pe seedha nahi jaata. Woh pehle kisi aise machine tak jaata hai jiska apna pakka pata hai, aur woh use aage bhejti hai.

Yeh ek suraksha ki tarah bhi kaam karta hai aur ek majboori bhi hai. Aur yeh bahut bada karobaari nateeja rakhta hai: **beech mein hamesha koi baithta hai, aur jo beech mein baithta hai woh sab kuch dekhta hai.**

ASLI DUNIYA SE EK EXAMPLE

1981 mein IPv4 tay hua tha, chaar arab pate ke saath. Us waqt internet pe kuch sau machinein thin. Chaar arab kabhi khatam na hone wali sankhya lagti thi.

31 saal baad, 2011 mein, sabse upar wale bandware ne aakhri bade blocks baant diye. Pate khatam ho gaye.

IPv6, jo 1998 mein hi taiyaar tha, aaj 26 saal baad bhi duniya ke lagbhag aadhe traffic tak hi pahuncha hai.

Yeh soch ne laayak baat hai. Ek behtar cheez, poori tarah taiyaar, samasya saaf, aur phir bhi 26 saal mein aadhi duniya. Wajah technical nahi hai. Wajah yeh hai ki sabko ek saath badalna padta hai, aur kisi ek ke badalne ka koi faayda nahi hota.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki IP address aapki jagah batata hai.**

Filmon mein “IP trace kar liya, ghar ka pata mil gaya” dikhaya jaata hai. Asliyat kaafi alag hai.

IP address batata hai ki aap kis network se jude ho, na ki aap kahan ho. Woh aapke internet dene wali company ka pata hota hai, jo aksar aapke sheher ya usse bade ilaake tak hi pahunchata hai. Aur mobile pe woh poore rajya ka kuch bhi ho sakta hai.

Aur NAT ki wajah se ek hi pate ke peeche kai sau log ho sakte hain.

Yeh galti tempting isliye hai ki “address” shabd hi ghar ka pata sujhata hai. Woh ghar ka pata nahi hai. Woh **network mein jagah** hai, zameen pe jagah nahi.

MAP PE

PATA	ek pata, ek jagah public bahar, private andar	← naya
NETWORK	value judav mein hai	
OS, PROCESS, LANGUAGE		
CPU, RAM/DISK, PROGRAM, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Aapke ghar mein chaar cheezein internet se judi hain aur ek hi public pata hai. Do phone ek saath ek hi website kholte hain. Jawab sahi phone pe kaise pahunchta hai?

Jawab: Router ek chhoti list rakhta hai.

Jab pehla phone bahar bhejta hai, toh router likh leta hai: “yeh baat-cheet andar wale phone 192.168.1.2 ki hai.” Doosre phone ke liye alag entry.

Jawab aata hai toh router list dekh kar sahi phone ko de deta hai.

Yeh Chapter 3.4 wali baat hai dobara: **jab ek cheez hai aur maangne wale kai, toh koi beech mein baith kar baantta hai, aur use yaad rakhna padta hai ki kaun kaun hai.**

2. (samajh check) IPv6 1998 se taiyaar hai aur behtar hai. 26 saal mein bhi poori duniya nahi aayi. Iss se aap kya seekhte ho?

Jawab: Ki technology ki duniya mein behtar hona kaafi nahi hota.

Badlav ki keemat sabko milkar chukani padti hai, lekin faayda tab tak kisi ko nahi milta jab tak sab na badlein. Toh har akela aadmi intezaar karna chunta hai.

Yeh Chapter 4.1 ka network effect hai, ulti shakal mein: jo cheez sabko jodti hai, wahi sabko rok bhi leti hai.

Aur yeh bahut aage tak jaata hai. Chapter 8.5 mein hum poochenge ki technology jeetati kyun hai, aur yeh un teen-chaar wajahon mein se ek hoga.

3. (jodne wala) Chapter 4.1 kehta tha ki value judav mein hai. Ab pata chala ki aapka phone bahar se pahuncha hi nahi ja sakta. Toh phir aapka phone judta kaise hai?

Jawab: Kisi beech wale ke through.

Aapka phone hamesha pehle bahar jaata hai, kisi aise server tak jiska pakka pata hai. Woh server aapke liye sandesh rakhta hai aur aage bhejta hai.

Iska seedha nateeja: **do phone kabhi seedha baat nahi karte.** Beech mein hamesha koi hota hai.

Aur jo beech mein hai, woh sab dekhta hai, sab rok sakta hai, aur sab band kar sakta hai. Yahan se do baatein nikalti hain jo aage aayengi: encryption kyun zaroori hai (Chapter 4.6), aur beech wali jagah itni keemti kyun hai (Part 6).

Chapter 4.3 [DEPTH]

Message ko tukdon mein todna

DEPTH chapter. Iske baad “internet slow hai,” “call kat gayi,” aur “video ruk-ruk kar chal raha hai” ek hi tasveer ke hisse ban jaate hain.

SAMVAAD

Madhav: Ek gaon se doosre gaon tak ek hi sadak hai. Sau log apna saaman bhejna chahte hain. Ek aadmi ka saaman bahut bada hai, poori sadak ghera leta hai, aur pahunchne mein do din lagenge.

Kabir: Baaki ninyanve log do din intezaar karenge.

Madhav: Achha lagta hai?

Kabir: Nahi. Chhote saaman wale pehle nikal jaane chahiye.

Madhav: Toh kya karoge?

Kabir: Bade saaman ko tukdon mein tod dunga. Har tukda alag bhejunga, aur beech mein doosron ka saaman bhi ja sakta hai.

Madhav: Ab problem yeh hai ki tukde alag-alag pahunchenge. Doosri taraf wale ko kaise pata chalega ki kya jodna hai?

Kabir: Har tukde pe likh dunga: kiska hai, aur kaunse number ka tukda hai.

Madhav: Aur agar tukda number 5 pehle pahunch gaya aur number 3 baad mein?

Kabir: Woh number dekh kar sahi order mein rakh lega.

Madhav: Aur agar tukda number 3 raaste mein kho gaya?

Kabir: Toh... woh dekhega ki 3 nahi aaya, aur dobara maangega.

Madhav: Kaise maangega? Use kaise pata chalega ki tukda kho gaya aur dheere nahi aa raha?

Kabir: Kuch der intezaar karega. Nahi aaya toh maan lega ki gaya.

Madhav: Ab ulta socho. Bhejne wale ko kaise pata chalega ki tukda pahuncha?

Kabir: Lene wala bataye ki mil gaya.

Madhav: Har tukde ke liye?

Kabir: Haan.

Madhav: Toh ab har tukde ke liye do baar safar. Ek jaane ka, ek jawab ka. Kya yeh dheema nahi ho jaayega?

Kabir: Ho jaayega. Toh... kuch tukde ek saath bhej doon, phir ek hi jawab maangun.

Madhav: Kitne ek saath?

Kabir: Pata nahi. Zyada bheje toh sadak jaam ho jaayegi.

Madhav: Toh kaise tay karoge?

Kabir: Dheere se badhaunga. Kaam chal raha hai toh aur bhejo. Kuch khone lage toh kam kar do.

Madhav: Tumne abhi internet ka sabse zaroori niyam khud bana diya. Ab aakhri sawal. Ek video call chal rahi hai. Ek tukda kho jaata hai. Kya karoge, dobara maangoge?

Kabir: Haan.

Madhav: Us tukde mein woh awaaz thi jo aadhe second pehle boli gayi thi. Dobara maangne mein aur aadha second lagega.

Kabir: Toh woh purani ho chuki hogi. Bekaar.

Madhav: Toh?

Kabir: Toh video call mein kho gaya toh chhod do. Aage badho.

Madhav: Aur file download mein?

Kabir: Wahan ek bhi tukda nahi chhod sakte. File toot jaayegi.

Madhav: Bas. Do bilkul alag zaroortein, aur isiliye do alag tareeke.

NAAM

Har chhote tukde ko **packet** kehte hain. Aam taur pe ek packet 1,500 byte se chhota hota hai.

Har packet pe likha hota hai:

kahan se aaya	(bhejne wale ka pata)
kahan jaana hai	(lene wale ka pata)
kaunse number ka	(taaki order banaya ja sake)
kis baat-cheet ka	(taaki alag baaton ke tukde na milein)
asli saaman	(data)

Aur woh do tareeke jo Kabir ne nikaale:

TCP	har tukda pahunchna zaroori
	└ lene wala har baar batata hai ki mila
	└ kho gaya toh dobara bheja jaata hai
	└ order theek kiya jaata hai
	└ file, message, website, bank
UDP	pahunche toh theek, na pahunche toh chalo aage
	└ koi jawab nahi maanga jaata
	└ kho gaya toh gaya
	└ tez, kyunki intezaar nahi
	└ video call, live stream, game

Aur woh aakhri cheez, dheere se badhana aur girne pe kam karna, use **congestion control** kehte hain.

Yeh internet ka sabse kam dikhne wala aur sabse zaroori hissa hai. Kisi ne kabhi tay nahi kiya ki kis machine ko kitni sadak milegi. Har machine khud koshish karti hai, aur girte hi peeche hat jaati hai. Sab milkar sadak baant lete hain, bina kisi manager ke.

Isiliye jab bahut log ek saath internet istemaal karte hain, toh sabko thoda thoda dheema milta hai, kisi ek ko poora band nahi milta.

ASLI DUNIYA SE EK EXAMPLE

Aap Netflix pe video dekhte ho. Woh kabhi kabhi dhundhla ho jaata hai aur phir saaf ho jaata hai. Ruk-ruk kar nahi chalta, sirf quality girti hai.

Woh jaan-boojh ke hai. App har waqt naap raha hai ki tukde kitni jaldi aa rahe hain. Jab sadak bhari hoti hai, toh woh chhoti quality wala video maangna shuru kar deta hai, jiske tukde chhote hote hain.

Aap dhundhla dekh lete ho. Rukna aapko bura lagta hai, dhundhlapan utna nahi.

Aur video call mein bilkul ulta chunav hota hai. Wahan woh tukde chhodta hai, quality nahi girata, kyunki der se aayi awaaz kisi kaam ki nahi.

Do alag product, ek hi problem, do ulte hal. Aur dono hal usi chunav se aaye jo Kabir ne khud kar liya tha: **kya purana data kaam ka hai ya nahi.**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **"internet slow hai" ko ek hi cheez samajhna.**

Do bilkul alag cheezein hoti hain, aur unke ilaaj bhi alag hain:

BANDWIDTH	ek second mein kitna data aa sakta hai (sadak kitni chaudi hai)
LATENCY	ek tukde ko jaane aur wapas aane mein kitna waqt (sadak kitni lambi hai)

Film download karne mein bandwidth mayne rakhti hai. Video call aur game mein latency mayne rakhti hai.

Isiliye ek bahut tez connection pe bhi call kharaab ho sakti hai, aur ek dheeme connection pe bhi theek chal sakti hai. Aur isiliye satellite internet mein bandwidth bahut ho sakti hai lekin game nahi khela ja sakta: signal ko upar jaakar wapas aana padta hai.

Yeh galti tempting isliye hai ki dono ka nateeja ek jaisa dikhta hai. Cheez dheere ho gayi. Lekin wajah alag hai, aur isliye hal bhi alag hai.

Aur yeh Chapter 2.3 wali baat hai, badi shakal mein: **doori speed se zyada mayne rakhti hai.**

MAP PE

PACKET	bade ko tukdon mein todo	← naya
TCP / UDP	pakka, ya tez	← naya
BANDWIDTH vs LATENCY		← naya
PATA, NETWORK		
OS, PROCESS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Ek online game mein aapka character ruk-ruk kar chalta hai. Aapka connection 500 Mbps ka hai. Kya problem hai?

Jawab: Bandwidth nahi, latency. Sadak chaudi hai lekin lambi.

Game mein har second kai baar chhoti si jaankari aani-jaani hoti hai, aur har baar jawab ka intezaar hota hai. Agar ek chakkar mein 200 millisecond lagte hain, toh chaudi sadak se koi faayda nahi.

Isiliye gamers ping dekhte hain, speed nahi.

Aur isiliye game ke server aapke sheher ke paas rakhe jaate hain. Woh doori kam karne ke liye hai, taakat badhane ke liye nahi.

2. (samajh check) Aap ek file download kar rahe ho aur beech mein internet 5 second ke liye chala jaata hai. File toot jaayegi?

Jawab: Nahi. TCP dobara maang lega.

Bhejne wale ko jawab nahi milega, toh woh un tukdon ko dobara bhejega. Download ruk kar phir chalu ho jaayega.

Ab wahi 5 second ek video call mein hote toh woh awaaz hamesha ke liye chali jaati, kyunki UDP dobara nahi maangta.

Ek hi ghatna, do alag nateeje, aur wajah sirf yeh ki kaunsa tareeka chuna gaya tha.

3. (jodne wala) Chapter 4.1 kehta tha ki network mein koi manager nahi hai. Ab pata chala ki sadak bhi koi nahi baant raha. Yeh kaise chal jaata hai?

Jawab: Kyunki har machine khud peeche hat jaati hai jab use dikhta hai ki tukde kho rahe hain.

Koi upar se nahi keh raha ki tum itna hi bhejo. Har ek badhata hai jab tak dard na ho, phir peeche hat jaata hai.

Aur yeh isiliye chalta hai kyunki lagbhag sab log wahi niyam maan rahe hain. Agar koi jaan-boojh ke peeche na hate, toh woh zyada le lega.

Yeh internet ke baare mein sabse zaroori baat hai, aur Chapter 4.7 mein hum ise poora karenge: **internet ek machine nahi hai. Woh ek sehmati hai.**

Chapter 4.4 [SPINE]

Rules pe pehle se agree karna

SAMVAAD

Madhav: Ek packet pahunch gaya. Usmein numbers hain. Lene wali machine ab kya kare?

Kabir: Padhe.

Madhav: Kaise? Chapter 1.4 yaad karo.

Kabir: Andar sirf numbers hain, matlab bahar se aata hai. Toh use pata hona chahiye ki inhe kaise padhna hai.

Madhav: Aur use kaise pata chalega?

Kabir: Bhejne wala bata de.

Madhav: Kis shakal mein batayega?

Kabir: Numbers mein hi.

Madhav: Aur un numbers ko kaise padhega?

Kabir: Hmm. Yeh toh ghoom raha hai.

Madhav: Bilkul ghoom raha hai. Toh iska ek hi hal hai. Batao.

Kabir: Dono ko pehle se pata hona chahiye. Baat shuru hone se pehle.

Madhav: Bas. Ek aisa niyam jo dono taraf pehle se maujood ho, aur jise koi bhi baat-cheet mein tay na kiya jaaye. Ab woh niyam kaisa dikhega? Ek chitthi socho. Usmein kya hoga?

Kabir: Upar kuch jaankari, phir asli baat.

Madhav: Aur padhne wale ko kaise pata chalega ki jaankari khatam hui aur asli baat shuru?

Kabir: Beech mein koi nishaan hoga.

Madhav: Aur asli baat kitni lambi hai, yeh kaise pata chalega?

Kabir: Ya toh upar likha ho, ya aakhir mein koi nishaan ho.

Madhav: Tumne ek poora protocol bana liya. Ab ek asli wala dekhte hain. Tum ek website kholte ho. Tumhara browser server se kya kehta hai?

Kabir: "Mujhe yeh page do."

Madhav: Aur usmein kya kya likhna padega?

Kabir: Kaunsa page. Aur shayad ki main kaun hoon.

Madhav: Aur kya batana chahoge?

Kabir: Ki main kaunsi bhasha samajhta hoon. Ki mera phone chhota hai ya bada.

Madhav: Aur server kya wapas bhejega?

Kabir: Page. Aur yeh bhi ki mila ya nahi mila.

Madhav: Kyun batayega ki mila ya nahi? Page bhej hi diya, kaafi nahi?

Kabir: Kyunki page na milna bhi ek jawab hai. Aur galti bhi ho sakti hai.

Madhav: Toh tumhe alag-alag jawab chahiye: mil gaya, nahi mila, tumhe ijaazat nahi hai, server kharab hai.

Kabir: Aur unhe numbers de dete honge, taaki chhota rahe.

Madhav: Bilkul. Ab aakhri sawal, aur yeh sabse zaroori hai. Yeh niyam kisne banaya?

Kabir: Kisi company ne?

Madhav: Agar ek company banati, toh baaki kyun maanteen?

Kabir: Nahi maanteen. Sabko milkar tay karna padega.

Madhav: Aur ek baar tay ho jaaye toh?

Kabir: Toh use badalna bahut mushkil hoga. Sabko ek saath badalna padega.

Madhav: Wahi baat jo IPv6 mein dekhi thi. Ab tum internet ka asli dhaancha samajhne lage ho.

NAAM

Aise pehle se tay niyam ko **protocol** kehte hain.

Protocol tay karta hai:

```
sandesh ki shakal kya hogi
kaunsa hissa jaankari hai, kaunsa asli baat
kaun pehle boleگا
jawab kaise dikheگا
galti kaise batayi jaayegi
```

Web ka protocol **HTTP** hai. Uska ek request kuch aisa dikhta hai:

```
GET /photo.jpg          ← kya chahiye
Host: example.com       ← kis jagah se
Accept-Language: hi     ← kaunsi bhasha
                        ← khaali line: jaankari khatam
```

Aur jawab:

```
200 mil gaya
404 woh cheez hai hi nahi
403 hai, lekin aapko ijaazat nahi
500 server ke andar kuch toot gaya
```

Ab woh baat jo iss chapter ko aage tak le jaati hai.

Protocol ke bina do ajnabi baat nahi kar sakte. Aur protocol koi bhej nahi sakta, kyunki bhejne ke liye bhi protocol chahiye.

Isliye har protocol pehle se maujood hona hi padta hai. Aur isliye protocol badalna lagbhag namumkin hai, kyunki sabko ek saath badalna padega.

Aur yahin se ek badi karobaari baat nikalti hai: **jo protocol tay karta hai, woh us poore ilaake ki shakal tay kar deta hai, hamesha ke liye.** Woh us cheez ka maalik nahi hota, lekin uske baad har koi uske hisaab se hi chalta hai.

ASLI DUNIYA SE EK EXAMPLE

HTTP 1991 mein banaya gaya, ek aadmi ne, ek research lab mein: Tim Berners-Lee.

Usne aur uske saath ke logon ne teen cheezein banayin: page likhne ka tareeka (HTML), page maangne ka tareeka (HTTP), aur page ka pata (URL).

Phir unhone woh sab kuch **kisi ko bhi, muft** de diya. CERN ne 1993 mein saaf likh kar de diya ki koi bhi ise bina paise ke istemaal kar sakta hai.

Ab Chapter 3.6 wale chaar wajahon pe lagao. Agar unhone iska paisa maanga hota, toh kya hota?

Log doosra tareeka bana lete, ya sab alag-alag tareeke banate aur koi kisi se baat na kar pata. Web ki poori value hi iss baat mein thi ki sab ek hi niyam maanein. Paisa maangna us value ko hi maar deta.

Toh muft dena daan nahi tha. Woh ek hi tareeka tha jisse woh cheez kaam kar sakti thi.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **”internet” aur “web” ko ek hi cheez samajhna.**

Internet woh jaal hai: pate, packet, taar. Web usme se ek cheez hai jo uske upar chalti hai, HTTP ke through.

Email alag protocol pe chalta hai. Video call alag pe. Aapke phone ke app aksar apne khud ke tareeke istemaal karte hain.

Yaani web internet ka ek kirayedaar hai, maalik nahi.

Yeh galti tempting isliye hai ki zyadatar log internet ko browser ke through hi dekhte hain, toh dono ek hi lagte hain.

Iska seedha nateeja: log samajh nahi paate ki ek app bina browser ke kaise chalta hai, ya ki internet band kiye bina sirf ek cheez kaise band ki ja sakti hai. Alag protocol, alag raaste, alag taale.

MAP PE

PROTOCOL	pehle se tay niyam badalna lagbhag namumkin	← naya
PACKET, TCP/UDP, PATA, NETWORK		
OS, PROCESS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Ek website kholi aur “404” aaya. Kya hua, aur kya yeh aapki galti hai ya server ki?

Jawab: Server ne kaha ki us pate pe koi cheez hai hi nahi.

Server theek chal raha hai. Usne aapki baat samajh li aur jawab bhi de diya. Bas woh cheez wahan nahi hai.

Yeh 500 se alag hai, jismein server khud toot gaya hota hai. Aur 403 se bhi alag, jismein cheez hai lekin aapko ijaazat nahi.

Yeh farak zaroori hai: **”nahi mila” aur “main toot gaya” do bilkul alag baatein hain**, aur unke ilaaj bhi alag hain.

2. (samajh check) Do log alag-alag desh mein hain, alag company ke phone, alag internet company. Phir bhi unka message pahunch jaata hai. Kaise?

Jawab: Kyunki dono taraf wahi protocol pehle se maujood hai.

Kisi ne beech mein tay nahi kiya. Woh niyam saalon pehle likhe gaye the aur dono ke phone unhi ke hisaab se bane hain.

Yeh internet ka sabse bada karnama hai: **koi milkar tay nahi karta, sab pehle se tay hai.**

Aur isiliye internet ka koi maalik nahi hai. Uske paas ek sarkar nahi hai, sirf kuch likhe hue kagaz hain jinhe sab maante hain.

3. (jodne wala) Chapter 3.6 mein hum ne poocha tha ki free cheezein kyun exist karti hain. HTTP ka muft hona un chaar wajahon mein se kaunsi hai?

Jawab: Teesri, aur sabse teekhi shakal mein: **neeche wali parat pe kabza.**

Lekin ek farak ke saath. Yahan kisi ne upar bechne ke liye neeche free nahi kiya. Yahan cheez ki value hi iss baat mein thi ki sab ek hi niyam maanein.

Toh yeh ek nayi baat sikhata hai: **kuch cheezein sirf tab kaam karti hain jab sab unhe maanein. Aisi cheezon ka daam maangna unhe maar dena hai.**

Yeh baat protocol, bhasha, naap-tol aur paise pe laagu hoti hai. Aur yeh yaad rakhna zaroori hai jab aap kuch banane ki soch rahe ho: kya aapki cheez ki value akele mein hai, ya sirf sehmati mein?

Chapter 4.5 [SPINE]

Numbers ki jagah naam

SAMVAAD

Madhav: Har machine ka ek pata hai, jaise `142.250.183.14`. Ab tumhe koi website kholni ho toh?

Kabir: Woh number yaad rakhna padega.

Madhav: Tumhe kitne numbers yaad hain?

Kabir: Do teen phone number. Bas.

Madhav: Aur tum roz kitni websites khologe?

Kabir: Bees, tees.

Madhav: Toh?

Kabir: Naam chahiye. Number ki jagah naam.

Madhav: Toh ek list bana lo. Naam ke saamne number.

Kabir: Theek hai.

Madhav: Woh list rakhoge kahan?

Kabir: Apne phone mein.

Madhav: Duniya mein lagbhag 35 crore naam hain. Aur roz naye jud rahe hain. Sabki list tumhare phone mein?

Kabir: Nahi. Bahut badi ho jaayegi. Aur purani bhi ho jaayegi.

Madhav: Toh?

Kabir: Ek jagah list rakh do. Sab wahan se poochein.

Madhav: Duniya ke saare sawal ek machine se? Ek second mein karodon sawal.

Kabir: Woh baith jaayegi.

Madhav: Aur agar woh ek machine band ho jaaye?

Kabir: Toh poori internet band. Naam kaam hi nahi karenge.

Madhav: Toh ek jagah bhi nahi chalega, har jagah bhi nahi chalega. Beech ka raasta socho.

Kabir: List ko baant do. Har machine ka apna hissa.

Madhav: Kaise baantoge? Naam toh koi bhi ho sakta hai.

Kabir: Naam mein hi kuch dhaancha daal do. Jaise pate mein hota hai: desh, sheher, gali, ghar.

Madhav: Peeche se aage, ya aage se peeche?

Kabir: Pehle bada, phir chhota. Jaise `.com` bada, uske andar `google`, uske andar `mail`.

Madhav: Bas. Ab har hisse ka apna rakhwala ho sakta hai. Jo `.com` sambhalta hai, use har `.com` ka pata nahi rakhna padta. Use sirf itna rakhna hai ki `google.com` kaun sambhalta hai.

Kabir: Aur woh phir aage bata dega.

Madhav: Toh ek sawal ke liye kitne kadam?

Kabir: Do teen. Har kadam pe list chhoti hoti jaayegi.

Madhav: Ab aakhri problem. Woh do-teen kadam har baar? Har website ke liye, har baar?

Kabir: Nahi. Jawab yaad rakh lo kuch der ke liye.

Madhav: Kitni der?

Kabir: Kuch ghante? Kyunki pate kabhi kabhi badalte hain.

Madhav: Aur agar tumne bees ghante rakh liya aur pata beech mein badal gaya?

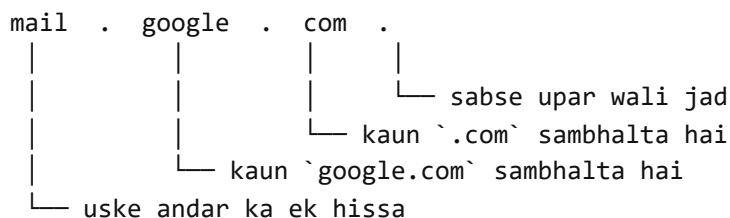
Kabir: Toh main galat jagah jaunga jab tak purani jaankari na hate.

Madhav: Isiliye jab koi website apna server badalti hai, toh kuch logon ke liye woh turant chalti hai aur kuch ke liye kuch ghante purani jagah pe hi jaati rehti hai.

NAAM

Us naam wali system ka naam hai **DNS**, yaani naam ko pate mein badalne wala jaal.

Naam ka dhaancha, peeche se aage:



Ek sawal ka safar:

1. tumhara phone poochta hai: mail.google.com kahan hai?
2. koi nahi jaanta toh jad se poocha jaata hai:
"`.com` kaun sambhaltta hai?"
3. `.com` wale se: "google.com kaun sambhaltta hai?"
4. google wale se: "mail.google.com ka pata kya hai?"
5. pata mil gaya. Ab yaad rakh liya jaata hai, kuch ghante ke liye.

Us yaad rakhne ko **cache** kehte hain, aur us waqt ki seema ko **TTL**.

Ab woh baat jo iss chapter ko bahut aage tak le jaati hai:

Naam ka maalik alag hota hai, machine ka maalik alag.

Aap ek naam kirye pe lete ho, saal ke hisaab se. Machine kahin aur hai. Agar aap machine badal do toh naam wahi rehta hai, bas uska ishara nayi jagah kar dete ho.

Aur iska ulta bhi sach hai, aur woh khatarnak hai: **jo naam ka rakhwala hai, woh use kisi aur jagah bhi bhej sakta hai.** Naam ki poori taakat ek list mein likhi ek line pe tiki hai, aur woh line kisi aur ke paas hai.

ASLI DUNIYA SE EK EXAMPLE

2021 mein Facebook, Instagram aur WhatsApp lagbhag chhe ghante ke liye poori duniya se gayab ho gaye.

Server chal rahe the. Data theek tha. Lekin ek galat badlav ki wajah se unke pate baantne wali jaankari internet se hat gayi. Naam ka jawab dene ka koi raasta hi nahi bacha.

Aur uska ek nateeja aur nikla jo dilchasp hai: unke apne office ke darwaze bhi unhi systems se chalte the, jo unhi naamon pe khade the. Log andar nahi ja paaye us jagah ko theek karne, jisne unhe bahar kar diya tha.

Yeh Chapter 0.4 ki baat hai, ek dum saaf: **jo cheez achhi tarah chalti hai woh dikhti nahi. Woh sirf toot ne pe dikhti hai.** Aur us din duniya ne dekha ki naam poore internet ki neev mein baitha hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki website ka naam khareedne se aap uske maalik ban jaate ho.**

Aap naam khareedte nahi. Aap use kirye pe lete ho, aksar saal ke hisaab se. Kiraya rukta hai toh naam kisi aur ko mil sakta hai.

Aur iska ek gehra nateeja hai. Poori company ka naam, uske saare link, uska email, uske customer ka bharosa, sab ek entry pe tike hote hain jiske aap maalik nahi hain.

Isliye purani companies apna naam bees-tees saal ke liye ek saath le leti hain. Woh kanjoosi ka ulta hai, aur woh isliye hai ki unhe pata hai ki neeche kya hai.

Yeh galti tempting isliye hai ki naam permanent lagta hai. Woh aapke card pe likha hai, aapke sign board pe hai. Lekin woh sirf ek list mein ek line hai, aur list kisi aur ki hai.

MAP PE

NAAM	insaan naam yaad rakhta, machine pata istemaal	← naya
CACHE	jawab kuch der yaad rakho	← naya
PROTOCOL, PACKET, PATA, NETWORK		
OS, PROCESS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Ek company apna server badalti hai. Kuch logon ke liye website turant nayi jagah khulti hai, kuch ke liye do ghante tak purani. Kyun?

Jawab: Cache ki wajah se.

Jinhone haal hi mein woh naam poocha tha, unke paas purana jawab abhi bhi pada hai, aur woh tab tak rahega jab tak uska waqt khatam na ho.

Isiliye jab koi company jagah badalne wali hoti hai, toh woh pehle us waqt ko chhota kar deti hai, kuch din pehle. Phir badlav karti hai. Phir waqt wapas bada kar deti hai.

Yeh cache ka aam sauda hai, jo aage Part 6 mein poora khulega: **cache tez banata hai aur purana bhi rakhta hai. Dono ek hi cheez ke do hisse hain.**

2. (samajh check) DNS ko ek hi jagah kyun nahi rakha gaya, jab woh zyada simple hota?

Jawab: Do wajahein, aur dono Chapter 4.1 se aati hain.

Ek: karodon sawal ek second mein. Koi ek machine woh nahi utha sakti.

Do: agar woh ek machine gir jaaye toh poora internet naam ke bina reh jaayega. Ek jagah rakhne ka matlab hai ek jagah todne se sab tootega.

Isiliye use pedh ki tarah baanta gaya, jahan har shakha apna hissa sambhalti hai.

Yeh dhaancha aage phir milega. **Jab kuch bahut bada ho jaaye, toh use baanto, aur baantne ka tareeka naam mein hi daal do.**

3. (jodne wala) Chapter 4.4 kehta tha ki jo protocol tay karta hai woh us ilaake ki shakal tay kar deta hai. DNS mein woh taakat kiske paas hai?

Jawab: Un logon ke paas jo sabse upar wali jad aur bade hisse sambhalte hain.

Woh yeh tay kar sakte hain ki naye tarah ke naam bane ya nahi, aur unke niyam kya honge.

Aur theory mein, woh kisi naam ko hata bhi sakte hain. Aisa hua bhi hai.

Yeh Chapter 4.8 ka rasta hai. Internet ka “koi maalik nahi hai” bilkul sach hai, aur phir bhi kuch jagahein aisi hain jahan taakat sach mein baithi hai. Naam un mein se sabse badi jagah hai.

Chapter 4.6 [SPINE]

Public road pe private baat

SAMVAAD

Madhav: Tumhara packet ghar se nikalta hai aur server tak pahunchta hai. Beech mein kitni machinein hoti hain?

Kabir: Pata nahi. Kuch.

Madhav: Aam taur pe das se bees. Aapki internet company ki, phir beech wale raaston ki, phir manzil wale ki. Kya har ek us packet ko padh sakti hai?

Kabir: Haan. Woh toh usmein se guzar raha hai.

Madhav: Toh tum jo bank ka password bhejte ho, woh bees ajnabi machinein dekh sakti hain.

Kabir: Toh use chhupana padega.

Madhav: Kaise?

Kabir: Ghuma do. Har akshar ko kuch aage khiska do. A ko D, B ko E.

Madhav: Ab lene wala use kaise kholega?

Kabir: Peeche khiskayega. Use pata hona chahiye ki kitna.

Madhav: Woh “kitna” hi chaabi hui. Ab problem batao.

Kabir: Use woh chaabi kaise pata chalegi?

Madhav: Bhej do.

Kabir: Toh beech wale bhi dekh lenge.

Madhav: Toh chaabi ko chhupa kar bhejo.

Kabir: Usko chhupane ke liye ek aur chaabi chahiye. Yeh phir ghoom raha hai.

Madhav: Yehi asli problem hai, aur do hazaar saal se yehi problem thi. Chaabi hamesha pehle se milni padti thi, aamne-saamne. Ab batao, kya aap us bank se kabhi mile ho jiska app aap chalte ho?

Kabir: Nahi.

Madhav: Toh 1970 tak yeh mumkin hi nahi tha. Ab main tumhe ek ajeeb sawal deta hoon. Kya aisa taala ho sakta hai jise koi bhi band kar sake, lekin khol sirf ek aadmi paaye?

Kabir: Hmm. Ek dabba jismein chitthi daali ja sakti ho lekin nikaali na ja sake. Post box.

Madhav: Bilkul. Post box ka muh sabke liye khula hai. Chaabi sirf postman ke paas.

Kabir: Toh woh bank sabko apna post box de de.

Madhav: Aur woh post box public ho, sabke saamne?

Kabir: Haan. Usse koi nuksaan nahi. Usmein daal hi sakte hain, nikaal nahi sakte.

Madhav: Toh ab bank ko chaabi bhejne ki zaroorat hai?

Kabir: Nahi. Woh apna post box sabko de sakta hai. Main usmein daal dunga, sirf woh kholega.

Madhav: Tumne abhi woh cheez nikaali jisne poore internet ka vyapar mumkin banaya. Ek aakhri sawal. Main tumhe ek post box deta hoon aur kehta hoon ki main bank hoon. Tumhe kaise pata ki main sach mein bank hoon?

Kabir: Pata nahi chalega.

Madhav: Toh?

Kabir: Koi teesra ho jise dono jaante hon, aur woh bataye ki yeh sach mein bank hai.

Madhav: Aur us teesre pe kaun bharosa karega?

Kabir: Woh pehle se sabke paas hona chahiye.

Madhav: Jaise protocol. Pehle se maujood, kyunki bheja nahi ja sakta.

NAAM

Wo do tarah ki chaabi wali cheez, uska naam hai **public key cryptography**.

PUBLIC KEY	sabko do. Isse band kiya ja sakta hai.
PRIVATE KEY	kisi ko mat do. Sirf isse khulta hai.

Aur woh teesra, jo batata hai ki koi sach mein wahi hai jo keh raha hai, use **certificate authority** kehte hain. Unke naam aapke phone aur browser mein pehle se bhare hote hain, jab woh bana tha.

Poori baat-cheet aise chalti hai:

1. tum server se juda, usne apna public key aur certificate diya
2. tumne certificate check kiya un naamon se jo pehle se tumhare paas the
3. theek nikla, toh tumne ek nayi chaabi banayi, sirf iss baat-cheet ke liye
4. woh nayi chaabi server ke public key se band karke bheji
5. ab dono ke paas ek hi chaabi hai, aur baaki saari baat usi se hoti hai

Aakhri kadam kyun? Kyunki public key wali cheez dheemi hai. Toh use sirf ek baar istemaal karte hain, ek aam chaabi bhejne ke liye. Uske baad tez wali chalti hai.

Yeh poori cheez **TLS** hai, aur jab HTTP uske andar chalta hai toh use **HTTPS** kehte hain. Woh taala jo aapke browser mein dikhta hai, wahi hai.

Aur ab woh baat jo log lagbhag hamesha galat samajhte hain, isliye ise saaf likh raha hoon:

Taala sirf yeh batata hai ki raaste mein koi padh nahi sakta. Woh yeh nahi batata ki doosri taraf wala imaandaar hai.

Ek thagne wali website pe bhi taala ho sakta hai, aur aksar hota hai. Usne bas yeh saabit kiya hai ki woh wahi naam hai jo likha hai. Ki woh naam bharosa laayak hai ya nahi, uska iss taale se koi lena dena nahi.

ASLI DUNIYA SE EK EXAMPLE

2015 tak zyadatar websites bina taale ke chalti thin, kyunki certificate mehnga tha aur lagana mushkil tha. Sirf bank aur bade sites hi lagate the.

Phir ek muft aur apne aap chalne wali certificate dene wali cheez shuru hui, Let's Encrypt. 2016 mein web ka lagbhag aadha traffic taale ke saath tha. Aaj lagbhag 95 pratishat se zyada hai.

Chapter 3.6 ke chaar wajahon pe lagao. Yeh kisne banaya aur kyun? Ise browser banane wali companies ne paisa diya. Unhe iska seedha paisa nahi mila.

Unka faayda yeh tha ki poora web surakshit ho gaya, aur unka product uspe khada hai. Neeche wali parat sabke liye muft kar do, taaki upar wali cheez chal sake.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **taale ko “yeh site surakshit hai” samajhna.**

Log taala dekh kar apna card number daal dete hain, yeh sochkar ki taale ka matlab bharosa hai.

Taale ka matlab sirf itna hai: raaste mein koi nahi padh sakta, aur doosri taraf wala wahi naam hai jo likha hai.

Ek thag ek naam khareed sakta hai, uspe taala laga sakta hai, aur aapka data poori suraksha ke saath apne paas mangwa sakta hai. Encryption ne uska kaam aur asaan hi kiya, kyunki ab aapka internet dene wala bhi nahi dekh sakta.

Yeh galti tempting isliye hai ki browser ne taale ka nishaan hi aisa banaya ki woh “sab theek hai” jaisa lagta hai. Woh sirf ek hi baat keh raha hai, aur log usse teen baatein sun lete hain.

Sahi soch:

taala kehta hai	→ raaste mein koi nahi padh raha
taala kehta hai	→ naam wahi hai jo dikh raha hai
taala NAHI kehta	→ yeh log imaandaar hain
taala NAHI kehta	→ aapka data unke paas surakshit hai

MAP PE

DO CHAABIYAN	ek band kare, ek khole	← naya
BHAROSA	teesra, pehle se maujood	← naya
NAAM, PROTOCOL, PACKET, PATA		
OS, PROCESS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Aap ek public wifi pe ho. Ek site pe taala hai, doosri pe nahi. Wifi ka maalik kya dekh sakta hai?

Jawab: Bina taale wali site pe: sab kuch. Har page, har shabd, har password.

Taale wali site pe: sirf yeh ki aap kis naam se baat kar rahe ho, aur kitna data aa-ja raha hai. Andar ki baat nahi.

Toh taala aapko chhupata nahi hai, woh aapki **baat** chhupata hai. Yeh farak mayne rakhta hai: koi ab bhi jaan sakta hai ki aap kis site pe gaye, bas yeh nahi ki wahan kya kiya.

2. (samajh check) Public key sabko diya jaata hai. Toh koi usse aapke sandesh khol kyun nahi sakta?

Jawab: Kyunki woh chaabi sirf band karne ke liye hai, kholne ke liye nahi.

Post box wale example mein: sabko us box ka muh dikhta hai, aur sab usmein daal sakte hain. Uska muh dekh kar koi andar ki chitthi nahi nikaal sakta.

Yeh ganit se aata hai. Kuch hisaab aise hote hain jinhe ek taraf karna aasan hai aur ulta karna lagbhag namumkin. Do bade numbers guna karna aasan hai. Nateeje se wapas woh do numbers nikaalna bahut mushkil.

3. (jodne wala) Chapter 4.4 kehta tha ki protocol pehle se maujood hona zaroori hai, kyunki use bheja nahi ja sakta. Bharosa uss baat ka kaunsa roop hai?

Jawab: Bilkul wahi roop.

Bharosa bheja nahi ja sakta. Agar main tumhe bhej kar batata ki mujh pe bharosa karo, toh woh bekaar hai, kyunki koi bhi wahi bhej sakta hai.

Isliye bharosa pehle se aana padta hai. Woh browser aur phone ke andar bhara jaata hai jab woh bante hain.

Aur yahan ek gehri baat hai: **poori internet ki suraksha ek chhoti si list pe khadi hai jo aapke device mein pehle se pad gayi thi.** Aapne woh list kabhi nahi dekhi. Aapne kabhi tay nahi kiya ki usmein kaun ho.

Chapter 4.7 [SPINE]

Internet actually hai kya

SAMVAAD

Madhav: Internet kahan hai?

Kabir: Hawa mein? Cloud mein?

Madhav: Tumhara packet hawa se guzarta hai?

Kabir: Nahi. Taar se. Wifi thodi door tak hawa mein, phir taar.

Madhav: Aur samundar paar?

Kabir: Samundar ke neeche taar hote hain.

Madhav: Bilkul. Lagbhag saara internet dhaatu aur kaanch ke taar hain, zameen aur samundar ke neeche. Ab agla sawal. Woh taar kiske hain?

Kabir: Internet company ke?

Madhav: Kaunsi? Duniya mein hazaaron hain.

Kabir: Toh sabke apne apne hisse honge.

Madhav: Aur woh hisse aapas mein jude kaise?

Kabir: Kisi ne unhe joda hoga.

Madhav: Kisne? Koi sarkar? Koi company?

Kabir: Pata nahi.

Madhav: Do companies apne taar aapas mein isliye jodti hain kyunki dono ko faayda hai. Mere users ko tumhare tak pahunchna hai, tumhare mere tak. Woh aapas mein baat karke tay kar leti hain.

Kabir: Toh koi bada plan nahi hai?

Madhav: Koi nahi. Har jodi apna hisaab lagati hai. Poora internet un hazaaron alag-alag samjhauton ka jod hai.

Kabir: Toh phir woh chalta kaise hai? Kisi ko poora naksha toh pata hona chahiye.

Madhav: Kisi ko nahi pata. Ab yeh dhyaan se socho. Tumhara packet nikalta hai aur use bees machinein paar karni hain. Kya pehli machine ko poora raasta pata hai?

Kabir: Hona chahiye.

Madhav: Zaroorat kya hai? Use sirf itna pata ho ki agla kadam kaunsa hai.

Kabir: Aur agli machine ko uska agla kadam.

Madhav: Bas. Koi bhi poora raasta nahi jaanta. Har ek sirf itna jaanta hai ki iss disha ke liye agla kaun hai.

Kabir: Aur agar beech ka koi taar toot jaaye?

Madhav: Kya hoga, socho.

Kabir: Uske padosi ko pata chal jaayega ki woh raasta nahi hai. Woh doosra raasta chun lega.

Madhav: Aur woh baat kaise phailegi?

Kabir: Har ek apne padosi ko batayega.

Madhav: Toh poora jaal khud ko theek kar leta hai, bina kisi upar wale ke. Ab batao, internet band karne ke liye kya karna padega?

Kabir: Poora... sab kuch band karna padega. Ek jagah band karne se woh raasta badal lega.

Madhav: Isiliye woh aise banaya hi gaya tha. Aur ab ek aakhri sawal. Agar koi maalik nahi hai, koi manager nahi hai, koi poora naksha nahi hai, toh internet ko ek cheez kya banata hai?

Kabir: Sirf yeh ki sab ek hi niyam maan rahe hain.

Madhav: Bas. Internet koi cheez nahi hai. Woh ek sehmati hai.

NAAM

Internet ka ek line mein matlab:

Alag-alag maalikon ke jaal, jo ek doosre se isliye jude hain ki dono ko faayda hai, aur jo sirf isliye ek saath kaam karte hain ki sab ek hi protocol maante hain.

Woh cheez jismein har machine sirf agla kadam jaanti hai, use **routing** kehte hain. Aur woh tareeka jisse padosi ek doosre ko batate hain ki kahan se kya pahuncha ja sakta hai, use **BGP** kehte hain.

Poori tasveer, neeche se upar:

TAAR	zameen aur samundar ke neeche, kisi ka apna maal
PATA	har machine ko ek nishaan (Ch 4.2)
PACKET	bada sandesh tukdon mein (Ch 4.3)
ROUTING	har machine sirf agla kadam jaanti hai
PROTOCOL	sab ek hi niyam maante hain (Ch 4.4)
NAAM	insaan ke liye naam, machine ke liye number (Ch 4.5)
TAALA	raaste mein koi padh na sake (Ch 4.6)

In saaton mein se ek bhi kisi ek maalik ka nahi hai.

Aur ab woh cheez jo iss poore part ka nichod hai:

Internet ka koi centre nahi hai, isliye use band karna mushkil hai. Lekin uske kuch jode aise hain jinpe bahut kuch tika hai, aur woh kuch haathon mein hain. Naam ki jad. Samundar ke neeche ke bade taar. Bharose wali list jo aapke phone mein pehle se hai.

”Koi maalik nahi hai” aur “taakat sab jagah barabar hai” do alag baatein hain. Pehli sach hai, doosri nahi.

ASLI DUNIYA SE EK EXAMPLE

Internet ki neev 1960 ke dashak mein padi thi, aur uske peeche ek khaas soch thi: aisa jaal banao jo tukde-tukde hone par bhi chalta rahe. Ek beech ka daftar nahi, kyunki beech ka daftar ek nishana hota hai.

Us soch ka nateeja aaj bhi dikhta hai. Jab samundar ke neeche ka koi bada taar toot jaata hai, aur woh har saal kai baar hota hai, toh poore desh ka internet band nahi hota. Woh dheema ho jaata hai, kyunki traffic lambe raaston se ghoom kar jaata hai.

Aur ulta bhi hua hai. Kuch mauke aise aaye jab kisi ek company ne galti se yeh keh diya ki “ye saare pate mere paas hain,” aur uske padosiyon ne maan liya, aur duniya ka bada hissa traffic galat jagah pahunch gaya.

Kyunki BGP mein padosi ek doosre pe bharosa karte hain. Koi upar wala jaanch nahi karta. Poora internet, aaj bhi, kaafi hadd tak iss baat pe khada hai ki log sach bolenge.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki internet kisi ek jagah hai.**

Log “cloud” sunkar aasman jaisa kuch sochte hain. Cloud kisi ki building mein rakhi machinein hain, kisi sheher mein, bijli aur AC ke saath. Aap uska pata Google Maps pe dekh sakte ho.

Iska seedha nateeja: log samajh nahi paate ki unka data kis desh mein pada hai, kaunse kanoon uspe lagte hain, aur ek building mein aag lagne se unki cheezein kyun ja sakti hain.

Aur doosri taraf ka galat samajh bhi utna hi aam hai: yeh sochna ki internet itna failaa hua hai ki uspe kisi ka bas nahi. Bas chalta hai. Poore desh ka internet band kiya ja chuka hai, kai baar, kyunki ek desh ke andar aksar gine-chune taar hote hain jo bahar jaate hain.

Sahi tasveer beech mein hai: **koi ek maalik nahi hai, lekin gine-chune jode hain jahan bahut kuch tika hai.**

MAP PE

INTERNET	koi maalik nahi har machine sirf agla kadam jaanti hai ek sehmati, ek cheez nahi	← naya
TAALA, NAAM, PROTOCOL, PACKET, PATA		
OS, PROCESS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Ek samundar ke neeche ka taar toot jaata hai. Poore desh ka internet band ho jaayega?

Jawab: Lagbhag kabhi nahi. Woh dheema ho jaayega.

Kyunki har machine ko sirf agla kadam pata hai, aur woh badla ja sakta hai. Traffic doosre raaste se ghoom kar jaata hai, jo lamba hai, isliye latency badh jaati hai.

Lekin agar ek desh mein sirf ek ya do taar bahar jaate hon, toh yeh sach mein poora band kar sakta hai. Isiliye ek desh ka internet kitna mazboot hai, yeh uske bahar jaane wale raaston ki ginti pe tikta hai.

2. (samajh check) Internet mein koi manager nahi hai. Toh kya kuch bhi band nahi kiya ja sakta?

Jawab: Kiya ja sakta hai, aur teen jagahon se aksar kiya jaata hai.

Naam: naam hata do, cheez pahunch se bahar ho jaati hai, chahe machine chal rahi ho.

Raasta: internet company apne yahan se us pate tak jaana band kar de.

Machine: jis company ke server pe cheez rakhi hai, woh use hata de.

Dhyaan do ki teeno mein se koi bhi “internet” ko band nahi kar raha. Har ek kisi ek jode ko kaat raha hai. Aur teeno jode kisi na kisi ke haath mein hain.

3. (jodne wala) Chapter 4.1 mein Kabir ne kaha tha ki network ki value judav mein hai. Ab yeh chapter kehta hai ki internet ek sehmati hai. Dono ko jodo.

Jawab: Judav sirf taar se nahi banta. Woh sehmati se banta hai.

Do machinein taar se judi ho sakti hain aur phir bhi ek doosre ko na samajh paayein, agar dono alag niyam maan rahi hon.

Toh asli judav protocol mein hai, dhaatu mein nahi.

Aur isse ek gehri baat nikalti hai jo Chapter 8.5 mein poori khulegi: **network wali duniya mein sabse badi taakat cheez banane wale ke paas nahi hoti. Woh us niyam ke paas hoti hai jise sab maan lete hain.**

Chapter 4.8 [DEPTH]

Internet ke rules kaun banata hai

DEPTH chapter. Yahan technology nahi, taakat hai. Agar aap yeh samajhna chahte ho ki paisa aur asar kahan baithta hai, toh yeh Part 4 ka sabse kaam ka chapter hai.

SAMVAAD

Madhav: Pichhle chapter mein tay hua ki internet ek sehmati hai. Ab sawal: woh sehmati likhi kisne?

Kabir: Kisi sarkar ne?

Madhav: Kaunsi? Internet 190 se zyada deshon mein hai.

Kabir: Toh United Nations jaisa kuch?

Madhav: Nahi. Ab main tumhe ek cheez batata hoon jo lagbhag kisi ko pata nahi hoti. Internet ke bade niyam ek aise samooch se aate hain jiski koi sadasyata nahi hai. Koi bhi aa sakta hai. Koi vote nahi hota.

Kabir: Toh faisla kaise hota hai?

Madhav: Log baat karte hain jab tak kaafi log sehmat na ho jaayein. Woh use “rough consensus” kehte hain.

Kabir: Aur agar koi na maane?

Madhav: Kuch nahi hota. Koi saza nahi hai. Koi force nahi hai.

Kabir: Toh log maante kyun hain?

Madhav: Yehi asli sawal hai. Socho. Agar tum apna alag protocol banao toh kya hoga?

Kabir: Mujhse koi baat nahi kar payega.

Madhav: Toh taakat kahan hai?

Kabir: Niyam mein nahi. Iss baat mein ki baaki sab kya maan rahe hain.

Madhav: Bas. Ab ek aur jagah dekhte hain. Naam ki jad kaun sambhalta hai?

Kabir: Woh toh koi hoga hi. Chapter 4.5 mein tha ki jad sabse upar hai.

Madhav: Ek sanstha hai jo yeh tay karti hai ki kaunse naye tarah ke naam banenge, aur kaun unhe sambhalega.

Kabir: Woh kiski hai?

Madhav: Kisi ki nahi, aur uski kahani dilchasp hai. Kai saal tak woh America ki sarkar ke ek theke pe chalti thi. Duniya ke baaki desh ko yeh pasand nahi tha. 2016 mein woh sarkar se alag ho gayi.

Kabir: Toh ab woh sabki hai?

Madhav: Ab woh ek aisi sanstha hai jismein sarkarein, companies aur aam log sab baithte hain. Behtar hai. Lekin dhyaan do ki woh jagah abhi bhi ek hi hai.

Kabir: Aur bharose wali list? Jo phone mein pehle se hoti hai?

Madhav: Achha sawal. Woh list kaun tay karta hai?

Kabir: Phone banane wala? Ya browser banane wala?

Madhav: Bilkul. Ek chhoti si sanstha hai jismein browser banane wali companies aur certificate dene wale baithte hain, aur woh niyam tay karte hain. Lekin aakhri faisla browser banane wale ka hai. Woh kisi ko bhi list se hata sakte hain.

Kabir: Aur agar hata dein?

Madhav: Us certificate dene wale ka karobaar khatam. Aisa hua hai, ek se zyada baar.

Kabir: Toh chaar-paanch companies duniya ka bharosa tay karti hain.

Madhav: Ab teen jagahein gin lo jahan asli taakat hai.

Kabir: Protocol likhne wale. Naam ki jad. Aur bharose wali list.

Madhav: Aur ek chauthi bhi hai, jo sabse purani hai. Woh taar kiske hain, aur woh kahan se guzarte hain.

NAAM

Chaar jagahein jahan internet ki taakat sach mein baithti hai:

1. PROTOCOL LIKHNE WALE
koi sadasyata nahi, koi vote nahi, sirf sehmati
taakat isliye hai ki baaki sab unke likhe niyam maan lete hain
2. NAAM KI JAD
ek sanstha tay karti hai ki naye naam kaise banenge
aur kaunsa desh ya sanstha kaunsa hissa sambhalega
3. BHAROSE WALI LIST
gine-chune browser aur phone banane wale
jo chaahein use bharose se bahar kar sakte hain
4. TAAR
samundar ke neeche ke bade taar, aur woh jagahein
jahan bade jaal aapas mein milte hain

Ab woh baat jo iss chapter ki asli seekh hai.

Chapter 4.7 mein tay hua tha ki internet ka koi maalik nahi hai. Woh sach hai. Lekin **”koi maalik nahi” ka matlab “koi taakat nahi” nahi hota.**

Taakat wahan chali jaati hai jahan sehmati zaroori hoti hai. Jahan sabko ek hi cheez maanni padti hai, wahan jo us cheez ko sambhalti hai, uske paas asar aa jaata hai, chahe uska koi kanooni haq na ho.

Aur ab Chapter 0.2 ke dials lagao. Ek protocol likhne wale ko kitna paisa milta hai?

Aksar bahut kam. Kai toh muft mein karte hain.

Lekin unka asar arbon logon pe padta hai. Toh unka leverage kalpana se bahar hai aur unka paisa lagbhag zero.

Yeh Chapter 0.1 ke नियम ko todta nahi. Value di gayi, aur bahut di gayi. Woh tokens mein wapass nahi aayi, kyunki unhone woh cheez banayi jise sirf muft rehne pe hi kaam karna tha.

Jo cheez sabki sehmati se chalti hai, uska daam koi nahi le sakta. Isliye us jagah taakat toh hoti hai, paisa nahi. Paisa uske upar wali parat pe baithta hai.

ASLI DUNIYA SE EK EXAMPLE

2011 mein ek certificate dene wali Dutch company, DigiNotar, hack ho gayi. Hamlawaron ne uske naam se nakli certificate banaye, jinmein Google ka bhi ek tha, aur unhe Iran mein logon ki baatein padhne ke liye istemaal kiya gaya.

Jab yeh pata chala, toh browser banane walon ne mil kar us company ko apni bharose wali list se hata diya.

Us company ke certificate duniya bhar mein turant bekaar ho gaye. Kuch hi hafton mein woh diwaaliya ho gayi.

Kisi adalat ne faisla nahi diya. Kisi sarkar ne band nahi kiya. Chahr-paanch browser banane walon ne apni ek list se ek naam hata diya, aur woh company khatam ho gayi.

Yeh saaf tarah dikhata hai ki asli taakat kahan baithi hai: kanoon mein nahi, us list mein jo pehle se sabke device mein pad chuki hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki technology neutral hai aur uske faisle technical hote hain.**

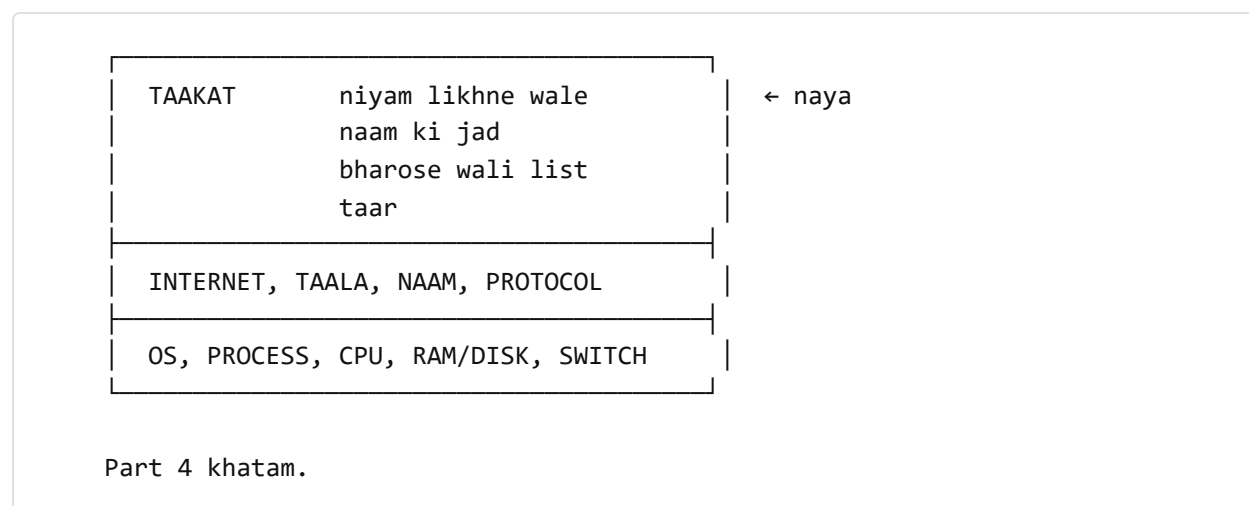
Log maante hain ki protocol isliye chuna gaya ki woh behtar tha, aur नियम isliye aise hain ki technically sahi hain.

Bahut se faisle technical the. Lekin bahut se nahi. Kaunse naam ban sakte hain, kaun certificate de sakta hai, taar kis desh se guzarte hain, kaunsi baat encrypt hogi aur kaunsi nahi. Yeh sab faisle hain, aur faisle koi leta hai.

Yeh galti tempting isliye hai ki technology gani t aur logic jaisi lagti hai, aur unmein raay ki jagah nahi hoti. Lekin kaunsa hisaab kiya jaaye, yeh hamesha ek chunav hai.

Aur agar aap yeh samajhna chahte ho ki paisa aur asar kahan jaate hain, toh yeh sabse kaam ki aadat hai: **jab bhi koi technology dekho, poocho ki iske niyam kaun tay karta hai, aur agar woh kal badal de toh kya hoga.**

MAP PE



PART 4 KA GATE

1. Aapka phone bahar se pahuncha nahi ja sakta. Toh WhatsApp ka message aapke phone pe pahunchta kaise hai? (*nahi aaya toh 4.2*)

Jawab: Aapka phone pehle bahar jaata hai aur ek server se juda rehta hai jiska pakka pata hai. Message us server pe aata hai aur woh use aage bhejta hai.

Do phone kabhi seedha baat nahi karte. Beech mein hamesha koi hota hai.

2. Ek video call mein awaaz kat jaati hai aur wapas aa jaati hai, lekin jo chhoot gaya woh dobara nahi milta. File download mein aisa nahi hota. Kyun? (*nahi aaya toh 4.3*)

Jawab: Do alag tareeke. Video call mein tukda kho jaaye toh chhod diya jaata hai, kyunki purani awaaz kisi kaam ki nahi. File mein har tukda dobara maanga jaata hai, kyunki ek bhi tukda gaya toh file toot jaayegi.

3. Ek site pe taala hai. Kya aapka data wahan surakshit hai? (*nahi aaya toh 4.6*)

Jawab: Taala sirf itna kehta hai ki raaste mein koi padh nahi raha, aur naam wahi hai jo dikh raha hai.

Woh us site ke maalik ke baare mein kuch nahi kehta. Thagne wali site pe bhi taala hota hai.

CHAT APP PE WAPAS

Aapka message packet mein tootta hai.
Har packet pe aapka pata aur manzil ka pata likha hai.
Woh bees ajnabi machinon se guzarta hai, aur koi bhi machine
poora raasta nahi jaanti.

Woh TCP se jaata hai, kyunki ek bhi shabd khona theek nahi.
Woh taale mein band hai, isliye raaste wale padh nahi sakte.
Woh seedha doosre phone pe nahi jaata, kyunki doosra phone
bahar se pahuncha hi nahi ja sakta. Woh ek server tak jaata hai.

Aur woh server ka naam ek list mein ek line hai, jo kisi aur ke paas hai.

Ab woh server. Usne aapka message le liya. Aur ab?

Usse woh sambhal kar kahan rakhega, aur agar aap chhe saal baad usmein se ek shabd dhoondho, toh woh kaise milega? Yeh Part 5 hai.

Chapter 5.1 [SPINE]

Bhoolne ki problem

SAMVAAD

Madhav: Server ne aapka message le liya. Ab use rakhna hai. Kahan rakhega?

Kabir: Disk pe. RAM mein rakhega toh bijli jaate hi gaya.

Madhav: Theek. Kis shakal mein?

Kabir: Ek file mein likh dega.

Madhav: Achha. Ek file, aur usmein har message ek line. Ab do sawal. Pehla: das crore message ke baad woh file kitni badi hogi?

Kabir: Bahut badi. Kuch GB.

Madhav: Aur agar usmein se ek khaas message dhoondhna ho?

Kabir: Poori file padhni padegi.

Madhav: Kitna waqt lagega?

Kabir: Chapter 2.3 ke hisaab se, disk dheemi hai. Kuch second, shayad zyada.

Madhav: Har sawal ke liye kuch second, aur ek second mein hazaaron sawal. Chalega?

Kabir: Nahi.

Madhav: Doosra sawal. Do log ek hi waqt pe us file mein likhte hain. Kya hoga?

Kabir: Dono ek hi jagah likh sakte hain. Ek doosre ke upar.

Madhav: Aur teesra. Likhte waqt bijli chali jaaye?

Kabir: Aadhi line likhi jaayegi. File kharaab.

Madhav: Toh ek simple file ke saath teen problem hain. Ginao.

Kabir: Dhoondhna dheema. Do log ek saath nahi likh sakte. Aur beech mein ruk jaaye toh sab kharaab.

Madhav: Ab ek chauthi bhi hai. Maan lo har message ke saath yeh bhi rakhna hai ki kisne bheja, kab bheja, kis group mein. Toh har line mein woh sab likhoge?

Kabir: Haan.

Madhav: Aur ek aadmi apna naam badal de? Uske purane das lakh message mein naam likha hai.

Kabir: Sab badalne padenge.

Madhav: Toh chauthi problem: ek hi cheez kai jagah likhi hai, aur badalne pe sab jagah badalni padti hai.

Kabir: Toh naam alag jagah rakh do. Message mein sirf ek number rakh do ki kaun tha.

Madhav: Ab tum sochne lage ho. Toh ab do file hain: ek logon ki, ek message ki. Aur message wali mein sirf ishara hai.

Kabir: Haan.

Madhav: Ab teen file, chaar file, das file. Aur unke beech ke ishare. Yeh sab kaun sambhalega?

Kabir: Mujhe khud likhna padega.

Madhav: Har app banane wale ko, har baar, shuru se?

Kabir: Nahi. Koi ek baar bana de, sab istemaal karein.

Madhav: Bas. Aur wahi hua.

NAAM

Woh alag program jo yeh saara kaam sambhalta hai, use **database** kehte hain.

Woh chaar problem jo Kabir ne khud nikaali, database unhi chaar ke liye maujood hai:

1. DHOONDHNA das crore mein se ek, turant
2. EK SAATH LIKHNA kai log ek waqt pe, bina ek doosre ko bigaade
3. BEECH MEIN RUKNA bijli jaaye toh aadha kaam na bache
4. DOHRAV ek cheez ek jagah, baaki jagah sirf ishara

Aur database khud kuch nayi cheez nahi hai. Woh ek program hai jo disk pe files hi likhta hai. Bas usne yeh chaar problem itni baar hal ki hain ki ab koi dobara nahi karta.

Yeh Chapter 3.1 wali baat hai, phir se. Kisi ne ek baar takleef uthayi, aur uske baad kisi ko nahi uthani padi.

Aur ab woh cheez jo iss poore part ka aadhaar hai:

Data ko rakhna aasan hai. Use wapas dhoondhna mushkil hai.

Kuch bhi likh dena sasta hai. Das crore cheezon mein se ek nikalna, woh asli kaam hai. Poora Part 5 lagbhag isi ek sawal ke aas-paas hai.

ASLI DUNIYA SE EK EXAMPLE

1970 mein IBM ke ek researcher, Edgar Codd, ne ek paper likha jismein usne kaha ki data ko tables mein rakha jaaye, aur unke beech ke rishte numbers se banaye jaayein, na ki us baat se ki file mein kya kahan pada hai.

Us waqt log data ko is baat se dhoondhte the ki woh disk pe kis jagah pada hai. Yaani program ko pata hona padta tha ki cheez kahan rakhi hai.

Codd ne kaha ki program ko yeh nahi pata hona chahiye. Program bataye ki **kya** chahiye, aur database khud dekhe ki woh **kahan** hai.

Yeh Chapter 0.4 wala abstraction hai, data pe laga hua. Aur woh ek soch aaj bhi lagbhag har database ke andar chal rahi hai, pachas saal baad.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki database ek jaadui cheez hai jo files se alag hai.**

Log “database” sunkar kuch bahut alag sochte hain. Woh disk pe files hi hain. Aap unhe kholo toh andar numbers milenge.

Farak yeh nahi hai ki woh kis cheez pe likhta hai. Farak yeh hai ki woh un chaar problem ka hal apne andar rakhta hai.

Iska seedha nateeja: log poochte hain “database istemaal karun ya file?” Sahi sawal yeh hai: “kya mujhe un chaar problem mein se koi bhi hai?” Agar aapko sirf kuch settings rakhni hain, ek file kaafi hai. Agar das crore cheezein hain aur sau log ek saath likh rahe hain, toh nahi.

Yeh galti tempting isliye hai ki database ka naam bada hai aur woh alag cheez lagta hai. Neeche wahi disk hai jo Chapter 2.4 mein thi.

MAP PE

DATABASE	dhoondhna, ek saath likhna beech mein rukna, dohrav	← naya
INTERNET, PROTOCOL, PATA, NETWORK		
OS, PROCESS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Aap ek chhota app bana rahe ho jismein sirf aapki apni 200 notes hain, aur sirf aap use karte ho. Database chahiye?

Jawab: Nahi. Ek file kaafi hai.

Chaaron problem dekho: 200 cheezein dhoondhna turant hai. Sirf aap likh rahe ho, toh takraav nahi. Beech mein rukna ek chhoti problem hai. Aur dohrav lagbhag nahi hai.

Database ki keemat tab chukayi jaati hai jab woh chaar problem asli hon. Warna woh sirf ek aur cheez hai jise sambhalna padta hai.

Yeh aage bar bar aayega: **har hal ki apni keemat hoti hai, aur woh keemat tabhi chukani chahiye jab problem asli ho.**

2. (samajh check) Ek aadmi apna naam badalta hai. Uske das lakh purane message mein naam likha hai. Do tareeke hain: sab jagah badal do, ya sirf ek jagah rakho aur baaki jagah ishara. Kaunsa behtar hai, aur kya kimat hai?

Jawab: Ishara behtar hai, lekin muft nahi hai.

Faayda: naam ek jagah hai, ek baar badla, sab jagah badal gaya.

Keemat: ab har baar message dikhaate waqt naam alag se laana padta hai. Yaani ek kaam do jagah se hota hai, jo dheema hai.

Isiliye kabhi kabhi log jaan-boojh ke dohrav rakhte hain, sirf speed ke liye. Aur phir unhe khud sambhalna padta hai ki dono jagah ek jaisa rahe.

Yeh iss poore part ka sabse aam sauda hai: sahi rakhna, ya tez rakhna.

3. (jodne wala) Chapter 2.4 mein RAM aur disk ka farak tha. Database uss farak ka kaunsa istemaal karta hai?

Jawab: Dono ka, ek saath.

Woh data disk pe rakhta hai, taaki bijli jaane pe bache. Lekin woh jo hissa abhi kaam mein aa raha hai, use RAM mein rakhta hai, taaki tez ho.

Aur likhte waqt woh pehle ek chhoti si “yeh hone wala hai” wali entry disk pe likhta hai, phir asli kaam karta hai. Isse beech mein bijli jaane pe woh baad mein dekh kar tay kar sakta hai ki kya poora hua tha aur kya nahi.

Yeh teesri problem ka hal hai, aur Chapter 5.4 mein hum ise poora kholenge.

Chapter 5.2 [SPINE]

Ek crore mein se ek dhoondna

SAMVAAD

Madhav: Ek kitaab hai, hazaar page ki. Usmein ek shabd dhoondhna hai. Kya karoge?

Kabir: Pehle page se padhna shuru karunga.

Madhav: Aurat se kitne page padhne padenge?

Kabir: Aadhe, aam taur pe. Paanch sau.

Madhav: Ab wahi kitaab, lekin peeche ek index hai. Ab?

Kabir: Index mein shabd dhoondhunga, woh page number dega, main seedha wahan jaunga.

Madhav: Index mein dhoondhne mein kitna waqt?

Kabir: Woh alphabet mein hai, toh jaldi.

Madhav: Kitni jaldi? Socho. Index mein das hazaar shabd hain. Tum beech mein khologe.

Kabir: Agar mera shabd usse pehle hai toh aadha index gaya. Phir uske beech mein.

Madhav: Har kadam pe kitna bachta hai?

Kabir: Aadha.

Madhav: Das hazaar se ek tak pahunchne mein kitne kadam?

Kabir: Baar baar aadha... das hazaar, paanch hazaar, dhai hazaar... lagbhag chaudah kadam.

Madhav: Aur bina index ke paanch hazaar kadam. Ab isse bada karo. Das lakh cheezein.

Kabir: Bina index: paanch lakh kadam. Index se... bees kadam ke aas-paas.

Madhav: Das crore?

Kabir: Bina index: paanch crore. Index se sattaais, attais kadam.

Madhav: Dhyaan do. Data das hazaar guna bada hua aur index wale kadam chaudah se sattaais ho gaye. Data lakhon guna badhta hai aur kaam mushkil se dugna hota hai.

Kabir: Yeh toh bahut bada farak hai.

Madhav: Yeh sabse bade farakon mein se ek hai jo poore computer ki duniya mein hai. Ab agla sawal. Agar index itna achha hai, toh har cheez ka index kyun nahi bana lete?

Kabir: Bana lo.

Madhav: Index kahan rakha jaayega?

Kabir: Disk pe. Woh bhi jagah lega.

Madhav: Aur jab ek nayi cheez jodi jaaye?

Kabir: Index mein bhi jodni padegi.

Madhav: Aur agar das index hain?

Kabir: Toh das jagah jodna padega. Har baar likhne mein das guna kaam.

Madhav: Toh index kya deta hai aur kya leta hai?

Kabir: Padhna tez karta hai, likhna dheema karta hai. Aur jagah leta hai.

Madhav: Ab batao, kaunse cheezon ka index banaoge?

Kabir: Jinse main sach mein dhoondhta hoon.

Madhav: Aur jinse kabhi nahi dhoondhta?

Kabir: Unka index bekaar hai. Sirf nuksaan hai.

NAAM

Us peeche wali list ka naam hai **index**.

Aur woh tareeka jismein har kadam pe aadha hissa chhod diya jaata hai, use **binary search** kehte hain, aur database uska ek bada roop istemaal karte hain jise **B-tree** kehte hain.

Sabse zaroori tasveer:

cheezein	bina index	index ke saath
1,000	~500 kadam	~10 kadam
10 lakh	~5,00,000	~20 kadam
10 crore	~5,00,00,000	~27 kadam
1 arab	~50 crore	~30 kadam

Baayan column data ke saath badhta hai. Daayan column lagbhag badhta hi nahi.

Yeh farak itna bada hai ki iske bina aaj ki koi bhi badi cheez chal hi nahi sakti. Google, bank, ticket, sab isi ek hisaab pe khade hain.

Aur uska sauda, jo Kabir ne nikaala:

INDEX DETA HAI	padhna lagbhag turant
INDEX LETA HAI	har likhne pe extra kaam aur disk pe jagah

Isliye ek aisa system jismein bahut padha jaata hai aur kam likha jaata hai, usmein bahut index hote hain. Aur jismein har second lakhon cheezein likhi jaati hain, usmein bahut kam.

ASLI DUNIYA SE EK EXAMPLE

Ek chhoti company ka app theek chalta hai jab uske paas das hazaar users hain. Woh badhta hai, aur ek din das lakh pe pahunch kar sab kuch dheema ho jaata hai.

Yeh sabse aam kahani hai jo har badhne wale product mein hoti hai, aur uski wajah lagbhag hamesha yehi hoti hai: kahin ek jagah poori list padhi ja rahi thi.

Das hazaar par woh 5,000 kadam the, jo itne tez the ki kisi ko dikha hi nahi. Das lakh par woh 5,00,000 kadam ho gaye.

Kuch nahi badla. Code wahi hai. Machine wahi hai. Sirf data bada ho gaya, aur ek aisa hissa jo badhne ke saath badhta tha, woh saamne aa gaya.

Isiliye yeh sawal hamesha poocha jaata hai: **jab yeh cheez sau guna badi hogi, tab kya hoga?** Kuch cheezein badhne pe wahi rehti hain, kuch dugni ho jaati hain, aur kuch sau guna. Farak wahin se aata hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki dheema hone ka ilaaj badi machine hai.**

Jab kuch dheema hota hai toh pehla khayal aata hai: bada server le lo.

Lekin agar aapka kaam data ke saath seedha badhta hai, toh dugni machine sirf dugna sambhalegi. Aur data das guna hone wala hai.

Index badalne se kaam **paanch lakh kadam se bees kadam** ho jaata hai. Koi machine woh farak nahi de sakti.

Yeh galti tempting isliye hai ki machine khareedna ek din ka kaam hai aur samajhna ki dheema kyun hai, woh mehnat hai. Aur kyunki machine badhane se thoda faayda hota bhi hai, toh lagta hai ki disha sahi thi.

Sahi soch:

pehle poocho	yeh kaam data ke saath kaise badhta hai?
	lagbhag wahi rehta hai, ya seedha badhta hai?
phir poocho	kya main use badalne ka tareeka badal sakta hoon?
aakhir mein	kya mujhe badi machine chahiye?

MAP PE

INDEX	har kadam pe aadha chhodo padhna tez, likhna dheema	← naya
DATABASE	chaar problem	
INTERNET, PROTOCOL, PATA		
OS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Ek app hai jismein har second das hazaar naye messages aate hain, aur log unhe kabhi kabhi dhoondhte hain. Bahut index rakhoge ya kam?

Jawab: Kam. Sirf woh jo sach mein chahiye.

Har index har likhne pe kaam badhata hai. Das hazaar likhne per second, das index, matlab ek lakh extra kaam har second.

Padhna kam hai, toh index ka faayda kam baar milta hai aur nuksaan har baar.

Ulta case: ek aisi jagah jahan data raat mein ek baar aata hai aur din bhar log dhoondhte hain. Wahan jitne index chaho.

2. (samajh check) Index se dhoondhna itna tez hai. Toh phir kuch cheezein aaj bhi dheemi kyun hain?

Jawab: Kyunki index tabhi kaam karta hai jab aap **usi cheez se** dhoondho jiska index bana hai.

Naam ka index hai aur aap naam se dhoondhoge toh turant. Aap tarikh se dhoondhoge toh index bekaar hai, poori list padhni padegi.

Aur kuch sawal aise hain jinka index ban hi nahi sakta. “Aisa har message dhoondho jismein kahin bhi yeh shabd ho” ek mushkil sawal hai, aur uske liye alag tarah ka index banana padta hai.

Yeh Book 2 mein wapas aayega. Wahan aap dekhoge ki “matlab se dhoondhna” ek bilkul nayi tarah ka index maangta hai.

3. (jodne wala) Chapter 2.3 kehta tha ki hisaab sasta hai aur data laana mehnga. Index uss baat se kaise judta hai?

Jawab: Index ka poora kaam hi yeh hai ki **kam data laana pade**.

Woh hisaab nahi bachata. Woh yeh bachata hai ki disk se kitne tukde uthane padenge.

Bina index: das lakh cheezein disk se uthao aur dekho.

Index ke saath: bees tukde uthao, phir seedha sahi jagah jao.

Toh index ek hisaab ka jugaad nahi hai. Woh doori ka jugaad hai. Aur wahi wajah hai ki uska faayda itna bada hai.

Chapter 5.3 [SPINE]

Memory ki alag shapes

SAMVAAD

Madhav: Ek school ka record rakhna hai. Har student ka naam, roll number, class, marks. Kaise rakhoge?

Kabir: Ek table. Har student ek line, har cheez ek column.

Madhav: Har student ke paas wahi cheezein hain?

Kabir: Haan. Sabka naam hai, sabka roll number hai.

Madhav: Ab ek doosra kaam. Ek online dukaan. Har cheez ka naam aur daam. Lekin ek shirt ka size hai, ek kitaab ka lekhak hai, ek phone ki battery hai.

Kabir: Toh table mein bahut saare column honge, aur zyadatar khaali rahenge.

Madhav: Aur kal ek nayi cheez aayi jismein ek naya gun hai?

Kabir: Naya column jodna padega. Poori table mein.

Madhav: Har baar?

Kabir: Yeh mushkil hai. Toh har cheez ko apna alag dabba de do, aur usmein jo hai woh likh do.

Madhav: Toh ab har cheez ki shakal alag ho sakti hai.

Kabir: Haan.

Madhav: Ab problem batao.

Kabir: Agar shakal tay hi nahi hai, toh main galti se kuch bhi daal sakta hoon. Daam ki jagah naam.

Madhav: Aur koi rokega?

Kabir: Nahi. Kyunki koi niyam hai hi nahi.

Madhav: Toh table mein sakhti hai aur badalna mushkil. Dabbon mein aazadi hai aur galti aasan. Ab ek teesra kaam. Ek social app. Kaun kiska dost hai. Kaise rakhoge?

Kabir: Ek table: dost A, dost B.

Madhav: Ab poocho: mere dost ke dost kaun hain?

Kabir: Do baar dekhna padega.

Madhav: Aur unke dost ke dost?

Kabir: Teen baar. Har baar table dobara padhni padegi.

Madhav: Aur chhe kadam door tak?

Kabir: Yeh bahut mehnga ho jaayega.

Madhav: Toh yahan kya chahiye?

Kabir: Aisa tareeka jismein rishte hi asli cheez hon, aur unpe chalna sasta ho.

Madhav: Ab teen alag kaam, teen alag shakal. Ek line mein batao ki shakal kis cheez se tay hoti hai.

Kabir: Iss baat se ki main kya sawal poochunga.

Madhav: Bas. Data ki shakal data se nahi aati. Woh sawal se aati hai.

NAAM

Teen badi shakalein, aur unke naam:

TABLE (SQL)

- |— har cheez ki shakal ek jaisi, pehle se tay
- |— rishte numbers se jode jaate hain
- |— sakhti: galat cheez daali nahi ja sakti
- |— badalna mushkil: shakal badlo toh sab badalna padta hai
- |— Postgres, MySQL, SQLite

DABBA (document / NoSQL)

- |— har cheez ki apni shakal
- |— ek cheez se judi saari jaankari ek hi jagah
- |— aazadi: naya gun jodna aasan
- |— khatra: koi niyam nahi, galti pakdi nahi jaati
- |— MongoDB, Firestore

RISHTA (graph)

- |— asli cheez rishte hain, cheezein nahi
- |— "iske iske iske dost" jaise sawal sasta
- |— kam istemaal hota hai, lekin jahan chahiye wahan koi vikalp nahi
- |— Neo4j

Aur ab woh baat jo iss chapter ki asli seekh hai:

Koi shakal behtar nahi hai. Shakal us sawal se aati hai jo aap sabse zyada poochoge.

Bank ko table chahiye, kyunki wahan sakhti sabse zaroori hai. Ek nayi app ko shayad dabbe theek lagein, kyunki uski shakal abhi tay hi nahi hai. Ek social app ko rishton wali shakal chahiye.

Aur bahut se asli systems mein ek se zyada saath chalte hain, alag alag kaam ke liye.

Ek chetavni, kyunki yeh galti bahut mehngi padti hai. Log aksar shakal isliye chunte hain ki koi nayi hai ya charcha mein hai. Shakal badalna baad mein sabse mushkil kaam ho jaata hai, kyunki poora data usi shakal mein pad chuka hota hai.

Language badalna hafte ka kaam hai. Data ki shakal badalna saal ka kaam hai.

ASLI DUNIYA SE EK EXAMPLE

2010 ke aas-paas “NoSQL” ki badi lehar aayi. Kaha gaya ki table wale database purane ho gaye aur ab sab dabbon mein jaayega.

Bahut si companies ne apna data table se hata kar dabbon mein daal diya.

Kuch saal baad unmein se kai wapas aa gayin. Wajah yeh nahi thi ki dabbe kharaab the. Wajah yeh thi ki unhe sakhti chahiye thi aur unhone aazadi chun li thi, aur uski keemat unhe tab pata chali jab data mein gadbad milne lagi.

Aur ab dilchasp hissa: table wale database ne dabbe wali sahulat apne andar jod li. Aaj Postgres mein aap ek column ke andar poora dabba rakh sakte ho.

Yeh Chapter 3.3 wali baat hai, phir se: **jo cheez pehle se sab jagah hai, use hataya nahi jaata. Woh nayi cheez ko apne andar le leti hai.**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki “SQL vs NoSQL” ek behtar-ghatiya wala sawal hai.**

Log poochte hain “kaunsa achha hai” jaise ek jeet raha ho.

Sahi sawal alag hai: **mujhe sakhti chahiye ya aazadi? Aur main kaunse sawal sabse zyada poochunga?**

Aur ek aur galti isi ke saath aati hai: yeh sochna ki NoSQL ka matlab tez hai. Woh tez isliye lagta hai kyunki woh kuch cheezein nahi karta, jaise poori jaanch aur poori guarantee. Woh speed muft nahi hai. Aapne kuch chhoda hai, aur aksar aapko pata nahi hota ki kya chhoda.

Yeh galti tempting isliye hai ki tools ki tulna aise hi ki jaati hai, jaise phone ki. Lekin yeh tools alag kaam ke liye hain, ek hi kaam ke alag darje nahi.

MAP PE

SHAKAL	table, dabba, rishta sawal se tay hoti hai	← naya
INDEX, DATABASE		
INTERNET, PROTOCOL, PATA		
OS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Ek hospital ka system bana rahe ho jismein mareez ka record, dawai, aur bill hai. Kaunsi shakal, aur kyun?

Jawab: Table.

Wajah sakhti hai. Ek dawai ki matra galat shakal mein nahi jaani chahiye. Ek bill ka number kabhi khaali nahi hona chahiye. Aap chahte ho ki galat data andar ja hi na sake.

Aur mareez, dawai, doctor, bill ke beech ke rishte tay hain aur badalte nahi.

Yeh Chapter 3.3 wala sawal hai dobara: **galti ki keemat kitni hai?** Jahan keemat zyada hai, wahan sakhti chuni jaati hai, chahe woh likhne mein dheemi ho.

2. (samajh check) Ek naya app bana rahe ho aur aapko abhi pata nahi ki usmein kaunsi cheezein rakhni hongii. Dabbe chunoge?

Jawab: Yeh sabse aam tark hai aur woh aadha sahi hai.

Sahi: shuruaat mein shakal badalti rehti hai, aur dabbe usmein aasan hain.

Aadha galat: shakal tab bhi maujood hoti hai. Woh sirf likhi nahi hoti. Woh aapke code mein bikhri hoti hai, aur jab teen saal baad koi naya aadmi aata hai, toh use kahin bhi nahi milti.

Toh sawal yeh nahi hai ki shakal hai ya nahi. Sawal yeh hai ki **shakal ek jagah likhi hai, ya sau jagah bikhri hai.**

3. (jodne wala) Chapter 5.1 mein chauthi problem “dohrav” thi. Dabbon wali shakal us problem ke saath kya karti hai?

Jawab: Woh use jaan-boojh ke wapas le aati hai.

Table wali shakal kehti hai: naam ek jagah rakho, baaki jagah ishara.

Dabbe wali aksar kehti hai: naam wahin rakh do jahan zaroorat hai, chahe woh das jagah ho.

Faayda: padhte waqt sab ek jagah mil jaata hai, doosri jagah jaana nahi padta. Tez.

Keemat: naam badla toh das jagah badalna padega, aur woh aapka kaam hai, database ka nahi.

Yeh wahi sauda hai jo poore Part 5 mein chal raha hai: **sahi rakhna, ya tez rakhna.** Har shakal ne uska ek alag jawab chuna hai.

Chapter 5.4 [DEPTH]

Jab do log ek hi waqt likhein

DEPTH chapter. Yeh woh problem hai jiski wajah se bank ka software itna mehnga hota hai, aur jiski wajah se ticket “sold out” hone ke baad bhi bik jaata hai.

SAMVAAD

Madhav: Ek bank ka khaata hai. Usmein hazaar rupaye hain. Do jagah se ek saath sau nikaale jaate hain. Kya hona chahiye?

Kabir: Aath sau bachne chahiye.

Madhav: Ab dekho ki machine kya karti hai. Nikaalne ka kaam teen kadam ka hai. Padho ki kitne hain. Ghatao. Wapas likho. Ab do log ek saath karein toh?

Kabir: Dono hazaar padhenge. Dono nau sau likhenge.

Madhav: Aur khaate mein?

Kabir: Nau sau. Do sau nikle aur sirf sau kate.

Madhav: Bank ne sau rupaye kho diye. Ab yeh ek din mein lakh baar hota hai. Kya karoge?

Kabir: Ek waqt mein sirf ek ko karne do.

Madhav: Kaise rokoge doosre ko?

Kabir: Taala laga do. Jab tak pehla khatam na ho, doosra intezaar kare.

Madhav: Theek. Ab ek naya kaam. Paise ek khaate se doosre mein bhejne hain. Kitne kadam?

Kabir: Ek se kam karo, doosre mein jodo. Do kadam.

Madhav: Pehla kadam ho gaya aur beech mein bijli chali gayi.

Kabir: Toh paise ek se kat gaye aur doosre mein aaye hi nahi. Gayab.

Madhav: Toh kya chahiye?

Kabir: Ya toh dono kadam hon, ya ek bhi na ho.

Madhav: Kaise?

Kabir: Pehle likh do kahin ki main yeh dono karne wala hoon. Phir karo. Beech mein ruk jaaye toh baad mein woh likha hua dekh kar tay kar lo.

Madhav: Kya tay karoge, poora karoge ya wapas karoge?

Kabir: Agar “main karne wala hoon” likha tha lekin “ho gaya” nahi likha, toh wapas kar do. Jaise kuch hua hi nahi.

Madhav: Tumne abhi database ka sabse zaroori hissa bana diya. Ab aakhri problem, aur yeh sabse tedhi hai. Do khaate hain. Ek aadmi A se B mein bhej raha hai. Doosra B se A mein. Dono ne apne pehle khaate pe taala laga liya.

Kabir: Ab pehla B ka taala maangega, jo doosre ke paas hai. Aur doosra A ka maangega, jo pehle ke paas hai.

Madhav: Aur?

Kabir: Dono intezaar karenge. Hamesha ke liye.

Madhav: Kya karoge?

Kabir: Kuch der baad ek ko chhod dena chahiye. Aur dobara koshish kare.

Madhav: Kaun chhodega? Dono ko lagta hai woh sahi hai.

Kabir: Koi niyam bana do. Jaise jo baad mein aaya woh chhode.

Madhav: Ya sabse aasan: taale hamesha ek hi order mein lo, chahe kaam kuch bhi ho. Toh do log kabhi ulte order mein phansenge hi nahi.

NAAM

Us “ya toh sab, ya kuch bhi nahi” wale ek kaam ko **transaction** kehte hain.

Woh chaar guarantee jo ek achha database deta hai:

SAB YA KUCH NAHI	aadha kaam kabhi nahi bachta (yahi Kabir ne banaya)
NIYAM TOOTE NAHI	jo shartein tay hain woh har haal mein sach rahein (khaata negative na ho, wagairah)
EK DOOSRE SE ALAG	do kaam ek saath chalein toh aise dikhein jaise baari baari hue hon
PAKKA HO JAAYE	"ho gaya" kaha toh bijli jaane pe bhi bacha rahe

Aur woh do problem jo iske saath aati hain:

TAALA	ek waqt mein ek. Surakshit, lekin dheema. Jitne zyada taale, utna zyada intezaar.
DONO ATAK JAANA	do kaam ek doosre ka taala maang rahe hain Hal: hamesha ek hi order mein taale lo, ya kuch der baad ek ko chhod do

Aur ab woh baat jo iss chapter ki asli seekh hai:

Yeh sab suraksha muft nahi hai. Uski keemat speed hai.

Ek database jo yeh chaaron cheezein pakki karta hai, woh us database se dheema hoga jo nahi karta. Isiliye kuch systems jaan-boojh ke inmein se kuch chhod dete hain.

Ek bank kabhi nahi chhodta. Ek app jo yeh gin raha hai ki kitne logon ne like kiya, woh aaram se chhod sakta hai. Ek like idhar udhar ho jaaye toh kisi ka kuch nahi jaata.

Sawal hamesha ek hi hai: galti ki keemat kya hai?

ASLI DUNIYA SE EK EXAMPLE

Concert ke ticket bikne pe ek jaani-pehchani ghatna hoti hai. Site kehti hai ki 100 ticket bache hain. Hazaaron log ek saath click karte hain. Kuch logon ko ticket mil jaata hai aur baad mein cancel ho jaata hai.

Wajah lagbhag hamesha wahi hai jo iss chapter mein hai. Ginti padhi gayi, ghatai gayi, wapas likhi gayi, aur beech mein koi aur bhi wahi kar raha tha.

Sahi hal taala hai. Lekin taala matlab ek waqt mein ek aadmi, aur hazaaron log line mein.

Toh companies aksar beech ka raasta chunti hain: ticket ko kuch minute ke liye rok do, poora taala mat lagao, aur agar ginti galat ho jaaye toh baad mein maafi maang lo.

Yeh galti nahi hai. Yeh ek chunav hai: **thodi galti, ya bahut intezaar**. Aur woh chunav technical nahi, karobaari hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki “database istemaal kar liya, ab sab surakshit hai.”**

Database yeh guarantee tab deta hai jab aap use sahi tareeke se maango. Agar aapne apna kaam ek transaction mein nahi rakha, toh database aapko kuch nahi bachayega. Woh aapke teen alag kaam ko teen alag kaam hi samjhega.

Aur doosri galti: yeh sochna ki yeh problem sirf paise wale systems mein hoti hai. Kahin bhi jahan do log ek hi cheez badalte hain, wahi problem hai. Ek hi document do log edit karein. Ek hi seat do log book karein. Ek hi saaman do log khareedein.

Yeh galti tempting isliye hai ki test karte waqt yeh kabhi nahi dikhta. Ek aadmi test karta hai, sab theek chalta hai. Problem sirf tab aati hai jab bahut log ek saath aayein, yaani tab jab aap kaamyab ho jaate ho.

Yeh kaamyabi ke saath aane wali bug hai. Woh sabse mehngi hoti hai kyunki woh sabse bure waqt pe aati hai.

MAP PE

TRANSACTION	sab ya kuch nahi	← naya
TAALA	ek waqt mein ek, dheema	← naya
SAUDA	thodi galti ya intezaar	
SHAKAL, INDEX, DATABASE		
INTERNET, OS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Ek app mein like ki ginti hai. Do log ek saath like karte hain aur ginti sirf ek badhti hai. Kya yeh theek karna chahiye?

Jawab: Shayad nahi.

Poocho: galti ki keemat kya hai? Ek like ki ginti ek kam. Kisi ka kuch nahi gaya.

Aur theek karne ki keemat? Har like pe taala, yaani har like pe intezaar, us cheez ke liye jo second mein hazaron baar hoti hai.

Isliye aise systems aksar ginti ko thoda galat rehne dete hain aur baad mein sudhaar lete hain.

Wahi baat bank ke khaate pe laagu karo aur jawab bilkul ulta ho jaata hai.

Technology wahi hai, faisla alag hai, kyunki galti ki keemat alag hai.

2. (samajh check) Do kaam ek doosre ka taala maang kar atak gaye. Aapne niyam banaya ki jo baad mein aaya woh chhod dega. Kya yeh poori tarah hal hai?

Jawab: Nahi, aur yeh dhyaan dene laayak hai.

Agar dono baar baar wahi koshish karein toh woh phir se takra sakte hain, baar baar.

Isliye asli systems mein chhodne ke baad thoda **alag-alag** intezaar kiya jaata hai, taaki dono ek saath wapas na aayein.

Aur behtar hal wahi hai jo Madhav ne kaha: taale hamesha ek hi order mein lo. Tab woh haalat ban hi nahi sakti. **Problem ko hal karne se behtar hai use hone hi na dena.**

3. (jodne wala) Chapter 5.2 kehta tha ki index padhna tez karta hai aur likhna dheema. Yeh chapter kehta hai ki taala surakshit karta hai aur dheema. Inmein kya saanjha hai?

Jawab: Dono ek hi shakal ke saude hain, aur poori kitaab unse bhari hui hai.

index	→	padhna tez		likhna dheema
taala	→	sahi		dheema
cache	→	tez		purana ho sakta hai
compression	→	chhota		kholne mein kaam
dabba	→	aazadi		galti aasan
table	→	sakhti		badalna mushkil

Ek bhi jagah aisi nahi hai jahan kuch muft mila ho.

Yeh iss kitaab ki sabse kaam ki aadat hai: **jab bhi koi cheez kuch deti dikhe, poocho ki woh badle mein kya le rahi hai.** Jawab hamesha hota hai. Agar nahi dikh raha, toh aapne dhoondha nahi hai.

Chapter 5.5 [SPINE]

Jo baar baar chahiye use paas rakhna

SAMVAAD

Madhav: Ek website ka home page hai. Ek second mein das hazaar log use kholte hain. Har baar database se poora data laaya jaata hai. Problem?

Kabir: Das hazaar baar wahi kaam. Bekaar.

Madhav: Kya karoge?

Kabir: Ek baar laao aur rakh lo. Agli baar wahi de do.

Madhav: Rakhoge kahan?

Kabir: RAM mein. Woh tez hai.

Madhav: Kitna tez? Chapter 2.3 se yaad karo.

Kabir: RAM lagbhag 200 kadam, disk das lakh. Toh paanch hazaar guna.

Madhav: Ab problem batao.

Kabir: Agar database mein data badal jaaye toh mera rakha hua purana ho jaayega.

Madhav: Toh kya karoge?

Kabir: Jab data badle tab rakha hua hata dun.

Madhav: Aur agar data das jagah rakha hai, das alag machinon pe?

Kabir: Toh das jagah hatana padega. Aur kisi ek pe na hata toh woh purana deta rahega.

Madhav: Aur agar tumhe pata hi na ho ki data badla?

Kabir: Toh main hamesha purana deta rahunga.

Madhav: Toh ek doosra tareeka socho. Sochne mein aasan.

Kabir: Waqt lagao. Paanch minute baad apne aap phenk do, chahe badla ho ya nahi.

Madhav: Ab uski keemat kya hai?

Kabir: Paanch minute tak log purana dekh sakte hain.

Madhav: Toh do tareeke hain: badalne pe hatao, ya waqt pe hatao. Pehla sahi hai lekin mushkil. Doosra aasan hai lekin thoda purana deta hai.

Kabir: Aur teesra? Dono?

Madhav: Aksar dono. Ab aakhri problem. RAM chhoti hai. Sab kuch usmein nahi aayega. Kya phenkoge?

Kabir: Jo sabse kam istemaal ho raha hai.

Madhav: Kaise pata chalega?

Kabir: Yaad rakhunga ki kis cheez ko aakhri baar kab maanga gaya. Sabse purani phenk dunga.

Madhav: Bas. Ab ek aakhri baat. Cache mein cheez na mile toh?

Kabir: Toh database se laani padegi. Yaani woh baar dheemi.

Madhav: Toh cache se hamesha faayda hota hai?

Kabir: Nahi. Agar har baar naya sawal aaye aur cache mein kabhi kuch na mile, toh cache sirf extra kaam hai.

Madhav: Toh cache kis cheez pe chalta hai?

Kabir: Iss baat pe ki wahi cheez baar baar maangi jaaye.

Madhav: Aur agar na maangi jaaye?

Kabir: Toh cache bekaar hai.

NAAM

Us paas rakhi hui copy ko **cache** kehte hain.

CACHE HIT	maangi hui cheez cache mein mil gayi. Tez.
CACHE MISS	nahi mili. Ab asli jagah se laani padegi. Dheemi + extra kaam.
HIT RATE	sau sawalon mein se kitne cache se mile

Cache kab phenka jaaye, uske do tareeke:

WAQT SE	paanch minute baad apne aap. Aasan, thoda purana de sakta hai.
BADLAV SE	jab asli data badle tab hatao. Sahi, lekin har jagah hatana padta hai.

Aur jab jagah bhar jaaye toh kya phenkein: aksar woh jo sabse lambe waqt se kisi ne nahi maanga.

Ab woh sabse zaroori baat, jise ek line mein rakh lo:

Cache tez isliye hai kyunki woh purana ho sakta hai. Woh do alag cheezein nahi hain. Woh ek hi cheez hai.

Agar cache kabhi purana na ho, toh use har baar jaanch karni padegi ki data badla ya nahi, aur phir woh cache hai hi nahi.

Isiliye har cache ke saath yeh sawal aata hai: **kitna purana chalega?** Share bazaar ke daam mein: ek second bhi nahi. Ek news article mein: paanch minute theek hai. Ek company ke “About Us” page mein: ek din bhi chalega.

Aur cache sirf ek jagah nahi hota. Aapke ek sawal ke raaste mein aksar chaar-paanch cache hote hain, aur unmein se koi bhi purana de sakta hai.

aapka browser	→	cache
raaste ka CDN	→	cache
server ki RAM	→	cache
database ki RAM	→	cache
CPU ki cache	→	cache (Chapter 2.3)

Isiliye “refresh karo, chal jaayega” itni baar kaam karta hai. Aap ek cache ko chhod kar aage ki taraf poochh rahe ho.

ASLI DUNIYA SE EK EXAMPLE

Jab aap koi video dekhte ho, woh aksar us company ke asli server se nahi aata. Woh aapke sheher ya aapke desh mein rakhi ek copy se aata hai.

Aisi copiyon ke jaal ko **CDN** kehte hain. Duniya bhar mein hazaaron jagah, popular cheezein pehle se rakhi hui.

Wajah Chapter 4.3 se aati hai: bandwidth badhayi ja sakti hai, latency nahi. Roshni se tez kuch nahi ja sakta. Agar video doosre mahadweep se aayega toh ek chakkar mein 200 millisecond lagenge, chahe sadak kitni bhi chaudi ho.

Toh hal doori kam karna hai, aur uska ek hi tareeka hai: cheez ko paas rakh do.

Aur iska ek gehra nateeja hai. **Internet ka bahut bada hissa asal mein cache hai.** Aap jo dekhte ho uska bada hissa asli jagah se nahi aata. Woh kisi copy se aata hai jo aapke paas rakhi hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **cache ko sirf ek speed ka jugaad samajhna.**

Log sochte hain cache lagane se cheez tez ho jaayegi, bina kisi keemat ke.

Keemat purana data hai, aur woh keemat hamesha chukani padti hai. Cache ka poora matlab hi yeh hai ki aap asli jagah se nahi pooch rahe.

Iska seedha nateeja: log ghanton yeh dhoondhte hain ki unka badlav dikh kyun nahi raha, jabki unhone data theek badal diya tha. Woh kisi cache mein atka hua hai, aur aksar unhe pata bhi nahi hota ki woh cache maujood hai.

Ek purani kahavat hai ki computer science mein sirf do mushkil cheezein hain: cache ko kab hataya jaaye, aur cheezon ke naam rakhna. Woh mazaak hai, lekin pehli baat sach hai.

Aur ek doosri galti isi ke saath aati hai: har jagah cache laga dena. Agar hit rate kam hai toh cache sirf nuksaan hai. Woh jagah leta hai, kaam badhata hai, aur purana data deta hai, sab bina kisi faayde ke.

MAP PE

CACHE	paas rakhi copy tez = purana ho sakta hai ek hi sikke ke do pehlu	← naya
TRANSACTION, SHAKAL, INDEX, DATABASE		
INTERNET, OS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Ek site pe har user ka apna alag page hai, aur har user din mein ek baar aata hai. Cache lagane se kya hoga?

Jawab: Lagbhag kuch nahi, aur thoda nuksaan.

Har sawal alag hai, toh cache mein kabhi kuch nahi milega. Hit rate lagbhag zero.

Aur har baar cache mein daalne ka extra kaam hoga, aur RAM bhi bharegi.

Cache **dohrav** pe chalta hai, bilkul waise jaise compression (Chapter 2.1). Dohrav nahi, toh faayda nahi.

Yeh poochna hamesha pehla kadam hai: **kya wahi cheez baar baar maangi ja rahi hai?**

2. (samajh check) Aapne apni website ka logo badla lekin purana hi dikh raha hai. Kya kya ho sakta hai?

Jawab: Raaste mein kisi bhi cache mein purana atka hai.

Aapka browser, ya CDN, ya server ki apni copy.

Isiliye website banane wale aksar file ke naam mein ek number jod dete hain, jaise

`logo-v2.png`. Naya naam matlab naya sawal, aur kisi cache mein woh hai hi nahi.

Yeh ek achhi chaal hai: **cache ko hatane ki koshish mat karo, use chakma do.**

3. (jodne wala) Chapter 4.5 mein DNS ke jawab bhi kuch ghante ke liye rakhe jaate the. Woh bhi cache tha. Dono mein saanjha kya hai?

Jawab: Dono mein wahi sauda hai, aur wahi problem.

DNS mein: pata paas rakh lo, tez ho jao, aur jab pata badle toh kuch ghante purana chalta rahe.

Yahan: data paas rakh lo, tez ho jao, aur jab data badle toh kuch der purana chalta rahe.

Aur dono mein wahi hal hai: waqt ki seema chhoti kar do badlav se pehle.

Yeh iss kitaab ki ek badi baat hai: **jab aap ek shakal pehchan lete ho, toh woh har jagah dikhne lagti hai.** Cache CPU mein hai, DNS mein hai, browser mein hai, database mein hai. Ek hi soch, chhe jagah.

Chapter 5.6 [SPINE]

Kaam ko line mein lagana

SAMVAAD

Madhav: Aap ek app pe photo upload karte ho. App ko chaar kaam karne hain: photo rakhna, uska chhota version banana, aapke doston ko batana, aur aapko “ho gaya” dikhana. Kitna waqt lagega?

Kabir: Photo rakhna jaldi. Chhota version banana thoda waqt lega. Batana bhi.

Madhav: Maan lo chaar second lagte hain. Aapko “ho gaya” kab dikhega?

Kabir: Chaar second baad.

Madhav: Aur aap tab tak kya kar rahe ho?

Kabir: Ghoomta hua circle dekh raha hoon.

Madhav: Kya un chaaron kaamon ke liye aapka intezaar zaroori hai?

Kabir: Photo rakhna toh zaroori hai. Warna woh gayab ho jaayegi.

Madhav: Aur chhota version?

Kabir: Woh baad mein bhi ban sakta hai.

Madhav: Doston ko batana?

Kabir: Woh bhi baad mein.

Madhav: Toh kya karoge?

Kabir: Photo rakh do, aur baaki kaam kahin likh do ki karne hain. Phir turant “ho gaya” dikha do.

Madhav: Aur baaki kaam kaun karega?

Kabir: Koi doosra program jo us list ko dekhta rahe.

Madhav: Ab batao user ko kya dikha. Aur sach kya hai.

Kabir: User ko “ho gaya” dikha, aur sach yeh hai ki aadha hua hai.

Madhav: Kya yeh jhooth hai?

Kabir: Thoda toh hai.

Madhav: Kya user ko farak padta hai?

Kabir: Agar chhota version do second baad ban jaaye toh nahi padta.

Madhav: Aur agar woh kabhi na bane?

Kabir: Toh padta hai. Toh us list mein rakhe kaam kabhi kho nahi sakte.

Madhav: Bas. Ab woh list. Do log ek saath usmein daalein toh?

Kabir: Wahi purani problem. Taala.

Madhav: Aur ek kaam do log utha lein?

Kabir: Toh do baar hoga. Photo ka chhota version do baar ban jaayega.

Madhav: Kya bura hai?

Kabir: Uss case mein bura nahi. Lekin agar kaam “paise bhejo” hota toh?

Madhav: Toh?

Kabir: Toh do baar paise chale jaate.

Madhav: Toh kaam ko aisa banana padega ki do baar hone se bhi nuksaan na ho.

Kabir: Kaise?

Madhav: Har kaam pe ek pehchaan number daal do, aur karne se pehle dekh lo ki yeh number pehle to nahi ho chuka.

NAAM

Us list ka naam hai **queue**, yaani ek kataar.

Aur us poore tareeke ka naam hai kaam ko **turant** aur **baad mein** mein baant dena:

TURANT (user ka intezaar)	jo bina hue user ko nateeja nahi mil sakta
BAAD MEIN (queue)	jo baad mein ho sakta hai

Yeh sabse aam aur sabse zyada asar wala tareeka hai kisi bhi cheez ko tez dikhane ka. Aur dhyaan do ki usne kaam kam nahi kiya. Chaaron kaam abhi bhi ho rahe hain. Bas user ab unka intezaar nahi kar raha.

Cheez tez nahi hui. Intezaar hat gaya.

Queue ke saath teen niyam aate hain, aur teeno Kabir ne khud chhue:

1. KAAM KHO NA SAKE
queue ko disk pe rakhna padta hai, RAM mein nahi
2. DO BAAR HO SAKTA HAI
isliye har kaam aisa hona chahiye ki dobara hone pe nuksaan na ho
(iske liye har kaam ko ek pehchaan number diya jaata hai)
3. KAAM FAIL HO SAKTA HAI
toh use dobara koshish karni chahiye, kuch der baad
aur agar baar baar fail ho toh use alag rakh do, taaki
woh baaki sab ko na roke

Teesri baat pe ek naam hai: jo kaam baar baar fail hote hain unhe ek alag jagah rakh diya jaata hai, jise **dead letter queue** kehte hain. Woh us chitthi jaisi hai jo na pahunch payi aur na wapas ja payi, aur post office ke ek kone mein rakh di gayi.

ASLI DUNIYA SE EK EXAMPLE

Jab aap kisi shopping app pe order karte ho, toh “order confirmed” turant dikh jaata hai.

Us waqt tak sirf ek cheez hui hai: aapka order likh liya gaya hai. Baaki sab kuch queue mein pada hai. Payment ki jaanch, saaman ka reservation, godaam ko sandesh, aapko email, delivery ka plan.

Isiliye kabhi kabhi aapko “order confirmed” ke das minute baad “sorry, saaman khatam ho gaya” wala message aata hai. Woh galti nahi hai. Woh iss design ka seedha nateeja hai.

Company ne jaan-boojh ke yeh chuna hai. Wajah karobaari hai: agar checkout mein aath second lagein toh log chhod kar chale jaate hain. Aath second ka intezaar us thodi si gadbad se zyada mehnga hai jo baad mein maafi maang kar theek ki ja sakti hai.

Yeh Chapter 5.4 wala sauda hai, phir se: **thodi galti, ya bahut intezaar.**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **”ho gaya” ko “sach mein poora ho gaya” samajhna.**

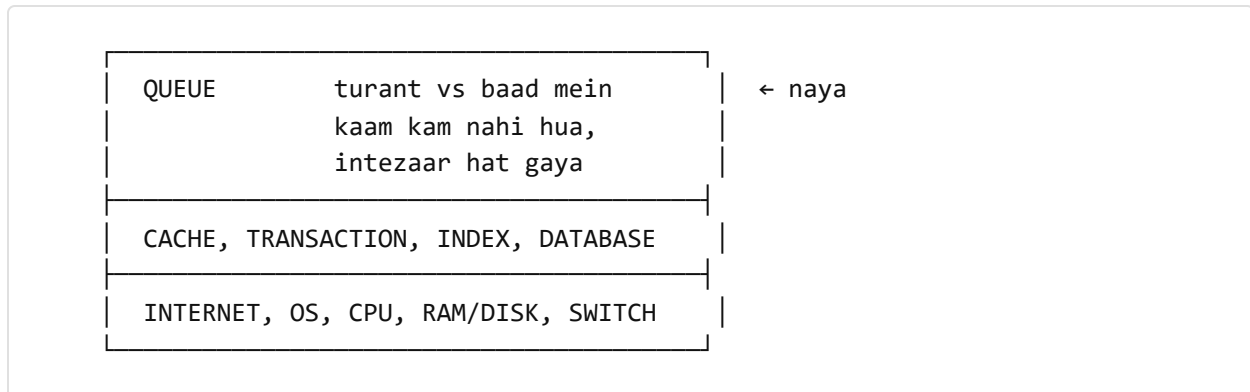
Aaj ke lagbhag har bade system mein “ho gaya” ka matlab hai “humne pakka likh liya hai ki yeh hoga.”

Yeh dhokha nahi hai, jab tak woh sach mein ho jaaye. Lekin iska matlab hai ki beech mein ek khidki hai jismein cheez aadhi hui padi hai.

Iska seedha nateeja: log samajh nahi paate ki paise “bhej diye” ke baad bhi kai ghante kyun lag jaate hain, ya order confirm hone ke baad cancel kyun ho jaata hai.

Aur banane walon ke liye ek doosri galti hai: queue daal dena aur yeh maan lena ki ab woh kaam ho hi jaayega. Queue kaam hone ki guarantee nahi hai. Woh sirf yeh guarantee hai ki kaam **yaad rakha jaayega**. Kya woh ho paayega, yeh ek alag sawal hai, aur usko koi na koi dekhta rehna chahiye.

MAP PE



SOCHNE KE LIYE

1. (samajh check) Ek email bhejne wala app hai. User “send” dabata hai. Kaunse kaam turant hone chahiye aur kaunse queue mein?

Jawab: Turant: email ko pakka likh lo, aur “bhej diya” dikha do.

Queue mein: asli email ko doosre server tak pahunchana, koshish karna, fail ho toh dobara koshish karna, attachment sambhalna.

Wajah: doosre server tak pahunchne mein kuch second se kuch ghante lag sakte hain, aur woh aapke control mein nahi hai. User ko utni der rokna bekaar hai.

Aur isiliye email kabhi kabhi der se pahunchte hain aur kabhi wapas aa jaate hain, “bhej diya” dikhne ke ghanton baad.

2. (samajh check) Ek kaam queue mein hai: “user ke khaate se sau रुपये काटो.” Woh galti se do baar utha liya jaata hai. Kya karoge?

Jawab: Har kaam ko ek pehchaan number do, aur karne se pehle dekho ki woh number pehle ho chuka hai ya nahi.

Yaani kaam ko aisa banao ki dobara chalane se kuch naya na ho.

Yeh soch bahut kaam ki hai: **maan lo ki har cheez do baar hogi, aur cheez ko aisa banao ki isse farak na pade.**

Kyunki bade systems mein cheezein sach mein do baar hoti hain. Network ka jawab kho jaata hai, koshish dobara hoti hai, aur pehli baar sach mein ho chuki thi.

3. (jodne wala) Chapter 4.3 mein UDP tha, jismein kho gaya toh chhod diya jaata hai. Queue usse kaise alag hai?

Jawab: Ulta hai, aur wajah wahi purani hai: kya purana kaam ab bhi kaam ka hai?

UDP: awaaz ka ek tukda kho gaya. Use dobara laane ka koi faayda nahi, kyunki woh purana ho chuka hai. Chhod do.

Queue: “paise kaato” wala kaam kho gaya. Woh purana nahi hota. Woh aaj bhi utna hi zaroori hai. Isliye use kabhi chhoda nahi ja sakta.

Toh ek hi sawal, do ulte jawab, aur farak sirf yeh hai ki **kaam ki keemat waqt ke saath girti hai ya nahi.**

Yeh sawal aage bar bar poochna kaam ka hai. Woh lagbhag hamesha yeh tay kar deta hai ki system ka design kaisa hoga.

Chapter 5.7 [SPINE]

Data kho jaaye toh kya

SAMVAAD

Madhav: Saara data ek disk pe hai. Woh disk kharaab ho jaati hai. Kya gaya?

Kabir: Sab kuch.

Madhav: Kya karoge?

Kabir: Copy rakhunga. Doosri disk pe.

Madhav: Kab copy karoge?

Kabir: Roz raat ko.

Madhav: Disk dopahar do baje kharaab hui. Kya gaya?

Kabir: Raat se dopahar tak ka sab kaam.

Madhav: Toh?

Kabir: Zyada baar copy karo. Har ghante.

Madhav: Toh ek ghanta jaayega. Aur agar ek second bhi na jaana chahiye?

Kabir: Toh har likhne ke saath hi doosri jagah bhi likhna padega.

Madhav: Ab woh dheema ho jaayega. Har kaam do jagah.

Kabir: Haan.

Madhav: Toh sauda kya hai?

Kabir: Jitna kam data khone ko taiyaar hoon, utna dheema aur mehnga.

Madhav: Ab ek doosri problem. Tumne copy bana li, usi kamre mein. Kamre mein aag lag gayi.

Kabir: Dono gaye. Copy door rakhni chahiye.

Madhav: Kitni door?

Kabir: Doosre sheher mein.

Madhav: Ab har likhne ke saath doosre sheher likhoge? Chapter 4.3 yaad karo.

Kabir: Latency. Har likhne mein safar ka waqt jud jaayega.

Madhav: Toh phir?

Kabir: Paas wali copy turant, door wali thodi der baad.

Madhav: Aur agar bade hadse mein paas wali dono chali jaayein?

Kabir: Toh door wali se thoda data kam milega.

Madhav: Bas. Ab teesri problem, aur yeh sabse tedhi hai. Tumne copy bana li, sab theek hai. Ab galti se ek command chal gayi jisne saara data mita diya.

Kabir: Copy se wapas le lunga.

Madhav: Copy turant banti hai, har likhne ke saath. Toh copy mein kya hai?

Kabir: Copy mein bhi mit gaya hoga.

Madhav: Kyunki copy ne kya kiya?

Kabir: Usne wahi kiya jo asli ne kiya. Woh galti bhi copy ho gayi.

Madhav: Toh copy kis cheez se bachati hai aur kis se nahi?

Kabir: Machine kharaab hone se bachati hai. Galti se nahi bachati.

Madhav: Toh galti se kya bachayega?

Kabir: Purana data. Kal ka, jo galti se pehle ka hai.

Madhav: Toh ab tumhare paas do bilkul alag cheezein hain, aur log unhe hamesha mila dete hain.

NAAM

Do alag cheezein, do alag kaam:

REPLICA (nakal)

- | har badlav turant doosri jagah bhi hota hai
- | machine kharaab ho toh doosri turant sambhal leti hai
- | galti se nahi bachati: galti bhi turant copy ho jaati hai
- | uska kaam hai: chalte rehna

BACKUP (purani copy)

- | kisi purane waqt ki tasveer, alag rakhi hui
- | galti, virus, ya kisi ke jaan-boojh ke mitane se bachati hai
- | wapas laane mein waqt lagta hai
- | uska kaam hai: waqt mein peeche jaana

Aur do naap jo har jagah istemaal hote hain:

KITNA DATA KHO SAKTA HAI	copy kitni baar banti hai
KITNI DER BAND REH SAKTA HAI	wapas aane mein kitna lagta hai

Dono ko zero ke kareeb laana mumkin hai, aur bahut mehnga hai. Isliye har company yeh tay karti hai ki uske liye kitna theek hai. Ek bank ke liye lagbhag zero. Ek chhote blog ke liye ek din bhi chalega.

Aur ab woh baat jo iss chapter ki asli seekh hai, aur jise log sabse zyada bhoolte hain:

Backup tab tak backup nahi hai jab tak aapne use wapas laakar dekha na ho.

Backup lena aasan hai. Woh chup-chaap chalta rehta hai aur koi kabhi nahi dekhta. Aur woh chup-chaap kharaab bhi ho sakta hai, mahinon tak, aur kisi ko pata nahi chalta.

Pata us din chalta hai jab zaroorat padti hai. Aur woh sabse bura din hota hai yeh pata karne ke liye.

ASLI DUNIYA SE EK EXAMPLE

2017 mein GitLab naam ki company mein ek engineer ne raat ko, thaka hua, ek command chalayi. Woh galat server pe chal gayi. Lagbhag 300 GB ka asli data mit gaya.

Unke paas paanch alag tarah ke bachaav the.

Unmein se chaar us waqt kaam nahi kar rahe the. Kuch chup-chaap band ho chuke the, kuch se banti hui files khaali thin, kuch kabhi jaanche hi nahi gaye the.

Paanchva ek chhe ghante purana snapshot tha, jo kisi ne kisi aur kaam ke liye banaya tha. Usi se woh bache. Chhe ghante ka data phir bhi gaya.

Sabse dhyaan dene wali baat: unhone poori ghatna sabke saamne likhi, live. Aur usse yeh saaf hua ki unke paas bachaav ki kami nahi thi. **Unke paas paanch bachaav the aur unmein se kisi ki jaanch nahi hoti thi.**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **replica ko backup samajhna.**

”Mera data cloud pe hai, do jagah copy hai, toh surakshit hai.”

Woh machine kharaab hone se surakshit hai. Woh aapki apni galti se bilkul surakshit nahi hai.

Agar aap galti se ek folder delete karo, toh woh dono jagah se delete hoga. Turant. Kyunki replica ka poora kaam hi yeh hai ki wahi kare jo asli ne kiya.

Iska seedha nateeja: log apni photos “cloud pe hain” samajh kar nishchint rehte hain, aur ek din galti se delete karke unhe pata chalta hai ki cloud ne woh galti bhi wafadari se copy kar di.

Isiliye achhe systems delete ko turant nahi karte. Woh use kuch din ke liye ek kone mein rakhte hain. Woh ek chhota sa backup hai, jaan-boojh ke banaya gaya, aapki apni galti ke liye.

Sahi soch:

machine kharaab	→	replica bachayega
building mein aag	→	door wala replica bachayega
aapki galti	→	sirf backup bachayega
virus / hamla	→	sirf purana backup bachayega

MAP PE

REPLICA	chalte rehna	← naya
BACKUP	waqt mein peeche jaana (jaanchi na ho toh backup hai hi nahi)	← naya
QUEUE, CACHE, TRANSACTION, INDEX		
INTERNET, OS, CPU, RAM/DISK, SWITCH		

Part 5 khatam.

PART 5 KA GATE

1. Ek app das hazaar users pe theek chal raha tha, das lakh pe dheema ho gaya. Code wahi hai. Kya hua? (*nahi aaya toh 5.2*)

Jawab: Kahin poori list padhi ja rahi hai. Woh kaam data ke saath seedha badhta hai.

Das hazaar pe woh dikha nahi. Das lakh pe woh saamne aa gaya. Ilaaj index hai, badi machine nahi.

2. Aapka data do jagah copy hota hai, turant. Aap galti se ek table delete kar dete ho. Bach jaaoge? (*nahi aaya toh 5.7*)

Jawab: Nahi. Replica ne woh delete bhi turant copy kar diya.

Aapko purana backup chahiye, jo galti se pehle ka ho.

3. Ek app “order confirmed” turant dikhata hai lekin das minute baad cancel ka message aata hai. Yeh bug hai? (*nahi aaya toh 5.6*)

Jawab: Nahi, yeh design hai. “Confirmed” ka matlab tha ki order pakka likh liya gaya. Baaki sab queue mein tha.

Company ne jaan-boojh ke thodi gadbad chuni, taaki intezaar na ho.

CHAT APP PE WAPAS

Aapka message ek database mein jaata hai, file mein nahi, kyunki chaar problem hain: dhoondhna, ek saath likhna, beech mein rukna, aur dohrav.

Use dhoondhne ke liye index hai, warna das crore mein se ek nikaalne mein paanch crore kadam lagte.

Bhejna ek transaction hai: message likha bhi jaaye aur "bheja gaya" bhi lage, ya dono mein se kuch bhi nahi.

Notification queue mein jaati hai, isliye aapko "sent" turant dikhta hai jabki doosre ke phone tak pahunchne mein waqt lagta hai.

Aapki purani chat cache mein hai, isliye scroll karna tez hai.

Aur woh sab do jagah copy hai, taaki machine kharaab ho toh chalta rahe. Lekin agar aap khud chat delete kar do, toh woh dono jagah se delete ho jaayegi.

Ab ek sawal jo abhi tak khula hai:

Yeh sab ek machine pe chal raha tha. Ab ek crore log ek saath aa jaayein toh? Yeh Part 6 hai.

Chapter 6.1 [SPINE]

Ek user se hazaar user tak

SAMVAAD

Madhav: Ek machine hai. Ek user aata hai, sawal poochta hai, jawab milta hai. Ek sawal mein sau millisecond lagte hain. Ek second mein kitne sawal sambhal sakti hai?

Kabir: Das.

Madhav: Ab sau log ek saath aa gaye. Kya hoga?

Kabir: Line lag jaayegi. Aakhri wale ko das second lagenge.

Madhav: Aur hazaar?

Kabir: Sau second. Log chale jaayenge.

Madhav: Toh kya karoge?

Kabir: Badi machine le lo.

Madhav: Achha. Do guna badi machine mili. Ab kitne?

Kabir: Bees per second.

Madhav: Chaar guna badi?

Kabir: Chalis.

Madhav: Ab sabse badi machine jo duniya mein milti hai. Uske baad?

Kabir: Uske baad kuch nahi.

Madhav: Toh ek deewar hai. Doosra raasta socho.

Kabir: Kai machinein lagao.

Madhav: Das machinein hain. User kis pe jaayega?

Kabir: Koi tay kare.

Madhav: Toh ek aur cheez chahiye jo saamne khadi ho aur baante.

Kabir: Haan.

Madhav: Ab woh khud kitne sambhal sakti hai?

Kabir: Woh bhi ek machine hai. Uski bhi seema hogi.

Madhav: Lekin uska kaam chhota hai. Woh sirf aage bhej rahi hai, sochna nahi pad raha. Toh woh ek lakh per second kar sakti hai jabki asli kaam wali sirf das.

Kabir: Toh theek hai.

Madhav: Ab ek problem. User pehle machine pe login karta hai. Agli baar woh teesri machine pe pahunchta hai. Kya hoga?

Kabir: Teesri ko pata hi nahi ki yeh kaun hai. Woh phir se login maangegi.

Madhav: Toh?

Kabir: Login ki jaankari machine mein nahi, kisi saanjhi jagah pe rakhni padegi.

Madhav: Kahan?

Kabir: Database mein. Ya kisi cache mein jo sabke liye ek ho.

Madhav: Bas. Aur ab yeh niyam ban jaata hai: machine ke andar kuch bhi aisa mat rakho jo sirf usi ke paas ho. Kyunki agli baar user kahin aur jaayega.

Kabir: Toh machinein khaali honi chahiye.

Madhav: Aur agar woh khaali hain, toh ek machine mar jaaye toh?

Kabir: Kuch nahi hoga. Baaki sambhal lengi.

Madhav: Aur agar aur log aa jaayein?

Kabir: Aur machinein jod do.

Madhav: Ab ek aakhri sawal. Das machinein hain, sab khaali. Sab kis cheez se data leti hain?

Kabir: Ek database se.

Madhav: Toh ab problem kahan chali gayi?

Kabir: Database pe. Ab woh sabka bojh utha raha hai.

Madhav: Bas. Tumne abhi wahi cheez dhoondh li jo har badhne wale system mein hoti hai. Aap bojh hatate nahi ho. Aap use khiska dete ho.

NAAM

Do tareeke bade hone ke:

UPAR JAANA (scale up)

- |— ek machine ko bada karo
- |— aasan: kuch nahi badalna padta
- |— mehnga: bade machine ka daam tez badhta hai
- |— ek deewar hai: sabse badi machine ke baad kuch nahi

BAGAL MEIN FAILNA (scale out)

- |— kai chhoti machinein
- |— sasta: chhoti machinein sasti hoti hain
- |— koi deewar nahi: aur jodte jao
- |— mushkil: ab sab kuch alag tareeke se banana padta hai

Woh saamne khadi cheez jo baantti hai, use **load balancer** kehte hain.

Aur woh niyam jo Kabir ne nikaala, woh sabse zaroori hai:

Har machine khaali honi chahiye. Yaani usmein aisi koi jaankari na ho jo sirf usi ke paas hai. Kyunki agli baar user kisi aur machine pe pahunchega.

Aisi machine ko **stateless** kehte hain, aur uske teen bade faayde hain:

ek machine mar jaaye	→	kisi ko farak nahi padta
bojh badhe	→	aur machinein jod do
kaam kam ho	→	machinein hata do

Aur ab woh baat jo iss poore Part ka dhaancha hai:

Bojh hatta nahi hai. Woh khiskta hai.

Aap web machinein badha lete ho, toh bojh database pe chala jaata hai. Database ko badha lete ho, toh bojh kisi ek table pe chala jaata hai. Usko baant do, toh bojh us jagah pe chala jaata hai jahan sab milte hain.

Har baar aapko naya sabse patla hissa dhoondhna padta hai. Kaam kabhi khatam nahi hota, sirf jagah badalti hai.

ASLI DUNIYA SE EK EXAMPLE

Twitter ke shuruaati saalon mein ek nishaan mashhoor ho gaya tha: ek whale ko chidiyan utha rahi hain. Woh tab dikhta tha jab site bojh se baith jaati thi, aur 2007 se 2009 tak woh bahut dikhta tha.

Unki shuruaati banawat ek hi bade database pe khadi thi. Jab tak users kam the, theek chala. Jab woh tez badhne lage, toh woh ek database sabse patla hissa ban gaya.

Unhe kai saal aur poori banawat badalni padi. Yeh naye code se nahi hua, banawat ke naye faisle se hua.

Aur yeh kahani lagbhag har badhne wale product mein dohrayi jaati hai. **Jo banawat sau users pe sahi hai, woh das lakh pe galat hoti hai. Aur woh galat isliye nahi hoti ki koi galti hui thi. Woh us waqt sahi thi.**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **pehle din se bade paimane ke liye banana.**

Log shuruaat mein hi woh sab lagate hain jo das lakh users ke liye chahiye: das machinein, load balancer, sab kuch baanta hua.

Uski keemat teen tarah se chukani padti hai: banane mein zyada waqt, chalane mein zyada kharcha, aur galti dhoondhna bahut mushkil, kyunki ab cheez kai jagah bikhri hai.

Aur zyadatar products kabhi das lakh users tak pahunchte hi nahi. Toh woh saari mehnat ek aisi problem ke liye thi jo kabhi aayi hi nahi.

Yeh galti templing isliye hai ki bade companies ke tareeke likhe hue milte hain aur woh samajhdaar lagte hain. Log woh tareeke uthate hain lekin unki wajah nahi uthate.

Sahi soch:

```
ab jo hai uske liye banao
lekin aisi cheezein mat karo jo aage badalna namumkin kar dein
```

Aur unmein sabse badi cheez wahi hai jo iss chapter mein aayi: machine ko khaali rakho. Woh aaj lagbhag kuch nahi maangta, aur kal sab kuch aasan kar deta hai.

MAP PE

BADA HONA	upar jaana, ya bagal	← naya
KHAALI MACHINE	jaankari machine mein	← naya
	nahi, saanjhi jagah pe	
BOJH KHISKTA HAI,	hatta nahi	
REPLICA, QUEUE, CACHE, DATABASE		
INTERNET, OS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Aapne das machinein lagayin aur site phir bhi dheemi hai. Kahan dekhoge?

Jawab: Us jagah jahan sab dus milti hain. Aksar database.

Das machinein sirf apna hissa baant rahi hain. Agar sabko ek hi jagah se data laana hai, toh woh jagah ab das guna bojh utha rahi hai.

Yeh iss chapter ka poora nichod hai: **jab aap ek jagah ka bojh baantte ho, toh woh agli jagah pe chala jaata hai.** Har baar naya sabse patla hissa dhoondhna padta hai.

2. (samajh check) Ek machine user ka login apne andar rakh leti hai kyunki woh tez hai. Kya problem hai?

Jawab: Agli baar user kisi doosri machine pe ja sakta hai, aur usko kuch pata nahi hoga.

Aur agar woh machine mar jaaye toh us par ke saare users ka login gaya.

Aur aap us machine ko hata nahi sakte, kyunki usmein kuch aisa hai jo sirf usi ke paas hai.

Ek chhoti si “speed ki chaal” ne teen aazadiyan cheen li: machine marne ki, hatane ki, aur nayi jodne ki.

3. (jodne wala) Chapter 3.4 mein OS ek manager tha jo baari baantta tha. Load balancer usse kaise alag hai aur kaise ek jaisa?

Jawab: Ek jaisa: dono baantte hain jab ek cheez hai aur maangne wale kai.

Alag: OS ek machine ke andar baantta hai, aur uske paas poori taakat hai. Load balancer machinon ke beech baantta hai, aur uske paas koi taakat nahi hai. Woh sirf aage bhejta hai.

Aur yeh farak mayne rakhta hai. OS ek program ko rok sakta hai. Load balancer ek machine ko nahi rok sakta, woh sirf usse door ja sakta hai.

Jitna aap failte ho, utni aapki taakat kam hoti jaati hai. Yeh Part 6 mein baar baar dikhega.

Chapter 6.2 [SPINE]

Woh machine jo kabhi band nahi hoti

SAMVAAD

Madhav: Chapter 4.2 mein tay hua tha ki aapka phone bahar se pahuncha nahi ja sakta, isliye beech mein koi hota hai. Woh beech wali machine kaisi honi chahiye?

Kabir: Hamesha chalu.

Madhav: Kyun? Aapka phone toh raat ko band ho jaata hai.

Kabir: Kyunki mujhe nahi pata ki kaun kab message bhejega. Agar woh us waqt band hui toh message kahin nahi jaayega.

Madhav: Toh “hamesha chalu” ek zaroorat hai. Ab batao, hamesha chalu rehna kitna mushkil hai?

Kabir: Bijli chahiye, internet chahiye. Bas.

Madhav: Bijli chali gayi?

Kabir: Battery honi chahiye.

Madhav: Battery khatam?

Kabir: Generator.

Madhav: Machine kharaab ho gayi?

Kabir: Doosri machine honi chahiye.

Madhav: Software update karna hai?

Kabir: Hmm. Update karne ke liye band karna padega.

Madhav: Toh update ke liye site band?

Kabir: Nahi. Ek machine band karo, doosri chalti rahe. Phir doosri.

Madhav: Achha. Ab batao, aap “hamesha chalu” kitna pakka kar sakte ho? Sau pratishat?

Kabir: Shayad nahi.

Madhav: Kyun nahi?

Kabir: Kuch bhi ho sakta hai. Bhookamp. Poore ilaake ki bijli. Kisi ki bahut badi galti.

Madhav: Toh aap sirf yeh keh sakte ho ki kitna kam band rahega. Chalo ginti karte hain. Saal mein kitne ghante hote hain?

Kabir: Lagbhag 8,760.

Madhav: 99 pratishat chalu ka matlab kitna band?

Kabir: Ek pratishat. Lagbhag 88 ghante. Teen din se zyada.

Madhav: 99.9?

Kabir: Lagbhag 9 ghante.

Madhav: 99.99?

Kabir: Lagbhag 53 minute.

Madhav: 99.999?

Kabir: Paanch minute. Poore saal mein.

Madhav: Ab socho. Saal bhar mein paanch minute. Uska matlab hai ki koi insaan uthkar theek nahi kar sakta, kyunki uthne mein hi paanch minute lag jaayenge.

Kabir: Toh sab kuch apne aap theek hona chahiye.

Madhav: Aur uski keemat?

Kabir: Bahut zyada hogi. Har cheez ki do copy, aur unhe apne aap sambhalne wala system.

Madhav: Ab batao, har cheez ko 99.999 banana chahiye?

Kabir: Nahi. Sirf usko jiska band hona sach mein mehnga ho.

NAAM

Us “kitna chalu raha” ko **uptime** kehte hain, aur log use naukon mein ginte hain.

99%	ek saal mein	~3.65 din band
99.9%		~8.8 ghante
99.99%		~53 minute
99.999%		~5 minute

Har naya nau daam ko lagbhag guna kar deta hai. 99.9 se 99.99 tak jaana aksar poore system ko dobara banana hota hai.

Aur woh cheez jiska matlab hai “band hone wali koi ek jagah na ho,” use **no single point of failure** kehte hain. Yaani koi bhi ek cheez aisi na ho jiske marne se sab ruk jaaye.

Ek machine ko hamesha chalu rakhne ke liye kya kya do baar chahiye:

bijli	→	do alag laain, battery, generator
internet	→	do alag company
machine	→	kam se kam do, alag jagah
disk	→	kai, ek kharaab ho toh chalta rahe
building	→	do alag sheher (bade systems mein)
aadmi	→	koi hamesha uplabdh ho

Aur ab woh baat jo iss chapter ki asli seekh hai:

”Hamesha chalu” ek naukri hai, koi feature nahi.

Woh ek baar bana kar chhoda nahi ja sakta. Koi hamesha dekh raha hota hai. Koi raat ko phone uthata hai. Koi har update ko aise karta hai ki beech mein kuch na ruke.

Aur isiliye “server pe daal do” itna sasta nahi hota jitna log sochte hain. Machine ka kiraya chhota hissa hai. **Use hamesha chalu rakhna asli kharcha hai**, aur woh kharcha zyadatar aadmiyon ka hai, machinon ka nahi.

ASLI DUNIYA SE EK EXAMPLE

2021 mein ek badi cloud company ki ek service kuch ghanton ke liye baith gayi. Uske saath duniya bhar ki hazaaron websites, apps aur cheezein baith gayin, kyunki woh sab usi ke upar khadi thin.

Un hazaaron companies mein se har ek ne apne system ko mazboot banaya tha. Har ek ke paas kai machinein thin, backup the, sab kuch tha.

Lekin sabke neeche ek hi cheez thi.

Yeh us baat ka sabse saaf udaharan hai jo iss chapter mein hai: **aapka system utna hi mazboot hai jitni uski sabse kamzor neeche wali parat**. Aur woh parat aksar aapki nahi hoti, aur aapko yaad bhi nahi hota ki woh wahan hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki “cloud pe hai isliye hamesha chalu rahega.”**

Cloud sirf kisi aur ki machinein hain. Woh bhi band hoti hain. Farak sirf yeh hai ki jab woh band hoti hain toh aapke saath hazaaron aur bhi band hote hain, aur aap kuch nahi kar sakte, sirf intezaar.

Doosri galti: yeh sochna ki zyada nau hamesha behtar hai.

99.99 ka matlab hai ki poore saal mein 53 minute se zyada band nahi. Uski keemat aksar poore system ko dobara banana hoti hai, aur log ko raat ko uthana hota hai.

Ek chhoti company ke liye 99.9 lagbhag hamesha kaafi hai. Woh saal mein 9 ghante hain, jo aksar update aur chhoti dikkaton mein hi chale jaate hain, aur kisi ko farak nahi padta.

Sahi sawal: **ek ghanta band rehne se kitna nuksaan hota hai?** Agar jawab “kuch khaas nahi” hai, toh naukon ke peeche bhagna sirf paisa jalana hai.

MAP PE

UPTIME	naukon mein ginti	← naya
	har nau daam guna karta	
KOI EK JAGAH	aisi na ho jo sab rok de	← naya
BADA HONA,	KHAALI MACHINE	
REPLICA,	QUEUE, CACHE, DATABASE	
INTERNET,	OS, CPU, RAM/DISK, SWITCH	

SOCHNE KE LIYE

1. (samajh check) Ek company kehti hai “hamara uptime 99.9% hai.” Woh saal mein kitni der band reh sakti hai, aur kya woh achha hai?

Jawab: Lagbhag 8.8 ghante saal mein, ya har mahine lagbhag 43 minute.

Achha hai ya nahi, yeh iss baat pe hai ki band hone se kya jaata hai.

Ek blog ke liye bahut achha. Ek shopping site ke liye theek. Ek bank ke card system ke liye bilkul nahi, kyunki 43 minute mein lakhon transaction fail ho jaayenge.

Naukon ka koi apna matlab nahi hai. Unka matlab band hone ki keemat se aata hai.

2. (samajh check) Aap sab kuch do baar rakhte ho: do machinein, do database, do internet connection. Kya ab aap surakshit ho?

Jawab: Un cheezon se, haan. Lekin do saval bache hain.

Ek: kya dono ek hi building mein hain? Kya dono ek hi bijli pe hain? Kya dono ek hi cloud company pe hain?

Do: kya woh doosri wali sach mein kaam karegi? Kabhi jaanchi hai?

Yeh Chapter 5.7 wali baat hai: **jo bachaav jaancha nahi gaya, woh bachaav hai hi nahi.** Bahut se systems mein doosri machine saalon se maujood thi aur kabhi chali hi nahi thi.

3. (jodne wala) Chapter 6.1 kehta tha ki bojh hatta nahi, khiskta hai. Uptime ke saath kya wahi hota hai?

Jawab: Haan, aur ek nayi shakal mein.

Aap ek machine ko do bana dete ho, toh woh problem hat jaati hai. Ab problem us cheez pe chali jaati hai jo tay karti hai ki kaunsi machine chal rahi hai.

Use do bana do, toh problem us cheez pe chali jaati hai jo un dono ke beech faisla karti hai.

Har baar aap ek jagah ki kamzori hatate ho aur ek nayi jagah bana dete ho.

Isiliye 100 pratishat kabhi mumkin nahi hota. **Har hal apne saath ek nayi, chhoti kamzori laata hai.** Aap use chhota kar sakte ho, khatam nahi.

Chapter 6.3 [SPINE]

Dikhne wala hissa, na dikhne wala hissa

SAMVAAD

Madhav: Aap ek shopping app kholte ho. Saaman ki list dikhti hai, daam dikhta hai, ek button hai. Yeh sab kahan chal raha hai?

Kabir: Mere phone pe.

Madhav: Aur daam kahan se aaya?

Kabir: Server se.

Madhav: Toh do jagah kaam ho raha hai. Ab ek sawal. Daam ka hisaab kahan hona chahiye, phone pe ya server pe?

Kabir: Phone pe hoga toh tez hoga.

Madhav: Achha. Toh main apne phone mein daam ka hisaab badal doon aur ek rupaye mein khareed loon?

Kabir: Nahi... phir toh server pe hona chahiye.

Madhav: Kyun? Ek line mein.

Kabir: Kyunki phone mera hai. Main usmein kuch bhi badal sakta hoon.

Madhav: Bas. Yeh iss chapter ka poora niyam hai. Ab dobara batao: kya kya phone pe ho sakta hai?

Kabir: Dikhana. Rang, layout. Scroll karna. Shayad ginti ki jaanch, jaise khaali box.

Madhav: Aur woh jaanch phone pe kyun, jab server bhi kar sakta hai?

Kabir: Kyunki turant dikhta hai. Server tak jaane mein waqt lagta hai.

Madhav: Toh woh jaanch phone pe hai. Ab main phone mein woh jaanch hata doon aur khaali form bhej doon. Kya hoga?

Kabir: Server ko phir se jaanchna padega.

Madhav: Toh jaanch do jagah?

Kabir: Haan. Phone pe user ki suvidha ke liye, aur server pe sach mein.

Madhav: Ab yeh yaad rakho, kyunki yahin log sabse badi galti karte hain. Phone wali jaanch **madad** hai. Server wali jaanch **suraksha** hai. Dono ek jaisi dikhti hain aur unka kaam bilkul alag hai.

Kabir: Samajh gaya.

Madhav: Ab ek aur sawal. Woh saaman ki list phone pe aayi. Kis shakal mein aayi?

Kabir: Data ki shakal mein.

Madhav: Ya server poora page bana ke bhej sakta tha, saja hua, taiyaar.

Kabir: Woh bhi ho sakta hai.

Madhav: Dono mein kya farak padta hai?

Kabir: Agar server poora bana ke bheje toh phone ka kaam kam hoga aur pehli baar jaldi dikhega.

Madhav: Aur agar sirf data bheje?

Kabir: Toh phone ko banana padega. Pehli baar dheemi, lekin uske baad har chhote badlav ke liye poora page nahi laana padega.

Madhav: Aur ek teesri baat. Agar server poora page banata hai, toh kya wahi data ek doosre app ko bhi de sakta hai?

Kabir: Nahi. Page toh browser ke liye hai.

Madhav: Aur agar sirf data deta hai?

Kabir: Toh koi bhi le sakta hai. Phone app, website, koi aur company.

Madhav: Toh ab dono ke faayde saaf hain, aur koi ek jawab nahi hai.

NAAM

FRONT (dikhne wala hissa)

- └─ user ke device pe chalta hai
- └─ dikhana, chhoona, turant jawab
- └─ user use badal sakta hai, isliye uspe bharosa nahi
- └─ kaam: anubhav

BACK (na dikhne wala hissa)

- └─ server pe chalta hai
- └─ asli faisla, asli hisaab, asli data
- └─ user use chhoo bhi nahi sakta, isliye yahi sach hai
- └─ kaam: sachai aur suraksha

Aur woh ek niyam jo kabhi nahi tootta:

Client pe kabhi bharosa mat karo.

Jo cheez user ke device pe chal rahi hai, use badla ja sakta hai. Chahe woh app ho, website ho, ya kuch bhi. Isliye har zaroori jaanch server pe dobara honi chahiye, chahe woh phone pe pehle ho chuki ho.

Do jaanch bekaar nahi hain. Woh do alag kaam kar rahi hain:

phone pe → user ko turant bata do, taaki use intezaar na ho
server pe → galat cheez andar aane hi mat do

Aur woh doosra sawal, page kahan bane:

SERVER PE BANE

- └─ pehli baar jaldi dikhta hai
- └─ purane aur dheeme device pe behtar
- └─ har chhote badlav pe server tak jaana padta hai

DEVICE PE BANE

- └─ pehli baar dheemi, uske baad tez
- └─ data alag milta hai, toh koi bhi use kar sakta hai
- └─ device pe zyada kaam, battery zyada

Aaj ke zyadatar bade systems dono karte hain: pehla page server se banaya hua, uske baad sab device pe.

ASLI DUNIYA SE EK EXAMPLE

Purane zamane mein har page server pe banta tha. Aap kuch bhi click karte, poora page dobara aata tha, aur screen safed hokar wapas aati thi.

2005 ke aas-paas ek naya tareeka phaila: page ko poora badalne ki jagah, sirf zaroori data mangwao aur page ka ek hissa badal do. Google Maps ne yeh mashhoor kiya, jahan aap naksha kheench sakte the aur page dobara nahi khulta tha.

Uske baad lehar itni door chali gayi ki lagbhag sab kuch device pe banne laga, aur log bhool gaye ki uski keemat kya hai. Websites bhaari ho gayin, purane phone pe atakne lagin, aur pehli baar khulne mein der lagne lagi.

Ab lehar wapas beech mein aa rahi hai. Pehla page server se, baaki device pe.

Yeh dhyaan dene laayak hai, kyunki yeh baar baar hota hai: **koi nayi cheez aati hai, log usse har jagah lagate hain, uski keemat dikhti hai, aur phir woh apni sahi jagah pe baith jaati hai.** Yeh kamzori nahi hai, yeh seekhne ka tareeka hai. Lekin agar aap beech mein aaye ho toh lagta hai ki sab log ek disha mein daud rahe hain.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki app ke andar ka code chhupa hua hai.**

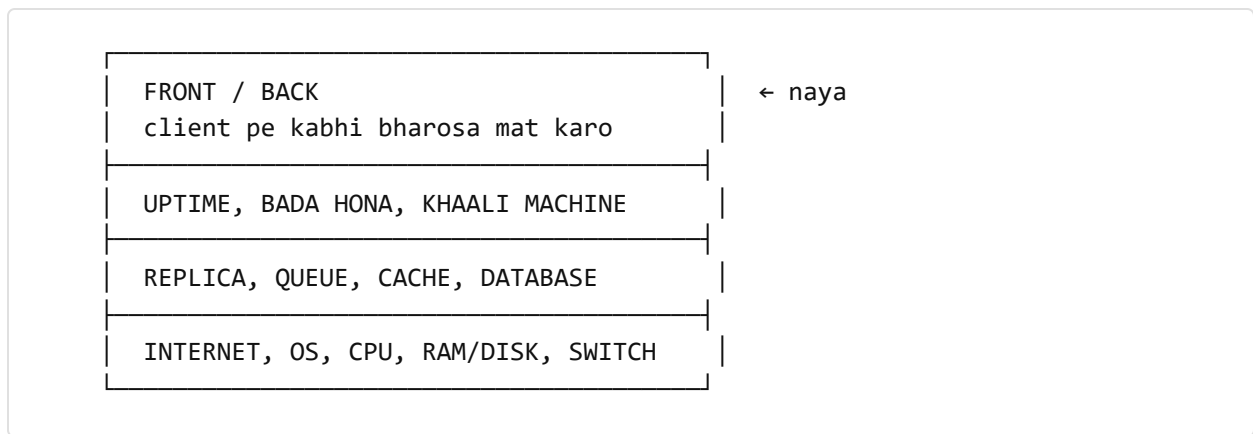
Log apne app ya website mein password, chaabiyen, ya zaroori niyam daal dete hain, yeh sochkar ki user use dekh nahi sakta.

User sab kuch dekh sakta hai. Woh app ko khol sakta hai, uske andar jhaank sakta hai, aur usmein se nikli har cheez padh sakta hai. Aisa hazaaron baar hua hai, aur aaj bhi roz hota hai.

Jo cheez user ke device pe hai, woh user ki hai.

Yeh galti tempting isliye hai ki code padhne mein mushkil dikhta hai, aur log maan lete hain ki mushkil ka matlab namumkin hai. Aur kyunki jab aap khud dekhte ho toh sab theek chalta hai, toh yeh khatra kabhi saamne nahi aata jab tak koi dhoondh na le.

MAP PE



SOCHNE KE LIYE

1. (samajh check) Ek app mein “sirf 18 saal se upar” wali jaanch hai. Woh phone pe hai. Kya problem hai?

Jawab: User use hata sakta hai.

Woh jaanch user ki madad ke liye theek hai, taaki use turant pata chale. Lekin woh suraksha nahi hai.

Agar us jaanch ka kanooni ya karobaari matlab hai, toh woh server pe honi chahiye, jahan user pahunch hi nahi sakta.

Yeh niyam har jagah lagta hai: **jo cheez sach mein zaroori hai, woh wahan honi chahiye jahan user ka haath na pahunchhe.**

2. (samajh check) Ek website pehli baar khulne mein 6 second leti hai lekin uske baad bahut tez chalti hai. Kya ho raha hai?

Jawab: Lagbhag poora page device pe banaya ja raha hai.

Pehli baar poora program utarna padta hai aur chalna padta hai, isliye der lagti hai. Uske baad sab kuch device pe hai, isliye har click turant hai.

Sauda saaf hai: **pehli baar mehngi, baad mein sasti.**

Aur woh sauda kis ke liye theek hai, yeh iss baat pe hai ki log ek baar aakar bahut kaam karte hain, ya ek baar aakar ek cheez dekh kar chale jaate hain. Ek dukaan ke liye ek jawab hai, ek news site ke liye doosra.

3. (jodne wala) Chapter 3.4 mein OS har app ko alag rakhta tha, aur app ko poori taakat nahi deta tha. Front aur back ka batwara usse kaise milta hai?

Jawab: Bilkul wahi soch hai, ek parat upar.

OS kehta hai: app ko taakat mat do, use maangne do, aur main tay karunga.

Server kehta hai: user ke device ko taakat mat do, use maangne do, aur main tay karunga.

Dono jagah wahi niyam hai: **jispe aapka control nahi hai, uspe bharosa mat karo.**

Yeh soch aage Book 2 mein phir milegi, jahan ek AI agent ke paas bhi apni koi taakat nahi hoti, aur har cheez kisi na kisi ke through jaati hai.

Chapter 6.4 [SPINE]

Do programs aapas mein baat kaise karte hain

SAMVAAD

Madhav: Aapke phone ka app server se data maangta hai. Ab yeh socho: woh kaise maangta hai?

Kabir: Chapter 4.4 wala protocol. Woh keh dega ki mujhe yeh chahiye.

Madhav: Protocol batata hai ki sandesh ki shakal kya hogi. Lekin yeh kaun tay karta hai ki kya kya maanga ja sakta hai?

Kabir: Server wala.

Madhav: Aur app banane wale ko kaise pata chalega?

Kabir: Server wale ko batana padega.

Madhav: Kis shakal mein?

Kabir: Ek list. “Yeh yeh cheezein maang sakte ho, aur is tarah maangna.”

Madhav: Ab agar server wala kal woh list badal de?

Kabir: Toh saare app toot jaayenge.

Madhav: Kitne app?

Kabir: Jo bhi use istemaal kar rahe hain. Ho sakta hai hazaaron.

Madhav: Aur woh app kiske hain?

Kabir: Doosre logon ke. Doosri companies ke.

Madhav: Toh ab server wale ke haath bandh gaye. Woh apni cheez badal nahi sakta.

Kabir: Kyunki uspe doosre khade hain.

Madhav: Toh kya karega jab use sach mein kuch badalna ho?

Kabir: Purana chalta rahe, aur naya alag se de de.

Madhav: Dono ek saath, kitne saal?

Kabir: Jab tak sab naye pe na aa jaayein. Shayad kabhi na aayein.

Madhav: Bas. Ab ek aur baat. Agar doosri companies aapke server se data le sakti hain, toh yeh aapke liye achha hai ya bura?

Kabir: Bura. Woh mera data le rahe hain.

Madhav: Ya achha. Socho.

Kabir: Achha... agar woh mere upar apna kuch banayein toh mere bina unka kaam nahi chalega.

Madhav: Aur unke users?

Kabir: Woh sab ghoomkar mere paas hi aayenge.

Madhav: Toh ek list, jo sirf technical cheez lagti hai, woh kya ban gayi?

Kabir: Ek raasta jisse doosre log mere upar khade ho jaate hain.

Madhav: Aur us raaste ka maalik kaun hai?

Kabir: Main.

Madhav: Aur agar main kal woh raasta band kar doon?

Kabir: Toh jo log mere upar khade the woh gir jaayenge.

Madhav: Yaad rakhna. Yeh aage bahut zaroori hoga.

NAAM

Us list ka naam hai **API**: woh tay-shuda tareeka jisse ek program doosre program se baat karta hai.

Farak samajhna zaroori hai:

PROTOCOL	sandesh ki shakal kya hogi (sabke liye ek, jaise HTTP)
API	kaunse sandesh maangne ki ijaazat hai, aur unka matlab kya hai (har service ka apna)

Protocol yeh hai ki chitthi kaise likhi jaaye. API yeh hai ki iss daftar mein kaunsi darkhwastein manzoor hoti hain.

Ek API kuch aisa dikhta hai:

GET	/users/42	→	us user ki jaankari do
GET	/users/42/messages	→	uske messages do
POST	/messages	→	naya message banao
DELETE	/messages/99	→	woh message hata do

Aur ab teen baatein jo API ko technical cheez se zyada bana deti hain.

Ek: API ek waada hai. Ek baar aapne diya, log uspe khade ho jaate hain. Use badalna unhe todna hai. Isliye badalne ki jagah aksar naya jod diya jaata hai aur purana chalta rehta hai, saalon tak.

Do: API ek darwaza hai. Jo aapke API pe kuch banata hai, woh aap pe nirbhar ho jaata hai. Uske users ghoomkar aapke paas aate hain.

Teen: API ek taala bhi hai. Jisne darwaza khola hai, woh use band bhi kar sakta hai. Aur jo log uspe khade the, woh gir jaate hain.

Yeh teesri baat ek asli karobaari chaal hai, aur woh baar baar chali gayi hai: pehle darwaza khol do taaki log aa jaayein aur aapke upar cheezein banayein, aur jab woh nirbhar ho jaayein tab shartein badal do.

ASLI DUNIYA SE EK EXAMPLE

2006 se 2012 tak Twitter ka API lagbhag khula tha. Uske upar hazaaron app bane, aur unmein se kuch Twitter ke apne app se behtar the. Woh khula darwaza hi ek badi wajah tha ki Twitter itna phaila.

Phir shartein sakht hoti gayin. Kitne users ho sakte hain, kya dikhaya jaa sakta hai, kya nahi. 2023 mein API lagbhag poori tarah paid kar diya gaya, aur daam itna rakha gaya ki chhoti cheezein chal hi nahi sakti thin.

Hazaaron app, jo saalon se chal rahe the, kuch hafton mein band ho gaye. Unka code theek tha. Unki machinein chal rahi thin. Bas darwaza band ho gaya.

Yeh Chapter 0.2 ki bhasha mein saaf hai: un app banane walon ka leverage kabhi unka tha hi nahi. Woh kisi aur ke darwaze pe khade the.

Aur yeh yaad rakhne laayak sabak hai: **kisi aur ke API pe banana Anil ki naali banana nahi hai. Woh kisi aur ki naali se paani lena hai.**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **API ko sirf ek technical cheez samajhna.**

Log sochte hain ki API bas ek raasta hai data laane ka, aur uska faisla technical hai.

Woh ek karobaari faisla hai. Kya khola jaaye, kise diya jaaye, kis daam pe, aur kab band kiya jaaye, yeh sab tay hota hai.

Iska seedha nateeja: log poora product kisi aur ke API pe bana dete hain, aur maan lete hain ki woh hamesha rahega. Uske do khatre hain: woh band ho sakta hai, aur woh mehnga ho sakta hai. Dono baar baar hote hain.

Yeh galti tempting isliye hai ki API istemaal karna bahut aasan hai. Ek line likho aur poori taakat mil jaati hai. Woh aasani hi khatra chhupa deti hai.

Sahi sawal, jab bhi aap kisi aur ke API pe kuch banao:

agar yeh kal band ho jaaye toh mera kya bachega?
 agar iska daam das guna ho jaaye toh?
 kya iske jaisa koi doosra hai?
 kya main uska hissa hoon ya uska mukabla?

Aakhri sawal sabse zaroori hai. Jo log platform ke mukable mein aa jaate hain, unka darwaza sabse pehle band hota hai.

MAP PE

API	waada, darwaza, taala	← naya
FRONT/BACK, UPTIME, BADA HONA		
REPLICA, QUEUE, CACHE, DATABASE		
INTERNET, PROTOCOL, OS, CPU, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Aapne ek API banaya aur sau companies use kar rahi hain. Aapko usmein ek galti dikhi jise theek karna hai, lekin theek karne se woh sau toot jaayengi. Kya karoge?

Jawab: Purana chalta rehne do, naya alag se do, aur logon ko waqt do.

Yeh lagbhag hamesha ka jawab hai, aur uski keemat yeh hai ki ab aap do cheezein sambhal rahe ho, shayad saalon tak.

Isiliye API banate waqt sabse zaroori kaam yeh hai ki **kam se kam waada karo**.

Har cheez jo aap dikhate ho, woh hamesha ke liye dikhani padegi.

Yeh Chapter 3.3 wali JavaScript ki baat hai: jo cheez phail gayi, use hataya nahi ja sakta.

2. (samajh check) Ek company apna API muft aur khula rakhti hai. Chapter 3.6 ke chaar wajahon mein se kaunsi ho sakti hai?

Jawab: Lagbhag hamesha teesri: neeche wali parat pe kabza.

Woh chahte hain ki log unke upar cheezein banayein, kyunki jitne zyada log unpe khade honge, utna hi unhe hataana mushkil hoga.

Muft dena shuruaat hai, seema nahi. Jab kaafi log nirbhar ho jaate hain, tab shartein badalti hain.

Yeh dhokha nahi hai, yeh ek jaani-pehchani chaal hai. Aur ise pehchan lena hi bachaav hai.

3. (jodne wala) Chapter 4.4 kehta tha ki jo protocol tay karta hai woh us ilaake ki shakal tay kar deta hai. API ke saath kya farak hai?

Jawab: Bada farak hai, aur wahi asli baat hai.

Protocol kisi ka nahi hota. HTTP ka koi maalik nahi hai, isliye koi use band nahi kar sakta.

API ka maalik hota hai. Woh use kabhi bhi badal ya band kar sakta hai.

Toh dono par khade hone mein zameen aasman ka farak hai:

protocol pe khade ho	→	koi aapko hata nahi sakta
API pe khade ho	→	ek company aapko hata sakti hai

Yeh sabse kaam ki baaton mein se ek hai jo iss kitaab mein hai. Jab bhi kuch banao, poocho: **main kis cheez pe khada hoon, aur uska maalik kaun hai?**

Chapter 6.5 [SPINE]

Kharidne ki jagah kiraye pe lena

SAMVAAD

Madhav: Aapko ek website chalani hai. Kitni machinein chahiye?

Kabir: Pata nahi. Ek se shuru karta hoon.

Madhav: Aur agar kal das hazaar log aa gaye?

Kabir: Aur machinein chahiye.

Madhav: Kitni der mein mil jaayengi, agar aap khareedte ho?

Kabir: Order karo, aaye, lagao. Kuch hafte.

Madhav: Aur woh das hazaar log kitni der rukenge?

Kabir: Kuch second.

Madhav: Toh khareedna kaam nahi karta. Ab ulta socho. Aapne das machinein pehle se khareed leen, aur log aaye hi nahi.

Kabir: Toh paisa gaya. Woh machinein khaali padi rahengi.

Madhav: Toh do taraf se nuksaan hai. Kam khareedo toh chook jaate ho, zyada khareedo toh paisa jalta hai. Aur pehle se pata nahi hota ki kitni chahiye.

Kabir: Toh kiraye pe le lo. Jitni chahiye utni.

Madhav: Kaun dega?

Kabir: Koi jiske paas bahut saari machinein hon.

Madhav: Woh itni machinein rakhega hi kyun?

Kabir: Kyunki bahut saare log usse maang rahe hain.

Madhav: Aur sab ek saath maangenge?

Kabir: Nahi. Kisi ka bojh subah zyada hoga, kisi ka raat ko. Kisi ka December mein, kisi ka June mein.

Madhav: Toh usko kitni machinein rakhni padengi? Sabke sabse bade bojh ka jod?

Kabir: Nahi. Usse kam. Kyunki sabke bade bojh ek waqt pe nahi aate.

Madhav: Bas. Wahi poora karobaar hai. Ab ek aur baat. Aapne ek machine kiraye pe li. Uska daam kya hoga, khareedne se zyada ya kam?

Kabir: Zyada hoga. Usko munafa bhi chahiye.

Madhav: Toh phir log kirya kyun lete hain?

Kabir: Kyunki jab zaroorat nahi hoti tab paisa nahi lagta. Aur turant milti hai.

Madhav: Aur agar aapko pata ho ki aapko poore saal, chaubees ghante, dus machinein chahiye hi chahiye?

Kabir: Toh khareedna sasta hoga.

Madhav: Kitna sasta? Aam taur pe teen se paanch guna.

Kabir: Itna zyada?

Madhav: Isliye jab company chhoti hoti hai toh kirya samajhdaari hai, aur jab bahut badi ho jaati hai toh kabhi kabhi wapas khareedne mein chali jaati hai. Ab aakhri sawal. Aapne apna saara system kisi ek kiraye wale pe daal diya. Kya mila aur kya gaya?

Kabir: Mila: mujhe machinein nahi sambhalni padeen. Gaya: ab main uspe nirbhar hoon.

Madhav: Aur woh daam badha de toh?

Kabir: Toh mujhe dena padega, ya sab kuch uthakar kahin aur le jaana padega.

Madhav: Kitni mehnat lagegi uthane mein?

Kabir: Bahut. Kyunki maine unki cheezein istemaal ki hongy jo kahin aur milti hi nahi.

NAAM

Kisi aur ki machinein kiraye pe lene ka naam hai **cloud**.

Uske teen darje hote hain, aur farak yeh hai ki aap kitna khud sambhalte ho:

KHAALI MACHINE

- | woh machine deta hai, baaki sab aap
- | sabse zyada control, sabse zyada kaam
- | aksar sabse sasta

TAIYAAR JAGAH

- | aap code deto ho, woh chalata hai
- | OS, update, bada hona woh sambhalta hai
- | beech ka daam

POORA KAAM

- | database, email, login: banaya banaya
- | aap sirf istemaal karte ho
- | sabse mehnga, sabse aasan

Cloud ka asli faayda machine ka daam nahi hai. Woh teen cheezein hain:

- | | |
|-----------------------|---|
| 1. TURANT MILNA | das machinein paanch minute mein |
| 2. JITNA CHAHIYE UTNA | raat ko kam, din mein zyada |
| 3. AADMI KAM | machine, bijli, AC, jaal: unka sar dard |

Aur uski keemat bhi teen hain:

- | | |
|---------------|--|
| 1. DAAM ZYADA | poore waqt chalne wali cheez ke liye 3-5 guna |
| 2. NIRBHARTA | unke daam, unke niyam, unka band hona |
| 3. FANS JAANA | jitni unki khaas cheezein istemaal ki,
utna nikalna mushkil |

Teesri baat pe ek naam hai: **lock-in**. Aur woh muft cheezon se shuru hota hai. Har aasan aur sasti cheez jo sirf ek jagah milti hai, woh ek aur dhaaga hai jo aapko wahin baandhta hai.

ASLI DUNIYA SE EK EXAMPLE

Dropbox ne apne shuruaati saalon mein sab kuch Amazon ke cloud pe rakha tha. Woh us waqt bilkul sahi faisla tha: unhe apna product banana tha, data centre nahi.

Jab woh bahut bade ho gaye, unhone apna khud ka system banaya aur apna zyadatar data usmein le aaye. Kai saal lage. Unhone bataya ki isse unhe do saal mein lagbhag 7.5 crore dollar ki bachat hui.

Ab dhyaan do ki dono faisle sahi the, alag alag waqt pe.

Shuru mein kirya sahi tha, kyunki tab paisa aur waqt banane mein lagna chahiye tha, na ki machinein sambhalne mein. Baad mein khareedna sahi tha, kyunki bojh pakka ho chuka tha aur bahut bada tha.

Sahi jawab company ke size aur waqt ke saath badalta hai. Yeh un chhote se sawalon mein se ek hai jinka jawab hamesha “depends” hota hai, aur woh “depends” ginta ja sakta hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki cloud sasta hai.**

Cloud lagbhag hamesha mehnga hai, per machine. Uska faayda daam mein nahi hai.

Uska faayda yeh hai ki aap sirf utna dete ho jitna istemaal karte ho, aur aapko machine sambhalne wale log nahi rakhne padte. Ek chhoti company ke liye woh do cheezein daam se kahin badi hain.

Lekin agar aapka bojh pakka hai aur poore waqt chalta hai, toh aap ek suvidha ke liye teen se paanch guna de rahe ho jiski aapko zaroorat hi nahi.

Doosri galti: yeh maan lena ki cloud pe hai toh hamesha chalega. Chapter 6.2 mein hum dekh chuke hain ki cloud bhi baithta hai, aur jab baithta hai toh aap kuch nahi kar sakte.

Teesri galti, aur yeh sabse mehngi hai: **bina soche unki khaas cheezein istemaal karte jaana.**

Har baar jab aap unki koi aisi cheez lete ho jo kahin aur nahi milti, toh nikalna thoda aur mushkil ho jaata hai. Ek din aapko pata chalta hai ki aap ja hi nahi sakte, aur us din aapke paas mol-bhav ki koi taakat nahi bachti.

MAP PE

CLOUD	kisi aur ki machinein turan, jitna chahiye mehnga + nirbharta	← naya
API, FRONT/BACK, UPTIME, BADA HONA		
REPLICA, QUEUE, CACHE, DATABASE		
INTERNET, OS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Ek app hai jismein din mein 10,000 log aate hain aur raat ko koi nahi. Kirya ya kharid?

Jawab: Kirya, aur saaf faisla hai.

Kyunki bojh ubhaar-chadhaav wala hai. Khareedoge toh aapko sabse bade waqt ke hisaab se khareedna padega, aur woh machinein raat bhar khaali padi rahengi.

Kirye mein aap raat ko machinein kam kar sakte ho aur us waqt ka paisa hi nahi dete.

Ubhaar-chadhaav wala bojh = kirya. Pakka aur poore waqt ka bojh = kharid.

Yeh sabse saaf niyam hai jo iss sawal pe lagta hai.

2. (samajh check) Ek cloud company aisi database deti hai jo bahut aasan hai aur sirf unke yahan milti hai. Aap use loge?

Jawab: Le sakte ho, lekin aankhein khuli rakh kar.

Faayda asli hai: kam kaam, tez shuruaat.

Keemat yeh hai ki us din se aapka jaana mushkil ho gaya. Aur woh mushkil dikhta nahi, kyunki jab tak aap ja nahi rahe tab tak koi dard nahi hota.

Ek achhi aadat: **poocho ki iska koi aisa vikalp hai jo har jagah milta hai?** Agar haan, toh sochna chahiye ki jo aasani mil rahi hai woh us aazadi se badi hai ya nahi.

3. (jodne wala) Chapter 6.4 kehta tha ki kisi aur ke API pe khada hona khatra hai. Cloud usse kaise judta hai?

Jawab: Woh wahi baat hai, sabse badi shakal mein.

API pe khade ho toh ek cheez kisi aur ki hai. Cloud pe khade ho toh **poori zameen** kisi aur ki hai.

Uske daam, uske niyam, uska band hona, sab aap pe seedha lagta hai.

Iska matlab yeh nahi ki cloud galat hai. Lagbhag har company usi pe khadi hai, aur wajah achhi hai.

Matlab sirf itna hai ki yeh ek chunav hai, majboori nahi. Aur har chunav ki keemat hoti hai, jo Chapter 5.4 se hum jaante hain.

Chapter 6.6 [DEPTH]

Sab ek saath aa jaayein toh kya tootta hai

DEPTH chapter. Yahan aap seekhoge ki systems kis tarah girte hain, aur kyun woh dheere-dheere kharaab nahi hote, balki ek jhatke mein baith jaate hain.

SAMVAAD

Madhav: Ek system ek second mein sau sawal sambhal sakta hai. Ninanve aane lage. Kya haal hoga?

Kabir: Theek chalega. Thoda dheema.

Madhav: Ek sau ek?

Kabir: Ek sawal line mein rukega.

Madhav: Aur agle second mein bhi ek sau ek?

Kabir: Do ruk jaayenge.

Madhav: Aur yeh ek ghante tak chalta rahe?

Kabir: Line badhti jaayegi. Teen hazaar se zyada.

Madhav: Aur line mein khade log kya kar rahe hain?

Kabir: Intezaar.

Madhav: Kitna intezaar?

Kabir: Aakhri wale ko... poori line ka waqt. Bahut zyada.

Madhav: Aur jo aadmi tees second intezaar kar chuka hai, woh kya karega?

Kabir: Chhod dega. Ya refresh karega.

Madhav: Refresh karega toh?

Kabir: Ek naya sawal aa jaayega. Line aur badhegi.

Madhav: Aur purana sawal?

Kabir: Woh bhi line mein hi hai. Woh bhi kaam karwayega, chahe ab uska jawab koi na dekhe.

Madhav: Toh ab dhyaan se dekho. Bojh sirf ek pratishat zyada tha. Kya hua?

Kabir: Log ruke, phir chhod kar dobara aaye, jisse bojh aur badh gaya, jisse aur log ruke.

Madhav: Toh system ka bojh kitna badha?

Kabir: Do guna, teen guna. Khud badhta chala gaya.

Madhav: Aur system ek pratishat zyada ke liye taiyaar tha?

Kabir: Ek pratishat ke liye toh tha. Teen guna ke liye nahi.

Madhav: Bas. Ab batao, yeh dheere-dheere kharaab hua ya achanak?

Kabir: Achanak. Ninanve pe sab theek tha, ek sau ek pe sab kuch baith gaya.

Madhav: Toh kya karoge?

Kabir: Line ko badhne hi mat do. Ek seema ke baad naye sawal mana kar do.

Madhav: User ko mana karna? Woh toh bura hai.

Kabir: Sabko dheere-dheere marne dene se behtar hai. Kam se kam kuch logon ko jawab toh milega.

Madhav: Bas. Yeh iss chapter ka sabse zaroori faisla hai. **Sabko thoda thoda dena, ya kuch ko poora dena.** Aur bade systems lagbhag hamesha doosra chunte hain.

NAAM

Woh haalat jismein bojh khud ko badhata jaata hai, use **cascading failure** kehte hain, yaani girti hui shrinkhla.

Uska dhaancha hamesha ek jaisa hota hai:

1. bojh seema ke thoda upar gaya
2. jawab dheema hua
3. log ne intezaar chhod kar dobara koshish ki
4. bojh aur badha
5. aur dheema hua
6. wapas 3 pe

Bojh 1% badha. System 100% baith gaya.

Aur uske hal, jo lagbhag har bade system mein hote hain:

LINE PE SEEMA	ek had ke baad naye sawal mana karo "abhi bahut bheed hai, thodi der baad"
PURANE SAWAL PHENK DO	agar aadmi ja chuka hai toh uska kaam mat karo
BACHAV KA SWITCH	jab ek hissa girne lage toh use turant band karo aur baaki chalne do (nahi toh woh sabko apne saath le jaayega)
THODI DER RUK KAR	dobara koshish karo, lekin sabhi ek saath nahi warna sab ek hi pal pe wapas aayenge

Aakhri wala dhyaan dene laayak hai. Agar sab log theek paanch second baad dobara koshish karein, toh woh ek saath aayenge aur system phir baith jaayega. Isliye har ek ko thoda alag intezaar diya jaata hai.

Aur ab woh baat jo iss chapter ki asli seekh hai:

System dheere-dheere kharaab nahi hote. Woh ek seema tak bilkul theek chalte hain aur uske baad ek jhatke mein baith jaate hain.

Isiliye "abhi toh sab theek chal raha hai" koi jaankari nahi hai. Aap seema se 20 pratishat neeche ho sakte ho ya 1 pratishat neeche, aur dono bilkul ek jaise dikhte hain.

ASLI DUNIYA SE EK EXAMPLE

Har saal kisi na kisi badi ticket ki bikri, sale, ya result ke din yeh dohraya jaata hai. Site pehle dheemi hoti hai, phir error dene lagti hai, phir poori tarah baith jaati hai.

Aur aksar company ne bojh ka andaza sahi lagaya hota hai. Unhone shayad das guna machinein bhi laga rakhi hoti hain.

Jo cheez tootti hai woh ginti nahi hoti, woh **shrinkhla** hoti hai. Log refresh karte hain, har refresh ek naya sawal hai, aur bojh us ginti se bahut aage nikal jaata hai jiska andaza lagaya gaya tha.

Isiliye achhe systems in dinon mein ek “line” wala page dikhate hain. Woh page kuch nahi karta, sirf aapko rokta hai. Woh bura anubhav hai, aur woh jaan-boojh ke chuna gaya hai, kyunki uska vikalp yeh hai ki kisi ko kuch na mile.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki zyada machinein lagane se yeh problem hal ho jaayegi.**

Machinein seema ko upar khiska deti hain. Woh shrinkhla ko nahi rokhti.

Agar aap das guna machinein laga do, toh aap das guna bojh sambhal loge. Aur jis din bojh gyarah guna hoga, us din bilkul wahi cheez hogi, sirf bade paimane pe.

Asli hal seema badhana nahi hai. Asli hal yeh hai ki **seema paar hone pe system ko sahi tarah se mana karna aana chahiye.**

Doosri galti: yeh sochna ki mana karna kharaab service hai.

Ek system jo dus hazaar logon ko turant mana kar deta hai aur nabbe hazaar ko theek se jawab deta hai, woh us system se kahin behtar hai jo ek lakh logon ko chhe minute intezaar karake sabko error deta hai.

Yeh galti tempting isliye hai ki mana karna dikhta hai aur dheema hona chhupa rehta hai. Manager ko “humne 10% users ko mana kiya” bura lagta hai aur “sab dheema chal raha tha” kam bura lagta hai, jabki doosra kaafi bura hai.

MAP PE

SHRINKHLA	bojh khud ko badhata hai	← naya
SEEMA	paar hone pe mana karo dheere nahi, jhatke mein	← naya
CLOUD, API, UPTIME, BADA HONA		
QUEUE, CACHE, DATABASE, INTERNET		

SOCHNE KE LIYE

1. (samajh check) Ek system fail hone lagta hai aur uske saare users ko “10 second baad dobara koshish karein” ka message milta hai. Kya problem hai?

Jawab: Sab log theek das second baad ek saath wapas aayenge.

System ko thodi der araam milega aur phir ek hi pal mein poora bojh wapas aa jaayega, shayad pehle se bhi bada.

Isliye har user ko thoda alag intezaar dena chahiye, jaise 8 se 15 second ke beech koi bhi.

Yeh chhoti si baat lagti hai aur woh kai bade systems ko bacha chuki hai. **Ek saath wapas aana khud ek hamla ban jaata hai.**

2. (samajh check) Aapka system 100 per second sambhal sakta hai aur abhi 60 aa rahe hain. Kya aap surakshit ho?

Jawab: Pata nahi, aur yeh iss chapter ki asli seekh hai.

60 par sab theek dikhta hai. 95 par bhi sab theek dikhta hai. Aapko dono mein farak nahi dikhega.

Aur bojh aksar thoda thoda nahi badhta. Ek tweet, ek khabar, ek sale, aur woh das guna ho jaata hai.

Isliye achhe systems mein yeh naapa jaata hai ki **abhi seema se kitna neeche hain**, na ki sirf yeh ki sab theek chal raha hai ya nahi.

3. (jodne wala) Chapter 5.6 mein queue thi, jismein kaam line mein lagta tha. Kya queue iss problem ko hal karti hai ya badhati hai?

Jawab: Dono, aur yeh dhyaan dene laayak hai.

Hal karti hai: user ko intezaar nahi karna padta, kyunki uska kaam line mein daal kar use “ho gaya” keh diya jaata hai.

Badhati hai: agar kaam aane ki raftaar hone ki raftaar se zyada rahi, toh line badhti jaayegi, hamesha. Woh chup-chaap badhti hai kyunki user ko dikhta hi nahi.

Aur ek din woh itni lambi ho jaati hai ki usmein pada kaam ab bekaar ho chuka hota hai. Log ko ek ghante purani notification milti hai.

Isliye har queue ke saath do cheezein dekhi jaati hain: **woh kitni lambi hai, aur kitni tezi se badh rahi hai.** Lambai se pata chalta hai ki abhi kya haal hai. Raftaar se pata chalta hai ki aage kya hoga.

Chapter 6.7 [SPINE]

Cheezein dheeri kyun lagti hain

SAMVAAD

Madhav: Ek button dabaya aur jawab aane mein do second lag gaye. Wajah kya kya ho sakti hai?

Kabir: Server dheema hoga.

Madhav: Aur?

Kabir: Internet dheema hoga.

Madhav: Aur?

Kabir: Bas.

Madhav: Chalo poore raaste pe chalte hain aur har jagah waqt naapte hain. Pehla kadam kya hai?

Kabir: Naam se pata nikalna. DNS.

Madhav: Kitna?

Kabir: Agar cache mein hai toh kuch nahi. Nahi toh shayad 50 millisecond.

Madhav: Phir?

Kabir: Server se judna. Aur taala lagana.

Madhav: Taala lagane mein kitne chakkar lagte hain?

Kabir: Kuch. Har chakkar mein aane-jaane ka waqt.

Madhav: Agar server doosre mahadweep mein hai toh ek chakkar lagbhag 150 millisecond ka hai. Teen chakkar?

Kabir: 450 millisecond. Aur abhi tak kuch maanga bhi nahi.

Madhav: Phir sawal jaata hai aur server sochta hai. Server kya karta hai?

Kabir: Database se poochta hai.

Madhav: Ek baar?

Kabir: Shayad kai baar. Pehle user, phir uske messages, phir har message ka bhejne wala.

Madhav: Agar sau messages hain aur har ek ke liye alag sawal?

Kabir: Sau sawal. Har ek mein thoda waqt.

Madhav: Har sawal 2 millisecond ka bhi ho toh?

Kabir: 200 millisecond, sirf yahan.

Madhav: Ab jawab wapas aata hai. Phir?

Kabir: Phone use dikhata hai. Photo utarti hai.

Madhav: Ab ginti karo. Kitne alag hisse the?

Kabir: Paanch chhe. Aur har ek mein sau-do sau millisecond.

Madhav: Ab batao, aapka pehla jawab kaisa tha? "Server dheema hoga."

Kabir: Woh sirf ek hissa tha. Aur shayad sabse chhota.

Madhav: Ab aakhri sawal. In sab mein kaunsa hissa aap sudhaar sakte ho, aur kaunsa nahi?

Kabir: Server ka hisaab sudhaar sakta hoon. Database ke sau sawal ek mein badal sakta hoon.

Madhav: Aur doosre mahadweep tak jaane ka waqt?

Kabir: Woh nahi badal sakta. Woh roshni ki speed hai.

Madhav: Toh kya karoge?

Kabir: Server paas le aaunga. Ya kam chakkar lagaunga.

NAAM

Kul waqt ka ek naam hai, aur uske hisse ginne ka bhi:

DNS	naam se pata	0 se 100 ms
JUDNA + TAALA	kuch chakkar	1 se 3 chakkar
SAWAL JAANA	ek chakkar	doori pe nirbhar
SERVER SOCHNA	hisaab + database	aksar sabse bada
JAWAB AANA	ek chakkar	doori pe nirbhar
DIKHANA	phone ka kaam	10 se 500 ms

Aur do cheezein jo alag hain aur log unhe mila dete hain, Chapter 4.3 se:

BANDWIDTH	sadak kitni chaudi	→ badhayi ja sakti hai
LATENCY	sadak kitni lambi	→ badhayi nahi ja sakti

Latency ki ek sakht seema hai: roshni. Ek chakkar Mumbai se New York lagbhag 150 millisecond ka hai, aur duniya ka koi paisa use kam nahi kar sakta.

Isliye speed ke do bilkul alag ilaaj hain:

KAAM KAM KARO	behtar hisaab, kam database sawal, cache
CHAKKAR KAM KARO	ya doori kam karo (aur yeh aksar zyada bada faayda deta hai)

Aur woh sabse aam galti jo har jagah milti hai, uska ek naam hai: **N+1**.

Ek sawal se sau cheezein aayin, aur phir har cheez ke liye ek alag sawal. Ek ki jagah ek sau ek sawal. Har ek chhota hai, aur unka jod bahut bada hai.

Ilaaj lagbhag hamesha wahi hai: **sau chhote sawal ek bade sawal mein badal do.**

ASLI DUNIYA SE EK EXAMPLE

Google ne apne experiments mein paya ki search ke nateeje 400 millisecond der se dikhane par log kam search karte the. Amazon ne bhi isi tarah dekha ki 100 millisecond ki der bikri par asar dalti thi.

Yeh numbers baar baar quote kiye jaate hain aur unka asli matlab yeh hai: **log der ko sochte nahi, mehsoos karte hain.**

Koi user yeh nahi kehta ki “yeh 400 millisecond dheema hai.” Woh bas thoda kam istemaal karta hai, thoda jaldi chhod deta hai, aur use khud pata nahi hota ki kyun.

Isiliye speed ek feature hai, ek technical baat nahi. Aur isiliye badi companies apne server duniya bhar mein rakhti hain, sirf latency ke liye.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **badi machine se dheemi cheez theek karne ki koshish.**

Agar aapka waqt safar mein ja raha hai, toh do guna tez CPU se lagbhag kuch nahi hoga. Server pehle hi khaali baitha intezaar kar raha tha.

Doosri galti: **average dekhna.**

Maan lo aapka average 200 millisecond hai. Sunne mein achha hai. Lekin ho sakta hai ki 90 pratishat logon ko 50 millisecond mile aur 10 pratishat ko 3 second.

Woh 10 pratishat aksar aapke sabse zyada istemaal karne wale log hote hain, kyunki unka data sabse zyada hai.

Isiliye ache systems mein average nahi dekha jaata. Woh dekha jaata hai jise “sabse bure 5 pratishat” kehte hain. **Average sabse bure anubhav ko chhupa deta hai, aur wahi anubhav log yaad rakhte hain.**

Teesri galti: yeh sochna ki dheemi cheez ek bug hai jo kabhi kabhi hoti hai. Aksar woh design hai jo har baar hoti hai, aur bas kisi ne naapi nahi.

MAP PE

WAQT KE HISSE	DNS, judna, safar, sochna, dikhana	← naya
N+1	sau chhote sawal	← naya
AVERAGE JHOOTH BOLTA HAI		
SHRINKHLA, CLOUD, API, UPTIME		
QUEUE, CACHE, DATABASE, INTERNET		

SOCHNE KE LIYE

1. (samajh check) Aapki site India mein banai gayi hai aur server bhi India mein hai. America se log kehte hain ki woh dheemi hai. Server ka hisaab tez karne se faayda hoga?

Jawab: Bahut kam.

Unka zyadatar waqt safar mein ja raha hai. Ek chakkar lagbhag 250 millisecond ka hai, aur judne mein kai chakkar lagte hain.

Ilaaj doori kam karna hai: cheezein unke paas rakh do (CDN, Chapter 5.5), ya wahan bhi ek server rakh do.

Jab problem doori ki ho, toh hisaab tez karne se kuch nahi milta.

2. (samajh check) Ek page 50 photos dikhata hai aur har photo alag se maangi jaati hai. Kya problem hai aur kya karoge?

Jawab: Yeh N+1 ka doosra roop hai. 50 alag chakkar.

Do ilaaj hain. Ek: sab ek saath maango, ek hi baar mein. Do: sirf woh photos maango jo abhi screen pe dikh rahi hain, baaki tab jab user neeche jaaye.

Doosra tareeka aksar behtar hai, kyunki bahut se log neeche jaate hi nahi. **Sabse tez kaam woh hai jo aap karte hi nahi.**

3. (jodne wala) Chapter 2.3 kehta tha ki CPU tez hai aur data laana dheema. Kya wahi baat yahan bhi hai?

Jawab: Bilkul wahi baat hai, sirf paimana badal gaya.

ek machine ke andar	RAM se data laana	~200 kadam
ek building ke andar	doosri machine se	~10 lakh kadam
duniya ke us paar	doosre mahadweep se	~50 crore kadam

Har baar hisaab sasta hai aur doori mehngi.

Yeh iss kitaab ki sabse bade dohraav wali baat hai. Woh chip ke andar bhi sach hai, machine ke andar bhi, aur duniya ke paimane pe bhi. **Jo bhi banao, sabse pehle poocho ki data kahan se aa raha hai.**

Chapter 6.8 [SPINE]

Yeh sab chalane ka kharcha kitna hai

SAMVAAD

Madhav: Ab ek sawal jo iss kitaab mein pehli baar poocha ja raha hai. Yeh sab chalane mein paisa kitna lagta hai?

Kabir: Pata nahi. Server mehngi hongi.

Madhav: Ek chhota server, jo ek chhoti site ke liye kaafi hai, mahine ka lagbhag paanch se bees dollar. Yeh sasta hai ya mehnga?

Kabir: Sasta.

Madhav: Ab ek text message rakhne ka kharcha. Ek message lagbhag 100 byte hai. Ek GB mein kitne messages?

Kabir: Ek GB matlab sau karod byte. Toh ek karod messages.

Madhav: Aur ek GB rakhne ka kharcha mahine ka lagbhag do rupaye.

Kabir: Toh ek karod messages, do rupaye mahina?

Madhav: Bas. Ab photos. Ek photo lagbhag 3 MB. Ek GB mein kitni?

Kabir: Lagbhag 300.

Madhav: Toh 300 photos ka kharcha utna hi hai jitna ek karod messages ka.

Kabir: Tees hazaar guna zyada.

Madhav: Ab video. Ek ghanta lagbhag 1 GB. Toh ek ghante ka video ek karod messages ke barabar hai.

Kabir: Yeh farak toh maine socha hi nahi tha.

Madhav: Isiliye har text wala app lagbhag muft chal sakta hai aur har video wala app paisa maangta hai. Ab ek aur cheez. Data rakhna sasta hai. Data **bhejna** kitna mehnga hai?

Kabir: Woh alag hai?

Madhav: Bilkul alag, aur yahi sabse badi jaal hai. Cloud pe data andar laana aksar muft hai. Bahar bhejna paisa leta hai, aur woh rakhne se kai guna mehnga hai.

Kabir: Kyun?

Madhav: Karobaari wajah hai. Andar aana sasta rakho taaki log aayein, aur bahar jaana mehnga rakho taaki jaana mushkil ho.

Kabir: Woh lock-in hai, Chapter 6.5 wala.

Madhav: Bilkul. Ab aakhri hissa, aur yeh sabse bada hai. Ek server mahine ka bees dollar ka hai. Aur ek engineer?

Kabir: Bahut zyada.

Madhav: India mein ek achha engineer mahine ka lagbhag ek se do lakh रुपये. Yaani hazaar se do hazaar dollar. Kitne servers ke barabar?

Kabir: Sau se do sau server.

Madhav: Toh agar aap ek engineer ka ek mahina lagakar server ka kharcha aadha kar dete ho, aur aapke paas das server hain, toh?

Kabir: Toh maine faayde se bees guna zyada kharch kar diya.

Madhav: Bas. Yeh sabse aam galti hai jo chhoti companies karti hain: machine ka paisa bachane ke liye aadmi ka waqt jala dena.

NAAM

Lagbhag mote andaze, taaki paimana dimaag mein baithe:

CHHOTA SERVER	mahine ke \$5 - \$20
BADA SERVER	mahine ke \$200 - \$2,000
DATA RAKHNA	~\$0.02 per GB per mahina
DATA BAHAR BHEJNA	~\$0.05 - \$0.12 per GB ← rakhne se kai guna
DOMAIN NAAM	saal ke \$10 - \$15
EK ENGINEER	mahine ke \$1,000 - \$10,000+

Ab isse teen niyam nikalte hain, aur teeno kaam ke hain.

Ek: likhawat lagbhag muft hai, dekhne-sunne wali cheezein mehngi.

1 karod text message	≈ 2 rupaye mahina
300 photos	≈ 2 rupaye mahina
1 ghanta video	≈ 2 rupaye mahina

Isiliye jo product likhawat pe khada hai woh aksar muft chal sakta hai, aur jo video pe khada hai woh kabhi muft nahi ho sakta.

Do: bhejna rakhne se mehnga hai.

Ek video jo ek baar rakha gaya aur das lakh baar dekha gaya, uska kharcha rakhne mein nahi, bhejne mein hai. Isiliye CDN (Chapter 5.5) sirf speed ke liye nahi hai, woh paise ke liye bhi hai.

Teen: aadmi ka waqt machine se bahut mehnga hai.

Yeh sabse zaroori niyam hai aur sabse zyada toda jaata hai. Ek engineer ka ek mahina sau server ke barabar hai. Isiliye “yeh cheez thodi mehngi hai lekin isse kaam aasan ho jaata hai” lagbhag hamesha sahi sauda hai, jab tak paimana bahut bada na ho jaaye.

Aur yahin Chapter 6.5 ka cloud wala sawal apne aap hal ho jaata hai. Cloud per machine mehnga hai. Lekin agar woh ek aadmi ka mahina bacha deta hai, toh usne apna daam kai baar chuka diya.

ASLI DUNIYA SE EK EXAMPLE

WhatsApp ke paas 2014 mein 55 log the aur lagbhag 45 crore users. Yeh Chapter 0.2 mein leverage ke udaharan mein aaya tha.

Ab usi ko kharche ke nazariye se dekho.

Unka product lagbhag poori tarah likhawat pe khada tha. Text message. Woh sabse sasti cheez hai jo koi bhi rakh aur bhej sakta hai.

Unhone jaan-boojh ke bahut kam cheezein rakhi thin. Messages phone pe pahunchte hi server se hata diye jaate the.

Agar unka product video pe khada hota, toh wahi 45 crore users ka kharcha kai hazaar guna zyada hota, aur 55 log usse kabhi nahi chala paate.

Woh 55 ki ginti sirf achhe engineering se nahi aayi. Woh product ki shakal se aayi. Jo cheez woh bech rahe the, uska kharcha hi lagbhag zero tha.

Yeh soch ne laayak hai jab aap kuch banane ki soch rahe ho: **aap kaunsi cheez rakh aur bhej rahe ho, aur uska kharcha kaise badhega jab log das guna ho jaayein?**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki technology ka kharcha machinon mein hai.**

Lagbhag har software company mein sabse bada kharcha log hain, machinein nahi. Aksar teen-chauthai se zyada.

Iska seedha nateeja: log mahinon lagakar machine ka kharcha aadha karte hain aur us mehnat mein us bachat se zyada kharch kar dete hain.

Doosri galti: **yeh maan lena ki “muft” wala product sach mein muft hai.**

Har muft cheez ka kharcha koi utha raha hai. Chapter 3.6 mein hum poochh chuke hain: **kaun utha raha hai, aur badle mein kya le raha hai?**

Aur jab aap dekhte ho ki ek product bahut mehngi cheez muft de raha hai, jaise unlimited video, toh do mein se ek baat sach hai: ya toh unhe kahin aur se bahut paisa mil raha hai, ya woh chal nahi payega.

Teesri galti: sirf aaj ka kharcha dekhna. Sahi sawal hai: **jab yeh sau guna bada hoga tab yeh kharcha kaise badhega?** Kuch kharche users ke saath seedha badhte hain aur kuch lagbhag nahi badhte. Farak wahin se aata hai ki koi karobaar chal sakta hai ya

nahi.

MAP PE

KHARCHA	likhawat sasti, video mehnga bhejna > rakhna aadmi >> machine	← naya
CLOUD, SHRINKHLA, API, UPTIME		
QUEUE, CACHE, DATABASE, INTERNET		

SOCHNE KE LIYE

1. (samajh check) Do app hain. Ek mein log text likhte hain, doosre mein video daalte hain. Dono ke ek lakh users hain. Kharche mein kitna farak hoga?

Jawab: Hazaaron guna.

Text wala app ek chhote server aur thodi si jagah mein chal jaayega. Uska kharcha mahine ka kuch hazaar rupaye ho sakta hai.

Video wala app har user ke har video ko rakhega aur har dekhne pe bhejega. Lakhon rupaye mahina, aasani se.

Aur isiliye unke karobaar bhi bilkul alag honge. Text wala app dher saare muft users rakh sakta hai. Video wale app ko shuru se hi paise ka koi raasta chahiye.

Product ki shakal uske karobaar ki shakal tay kar deti hai.

2. (samajh check) Aapka cloud ka bill dekhne pe pata chalta hai ki sabse bada hissa “data transfer” hai, machinein nahi. Kya ho raha hai?

Jawab: Aap bahut data bahar bhej rahe ho, aur woh rakhne se kai guna mehnga hai.

Aam wajah: bade photos ya video har baar server se seedha bheje ja rahe hain, bina kisi paas wali copy ke.

Ilaaj CDN hai. Woh cheez ek baar bahar bhejta hai aur uske baad copies se deta hai, jo sasta bhi hai aur tez bhi.

Yeh dhyaan dene laayak hai: **ek hi hal aksar do alag problem solve karta hai.** CDN Chapter 5.5 mein speed ke liye aaya tha, aur yahan paise ke liye wapas aa gaya.

3. (jodne wala) Chapter 0.2 ka formula tha: size x scarcity x leverage. Iss chapter ke numbers usmein kahan lagte hain?

Jawab: Woh leverage wale dial ko naapne ka tareeka dete hain.

Ravi ka kaam har baar khatam ho jaata tha. Anil ki naali ek baar bani aur chalti rahi.

Software mein woh sawal aise dikhta hai: **jab ek aur user aata hai, toh mera kharcha kitna badhta hai?**

Agar lagbhag zero badhta hai, jaise text wale app mein, toh leverage bahut upar hai. Ek baar banao, das lakh ko do.

Agar har user ke saath seedha badhta hai, jaise video ya kisi aadmi ki mehnat wale kaam mein, toh leverage kam hai, chahe cheez technology hi kyun na ho.

Aur yahin Chapter 0.3 ki chetavni sach nikalti hai: **level 3 pe hone se leverage apne aap nahi milta.** Woh iss baat se aata hai ki agla user aapko kya kharcha deta hai.

Chapter 6.9 [SPINE]

Part 0 ka daava, test kiya gaya

SAMVAAD

Madhav: Chapter 0.3 mein tumne ek daava kiya tha. Paisa hamesha kahan se ghusta hai?

Kabir: Level 1 pe. Jab ek insaan apni zaroorat ke liye deta hai.

Madhav: Ab hum chhe part se technology dekh rahe hain. Chalo us daave ko todne ki koshish karte hain. Pehla: ek company jo sirf doosri companies ko bechti hai. Cloud company. Uske customer insaan nahi hain.

Kabir: Lekin unke customer ki bhi customer hongii... aur aakhir mein koi insaan hoga.

Madhav: Kitne kadam?

Kabir: Do, teen, shayad paanch.

Madhav: Aur agar chain kabhi kisi insaan tak na pahunche?

Kabir: Toh paisa aa hi kahan se raha hai? Kisi ne toh diya hoga.

Madhav: Achha jawab. Daava bacha. Doosra: Linux. Woh muft hai. Koi paisa nahi deta.

Kabir: Lekin uspe kaam karne wale logon ko companies tankhwah deti hain. Aur woh companies apne customers se leti hain.

Madhav: Toh paisa hai, bas woh Linux se hokar nahi jaata.

Kabir: Haan. Woh uske aas-paas se jaata hai.

Madhav: Toh daava tootta hai ya nahi?

Kabir: Nahi tootta. Lekin woh yeh nahi batata ki paisa kiske paas rukega.

Madhav: Yeh important baat hai. Aage rakho. Teesra: ek company jo paisa nahi kamati, sirf nivesh se chalti hai. Kai saal tak.

Kabir: Woh paisa toh asli hai.

Madhav: Kahan se aaya?

Kabir: Kisi nivesh karne wale se.

Madhav: Aur woh use kyun de raha hai?

Kabir: Kyunki use lagta hai ki aage log paisa denge.

Madhav: Toh?

Kabir: Toh woh aage aane wale level 1 ke paise pe daanv laga raha hai.

Madhav: Aur agar woh kabhi na aaye?

Kabir: Toh company band ho jaayegi.

Madhav: Toh daava tootta nahi, bas usmein waqt jud jaata hai. Ab chautha, aur yeh sabse tedha hai. Ek muft app jo aapka data bechta hai.

Kabir: Wahan main paisa nahi de raha. Main data de raha hoon.

Madhav: Toh paisa kaun de raha hai?

Kabir: Vigyapan dene wala.

Madhav: Woh kyun de raha hai?

Kabir: Taaki main uska saaman khareedun.

Madhav: Aur jab tum khareedte ho toh paisa kahan se aaya?

Kabir: Meri jeb se.

Madhav: Toh chain kahan se shuru hui?

Kabir: Mujhse hi. Bas main beech mein hi nahi tha, main dono siron pe tha.

Madhav: Daava zinda hai. Ab tum mujhe batao ki iss poori jaanch se kya seekha.

Kabir: Ki paisa hamesha kisi insaan ki zaroorat se aata hai, lekin woh raasta bahut lamba aur ghooma hua ho sakta hai. Aur jitna lamba raasta, utna mushkil dekhna ki kaun sach mein de raha hai.

NAAM

Daava wahi rehta hai, aur ab woh mazboot hai kyunki usne chaar hamle jhel liye:

Paisa system mein sirf ek jagah se ghusta hai: jab ek insaan apni zaroorat ke liye deta hai. Baaki har cheez us paise ka batwara hai.

Lekin jaanch se teen nayi baatein nikli hain, jo daave mein nahi thin.

Ek: raasta lamba ho sakta hai.

seedha	insaan	→	aap
ek kadam	insaan	→	company → aap
kai kadam	insaan	→	company → cloud → chip → aap
ghooma hua	insaan	→	vigyapan → app → aap (aur wahi insaan doosre sire pe bhi)
udhaar pe	insaan	→	(aage kabhi) → nivesh → aap

Do: raasta lamba hone ka matlab hai ki aapko dikhta nahi ki kaun de raha hai. Aur jab aapko nahi dikhta, toh aap yeh bhi nahi dekh paate ki woh kab rukega. Chip banane wale ko pata nahi hota ki uska paisa kis insaan ki kaunsi zaroorat se aa raha hai.

Teen, aur yeh sabse zaroori hai: paisa kahan se aata hai aur kahan rukta hai, yeh do alag sawal hain.

Yeh daava sirf pehle sawal ka jawab deta hai. Doosre ka jawab Chapter 0.2 se aata hai: value, scarcity, leverage.

Linux mein paisa hai, bahut hai. Woh Linux mein nahi rukta. Protocol likhne walon ne bahut value di aur bahut kam paisa liya (Chapter 4.8). Cloud company ke paas paisa rukta hai, kyunki jo cheez sabko chahiye aur bahut kam log de sakte hain, wahi rokti hai.

Paisa neeche se aata hai aur wahan rukta hai jahan scarcity hoti hai.

ASLI DUNIYA SE EK EXAMPLE

Ek chip banane wali company ke paas aaj duniya ki lagbhag saari AI wali chips ka bahut bada hissa hai.

Uska paisa kahan se aata hai? Woh AI companies ko bechti hai. Woh companies apni cheezein doosri companies ko bechti hain. Woh companies apne product aam logon ko bechti hain. Aur woh log apni zaroorat ke liye paisa dete hain.

Chaar kadam. Har kadam pe kuch paisa rukta hai.

Ab sawal yeh hai ki sabse zyada kahan rukta hai. Jawab wahan hai jahan sabse kam vikalp hain.

Aur yahi wajah hai ki us chip company ke paas itna paisa ruk raha hai. Uske product ka koi aasan vikalp nahi hai. Baaki har kadam pe kai vikalp hain.

Chapter 0.2 ka doosra dial, poore paimane pe. **Scarcity tay karti hai ki paisa kahan rukta hai, chahe woh kahin aur se aaya ho.**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki “main sirf companies ko bechta hoon, toh aam log mere liye mayne nahi rakhte.”**

Har chain aakhir mein kisi insaan pe khatm hoti hai. Agar woh insaan rukta hai, toh poori chain rukti hai, chahe aap us se paanch kadam door ho.

Iska seedha nateeja: log samajh nahi paate ki jab mandi girti hai toh unka karobaar kyun gir jaata hai, jabki unka aur aam logon ka koi seedha rishta nahi hai.

Doosri galti, aur yeh ulti hai: **yeh sochna ki jo cheez muft di ja rahi hai uska koi kharcha nahi hai.**

Har muft cheez ka kharcha koi utha raha hai. Sawal sirf yeh hai ki kaun, aur badle mein kya. Chapter 3.6 aur 6.8 dono yeh sawal poochh chuke hain, aur woh hamesha jawab deta hai.

MAP PE

PAISA	neeche se aata hai scarcity pe rukta hai raasta lamba ho sakta hai	← naya
KHARCHA, CLOUD, API, UPTIME		
QUEUE, CACHE, DATABASE, INTERNET		
OS, CPU, RAM/DISK, SWITCH		

Part 6 khatam.

PART 6 KA GATE

1. Aapne das machinein lagayin aur site phir bhi dheemi hai. Sabse pehle kahan dekhoge? (*nahi aaya toh 6.1*)

Jawab: Us jagah jahan sab das milti hain, aksar database. Bojh hata nahi, khisak gaya.

2. Aapka system 100 per second sambhal sakta hai aur 101 aa rahe hain. Woh thoda dheema hoga ya baith jaayega? (*nahi aaya toh 6.6*)

Jawab: Baith sakta hai. Log intezaar chhod kar dobara koshish karenge, jisse bojh khud ba khud badhta jaayega.

Systems dheere-dheere kharaab nahi hote. Woh ek seema tak theek chalte hain aur phir jhatke mein girte hain.

3. Ek app 45 crore users ko 55 logon se chala. Do wajahein batao. (*nahi aaya toh 6.8, 0.2*)

Jawab: Ek: uska product likhawat pe khada tha, jo sabse sasti cheez hai rakhne aur bhejne mein.

Do: uska leverage bahut upar tha. Har naya user lagbhag koi naya kharcha nahi laata tha.

CHAT APP PE WAPAS

Aapka chat app ab ek machine pe nahi chalta. Woh kai machinon pe chalta hai, aur har machine khaali hai, taaki koi bhi kisi ka kaam kar sake.

Saamne ek cheez khadi hai jo baantti hai. Peeche ek database hai jisme ab poora bojh hai, kyunki bojh hata nahi, khisak gaya.

Aapke phone wala hissa sirf dikhata hai. Sach hamesha server pe tay hota hai, kyunki aapke phone pe bharosa nahi kiya ja sakta.

Woh machinein kirye pe hain, kisi aur ki hain, kisi doosre sheher mein hain. Woh tez isliye lagti hain kyunki aapke messages aapke paas ki ek copy se aate hain, asli jagah se nahi.

Aur uska poora kharcha lagbhag kuch bhi nahi hai, jab tak log sirf likhawat bhejte rahein. Jis din video shuru hua, sab badal jaayega.

Ab ek sawal jo iss poori kitaab mein abhi tak nahi poocha gaya:

Yeh sab kisi ne banaya. Banaya kaise? Aur usmein galtiyan kyun rehti hain? Yeh Part 7 hai.

Chapter 7.1 [SPINE]

Software mein bug hote hi kyun hain

SAMVAAD

Madhav: Ek pul banaya jaata hai. Woh kitni baar girta hai?

Kabir: Lagbhag kabhi nahi.

Madhav: Ek app banaya jaata hai. Usmein kitni galtiyan hoti hain?

Kabir: Hamesha kuch na kuch.

Madhav: Toh pul banane wale software banane walon se behtar hain?

Kabir: Shayad. Ya unka kaam alag hai.

Madhav: Chalo dekhte hain. Ek pul kitni alag haalaton mein hota hai?

Kabir: Wahi ek pul hai. Wazan badalta hai, mausam badalta hai.

Madhav: Kitni haalatein? Sau? Hazaar?

Kabir: Shayad.

Madhav: Ab ek chhota sa program. Usmein sirf teen cheezein hain: user ka naam, umar, aur ek button. Kitni haalatein?

Kabir: Naam kuch bhi ho sakta hai. Umar bhi. Toh bahut zyada.

Madhav: Naam khaali ho sakta hai? Bahut lamba? Usmein emoji ho sakta hai? Doosri bhasha?

Kabir: Haan sab.

Madhav: Umar negative ho sakti hai? Zero? Ek lakh?

Kabir: Agar koi rok na ho toh haan.

Madhav: Aur button do baar dab sakta hai? Beech mein internet ja sakta hai? Do log ek saath kar sakte hain?

Kabir: Haan, haan, haan.

Madhav: Ab ginti karo.

Kabir: Yeh toh anginat ho gaya.

Madhav: Teen cheezon wale program mein. Ab ek asli app mein sau cheezein hoti hain, jo ek doosre se judi hain.

Kabir: Toh use poora jaancha hi nahi ja sakta.

Madhav: Bas. Yeh iss chapter ka poora jawab hai. Ab dhyaan do ki yeh koi kaushal ki kami nahi hai. Ab batao, banane wala kya kar sakta hai?

Kabir: Woh sab jaanch nahi sakta. Toh use chunna padega ki kya jaanche.

Madhav: Kis aadhaar pe chunega?

Kabir: Jo cheezein sabse zyada hoti hain. Aur jinme galti ki keemat sabse zyada hai.

Madhav: Aur baaki?

Kabir: Baaki mein galtiyan rahengi, aur woh kabhi na kabhi milengi.

Madhav: Toh software mein galti hona kya hai: ek nakaami, ya ek sachai?

Kabir: Sachai.

NAAM

Galti ko **bug** kehte hain. Naam ki kahani mashhoor hai: 1947 mein ek machine mein sach mein ek keeda fansa mila tha, aur us kaagaz pe likh diya gaya tha “first actual case of bug being found.”

Lekin naam se zyada zaroori wajah hai. Do wajahein, aur dono Kabir ne chhui:

Ek: haalaton ki ginti bahut badi hai.

pul	~ hazaar haalatein
chhota program	~ anginat
bada app	~ anginat, aur woh ek doosre se judi hain

Aur “anginat” ka matlab hai ki poora jaanchna namumkin hai. Yeh mushkil nahi hai, yeh namumkin hai.

Do: software badalta rehta hai.

Ek pul ban kar khada rehta hai. Ek app har hafte badalta hai. Aur har badlav ek nayi jagah hai jahan kuch toot sakta hai, aksar wahan jahan kisi ne dekha bhi nahi tha.

Bug ke teen aam ghar:

SEEMA PE	khaali, zero, negative, bahut bada, pehla, aakhri (sabse zyada bug yahin hote hain)
EK SAATH	do log ek hi waqt pe (Chapter 5.4) ya kaam beech mein ruk jaana
JOD PE	do hisse alag alag theek hain aur milkar galat kaam karte hain

Teesra sabse mushkil hai. Ek aadmi ne apna hissa theek banaya, doosre ne apna. Dono sahi hain. Aur unke beech ki maanyata alag thi.

ASLI DUNIYA SE EK EXAMPLE

1999 mein NASA ka ek yaan Mars pe pahunchte hi kho gaya. Kaaran yeh nikla ki ek team ne apna hisaab ek naap mein diya aur doosri team ne use doosre naap mein padha.

Dono ka code sahi tha. Dono ne apna kaam theek kiya. Kisi ne likha hi nahi tha ki naap kaunsa hai, kyunki dono ko lagta tha ki yeh saaf hai.

Yeh teesre tarah ka bug hai, aur woh sabse mehnga hota hai. Woh kisi ek hisse mein nahi hota. Woh do hisson ke **beech** mein hota hai, aur uske liye koi zimmedar nahi hota.

Isiliye API (Chapter 6.4) itna zaroori hai. Woh sirf ek technical cheez nahi hai. Woh woh jagah hai jahan do hisson ki maanyataayein likh di jaati hain, taaki woh maanyataayein sirf kisi ke dimaag mein na rahein.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki bug ka matlab hai ki banane wala laaparwaah tha.**

Kuch bug laaparwaahi se aate hain. Zyadatar nahi.

Zyadatar bug un haalaton se aate hain jinke baare mein kisi ne socha hi nahi, kyunki sochne ke liye anginat haalatein thin.

Iska seedha nateeja: log software banane ka waqt bahut kam aankte hain. “Yeh toh chhota kaam hai” tab kaha jaata hai jab sirf seedha raasta dekha jaata hai. Aur asli waqt seedhe raaste mein nahi jaata, un sau tedhe raaston mein jaata hai jinka pata baad mein chalta hai.

Doosri galti: **yeh maan lena ki bug khatam kiye ja sakte hain.**

Woh kam kiye ja sakte hain. Zaroori jagahon pe lagbhag khatam kiye ja sakte hain, aur uski keemat bahut hoti hai: jahan galti se jaan jaati hai, wahan ek line code likhne mein ghante lagte hain aur uska daam sau guna hota hai.

Har baaki jagah pe, sawal yeh nahi hai ki bug hon ya na hon. Sawal yeh hai ki **kaunse bug bardaasht kiye ja sakte hain.** Aur woh ek karobaari faisla hai, technical nahi.

MAP PE

BUG	haalatein anginat hain poora jaanchna namumkin seema pe, ek saath, jod pe	← naya
KHARCHA, CLOUD, API, UPTIME		
QUEUE, CACHE, DATABASE, INTERNET		

SOCHNE KE LIYE

1. (samajh check) Ek form hai jismein sirf ek khaana hai: “apni umar likho.” Kitni galat cheezein ho sakti hain?

Jawab: Bahut zyada, aur ginne se hi samajh aata hai ki problem kya hai.

Khaali. Zero. Negative. Dashamlav. Bahut bada number. Akshar. Emoji. Space. Doosri bhasha ke ank. Bahut lamba text jo memory bhar de. Aisa text jo program ke liye kuch aur matlab rakhta ho.

Ek khaane mein pandrah se zyada. Ab sau khaane wala app socho.

Isliye “chhota feature” jaisi koi cheez nahi hoti. Har feature apne saath ek anginat duniya laata hai.

2. (samajh check) Do team alag alag hisse banati hain. Dono ka code sahi hai. Phir bhi milkar cheez toot jaati hai. Kya hua?

Jawab: Unki maanyataayein alag thin, aur woh kisi ne likhi nahi thin.

Ek ne socha ki tarikh iss shakal mein aayegi, doosre ne doosri shakal mein. Ek ne socha ki khaali ho sakta hai, doosre ne socha kabhi nahi.

Yeh sabse mehnga bug hai kyunki jaanch dono taraf paas ho jaati hai.

Ilaaj: **maanyataayein likh do, aur unhe jaancho.** Yahi API ka asli kaam hai. Woh do dimaagon ke beech ka anumaan ek kaagaz pe le aata hai.

3. (jodne wala) Chapter 3.2 kehta tha ki kuch galtiyen compiler pakad leta hai aur kuch sirf chalne pe dikhti hain. Ab woh baat poori karo.

Jawab: Compiler shakal ki galtiyen pakad sakta hai: galat likha, do cheezein jo jud hi nahi sakti.

Woh **matlab** ki galtiyen nahi pakad sakta: aapne jodne ki jagah ghata diya, aapne galat cheez ki jaanch ki, aapne ek haalat sochi hi nahi.

Isliye “compile ho gaya” ka matlab hai ki program chal sakta hai, na ki yeh ki woh sahi hai.

Aur isiliye jaanch ek alag kaam hai, jo agle chapter ka vishay hai.

Chapter 7.2 [SPINE]

Galti pehle dhoondhna

SAMVAAD

Madhav: Pichhle chapter mein tay hua ki poora jaanchna namumkin hai. Toh phir jaanchne ka faayda kya?

Kabir: Kuch galtiyan toh milengi.

Madhav: Kaunsi? Woh jo aapne socheen, ya woh jo aapne nahi socheen?

Kabir: Jo maine socheen.

Madhav: Aur bug aksar wahan hote hain jo aapne nahi socha. Toh phir?

Kabir: Toh jaanch bekaar hai?

Madhav: Ulta socho. Aapne ek galti dhoondi aur theek ki. Kal aap kuch aur badalte ho aur wahi galti wapas aa jaati hai. Kya hoga?

Kabir: Mujhe dobara dhoondhni padegi.

Madhav: Aur agar aapne uski jaanch likh kar rakh di hoti?

Kabir: Toh woh turant pakad leti.

Madhav: Toh jaanch ka asli kaam kya hai?

Kabir: Nayi galtiyan dhoondhna nahi. Purani galtiyan wapas aane se rokna.

Madhav: Bas. Yeh sabse zaroori baat hai aur lagbhag sab log ise ulta samajhte hain. Ab agla sawal. Ek app mein hazaar jaanch likhi hain. Woh kab chalti hain?

Kabir: Jab main chalaun.

Madhav: Aur agar main bhool jaaun?

Kabir: Toh nahi chalengi.

Madhav: Toh?

Kabir: Apne aap chalni chahiye. Har badlav pe.

Madhav: Kaun chalayega?

Kabir: Koi machine, jo dekhti rahe ki kuch badla ya nahi.

Madhav: Aur agar koi jaanch fail ho?

Kabir: Toh woh badlav aage na jaaye.

Madhav: Ab teen tarah ki jaanch hoti hain. Ek: ek chhote hisse ko akele jaancho. Doosri: do hisson ko milakar. Teesri: poora system, jaise ek asli user chala raha ho. Kaunsi tez hai?

Kabir: Pehli. Woh sabse chhoti hai.

Madhav: Kaunsi sach ke sabse kareeb hai?

Kabir: Teesri. Woh asli istemaal jaisi hai.

Madhav: Toh sirf teesri kyun na likhein?

Kabir: Woh dheemi hogi. Aur agar fail ho toh pata nahi chalega ki kahan galti hai.

Madhav: Aur sirf pehli kyun na likhein?

Kabir: Toh har hissa alag se theek hoga aur milkar phir bhi toot sakta hai. Pichhla chapter.

Madhav: Toh?

Kabir: Teeno chahiye. Chhoti wali bahut saari, badi wali kam.

NAAM

Teen darje, aur unka aam anupaat:

CHHOTI JAANCH (unit)

- |— ek chhota hissa, akela
- |— bahut tez, second ke hisse mein hazaaron
- |— fail ho toh turant pata ki kahan
- |— sabse zyada, shayad 70%

JOD KI JAANCH (integration)

- |— do ya teen hisse milkar
- |— dheemi
- |— woh galtiyan pakadti hai jo beech mein hoti hain
- |— beech ki ginti, shayad 20%

POORI JAANCH (end to end)

- |— asli user jaisa poora raasta
- |— sabse dheemi, aur sabse zyada tootne wali
- |— sach ke sabse kareeb
- |— sabse kam, shayad 10%

Aur woh machine jo har badlav pe apne aap sab chalati hai, use **CI** kehte hain.

Ab woh do baatein jo iss chapter ka nichod hain.

Ek: jaanch ka asli kaam bharosa hai, khoj nahi.

Bina jaanch ke, har badlav ek daanv hai. Aap chhota badlav karne se bhi darte ho, kyunki pata nahi kya toot jaaye. Toh aap kam badalte ho, aur software dheere dheere jamm jaata hai.

Jaanch ke saath aap nirbhay ho kar badal sakte ho. **Jaanch aapko bug se nahi bachati. Woh aapko darr se bachati hai.**

Do: jaanch ki bhi keemat hai.

Har jaanch likhni padti hai, sambhalni padti hai, aur jab aap jaan-boojh ke kuch badalte ho toh use bhi badalna padta hai.

Isliye har cheez ki jaanch likhna bhi galat hai. Wahan likho jahan:

galti ki keemat zyada ho
 woh hissa baar baar badalta ho
 usmein pehle bhi galti nikal chuki ho

Aakhri wali sabse kaam ki hai: **jahan ek bug mila, wahan aur bhi honge**. Bug jhund mein rehte hain.

ASLI DUNIYA SE EK EXAMPLE

Knight Capital naam ki ek trading company ne 2012 mein 45 minute mein lagbhag 44 crore dollar kho diye.

Wajah yeh thi ki naya code aath machinon pe daalna tha aur ek machine chhoot gayi. Us ek machine pe purana code chalta raha, aur uska ek purana hissa naye code ke saath milkar galat kaam karne laga.

Yeh Chapter 7.1 wala teesra bug hai: naya sahi tha, purana sahi tha, aur woh saath mein galat the.

Aur ismein ek aur sabak hai jo agle do chapter ka vishay hai: galti code mein nahi thi. Galti **daalne ke tareeke** mein thi. Ek aadmi ne haath se aath machinon pe copy kiya aur ek chhoot gayi.

Company us hafte lagbhag khatam ho gayi.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki 100 pratishat jaanch ka matlab hai ki bug nahi honge**.

Ek maap hai jo batata hai ki jaanch ne code ki kitni lines chalayi. Log usse 100 pratishat karne mein lag jaate hain.

Lekin har line chal jaana aur har **haalat** ka jaancha jaana do alag baatein hain. Aap ek line ko chala sakte ho aur phir bhi us haalat ko na jaanchein jismein woh galat hoti hai.

Toh 100 pratishat ka matlab hai “har line ek baar chali,” na ki “sab kuch sahi hai.”

Doosri galti: **poori jaanch pe zyada bharosa karna**.

Poori jaanch sach ke kareeb hai lekin woh dheemi hai aur bina wajah bhi fail hoti rehti hai: network dheema tha, page thodi der se aaya, kuch aur badal gaya.

Aur jab jaanch bina wajah fail hone lagti hai, toh log use dekhna band kar dete hain. Aur us din se woh jaanch hai hi nahi.

Ek jaanch jispe bharosa na ho, woh jaanch nahi hai. Woh sirf shor hai.

MAP PE

JAANCH	chhoti, jod, पूरी	← naya
CI	kaam: darr hataana har badlav pe apne aap	← naya
BUG, KHARCHA, API, UPTIME		
QUEUE, CACHE, DATABASE, INTERNET		

SOCHNE KE LIYE

1. (samajh check) Ek team kehti hai “hamare paas 5,000 jaanch hain” aur unka software phir bhi baar baar toot ta hai. Kya ho sakta hai?

Jawab: Kai sambhavnayein, aur teeno aam hain.

Ek: jaanch un jagahon pe hain jahan galti hoti hi nahi, aur jahan hoti hai wahan nahi hain.

Do: sab chhoti jaanch hain aur jod ki koi nahi. Har hissa akela theek hai.

Teen: jaanch fail hoti hai lekin log use nazarandaaz karte hain, kyunki woh bina wajah bhi fail hoti rehti hai.

Ginti kabhi jawab nahi hoti. 5,000 jaanch ka matlab kuch nahi hai. Sawal yeh hai ki woh kahan hain aur unpe bharosa hai ya nahi.

2. (samajh check) Aapko ek chhota badlav karna hai lekin darr lag raha hai ki kuch toot jaayega. Iska kya matlab hai?

Jawab: Ki wahan jaanch nahi hai, ya uspe bharosa nahi hai.

Aur woh darr apne aap mein ek nuksaan hai, chahe kuch toote ya na toote. Kyunki us darr ki wajah se aap sudhaar nahi karte, safai nahi karte, aur software dheere dheere jamm jaata hai.

Isiliye purane systems mein log kehte hain “isse haath mat lagao.” Woh code kharaab nahi hota. Bas uske aas-paas koi jaal nahi hota, isliye koi kudne ki himmat nahi karta.

3. (jodne wala) Chapter 6.6 mein systems ek seema par jhatke mein girte the. Kya jaanch us tarah ki galti pakad sakti hai?

Jawab: Aam jaanch bilkul nahi pakadti, aur yeh dhyaan dene laayak hai.

Kyunki jaanch aksar ek user, ek sawal, ek waqt pe chalti hai. Aur woh galti tab dikhti hai jab hazaaron log ek saath aayein.

Uske liye alag tarah ki jaanch hoti hai, jismein jaan-boojh ke bahut bojh daala jaata hai, aur dekha jaata hai ki cheez kahan girti hai.

Aur ek aur tareeka hai jo bade systems istemaal karte hain: **jaan-boojh ke apni hi cheezein todna**, chalte hue system mein, taaki pata chale ki woh sach mein sambhal payegi ya nahi. Kyunki Chapter 5.7 wali baat yahan bhi lagti hai: **jo bachaav jaancha nahi gaya, woh bachaav hai hi nahi.**

Chapter 7.3 [SPINE]

Har badlav ka record

SAMVAAD

Madhav: Aap ek file pe kaam kar rahe ho. Kal aapne kuch badla aur aaj sab kuch toot gaya. Kya karoge?

Kabir: Wapas purana kar dunga.

Madhav: Purana kahan hai?

Kabir: Maine copy bana li thi. `file-purani.txt`.

Madhav: Aur uske pehle wali?

Kabir: `file-purani-2.txt` ?

Madhav: Aur teen hafte pehle wali? Jo us din chal rahi thi jab sab theek tha?

Kabir: Woh nahi hai.

Madhav: Toh pehla sawal: har haalat kabhi bhi wapas mil sake. Kya rakhna padega?

Kabir: Har badlav.

Madhav: Poori file har baar?

Kabir: Woh bahut jagah legi.

Madhav: Toh?

Kabir: Sirf farak rakhun. Chapter 2.1 wali compression jaisi baat.

Madhav: Bas. Ab doosra sawal. Do log ek hi file pe kaam kar rahe hain. Kya hoga?

Kabir: Wahi purani problem. Ek doosre ke upar likh denge.

Madhav: Chapter 5.4 mein iska hal taala tha. Kya yahan taala theek hai?

Kabir: Toh ek aadmi kaam kare aur doosra baitha rahe. Ghanton tak.

Madhav: Toh?

Kabir: Dono kaam karein, apni apni copy pe, aur baad mein mila lein.

Madhav: Milane mein kya hoga?

Kabir: Agar dono ne alag jagah badla toh dono jud jaayenge.

Madhav: Aur agar dono ne ek hi line badli?

Kabir: Toh machine tay nahi kar sakti. Kisi insaan ko dekhna padega.

Madhav: Bas. Ab teesra sawal, aur yeh sabse kaam ka hai. Aapko ek nayi cheez try karni hai, jo shayad kaam na kare. Aur asli cheez chalti rehni chahiye.

Kabir: Toh main ek alag copy pe kaam karunga.

Madhav: Aur agar woh kaam kar gayi?

Kabir: Toh use asli mein mila dunga.

Madhav: Aur nahi kar gayi?

Kabir: Toh phenk dunga. Asli ko kuch nahi hua.

Madhav: Ab batao, isse kya mila?

Kabir: Main koshish kar sakta hoon bina kuch todne ke darr ke.

Madhav: Aur jab darr nahi hota toh?

Kabir: Toh main zyada koshish karta hoon.

NAAM

Us poori cheez ka naam hai **version control**, aur jo lagbhag sab log istemaal karte hain uska naam hai **git**.

Uske teen bade kaam wahi hain jo Kabir ne nikaale:

- | | |
|------------|---|
| 1. ITIHAAS | har badlav ka record, hamesha ke liye
kabhi bhi kisi bhi purani haalat pe wapas |
| 2. MILAANA | kai log alag alag kaam karein, phir mila lein
jahan takraav ho wahan insaan tay kare |
| 3. SHAKHA | nayi cheez alag jagah try karo
chal gayi toh mila do, nahi toh phenk do |

Har record ke saath yeh bhi likha hota hai: kisne badla, kab badla, aur kyun badla.

Woh teesri baat sabse zyada nazarandaaz hoti hai aur sabse zyada kaam aati hai. Do saal baad jab koi poochta hai “yeh line yahan kyun hai,” toh jawab record mein hota hai.

Aur ab woh baat jo iss chapter ki asli seekh hai:

Version control ka asli faayda purani cheez wapas laana nahi hai. Woh himmat hai.

Jab aap jaante ho ki kuch bhi wapas kiya ja sakta hai, toh aap koshish karte ho. Aap purani cheezein saaf karte ho. Aap badi cheezein badalte ho.

Bina uske, har badlav ek daanv hai, aur log daanv nahi lagate. Woh chhote chhote patch lagate hain, aur software dheere dheere aisa ho jaata hai jise koi chhoona nahi chahta.

Yeh wahi baat hai jo Chapter 7.2 mein jaanch ke baare mein thi. Dono ek hi cheez dete hain: **jaal, taaki aap kood sako.**

ASLI DUNIYA SE EK EXAMPLE

Git 2005 mein Linus Torvalds ne banaya tha, wahi jisne Linux banaya (Chapter 3.6).

Wajah yeh thi ki Linux pe duniya bhar mein hazaaron log kaam karte the, jo ek doosre ko jaante nahi the aur ek doosre pe bharosa nahi karte the. Us waqt ka koi bhi tool itne logon ke liye nahi bana tha.

Usne use lagbhag do hafte mein likha.

Aaj woh lagbhag har software company mein hai, aur woh muft hai.

Ab Chapter 3.6 ke chaar wajahon pe lagao: usne ise isliye banaya ki use khud chahiye tha aur mil nahi raha tha. Chauthi wajah. Aur uska asar aaj lagbhag har programmer pe roz padta hai.

Chapter 0.2 ke dials: leverage kalpana se bahar, paisa lagbhag zero. Wahi shakal jo Chapter 4.8 mein protocol likhne walon ki thi.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki version control sirf code ke liye hai.**

Woh kisi bhi aisi cheez ke liye hai jo badalti hai aur jiska itihaas mayne rakhta hai: likhawat, hisaab, settings, kaanooni kaagaz, kitaab ke chapters.

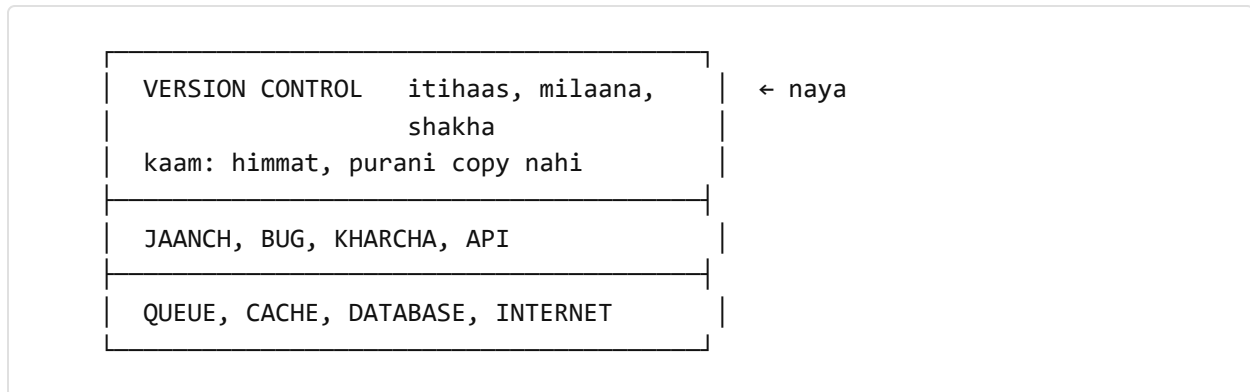
Ismein bhi ek sachai chhupi hai. Zyadatar log apne kaam ki koi asli itihaas nahi rakhte. Unke paas `final.doc`, `final-2.doc`, `final-FINAL.doc` hote hain, jo wahi purana tareeka hai jo Kabir ne shuruaat mein bataya tha aur jo turant toot gaya tha.

Doosri galti: **git ko bas ek jagah samajhna jahan file rakhi jaati hain.**

File rakhna uska sabse chhota kaam hai. Uska asli kaam yeh yaad rakhna hai ki cheezein **kaise badli**, aur usse aap sawal poochh sakte ho: yeh kab toota? Kis badlav se toota? Us waqt kya soch thi?

Aur ek teesri, jo shuruaat mein sabko hoti hai: yeh sochna ki har badlav ek bada kaam hona chahiye. Chhote, baar baar kiye gaye records se itihaas kaam ka rehta hai. Ek bade record mein pachas cheezein ek saath hoti hain, aur uske baad aap yeh nahi dhoondh sakte ki kaunsi cheez se kya toota.

MAP PE



SOCHNE KE LIYE

1. (samajh check) Ek app do mahine se theek chal raha tha aur aaj achanak ek purana feature toot gaya. Aapke paas poora itihahas hai. Kaise dhoondhoge?

Jawab: Beech se dhoondho.

Do mahine ke beech ka koi ek din chuno aur dekho ki us din cheez chalti thi ya nahi. Agar chalti thi toh galti uske baad hui, warna pehle. Phir usi tarah aadha karte jao.

Sau badlaon mein se galat wala saat kadam mein mil jaayega.

Yeh Chapter 5.2 wali binary search hai, ab code ke itihahas pe lagi hui. **Ek hi soch, nayi jagah.** Git mein iske liye ek banaya banaya hukum bhi hai.

2. (samajh check) Do log ek hi line badalte hain aur machine tay nahi kar pati. Kya yeh ek problem hai jise theek karna chahiye?

Jawab: Nahi. Yeh sahi vyavhaar hai.

Machine ko pata nahi hai ki kaunsa badlav sahi hai. Sirf woh do log jaante hain ki woh kya kar rahe the.

Agar machine apne aap koi ek chun leti, toh woh chup-chaap kisi ka kaam mita deti, aur kisi ko pata bhi na chalta.

Rukna aur poochna, chup-chaap galat kar dene se hamesha behtar hai. Yeh soch poore software mein baar baar dikhti hai.

3. (jodne wala) Chapter 7.2 mein jaanch ka asli kaam “darr hataana” tha. Version control ka bhi wahi hai. Toh dono mein farak kya hai?

Jawab: Dono jaal hain, lekin alag khatre ke liye.

Jaanch batati hai ki abhi kuch toota hai ya nahi. Woh aage ki taraf dekhti hai.

Itihaas batata hai ki kab toota aur kis wajah se, aur wapas jaane deta hai. Woh peeche ki taraf dekhta hai.

Ek ke bina doosra aadha hai. Jaanch ke bina aapko pata hi nahi chalega ki kuch toota. Itihaas ke bina aap jaan toh loge ki toota hai, lekin na wajah milegi na wapas ja paoge.

Isiliye har team mein dono hote hain, aur dono ek hi cheez ke liye hain: **badalne ki aazadi.**

Chapter 7.4 [DEPTH]

”Mere machine pe toh chal raha tha”

DEPTH chapter. Yeh software banane ka sabse purana mazaak hai, aur uska hal poore internet ki shakal badal chuka hai.

SAMVAAD

Madhav: Ek program aapke laptop pe theek chalta hai. Server pe daala aur woh nahi chala. Wajah kya ho sakti hai?

Kabir: Alag machine hai.

Madhav: Kya alag hai? Ginti karo.

Kabir: OS alag ho sakta hai. Version alag ho sakta hai.

Madhav: Aur?

Kabir: Jo doosri cheezein program istemaal karta hai, unke version alag ho sakte hain.

Madhav: Aur?

Kabir: Settings alag ho sakti hain. Files alag jagah ho sakti hain.

Madhav: Aur ek aur, jo sabse zyada hoti hai. Aapke laptop pe kuch aisa hai jo aapne mahine pehle lagaya tha aur aap bhool gaye.

Kabir: Aur server pe woh nahi hai.

Madhav: Ab batao, kitni cheezein ek jaisi honi chahiye?

Kabir: Sab.

Madhav: Toh?

Kabir: Toh main ek list bana lun ki kya kya chahiye.

Madhav: Aur agar list adhoori ho? Kyunki jo cheez aap bhool gaye ho, woh list mein bhi nahi hogi.

Kabir: Toh list bekaar hai.

Madhav: Ek doosra tareeka socho. List banane ki jagah, kya bheja ja sakta hai?

Kabir: Poori machine bhej doon?

Madhav: Kaise?

Kabir: Ek poori machine ki copy... woh toh bahut bhaari hoga. Poora OS.

Madhav: Bhaari hai, aur log ne pehle wahi kiya tha. Ab socho: kya poora OS bhejna zaroori hai?

Kabir: Nahi. Server pe toh pehle se ek OS hai.

Madhav: Toh kya chahiye?

Kabir: Sirf mera program aur woh cheezein jo woh istemaal karta hai. Baaki neeche wala hissa saanjha ho sakta hai.

Madhav: Aur agar do program ko alag alag version chahiye ek hi cheez ka?

Kabir: Toh dono apna apna leke chalein, aur ek doosre ko na dekh sakein.

Madhav: Yeh kaunsi soch hai? Kahan dekh chuke ho?

Kabir: Chapter 3.4. OS har app ko alag rakhta hai.

Madhav: Bas. Wahi soch, ek parat upar. Ab ek aakhri sawal. Agar poora dabba ek hi jaisa chal sakta hai kahin bhi, toh usse aur kya kya mumkin hua?

Kabir: Main use kahin bhi utha kar rakh sakta hoon.

Madhav: Aur agar bojh badhe?

Kabir: Toh usi dabbe ki das copy chala doon.

Madhav: Aur ek mar jaaye?

Kabir: Nayi chala dun. Woh bilkul waisi hi hogi.

Madhav: Ab tumne dekha ki yeh sirf ek jhanjhat ka hal nahi tha. Isne poori Part 6 wali duniya ko aasan bana diya.

NAAM

Us dabbe ka naam hai **container**, aur sabse mashhoor tareeke ka naam **Docker** hai.

Farak samajhna zaroori hai:

POORI MACHINE KI COPY (virtual machine)

- ├─ poora OS andar
- ├─ bhaari: GB mein
- ├─ shuru hone mein minute
- └─ poora alagav

DABBA (container)

- ├─ sirf program aur uski cheezein
- ├─ neeche ka OS saanjha
- ├─ halka: MB mein
- ├─ shuru hone mein second
- └─ kaafi alagav, poora nahi

Aur woh cheez jo iss chapter ki asli seekh hai:

Container ne “chalta kaise hai” ko “chalta kahan hai” se alag kar diya.

Pehle ek program aur ek machine bandhe hue the. Program us machine ke liye banaya jaata tha.

Ab program ek dabbe mein hai jo kahin bhi chal sakta hai. Machine sirf ek jagah hai jahan dabba rakha ja sakta hai.

Aur isse teen cheezein apne aap aa gayin, jo Part 6 mein bahut mehnat maangti thin:

BADA HONA	dus copy chala do (Chapter 6.1)
BADALNA	purana dabba hatao, naya rakho, sekend mein
CHALTE RAHNA	ek mar gaya toh naya utha do (Chapter 6.2)

Aur jab dabbe hazaaron ho jaate hain, toh unhe koi sambhalta hai: kaun kahan chalega, kaun mar gaya, kaun naya chahiye. Us kaam ke liye ek aur parat hai, jiska sabse aam naam **Kubernetes** hai.

Ab dhyaan do ki yeh kya hua. **Ek manager, jo dabbon ko machinon pe baantta hai.**

Chapter 3.4 mein OS programs ko ek machine pe baant raha tha. Ab wahi soch machinon ke poore jhund pe lag gayi.

Ek hi shakal, teesri baar.

ASLI DUNIYA SE EK EXAMPLE

2013 mein Docker aane se pehle, ek naye engineer ko apne laptop pe kaam shuru karne mein aksar do se paanch din lagte the. Ek lambi list hoti thi, aur usmein se aadhi cheezein purani ho chuki hoti thin.

Aaj wahi kaam aksar ek hukum aur kuch minute ka hota hai.

Aur uska asar sirf suvidha nahi tha. Chapter 6.8 yaad karo: **aadmi ka waqt machine se bahut mehnga hai.** Har naye aadmi ke teen din bachana, har baar, poori company mein, ek bada aankda ban jaata hai.

Iska ek aur asar hua jo shayad bada tha. Jab ek program kahin bhi ek jaisa chal sakta hai, toh cloud companies ko badalna aasan ho gaya. Chapter 6.5 wala lock-in thoda kamzor ho gaya.

Aur isiliye cloud companies ne apni khud ki aur upar wali cheezein banayin, jo sirf unke yahan chalti hain. Lock-in hataya nahi gaya, woh ek parat upar chala gaya.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **container ko ek poori machine samajhna.**

Container ke andar apna OS nahi hota. Woh neeche wale OS ka istemaal karta hai. Isliye woh halka hai, aur isliye uska alagav utna sakht nahi hai jitna log sochte hain.

Iska seedha nateeja: log maan lete hain ki container ke andar ki cheez पूरी तरह band hai aur bahar kuch nahi kar sakti. Woh kaafi alag hai, lekin पूरी तरह nahi. Jahan suraksha sach mein zaroori hai, wahan पूरी machine ki copy aaj bhi istemaal hoti hai.

Doosri galti: **yeh sochna ki container se scaling apne aap ho jaayegi.**

Container ne dabba ek jaisa bana diya. Usne Chapter 6.1 wala niyam nahi badla: machine khaali honi chahiye. Agar aapka program apne andar jaankari rakhta hai, toh dus copy chalane se woh dus alag jhoot bolengi.

Container ne cheezein le jaana aasan kiya. Usne banawat ke faisle nahi badle.

MAP PE

CONTAINER	program + uski cheezein	← naya
MANAGER	ab machinon ke jhund pe (teesri baar wahi shakal)	
VERSION CONTROL, JAANCH, BUG		
CLOUD, API, UPTIME, BADA HONA		
OS, CPU, RAM/DISK, SWITCH		

SOCHNE KE LIYE

1. (samajh check) Do program ek hi machine pe chalne hain, aur dono ko ek hi cheez ke alag version chahiye. Bina container ke kya hoga?

Jawab: Ek hi version lag sakta hai, toh ek program tootega.

Aur aksar aisa hota tha ki ek program ke liye version badalne se doosra toot jaata tha, jise koi mahinon baad dhoondhta tha.

Container mein dono apna apna leke chalte hain aur ek doosre ko dekhte bhi nahi.

Yeh Chapter 3.4 ka “alag rakhna” hai, ek parat upar lagaya hua. **Ek hi problem, ek hi hal, naya paimana.**

2. (samajh check) Aapne apna program container mein daal diya aur uski das copy chala di. Ab log ki login baar baar chali jaati hai. Kya hua?

Jawab: Program apne andar login ki jaankari rakh raha hai.

Har copy apna alag rakh rahi hai, aur user har baar kisi doosri pe pahunch jaata hai.

Container ne ise theek nahi kiya, kyunki yeh container ki problem thi hi nahi. Yeh Chapter 6.1 ka niyam hai: **machine khaali honi chahiye.**

Sabak: **ek naya tool purane faisle theek nahi karta. Woh unhe sirf pehle saamne le aata hai.**

3. (jodne wala) Iss kitaab mein ab teen jagah “manager” aa chuka hai. Teeno ginao aur batao ki saanjha kya hai.

Jawab:

Ch 3.4	OS	programs ko ek machine pe baantta hai
Ch 6.1	load balancer	users ko machinon pe baantta hai
Ch 7.4	Kubernetes	dabbon ko machinon ke jhund pe baantta hai

Saanjha: jab ek cheez hai aur maangne wale kai, tab koi beech mein baithkar baantta hai, aur usse yaad rakhna padta hai ki kaun kahan hai.

Aur har baar wahi sawal aate hain: agar manager mar jaaye toh? Agar do maangne wale atak jaayein toh? Agar ek maangne wala sab kuch le le toh?

Yeh iss kitaab ki sabse badi baat hai. Aap 112 cheezein yaad nahi kar rahe. Aap dus shakalein seekh rahe ho jo baar baar dikhti hain.

Chapter 7.5 [SPINE]

Estimate hamesha galat kyun hota hai

SAMVAAD

Madhav: Ek kaam hai. Kisi se poocho kitna waqt lagega. Jawab kya milega?

Kabir: Do din.

Madhav: Aur asli mein?

Kabir: Paanch. Ya dus.

Madhav: Yeh hamesha hota hai, har jagah, har company mein. Toh log seekhte kyun nahi?

Kabir: Shayad woh jaan-boojh ke kam batate hain.

Madhav: Kabhi kabhi. Lekin woh log bhi galat hote hain jo apne liye anumaan lagate hain aur jinpe koi dabaav nahi hai. Toh wajah kuch aur hai. Chalo dhoondhte hain. Jab aap “do din” kehte ho, toh aap kis cheez ka waqt gin rahe ho?

Kabir: Kaam karne ka.

Madhav: Kaunsa kaam? Jo आपको दिख रहा है, या जो दिख नहीं रहा?

Kabir: जो दिख रहा है.

Madhav: Aur Chapter 7.1 mein humne kya dekha tha? Ek chhote se kaam mein kitni haalatein thin?

Kabir: Anginat. Aur zyadatar dikhti nahi.

Madhav: Toh aap kis cheez ka anumaan laga rahe the?

Kabir: Sirf seedhe raaste ka. Jahan sab kuch theek chalta hai.

Madhav: Aur asli waqt kahan jaata hai?

Kabir: Un raaston mein jinka pata baad mein chalta hai.

Madhav: Bas. Ab doosri baat. Aap dus chhote kaam ginte ho, har ek do ghante ka. Kul bees ghante. Sahi?

Kabir: Haan.

Madhav: Aur beech mein? Ek kaam se doosre pe jaane mein? Kisi se poochne mein? Kisi ke jawab ka intezaar karne mein?

Kabir: Woh maine gina hi nahi.

Madhav: Aur agar aapko kisi doosre insaan ka jawab chahiye, jo do din baad milta hai?

Kabir: Toh mera kaam do ghante ka hai aur do din lag gaye.

Madhav: Toh dusra kaaran kya hai?

Kabir: Intezaar. Jo kisi ke kaam mein nahi ginta lekin waqt leta hai.

Madhav: Ab teesri baat, aur yeh sabse gehri hai. Ek kaam jise aap pehle kar chuke ho, uska anumaan kaisa hota hai?

Kabir: Kaafi sahi.

Madhav: Aur jo aapne kabhi nahi kiya?

Kabir: Bahut galat.

Madhav: Kyun?

Kabir: Kyunki mujhe pata hi nahi ki usmein kya kya hai.

Madhav: Toh anumaan asal mein kis cheez ka naap hai?

Kabir: Iss baat ka ki main kitna jaanta hoon.

Madhav: Bas. **Anumaan waqt ka naap nahi hai. Woh gyaan ka naap hai.** Aur isiliye woh us kaam mein sabse galat hota hai jo sabse naya ho, jahan uski sabse zyada zaroorat hoti hai.

NAAM

Teen kaaran, aur teeno alag hain:

1. JO DIKHTA NAHI
aap seedhe raaste ka anumaan lagate ho
asli waqt tedhe raaston mein jaata hai (Chapter 7.1)
2. INTEZAAR
kaam do ghante ka, jawab ka intezaar do din
yeh kisi ke ghante mein nahi ginta
3. GYAAN
jaana-pehchana kaam: anumaan theek
naya kaam: anumaan bahut galat
aur jo sabse zaroori kaam hote hain woh aksar naye hote hain

Do cheezein jo iss se nikalti hain aur kaam aati hain:

Ek: anumaan ek range hai, ek number nahi.

”Do din” jhoot hai. ”Do se aath din, aur agar woh purani cheez tedhi nikli toh do hafte” sach hai.

Log ek number maangte hain kyunki uspe plan banana aasan hai. Aur ek number dene ka matlab hai ki aap apni anischitta chhupa rahe ho, jo baad mein kisi aur ke sar phootegi.

Do: sabse achha ilaaj anumaan sudhaarna nahi hai. Woh kaam chhota karna hai.

Ek do-mahine ka kaam kabhi sahi nahi aanka jaayega. Ek do-din ka kaam aksar aank liya jaayega.

Isliye achhi teamein bade kaam ko chhote tukdon mein todte hain, aur har tukda apne aap mein kuch kaam ka hota hai. Woh isliye nahi ki chhota anumaan zyada sahi hai. Woh isliye ki jaldi pata chal jaata hai ki hum kitne galat the.

Aur ek teesri baat, jo Chapter 7.1 se seedhi aati hai: **jo kaam ek purani cheez ko chhota hai, uska anumaan hamesha sabse galat hota hai**, kyunki us purani cheez mein kya kya hai, yeh kisi ko poora nahi pata.

ASLI DUNIYA SE EK EXAMPLE

Sydney Opera House ka anumaan tha: saat lakh dollar aur chaar saal.

Woh bana das saal baad, aur uska kharcha lagbhag pandrah guna se zyada nikla.

Yeh software nahi tha. Yeh 1950 ke dashak ki baat hai.

Aur wajah wahi teeno thin: kisi ne aisi shakal pehle banayi hi nahi thi (gyaan), design banate banate badalta raha (jo dikhta nahi), aur kai alag logon ke faisle ek doosre ka intezaar karte rahe (intezaar).

Yeh iss chapter ki sabse zaroori baat hai: **yeh software ki bimari nahi hai**. Yeh har us kaam ki shakal hai jo pehle kabhi nahi kiya gaya.

Jo kaam dohraya jaata hai, uska anumaan sahi lagta hai. Ek jaisi sau building banao aur sauvi ka anumaan lagbhag theek hoga.

Software mein anumaan isliye zyada galat lagta hai kyunki **software mein dohraya jaane wala kaam hota hi nahi**. Agar woh dohraya ja sakta hai, toh usse copy kar liya jaata hai. Jo bacha rehta hai woh hamesha naya hota hai.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh maan lena ki anumaan behtar tajurbe se sahi ho jaayega**.

Tajurba jaane-pehchane kaam ka anumaan sudhaarta hai. Naye kaam ka nahi. Aur tees saal ka tajurba rakhne wale log bhi naye kaam mein bahut galat hote hain.

Iska seedha nateeja: companies anumaan sahi karwane ki koshish mein bahut waqt jalati hain, jab woh waqt kaam chhota karne mein lagta toh zyada faayda hota.

Doosri galti: **buffer daal dena aur maan lena ki problem hal ho gayi**.

Aap dus din ka anumaan pandrah kar dete ho. Aur woh kaam bees din leta hai. Kyunki jo cheez aapko dikh nahi rahi thi, woh 50 pratishat jodne se dikhne nahi lagti.

Teesri galti, aur yeh sabse mehngi hai: **anumaan ko waada samajhna**.

Anumaan yeh hai ki aaj kitna pata hai. Waada yeh hai ki kya hoga. Jab dono ko ek maan liya jaata hai, toh log anumaan hi dena band kar dete hain, ya bahut bade dene lagte hain taaki phanse na. Aur us din se aapke paas asli jaankari aani band ho jaati hai.

MAP PE

ANUMAAN	gyaan ka naap, waqt ka nahi range do, number nahi kaam chhota karo	← naya
CONTAINER, VERSION CONTROL, JAANCH, BUG		
CLOUD, API, UPTIME, DATABASE, INTERNET		

Part 7 khatam.

PART 7 KA GATE

1. Ek team kehti hai “hamare code mein koi bug nahi hai, humne poori jaanch ki hai.”
Kya yeh mumkin hai? (*nahi aaya toh 7.1*)

Jawab: Nahi. Haalatein anginat hain, isliye poora jaanchna namumkin hai.

Jaanch se yeh saabit hota hai ki jo aapne socha woh theek chalta hai. Jo aapne nahi socha, uske baare mein woh kuch nahi kehti.

2. Program aapke laptop pe chalta hai, server pe nahi. Sabse aam wajah? (*nahi aaya toh 7.4*)

Jawab: Do machinon pe kuch alag hai, aur aksar woh cheez hai jo aapne mahine pehle lagayi thi aur bhool gaye.

Hal container hai: program apni saari cheezein saath leke chalta hai.

3. Ek kaam ke liye “do din” bataya gaya aur usmein do hafte lage. Kaun galat tha? (*nahi aaya toh 7.5*)

Jawab: Shayad koi nahi. Anumaan waqt ka naap nahi hai, gyaan ka naap hai.

Do din ka matlab tha “jitna mujhe dikh raha hai woh do din ka hai.” Baaki bara din un cheezon mein gaye jo us waqt dikh hi nahi rahi thin.

CHAT APP PE WAPAS

Aapke chat app mein bug hain. Abhi bhi. Yeh nakaami nahi hai, kyunki uski haalatein anginat hain aur unhe poora jaancha ja hi nahi sakta.

Uske paas jaanch hai, taaki purani galtiyen wapas na aayein. Uska poora itihaas rakha hai, taaki koi bhi badlav wapas liya ja sake. Aur woh ek dabbe mein chalta hai, taaki har machine pe bilkul ek jaisa rahe.

Aur jab kisi ne poocha tha ki isse banane mein kitna lagega, jawab galat tha. Woh hamesha galat hota hai, kyunki woh sirf us hisse ka anumaan tha jo us waqt dikh raha tha.

Ab sirf ek cheez bachi hai. Aapne har parat alag alag dekh li.

Ab unhe ek saath rakh kar dekhte hain, ek asli product mein. Yeh Part 8 hai, aur woh iss kitaab ka aakhri part hai.

Chapter 8.1 [SPINE]

Ek asli product ko kholkar dekhna

Yeh ek jodne wala chapter hai. Ismein koi nayi cheez nahi hai. Har parat jo aapne alag alag dekhi, woh ab ek saath hai.

SAMVAAD

Madhav: Poori kitaab hum ek chat app saath leke chale hain. Ab use kholte hain. Aap ek message likhte ho aur “send” dabate ho. Shuru se batao, kadam ke hisaab se.

Kabir: Mera message akshar nahi hai. Woh numbers hain.

Madhav: Kaunse chapter?

Kabir: 1.4. Aur unhe padhne ka tareeka Unicode hai, kyunki matlab andar nahi hota.

Madhav: Aage.

Kabir: App khud bhi numbers hai. Woh nirdesh hain jo koi insaan ne likhe the, kisi language mein, aur ek translator ne unhe numbers banaya.

Madhav: Woh app abhi kya hai?

Kabir: Ek process. OS ne use RAM mein jagah di aur baari di. Use lagta hai poora phone uska hai, jo jhooth hai.

Madhav: Ab message bahar jaata hai.

Kabir: Woh tukdon mein tootta hai. Har tukde pe mera pata aur manzil ka pata. Woh TCP se jaata hai, kyunki ek bhi shabd khona theek nahi.

Madhav: Manzil ka pata mila kaise?

Kabir: Naam se. DNS ne naam ko number mein badla, aur woh jawab kuch ghante ke liye rakha bhi gaya.

Madhav: Raaste mein koi padh sakta hai?

Kabir: Nahi. Taala laga hai. Do chaabiyen wali cheez. Lekin taala sirf itna kehta hai ki raaste mein koi nahi padh raha, na ki doosri taraf wale imaandaar hain.

Madhav: Woh doosre phone pe seedha jaata hai?

Kabir: Nahi. Mera phone bahar se pahuncha hi nahi ja sakta. Woh ek server pe jaata hai jiska pakka pata hai.

Madhav: Wahan kya milta hai?

Kabir: Ek machine jo kabhi band nahi hoti. Aur woh akeli nahi hai, kai hain, aur saamne ek cheez khadi hai jo baantti hai.

Madhav: Woh machinein khaas hain?

Kabir: Nahi. Woh khaali hain. Unmein kuch aisa nahi rakha jo sirf usi ke paas ho, warna agli baar main kisi doosri pe pahunch jaunga.

Madhav: Server ne message le liya. Ab?

Kabir: Woh database mein jaata hai, file mein nahi, kyunki chaar problem hain: dhoondhna, ek saath likhna, beech mein rukna, dohrav.

Madhav: Aur woh likhna ek transaction hai?

Kabir: Haan. Ya message likha bhi jaaye aur “bheja gaya” bhi lage, ya dono mein se kuch bhi nahi.

Madhav: Doosre bande ko notification kab jaati hai?

Kabir: Woh queue mein jaati hai. Mujhe “sent” turant dikh jaata hai, jabki asli kaam abhi baaki hai.

Madhav: Aur mera purana chat itna tez kyun khulta hai?

Kabir: Index ki wajah se, aur cache ki wajah se. Cache tez isliye hai kyunki woh purana ho sakta hai. Woh do alag baatein nahi hain.

Madhav: Yeh sab kiske machinon pe chal raha hai?

Kabir: Kisi aur ke. Kirye pe. Kisi doosre sheher mein. Aur jitni unki khaas cheezein istemaal ki gayi hain, utna wahan se nikalna mushkil hai.

Madhav: Iska kharcha?

Kabir: Lagbhag kuch bhi nahi, jab tak sirf likhawat hai. Ek karod messages do rupaye mahina. Jis din video shuru hoga, sab badal jaayega.

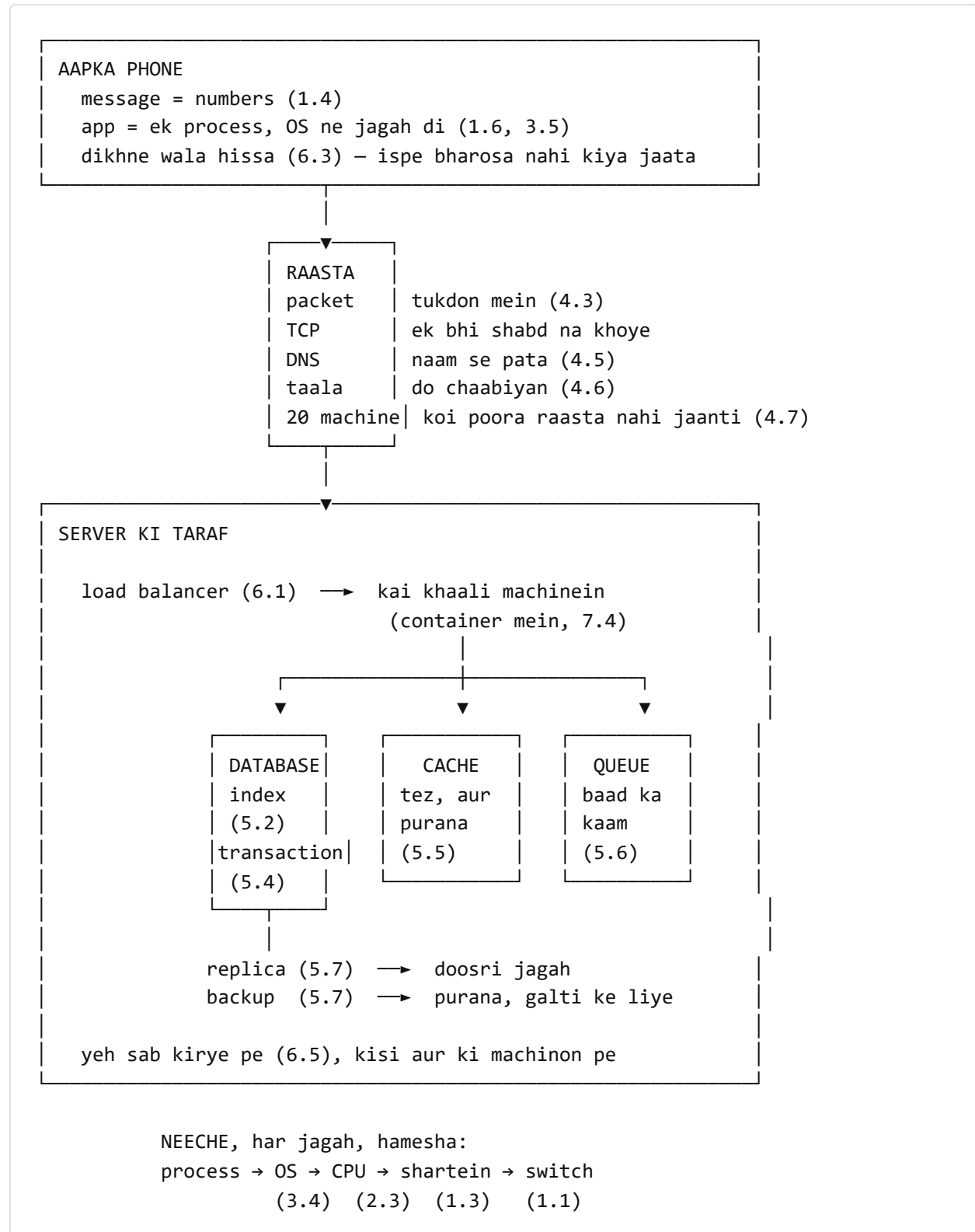
Madhav: Aur yeh sab kisne banaya?

Kabir: Kisi ne, aur usmein bug hain, abhi bhi. Uske paas jaanch hai taaki purani galtiyan wapas na aayein, poora itihaas hai taaki wapas ja sake, aur woh ek dabbe mein chalta hai taaki har jagah ek jaisa rahe.

Madhav: Aur uske andar kitna hissa kisi aur ka likha hua hai?

Kabir: Shayad aadhe se zyada. Muft mein. Kyunki neeche wali parat sabne milkar banayi hai.

POORA NAKSHA



EK BAAT JO AB DIKHNI CHAHIYE

Poore naksh e mein ek bhi cheez aisi nahi hai jo kisi problem ke bina maujood ho.

database	kyunki file mein dhoondhna dheema tha
index	kyunki poori list padhna dheema tha
cache	kyunki wahi cheez baar baar maangi ja rahi thi
queue	kyunki user ko intezaar nahi karna chahiye
kai machinein	kyunki ek machine ki seema thi
load balancer	kyunki kai machinein thin toh koi baante
container	kyunki har machine thodi alag thi
taala	kyunki raaste mein bees ajnabi the
DNS	kyunki koi numbers yaad nahi rakh sakta
packet	kyunki ek bada sandesh sadak rok leta tha

Har cheez ek dard ka jawab hai. **Agar dard nahi hai, toh us cheez ki zaroorat nahi hai.**

Yeh iss kitaab ka sabse kaam ka aujaar hai. Jab bhi koi keh de ki “aapko yeh cheez lagani chahiye,” ek hi sawal poochna kaafi hai:

Yeh cheez kis dard ke liye bani thi, aur kya mujhe woh dard hai?

SOCHNE KE LIYE

1. (samajh check) Iss chat app se ek cheez hatao: cache. Kya hoga?

Jawab: Sab kuch chalega, bas dheema.

Har purana chat database se aayega. Woh kaam karega, aur woh das lakh users pe database ko baitha dega.

Aur ek cheez behtar ho jaayegi: kabhi purana data nahi dikhega.

Har cheez hatai ja sakti hai. Sawal yeh hai ki uske hatane se kya milta hai aur kya jaata hai.

2. (samajh check) Ek dost kehta hai ki uska naya app “Kubernetes, cache, queue aur teen database” istemaal karta hai. Uske sau users hain. Aap kya poochoge?

Jawab: “In sab ne kaunsa dard hataya?”

Sau users pe unmein se lagbhag koi bhi dard maujood nahi hai. Ek machine, ek database, aur bas.

Har cheez ki keemat hai: banane ka waqt, chalane ka kharcha, aur sabse bada, galti dhoondhna mushkil ho jaana.

Yeh Chapter 6.1 wali baat hai: **pehle din se bade paimane ke liye banana, ek aisi problem hai karna hai jo shayad kabhi aayegi hi nahi.**

3. (design sawal, iss part ka) Ek product chuno jo aap roz istemaal karte ho. Uska yahi naksha khud banao, upar se neeche tak. Jahan pata na ho wahan sawaliya nishaan lagao.

Jawab: Iska sahi jawab aapka apna naksha hai. Use `WORKBOOK.md` mein likho.

Do cheezein dhyaan mein rakhna.

Ek: sawaliya nishaan sabse kaam ki cheez hain. Woh batate hain ki aapki abstraction ki deewar ab kahan hai (Chapter 0.4). Woh deewar ab pehle se bahut neeche honi chahiye.

Do: har hisse ke saamne likho ki woh kis dard ke liye hai. Agar dard nahi likh paate, toh do mein se ek baat hai: ya woh hissa wahan hai hi nahi, ya aapne use abhi tak samjha nahi.

Chapter 8.2 [SPINE]

Kya banana aasan hai, kya mushkil

SAMVAAD

Madhav: Ab aap naksha samajh gaye. Lekin design karne ke liye ek aur cheez chahiye. Do kaam batata hoon, aap batao kaunsa bada hai. Ek: login banana. Do: search banana.

Kabir: Dono ek jaise lagte hain.

Madhav: Login lagbhag ek hafte ka kaam hai. Achha search kai mahine ka.

Kabir: Kyun?

Madhav: Login mein kya karna hai?

Kabir: Naam aur password lo, milao, andar aane do.

Madhav: Aur search mein?

Kabir: Shabd lo, aur... jo cheezein us se milti hain woh do.

Madhav: “Milti hain” ka matlab kya hai? Agar main “kitab” likhun aur cheez mein “kitaab” likha ho?

Kabir: Woh bhi milna chahiye.

Madhav: Aur “book” likha ho?

Kabir: Shayad woh bhi.

Madhav: Aur agar sau cheezein milein, toh pehle kaunsi dikhegi?

Kabir: Jo sabse zyada milti hai.

Madhav: “Sabse zyada” kaise naapoge?

Kabir: Hmm. Pata nahi.

Madhav: Bas. Yahi farak hai. Login mein sahi jawab ek hai aur woh saaf hai. Search mein “sahi jawab” ki koi paribhasha hi nahi hai.

Kabir: Toh mushkil kaam woh hai jismein sahi jawab tay nahi hota.

Madhav: Yeh ek pehchaan hai. Aur bhi hain. Ek aur jodi. Ek: photo upload karna. Do: video call banana.

Kabir: Video call mushkil hoga. Live hai.

Madhav: Kyun mushkil hai?

Kabir: Kyunki der nahi ho sakti. Chapter 4.3, purani awaaz kisi kaam ki nahi.

Madhav: Toh doosri pehchaan kya hui?

Kabir: Jo cheez asli waqt mein honi chahiye woh mushkil hai.

Madhav: Ek aur. Ek: apni site pe payment lena. Do: apna khud ka payment system banana.

Kabir: Doosra bahut mushkil. Usmein bank, kanoon, suraksha sab hai.

Madhav: Toh teesri pehchaan?

Kabir: Jismein paisa ya kanoon jud jaaye woh bahut mushkil ho jaata hai.

Madhav: Aur aakhri. Ek: naya app banana. Do: ek purane app mein ek chhota feature jodna, jo dus saal purana hai aur jise bees log likh chuke hain.

Kabir: Naya banana aasan hoga.

Madhav: Kyun? Naye mein toh sab kuch shuru se banana hai.

Kabir: Lekin purane mein mujhe pata hi nahi ki kya kahan hai, aur kya toot jaayega.

Madhav: Bas. Chauthi pehchaan, aur woh sabse zyada nazarandaaz hoti hai: **purani cheez ko chhoona nayi banane se mehnga hota hai.**

CHAR PEHCHAANEIN

1. SAHI JAWAB SAAF HAI YA NAHI
saaf → aasan (login, hisaab, form, list)
dhundhla → mushkil (search, sujhav, "achha" kya hai)
2. WAQT KI SAKHTI
baad mein chalega → aasan
turant chahiye → mushkil (call, game, live)
3. PAISA YA KANOON
nahi juda → aasan
juda hua → bahut mushkil (payment, sehat, bachche ka data)
4. NAYA YA PURANA
khaali kaagaz → aasan
purani cheez → mushkil, aur anumaan hamesha galat (Ch 7.5)

MOTA ANDAZA, EK AADMI KE LIYE

Yeh sirf paimana dene ke liye hai. Har aankda haalat pe badalta hai, lekin **aapas ka anupaat** aksar sach rehta hai.

KUCH DIN

- └─ ek form, jo data leta hai aur database mein rakhta hai
- └─ ek list dikhana
- └─ login, agar kisi bane banaye ka istemaal karo
- └─ ek chhoti website
- └─ email bhejna, kisi bani banayi service se

KUCH HAFTE

- └─ login khud banana (theek se, suraksha ke saath)
- └─ payment lena, kisi bani banayi service se
- └─ file aur photo upload
- └─ notification bhejna
- └─ ek chhota mobile app
- └─ ek admin panel

KUCH MAHINE

- └─ achha search
- └─ kisi purane bade system mein ek nayi cheez jodna
- └─ ek app jo offline bhi chale aur baad mein mila le
- └─ theek se scale karna (Part 6 ki cheezein)
- └─ ek aisa system jo kai company istemaal karein

EK SAAL YA ZYADA

- └─ apna payment system
- └─ video call ya live stream, apna
- └─ apna database ya search engine
- └─ kuch aisa jismein kanoon ki sakhti ho
- └─ kuch aisa jo hazaron logon ke saath ek saath badle

Aur ek cheez jo iss poori list se zyada zaroori hai:

Har list mein neech wali cheezein aksar isliye mushkil nahi hain ki unka hisaab mushkil hai. Woh isliye mushkil hain ki unke andar anginat haalatein hain, ya unka “sahi” tay nahi hai.

Chapter 7.1 wali baat, ab design ke kaam mein lagi hui.

EK AUR CHEEZ: KHAREEDO YA BANA O

Har cheez ke liye ek aur sawal hai:

KHAREEDO / ISTEMAAL KARO jab

- └─ woh cheez aapka product nahi hai
- └─ woh cheez sab ke liye ek jaisi hai (login, payment, email)
- └─ uski galtiyan mehngi hain aur doosron ne unhe already jhel liya hai

KHUD BANAJO jab

- └─ wahi cheez aapka product hai
- └─ ya woh cheez aapke liye khaas tarah se alag hai
- └─ ya uspe nirbhar hona bahut khatarnak hai (Ch 6.4)

Zyadatar log ulta karte hain. Woh login khud banate hain, jo unka product nahi hai, aur woh khaas cheez khareed lete hain jo unka product hai.

ASLI DUNIYA SE EK EXAMPLE

Instagram 2012 mein ek arab dollar mein bika. Us waqt uske paas 13 log the.

Unhone bahut kuch khud nahi banaya. Server kirye pe the, bahut si cheezein bane banaye tukdon se thin, aur unka apna kaam kaafi chhota tha.

Unka asli kaam do cheezon mein tha: photo pe filter, aur ek aisa feed jo tez khule.

Baaki sab kuch woh tha jo kisi aur ne bana rakha tha.

Yeh design ki taakat hai. **Woh yeh nahi tha ki unhone bahut kuch banaya. Woh yeh tha ki unhone bahut kam banaya, aur sahi cheez banayi.**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **cheez ke dikhne se uski mushkil ka andaza lagana.**

Ek button jo kehta hai “sab dost bulao” bahut chhota dikhta hai. Uske peeche contact ki ijaazat, alag alag phone ke alag niyam, email bhejna, spam se bachna, aur duplicate sambhalna hota hai.

Aur ek page jo bahut bhara hua dikhta hai, woh aksar ek list hoti hai jo do din mein ban jaati hai.

Jo dikhta hai woh kaam ka naap nahi hai. Kaam un haalat mein hai jo dikhti nahi.

Doosri galti: **”Al se ab yeh sab aasan ho gaya.”**

Al ne kuch cheezein sach mein bahut sasti kar di hain, khaaskar woh jo pehle likhi ja chuki hain aur jinka sahi jawab saaf hai.

Woh chaar pehchaanein nahi badli hain. Jo cheez isliye mushkil thi ki uska “sahi” tay nahi tha, woh aaj bhi mushkil hai. Jo cheez isliye mushkil thi ki uske andar paisa aur kanoon hai, woh aaj bhi mushkil hai. Aur purani cheez ko chhoona aaj bhi mehnga hai.

Al ne likhne ki keemat giraayi hai. Usne samajhne ki keemat nahi giraayi. Aur design samajhne ka kaam hai.

SOCHNE KE LIYE

1. (samajh check) Ek aadmi kehta hai “main WhatsApp jaisa app banaunga, do mahine mein.” Chaar pehchaanon pe chalao.

Jawab: Messages bhejna aur dikhana: sahi jawab saaf hai, waqt ki sakhti kam hai. Woh hissa sach mein kuch hafton mein ban sakta hai.

Lekin: end to end taala (suraksha, jahan galti bahut mehngi hai), video call (asli waqt), lakhon logon pe chalna (Part 6), aur har phone pe kaam karna.

Toh “WhatsApp jaisa” ek chhota kaam hai aur “WhatsApp” ek bahut bada kaam hai. Farak sirf paimane ka nahi hai, woh chaaron pehchaanon ka hai.

Aur yahi sawal sahi tareeka hai kisi bhi bade idea ko naapne ka: **usmein se woh chhota hissa nikaalo jo pehli teen pehchaanon mein aasan hai.**

2. (samajh check) Ek team ne apna khud ka login system banaya. Woh chal raha hai. Kya yeh sahi faisla tha?

Jawab: Aksar nahi.

Login sabke liye ek jaisa hai, kisi ka product nahi hai, aur usmein galti ki keemat bahut zyada hai. Yeh teeno “khareedo” wale niyam hain.

”Chal raha hai” kaafi nahi hai. Login ki galtiyan chalne mein nahi dikhti. Woh us din dikhti hain jab koi dhoondh leta hai.

Ek apwaad hai: agar aapka product hi login ya pehchaan hai, toh khud banana hi sahi hai.

3. (jodne wala) Chapter 7.5 kehta tha ki anumaan gyaan ka naap hai. Yeh chapter chaar pehchaanein deta hai. Dono ko jodo.

Jawab: Chaar pehchaanein aapko batati hain ki aapka gyaan kitna kam hai.

Jaha sahi jawab saaf hai, wahan aap jaante ho ki kya banana hai, toh anumaan theek hoga.

Jaha sahi jawab dhundhla hai, wahan aapko yeh bhi nahi pata ki aap kya bana rahe ho, isliye anumaan lagbhag bekaar hai.

Toh yeh chaar pehchaanein ek aur kaam bhi karti hain: **woh batati hain ki aapke anumaan par kitna bharosa kiya jaa sakta hai.**

Ek “kuch din” wala kaam aksar sach mein kuch din ka hota hai. Ek “kuch mahine” wala kaam kuch bhi ho sakta hai.

Chapter 8.3 [SPINE]

Ek spec kaisa dikhta hai

SAMVAAD

Madhav: Ab aapko kuch banana hai. Aap khud code nahi likhoge. Aap kisi ko batayenge, ya AI se karwayenge. Kya doge unhe?

Kabir: Bataunga ki kya banana hai.

Madhav: “Ek chat app banao.” Bas?

Kabir: Nahi. Aur detail chahiye.

Madhav: Kitni? Chalo dekhte hain. “Users ek doosre ko message bhej sakein.” Kya yeh kaafi hai?

Kabir: Nahi. Message kitna lamba? Photo bhej sakte hain?

Madhav: Achha. Aur?

Kabir: Purane message dikhenge? Kitne?

Madhav: Aur?

Kabir: Delete kar sakte hain? Sirf apna, ya doosre ka bhi?

Madhav: Aur ab sabse zaroori wala sawal, jo lagbhag koi nahi poochta. Agar bhejne mein galti ho jaaye toh kya dikhega?

Kabir: Woh... maine socha hi nahi.

Madhav: Aur agar internet beech mein chala jaaye?

Kabir: Message ruk jaayega. Aur baad mein chala jaayega?

Madhav: Ya kho jaayega. Kaunsa chahiye?

Kabir: Ruk kar baad mein jaana chahiye.

Madhav: Toh use kahin rakhna padega. Aur user ko kya dikhega us beech mein?

Kabir: Ek nishaan ki abhi bheja nahi gaya.

Madhav: Ab dekho. Humne abhi tak sirf ek cheez ki baat ki hai, message bhejna. Aur usmein paanch faisle nikal aaye.

Kabir: Aur woh paanch mein se teen galtiyon ke baare mein the.

Madhav: Bas. **Ek spec ka aadha hissa yeh hota hai ki jab cheezein galat ho jaayein tab kya ho.** Aur woh aadha hissa hi hamesha chhoota hai.

Kabir: Aur banane wala use khud tay kar leta hai.

Madhav: Aur uska tay kiya hua aapke tay kiye hue se alag hoga. Aur woh baad mein pata chalega. Ab aakhri baat. Kaise pata chalega ki kaam poora ho gaya?

Kabir: Jab woh chalne lage.

Madhav: “Chalne lage” kya hai? Aap kaise jaanchoge?

Kabir: Mujhe pehle se likhna padega ki kya kya hona chahiye.

Madhav: Toh spec ka aakhri hissa kya hai?

Kabir: Yeh ki kaise pata chalega ki ho gaya.

EK SPEC KE PAANCH HISSE

1. YEH KYUN
kis insaan ki kaunsi problem hal ho rahi hai
agar yeh nahi likh sakte, toh baaki sab bekaar hai
2. YEH KYA KARTA HAI
seedha raasta, kadam ke hisaab se
user kya karta hai, system kya karta hai
3. YEH KYA NAHI KARTA
jo cheezein jaan-boojh ke nahi ki ja rahi
(yeh hissa jitna bada, utna spec achha)
4. JAB GALAT HO TAB KYA
internet gaya, do log ek saath, data galat,
cheez mili nahi, ijaazat nahi
← yahi sabse zyada chhoota hai aur sabse zyada mayne rakhta hai
5. KAISE PATA CHALEGA KI HO GAYA
woh cheezein jo jaanchi ja sakti hain
"achha lagta hai" jaanch nahi hai

EK POORA CHHOTA SPEC

Chat app ka ek hissa, poora likha hua:

YEH KYUN

User apna purana message dhoondhna chahta hai. Abhi use haath se scroll karna padta hai, jo mahinon purane chat mein lagbhag namumkin hai.

YEH KYA KARTA HAI

1. User ek shabd likhta hai
2. System uske apne saare chats mein woh shabd dhoondhta hai
3. Nateeje naye se purane ke kram mein dikhte hain
4. Har nateeje mein message aur uske aas-paas ka thoda hissa
5. Nateeje pe click karo toh us chat mein us jagah pahunch jao

YEH KYA NAHI KARTA

- photo aur video ke andar nahi dhoondhta
- group chat mein doosron ke message nahi dhoondhta
- "milte julte" shabd nahi dhoondhta, sirf wahi shabd
- hindi aur angrezi ke beech tarjuma nahi karta

JAB GALAT HO TAB KYA

- kuch nahi mila → "kuch nahi mila" dikhao, khaali screen nahi
- internet nahi hai → jo phone pe pehle se hai usmein dhoondho, aur batao ki adhoora hai
- search dheemi hai → 2 second baad "dhoondh raha hoon" dikhao, aur rok bhi sako
- shabd bahut chhota → ek akshar pe mat dhoondho, kam se kam do

KAISE PATA CHALEGA KI HO GAYA

- 10,000 message wale chat mein search 1 second se kam
- ek aisa shabd jo teen chats mein hai, teeno dikhein
- nateeje pe click karne se sahi message pe pahuncho
- internet band karke bhi kuch nateeje aayein
- ek akshar likhne pe kuch na ho

DO BAATEIN JO ISS SPEC MEIN CHHUPI HAIN

Ek: "yeh kya nahi karta" wala hissa sabse taakatwar hai.

Usne ek anginat kaam ko ek karne laayak kaam bana diya. Bina uske, "search banao" ka matlab hai photo ke andar dhoondhna, tarjuma karna, milte julte shabd samajhna, aur woh saal bhar ka kaam ho jaata hai.

Har cheez jo aap saaf-saaf hata dete ho, woh kaam ko chhota karti hai.

Do: “jab galat ho tab kya” wala hissa aapke faisle hain, banane wale ke nahi.

Agar aap nahi likhoge, toh woh kuch na kuch chunega. Aur uska chunav aksar sabse aasan wala hoga: khaali screen, koi message nahi, ya chup-chaap fail.

ASLI DUNIYA SE EK EXAMPLE

Jab aap kisi AI se kuch banwane ki koshish karte ho, toh yeh farak turant dikhta hai.

“Ek chat app banao” se kuch aisa milta hai jo dikhta toh chat app jaisa hai, aur usmein woh saare faisle kisi aur ne kar liye hain: kya nahi hoga, galat hone pe kya dikhega, kitna bada chalega.

Wahi baat pooray spec ke saath karo, aur nateeja bilkul alag hota hai.

Aur yeh Chapter 8.2 ki baat ko poora karta hai: **AI ne likhne ki keemat giraayi hai. Isliye ab woh hissa jo bacha hai, yaani yeh tay karna ki kya likhwana hai, pehle se zyada mayne rakhta hai, kam nahi.**

Jo aadmi spec likh sakta hai, uske liye AI ek bahut bada guna hai. Jo nahi likh sakta, uske liye AI woh cheez banata hai jo usne maangi thi, aur woh cheez usse chahiye hi nahi thi.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki spec lambi honi chahiye.**

Ek achha spec chhota hota hai aur usmein faisle hote hain. Ek bura spec lamba hota hai aur usmein bas baatein hoti hain.

Agar aapki spec mein “system fast hona chahiye” likha hai, toh usmein kuch bhi nahi likha. “10,000 message mein 1 second se kam” ek faisla hai.

Doosri galti: **seedhe raaste ko poora kaam samajhna.**

Zyadatar spec sirf hissa 2 likhte hain, aur wahi sabse chhota hissa hai. Asli kaam hisse 3 aur 4 mein hai.

Teesri galti: **yeh maan lena ki spec ek baar mein likh li jaayegi.**

Aap likhna shuru karoge aur aadhe sawal aapko tab dikhenge jab aap likh rahe honge. Yehi spec ka asli faayda hai. **Woh sochne pe majboor karti hai, kaagaz pe, code se pehle.**

SOCHNE KE LIYE

1. (samajh check) Ek spec mein likha hai “user apni profile photo badal sakta hai.” Paanch sawal poochho jo iss mein nahi likhe.

Jawab: Kuch aam wale:

Photo ka size kitna bada ho sakta hai? Kaunse tarah ki file? Kya woh kaati jaayegi ya chhoti ki jaayegi? Purani photo hat jaayegi ya rakhi jaayegi? Agar upload beech mein ruk jaaye toh? Agar photo aapatti-janak ho toh? Kitni baar badal sakta hai? Kya doosron ko turant dikhegi ya cache mein purani rahegi?

Aakhri wala dhyaan dene laayak hai, kyunki woh Chapter 5.5 se aata hai. **Naksha jaante ho, toh spec ke sawal apne aap dikhne lagte hain.**

2. (samajh check) Aap “yeh kya nahi karta” wale hisse mein das cheezein daal dete ho. Kya yeh spec ko kamzor karta hai?

Jawab: Ulta. Woh use bahut mazboot karta hai.

Har cheez jo aap hata dete ho, woh kaam ko chhota, saaf, aur poora hone laayak banati hai.

Aur woh bahas ko pehle le aati hai. Agar kisi ko lagta hai ki hatai gayi cheez zaroori hai, toh woh abhi bolega, na ki jab kaam ho chuka ho.

Spec ka sabse mehnga hissa woh hai jo likha hi nahi gaya, kyunki sabne maan liya ki woh saaf hai.

3. (jodne wala) Chapter 7.1 kehta tha ki bug un haalat mein hote hain jinke baare mein kisi ne socha nahi. Spec ka chautha hissa usse kaise judta hai?

Jawab: Woh seedha uska ilaaj hai.

Chautha hissa aapko majboor karta hai ki galat haalat ko pehle sochein, banane se pehle, na ki tab jab woh asli mein ho jaayein.

Woh saari haalatein nahi pakadta, kyunki woh anginat hain. Lekin woh sabse aam wali pakad leta hai: internet gaya, cheez mili nahi, ijaazat nahi, do log ek saath, data galat.

Aur woh ek aur kaam bhi karta hai: **jo haalatein aap likh dete ho, unka jawab ab ek faisla hai, ek durghatna nahi.**

Chapter 8.4 [DEPTH]

Asli zindagi mein cheezein kaise tootti hain

DEPTH chapter. Ismein sirf ghatnaayein hain. Har ek kisi na kisi chapter ki baat ko asli duniya mein saabit karti hai.

SAMVAAD

Madhav: Ab tak humne dekha ki cheezein kaise chalti hain. Ab dekhte hain ki woh kaise girti hain. Ek andaza lagao: bade systems mein sabse zyada nuksaan kis wajah se hota hai? Hacker? Machine kharaab hona?

Kabir: Hacker.

Madhav: Nahi. Sabse zyada nuksaan un badlaon se hota hai jo khud us company ke logon ne kiye.

Kabir: Apne hi log?

Madhav: Ek settings ki line badli. Ek naya version daala. Ek command galat jagah chali. Yeh sabse aam wajah hai, har saal, har jagah.

Kabir: Toh badlav hi khatra hai.

Madhav: Aur badlav zaroori bhi hai. Toh sawal yeh nahi hai ki badlav rok do. Sawal yeh hai ki badlav ko kaise kiya jaaye ki galti chhoti rahe.

Kabir: Kaise?

Madhav: Do नियम. Pehla: ek saath sab jagah mat badlo. Thodi jagah badlo, dekho, phir aage badho.

Kabir: Aur agar galat nikla?

Madhav: Doosra नियम: wapas jaane ka raasta hamesha taiyaar rakho, badlav se pehle.

Kabir: Aur agar wapas jaane ka raasta hi kaam na kare?

Madhav: Toh woh raasta hai hi nahi. Chapter 5.7 wali baat: jo jaancha nahi gaya woh hai hi nahi.

CHAAR GHATNAAYEIN

1. Ek naam gayab hua, aur duniya ruk gayi

2021 mein ek badi social media company ke saare product lagbhag chhe ghante ke liye poori duniya se gayab ho gaye.

Server chal rahe the. Data theek tha. Lekin ek badlav ke baad unke pate baantne wali jaankari internet se hat gayi, aur naam ka jawab dene ka koi raasta nahi bacha.

Aur ek nateeja aur nikla: unke office ke darwaze bhi unhi systems se chalte the. Log andar nahi ja paaye us cheez ko theek karne jisne unhe bahar kar diya tha.

Chapter 4.5 naam poore internet ki neev mein hai
Chapter 6.2 koi ek jagah aisi na ho jo sab rok de

2. Aath machinein, saat pe naya code

2012 mein ek trading company ne naya code daala. Aath machinein thin, saat pe naya gaya, ek chhoot gayi.

Us ek machine pe purana code chalta raha, aur uska ek bahut purana hissa naye code ke saath milkar galat kaam karne laga.

45 minute mein lagbhag 44 crore dollar chale gaye. Company us hafte lagbhag khatam ho gayi.

Chapter 7.1 galti hisson ke BEECH mein thi, kisi ek hisse mein nahi
Chapter 7.4 haath se daalna hi asli galti thi

3. Paanch bachaav, chaar bekaar

2017 mein ek company ke engineer ne raat ko, thake hue, ek command galat server pe chala di. 300 GB asli data mit gaya.

Unke paas paanch alag bachaav the. Chaar us waqt kaam nahi kar rahe the: kuch chup-chaap band ho chuke the, kuch se banti hui files khaali thin, kisi ki kabhi jaanch nahi hui thi.

Paanchva ek chhe ghante purana snapshot tha, jo kisi ne kisi aur kaam ke liye banaya tha. Usi se woh bache.

Chapter 5.7 jo bachaav jaancha nahi gaya, woh bachaav hai hi nahi

4. Ek cloud service baithi, hazaaron site baith gayin

Ek badi cloud company ki ek service kuch ghante ke liye baith gayi. Uske saath duniya bhar ki hazaaron websites aur apps baith gayin.

Un hazaaron mein se har ek ne apna system mazboot banaya tha: kai machinein, backup, sab kuch.

Lekin sabke neeche ek hi cheez thi.

Chapter 6.2 aapka system utna hi mazboot hai jitni uski
 sabse kamzor NEECHE WALI parat
Chapter 6.5 cloud ki nirbharta

CHAARON MEIN SAANJHA KYA HAI

KISI MEIN BHI HACKER NAHI THA
KISI MEIN BHI MACHINE KHARAAB NAHI HUI
KISI MEIN BHI CODE MEIN GALTI NAHI THI

Chaaron mein:

1. badlav apne hi logon ne kiya
2. woh cheez jo toot ti hai woh aksar HISSON KE BEECH hoti hai
3. bachaav maujood tha aur kaam nahi kar raha tha
4. neeche wali parat pe nirbharta, jise koi yaad nahi rakhta

Aur ek aur baat, jo sabse zaroori hai: **har ghatna mein cheez ek jhatke mein giri, dheere-dheere nahi.** Chapter 6.6 wali baat, asli duniya mein.

DO NIYAM JO HAR JAGAH KAAM KARTE HAIN

1. Thoda badlo, dekho, phir aage badho.

Naya code pehle ek pratishat logon ko do. Kuch der dekho. Theek lage toh das pratishat. Phir sab.

Iska naam alag alag jagah alag hai, lekin soch ek hi hai: **agar kuch galat hai toh woh ek pratishat ko dikhe, sabko nahi.**

2. Wapas jaane ka raasta pehle banao, badlav se pehle.

Aur woh raasta ek button hona chahiye, ek plan nahi. Kyunki jab cheez giri hui hoti hai, tab kisi ke paas plan padhne ka waqt nahi hota.

Aur teesra, jo in dono se nikalta hai: **jab kuch toota ho, tab sabse pehle use theek karo, phir dhoondho ki kyun toota tha.** Log ulta karte hain, aur system tab tak toota rehta hai jab tak wajah na mil jaaye.

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki galti kisi aadmi ki thi.**

Har ghatna ke baad ek aadmi milta hai jisne button dabaya tha. Aur woh sabse bekaar jawab hai.

Sawal yeh nahi hai ki usne button kyun dabaya. Sawal yeh hai ki **ek aadmi ke ek button se poora system kaise gir sakta tha.**

Agar ek aadmi ki ek galti se 44 crore dollar ja sakte hain, toh problem us aadmi mein nahi hai. Problem us system mein hai jismein aisi galti mumkin thi.

Isiliye achhi teamein ghatna ke baad aadmi ka naam nahi dhoondhti. Woh yeh dhoondhti hain ki kaunsi cheez ne is galti ko itna aasan bana diya, aur use kaise mushkil banaya jaaye.

Doosri galti: **yeh sochna ki yeh sirf badi companies ke saath hota hai.**

Yeh chhoti jagahon pe zyada hota hai. Bade systems ki ghatnaayein sirf isiliye mashhoor hoti hain ki woh dikhti hain, aur unhe likh diya jaata hai.

SOCHNE KE LIYE

1. (samajh check) Aapki team naya code hafte mein ek baar, shukravaar shaam ko daalti hai. Kya problem hai?

Jawab: Do.

Ek: hafte bhar ke badlav ek saath jaate hain. Kuch galat ho toh pata hi nahi chalega ki kis badlav se hua. Chhote badlav dhoondhna aasan banate hain.

Do: shukravaar shaam ka matlab hai ki agar kuch toota toh koi maujood nahi hoga.

Isiliye kai jagah “shukravaar ko kuch mat daalo” ek niyam hai. Aur behtar jagahon pe badlav itne chhote aur itne aam ho jaate hain ki din ka farak hi khatam ho jaata hai.

2. (samajh check) Ek ghatna ke baad manager kehta hai “aage se dhyaan rakhna.” Kya yeh kaafi hai?

Jawab: Nahi, aur woh lagbhag bekaar hai.

”Dhyaan rakhna” kisi cheez ko nahi badalta. Agla aadmi bhi dhyaan rakh raha tha.

Kaam ka jawab yeh hai: **us galti ko namumkin ya mushkil kaise banaya jaaye?**

Command chalane se pehle poochha jaaye? Asli server pe alag rang dikhe? Ek aadmi akela na kar sake? Wapas jaane ka button ho?

Yaad rakhne ki jagah dhaancha badlo. Yaddasht har baar fail hoti hai. Dhaancha nahi.

3. (jodne wala) Chaaron ghatnaon mein bachaav maujood tha. Teen mein woh kaam nahi kar raha tha. Yeh iss kitaab ki kaunsi baat dohrata hai?

Jawab: Chapter 5.7: **jo jaancha nahi gaya, woh hai hi nahi.**

Aur uska ek gehra roop yeh hai: bachaav ke saath ek khaas problem hai. Woh chup-chaap kharaab hota hai.

Ek website toot jaaye toh turant pata chalta hai. Ek backup toot jaaye toh saalon tak kisi ko pata nahi chalta, kyunki woh sirf us din kaam aata hai jab zaroorat padti hai.

Isiliye bade systems mein backup ko baar baar **wapas laakar** dekha jaata hai, apne aap, aur agar woh na aaye toh alarm bajta hai.

Jo cheez sirf bure din kaam aati hai, use achhe din jaancha jaana chahiye.

Chapter 8.5 [SPINE]

Ek technology jeetati kyun hai, doosri marti kyun

SAMVAAD

Madhav: Iss kitaab mein aapne kai baar dekha ki ek behtar cheez haar gayi. Ginao.

Kabir: IPv6 behtar tha aur 26 saal mein bhi aadhi duniya tak nahi pahuncha. JavaScript das din mein bani aur jeet gayi. Aur log behtar messaging app pe nahi jaate.

Madhav: Toh “behtar” kaafi nahi hai. Phir kya kaafi hai? Ek-ek karke dekhte hain. IPv6 kyun nahi phaila?

Kabir: Kyunki sabko ek saath badalna padta, aur akele badalne se kisi ko faayda nahi tha.

Madhav: JavaScript kyun jeeti?

Kabir: Kyunki woh browser ke andar pehle se thi. Jo wahan tha, wahi jeeta.

Madhav: Messaging app?

Kabir: Kyunki mere log wahan hain. Akele jaane se main akela ho jaunga.

Madhav: Ab teeno ko dekho. Kya saanjha hai?

Kabir: Teeno mein faisla akele ka nahi tha. Woh sabka milkar tha.

Madhav: Bas. Ab ek chauthi wajah bhi hai. Chapter 3.3 mein humne dekha tha ki Python AI mein kyun hai. Yaad hai?

Kabir: Kyunki uske aas-paas saara saaman pehle se bana hua tha.

Madhav: Aur woh saaman kyun bana?

Kabir: Kyunki kuch log use istemaal kar rahe the.

Madhav: Toh?

Kabir: Toh jo aage nikal gaya, woh aur aage nikalta gaya.

Madhav: Aur ab paanchvi, jo sabse kam dikhti hai. Do cheezein hain, dono barabar achhi. Ek ko chalane mein aur badalne mein aasani hai, doosri technically behtar hai lekin uske saath kaam karna dukhdayi hai. Kaunsi jeetegi?

Kabir: Pehli.

Madhav: Kyun? Log toh behtar cheez chahte hain.

Kabir: Kyunki jo cheez istemaal karne mein aasan hai use zyada log istemaal karenge, aur phir woh aage nikal jaayegi.

Madhav: Aur ab aakhri sawal, jo aapke apne kaam ka hai. Agar aap kuch banate ho, toh in paanch mein se aap kis pe kaam kar sakte ho?

Kabir: Behtar banane pe... jo shayad sabse kam kaam ka hai.

Madhav: Aur baaki chaar?

Kabir: Woh log ke aapas ke rishton pe hain, cheez pe nahi.

PAANCH WAJAHEIN

1. PEHLE SE WAHAN HONA
jo jagah pe pehle se maujood tha, woh jeet jaata hai
(JavaScript browser mein tha)
2. SABKO EK SAATH BADALNA PADE
agar badlav ka faayda tabhi mile jab sab badlein,
toh koi nahi badlega
(IPv6, 26 saal)
3. LOG WAHAN HAIN
network wali cheezon mein aap cheez nahi chunte,
aap yeh chunte ho ki kiske saath rahoge
(Chapter 4.1)
4. AAS-PAAS KA SAAMAN
jo aage nikla, uske aas-paas cheezein banti hain,
jisse woh aur aage nikal jaata hai
(Python aur AI)
5. ISTEMAAL KARNE MEIN AASANI
dukhdayi behtar cheez se aasan theek cheez aksar jeet jaati hai

Aur ab woh baat jo iss chapter ki asli seekh hai:

Paanchon mein se ek bhi “cheez kitni achhi hai” nahi hai.

Achha hona ek shart hai. Ek cheez itni buri nahi honi chahiye ki kaam hi na kare. Uske baad, jeet lagbhag poori tarah in paanch pe tay hoti hai.

Yeh nirasha ki baat lag sakti hai, aur woh nahi hai. Iska ulta matlab yeh hai: **agar aap in paanch ko samajh lo, toh aap unpe kaam kar sakte ho.**

behtar banana	→	sabse mushkil, sabse kam asar
aasan banana	→	aksar behtar banane se zyada asar
kisi jagah ghus jana	→	jahan log pehle se hain
chhote group se shuru	→	jahan poora group ek saath aa sake
saaman banane dena	→	doosron ko apne upar banane dena (Ch 6.4)

Teesra aur chautha wala Chapter 4.1 ki us problem ka jawab hai jo tab khuli chhodi gayi thi: **network wali cheez shuru kaise hoti hai jab uski shuruaati value zero hai?**

Jawab: poore bazaar se mat ladho. Ek chhota group dhoondho jo poora ek saath aa sake, jahan network aandar hi poora ho jaaye. Ek college, ek company, ek sheher, ek khaas kaam. Wahan value zero se turant upar chali jaati hai, kyunki sab jo aapko chahiye wahi hain.

ASLI DUNIYA SE EK EXAMPLE

Facebook ne ek college se shuru kiya. Sirf ek. Phir kuch aur college.

Yeh dheema tareeka lagta hai. Woh sabse tez tareeka tha, aur wajah ganit hai.

Ek college mein do hazaar log hain. Unmein se sab ek doosre ko jaante hain ya jaan sakte hain. Toh jaise hi kuch sau log aaye, un logon ke liye network **poora** ho gaya. Unka har dost wahan tha.

Ab isko ulta karo. Agar woh poori duniya mein ek saath khulte, toh ek karod log aane par bhi kisi ke liye uske apne das dost wahan hone ka koi bharosa nahi tha. Woh ek karod log ek khaali kamre jaisa lagta.

Yeh Chapter 4.1 ka ganit hai, ulta karke istemaal kiya gaya: **rishte ginti se tez badhte hain, isliye chhoti jagah mein poori bhar jaana bade bazaar mein bikhre rehne se behtar hai.**

YAHAN LOG KYA GALAT SAMAJHTE HAIN

Sabse aam galti: **yeh sochna ki behtar product jeetega.**

Yeh sabse mehngi galti hai jo kuch banane wale karte hain. Woh saal lagate hain kuch behtar banane mein, aur phir hairan hote hain ki log kyun nahi aaye.

Log isliye nahi aate ki woh moorkh hain. Woh isliye nahi aate ki unke liye badlav ki keemat aapke behtar hone se zyada hai. Unhe apne log chhodne padenge, apni aadat badalni padegi, apna purana data le jaana padega.

Aapka behtar hona unki keemat se bada hona chahiye, aur woh keemat aksar bahut badi hoti hai.

Doosri galti, ulti taraf se: **yeh maan lena ki jab bada khilaadi maujood hai toh kuch naya ho hi nahi sakta.**

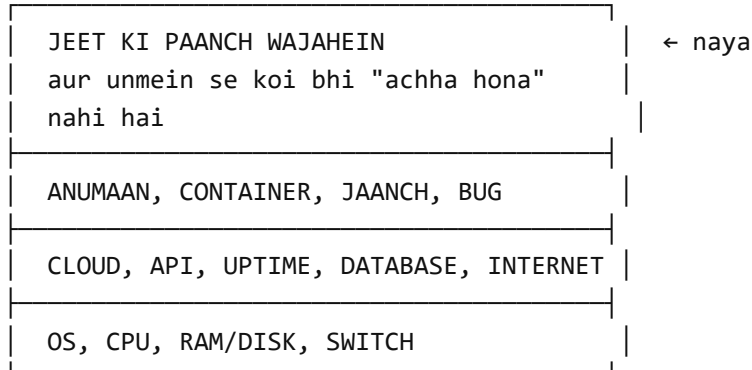
Naye jeet te hain, aur woh aksar in raaston se aate hain:

naya paimana	jab kuch itna sasta ho jaaye ki naye kaam khul jaayein
naya raasta	jab log kisi nayi jagah pe pahunch jaayein (jaise sab desktop se phone pe aa gaye)
purane ka bojh	jab bade khilaadi apne purane waadon mein bandhe hon (Chapter 6.4) aur mud na sakein
chhoti jagah	jahan bada khilaadi dhyaan nahi de raha

Teesri galti: **yeh sochna ki jo aaj jeet raha hai, uska jeetna kisi vajah se tay tha.**

Aksar woh timing thi. Wahi cheez do saal pehle marti aur do saal baad late hoti.

MAP PE



SOCHNE KE LIYE

1. (samajh check) Aap ek nayi cheez bana rahe ho jo tabhi kaam ki hai jab bahut log uspe hon. Kahan se shuru karoge?

Jawab: Ek chhoti, poori jagah se.

Ek college, ek daftar, ek sheher, ek khaas kaam, ek ilaaka. Aisi jagah jahan aapke shuruaati users ke liye **saare** log wahin hon.

Poore bazaar mein bikhre hue das hazaar users se ek jagah ke poore do hazaar behtar hain, kyunki dusre mein network poora hai aur pehle mein khaali.

Aur woh jagah chhoti hona ek nuksaan nahi hai. Woh hi ek matra tareeka hai.

2. (samajh check) Ek bade khilaadi ke saamne aapka product har tarah se behtar hai. Log phir bhi nahi aa rahe. Kya soch rahe ho?

Jawab: Badlav ki keemat naapo.

Unhe kya chhodna padega? Apne log? Apna purana data? Apni aadat? Apne doosre systems se jo jud gaya hai woh?

Aur phir poocho: kya mera behtar hona un sab se bada hai?

Aur ek kaam ka raasta: **keemat kam karo, faayda badhane ki jagah**. Purana data le aane do. Dono jagah kaam karne do. Ek hissa aisa banao jo unki purani cheez ke saath chale.

3. (jodne wala) Chapter 0.2 ke teen dials the: size, scarcity, leverage. Yeh paanch wajahein kis dial pe lagti hain?

Jawab: Zyadatar leverage pe, aur wahi sabse kaam ki baat hai.

Jo cheez pehle se jagah pe hai, jisme log hain, aur jiske aas-paas saaman bana hai, woh har naye user se aur mazboot hoti jaati hai bina uske maalik ki mehnat ke.

Yeh Chapter 4.1 wala leverage ka teekha roop hai: **ek baar banao, aur phir har naya aane wala use aur bada kar de**.

Aur isliye yeh paanch wajahein sirf technology ki baat nahi hain. Woh yeh batati hain ki technology mein sabse bada leverage kahan banta hai, aur woh cheez ki quality mein kabhi nahi banta.

Chapter 8.6 [SPINE]

Book 1 ka check

SAMVAAD

Madhav: Chaar hafte pehle tumne kaha tha ki har hafte kuch naya sunte ho jo baaki sabko pehle se pata tha.

Kabir: Haan.

Madhav: Ab ek khabar padhta hoon. “Ek nayi database company ne kaha hai ki unka system purane se sau guna tez hai kyunki woh sab kuch memory mein rakhta hai.” Kya sawal poochoge?

Kabir: Memory mein rakhta hai matlab RAM mein. Toh bijli jaane pe kya hoga?

Madhav: Aur?

Kabir: RAM mehngi hai aur chhoti hai. Toh kitna data aa sakta hai?

Madhav: Aur?

Kabir: Aur “sau guna tez” kis kaam mein? Padhne mein ya likhne mein?

Madhav: Chaar hafte pehle tum kya poochte?

Kabir: Kuch nahi. Main bas maan leta ki woh tez hai.

Madhav: Ek aur. “Ek nayi cheez aayi hai jo aapke code ko apne aap kai machinon pe chala deti hai.”

Kabir: Toh mera code khaali hona chahiye, warna das copy das alag jhoot bolengi.

Madhav: Aur?

Kabir: Aur bojh hat ta nahi, khiskta hai. Toh database ka kya?

Madhav: Ek aakhri. “Ek naya protocol aaya hai jo internet ko dugna tez kar dega.”

Kabir: Sabko ek saath badalna padega?

Madhav: Haan.

Kabir: Toh shayad woh das saal lagega, ya kabhi nahi aayega. IPv6 ki tarah.

Madhav: Ab batao, kya badla?

Kabir: Pehle har khabar ek nayi cheez thi. Ab woh ek jaani-pehchani jagah pe aa kar baithti hai.

Madhav: Aur jab kuch aisa aaye jo sach mein naya ho?

Kabir: Toh mujhe pata chal jaayega ki woh sach mein naya hai. Kyunki uske liye meri jagah nahi hogi.

Madhav: Aur woh sabse zaroori cheez hai jo yeh kitaab de sakti thi.

AB AAPKE PAAS KYA HAI

- PART 0 paisa value ka record hai
 $\text{pay} = \text{size} \times \text{scarcity} \times \text{leverage}$
 chaar levels, aur technology wala level chhupa hai
- PART 1 ek switch, aur usse har cheez
 machine badalti nahi, sirf numbers badalte hain
- PART 2 sab kuch numbers hai, aur matlab bahar se aata hai
 CPU tez hai, data laana dheema
- PART 3 language ek insaani suvidha hai
 OS ek manager hai jo baantta hai aur alag rakhta hai
- PART 4 value judav mein hai, machine mein nahi
 internet ek sehmati hai, ek cheez nahi
- PART 5 rakhna aasan, dhoondhna mushkil
 har cheez ka ek sauda hai, hamesha
- PART 6 bojh hat ta nahi, khiskta hai
 systems jhatke mein girte hain, dheere nahi
- PART 7 haalatein anginat hain, poora jaanchna namumkin
 anumaan gyaan ka naap hai, waqt ka nahi
- PART 8 har cheez kisi dard ka jawab hai
 jeet cheez ki quality se nahi aati

AAKHRI CHECK

Yeh Book 1 ka gate hai. Har sawal do ya teen parton se aata hai.

1. Aapki site das hazaar users pe theek chal rahi thi aur das lakh pe baith gayi. Do bilkul alag wajahein batao, aur har ek ke liye ek alag ilaaj.

Jawab:

Ek: kahin poori list padhi ja rahi hai. Kaam data ke saath seedha badhta hai. Das hazaar pe woh dikha nahi, das lakh pe saamne aa gaya. Ilaaj index hai (5.2), badi machine nahi.

Do: seema paar ho gayi aur shrinkhla chal padi. Log intezaar chhod kar dobara koshish kar rahe hain, jisse bojh khud badh raha hai (6.6). Ilaaj seema pe mana karna hai, machine badhana nahi.

Farak pehchanne ka tareeka: pehli mein cheez dheere dheere dheemi hui hogi jaise data badha. Doosri mein woh ek din achanak giri hogi.

2. Ek dost kehta hai “maine apna data do jagah rakha hai, ab woh surakshit hai.” Kya kahoge?

Jawab: Ki woh machine kharaab hone se surakshit hai, apni galti se bilkul nahi (5.7).

Replica wahi karta hai jo asli karta hai. Aap galti se kuch mita do toh woh dono jagah mit jaayega, turant.

Uske liye purana backup chahiye. Aur woh backup tabhi backup hai jab use wapas laakar dekha gaya ho.

3. Ek naya app banane wala aapse poochta hai: “search feature banane mein kitna waqt lagega?” Aap kya poochoge, jawab dene se pehle?

Jawab: Kam se kam yeh:

Kis cheez mein dhoondhna hai, sirf likhawat ya photo bhi? Milte-julte shabd chalenge ya bilkul wahi? Kitna data hai? Nateeje kis kram mein aayein, aur “sahi kram” ka matlab kya hai? Galat likhne pe kya ho? Kuch na mile toh kya dikhe?

Aur phir batana ki yeh ek range hai, ek number nahi (7.5), kyunki search un kaamon mein hai jinka “sahi jawab” saaf nahi hota (8.2).

4. Ek khabar: “nayi chip 40 pratishat tez hai.” Doosri khabar: “ek nayi cheez ne AI chalane ka kharcha 30 guna kam kar diya.” Kaunsi zyada mayne rakhti hai?

Jawab: Doosri, aur bahut zyada (2.2).

40 pratishat ek hi paimane ke andar ka sudhaar hai. Jo cheezein pehle ho sakti thin woh thodi behtar ho gayin.

30 guna sasta hone ka matlab hai ki woh sab kuch jo pehle mehnga hone ki wajah se banaya hi nahi ja sakta tha, ab ban sakta hai. Woh paimana badal deta hai.

Pratishat mein badlav khabar hai. Guna mein badlav badlav hai.

5. Aap ek product design kar rahe ho. Uska ek hissa yeh hai ki users video daal sakein. Kis cheez pe sabse pehle sochoge?

Jawab: Kharche pe (6.8).

Ek ghanta video lagbhag ek karod text message ke barabar hai. Aur bhejna rakhne se kai guna mehnga hai.

Toh video wala product kabhi bhi text wale product jaisa muft nahi ho sakta. Uska poora karobaari dhaancha alag hoga: shuru se paise ka raasta chahiye, ya bahut sakht seemayein.

Aur yeh technical faisla nahi hai. Yeh sabse bada karobaari faisla hai jo iss product mein hai.

AAP AB KYA KAR SAKTE HO

Kitaab ke shuru mein teen vaade the. Ab unhe check karo.

Ek: koi bhi nayi technology sun kar use naksh e pe rakh dena.

Aap yeh kar sakte ho. Teen sawal kaafi hain:

1. Yeh kis parat pe hai?
switch, ya language, ya network, ya data, ya serving,
ya banane ka tareeka
2. Yeh kis dard ke liye bani thi?
Agar dard nahi bata sakte, toh shayad woh cheez marketing hai
3. Iska sauda kya hai?
Har cheez kuch deti hai aur kuch leti hai
Agar nahi dikh raha toh aapne dhoondha nahi

Do: ek product kaagaz pe design karna.

Aap yeh kar sakte ho. Chapter 8.1 ka naksha, 8.2 ki chaar pehchaanein, aur 8.3 ke paanch hisse. Woh teen milkar ek asli spec ban jaate hain.

Teen: AI se woh nikaalna jo expert nikaalte hain.

Yeh adhoora hai, aur woh jaan-boojh ke hai. Iska aadha hissa Chapter 8.3 mein aa chuka hai: jo aadmi spec likh sakta hai, uske liye AI ek bahut bada guna hai.

Baaki aadha Book 2 hai, kyunki uske liye yeh jaanna zaroori hai ki AI andar se kaam kaise karta hai, aur woh kya nahi kar sakta.

AB AAGE

Book 1 khatam.

Aapne sabse neeche se shuru kiya, ek switch se, aur upar chadhte gaye. Aur ab neeche wali poori parat, jo Chapter 0.4 mein “sab chhupa hua” likhi thi, woh khuli hui hai.

Book 2, The Thinking Machine, wahin se shuru hogi jahan Book 1 ne chhoda: ek machine jo sirf numbers rakhti hai aur shartein chalati hai. Aur usmein hum poochenge ki us machine se woh cheez kaise bani jo baat karti hai.

Us kitaab ka har chapter iss kitaab pe khada hai. Model ek file hai numbers ki. Woh ek machine pe chalta hai jismein RAM aur chip hai. Woh ek network se aata hai. Uske jawab cache mein rakhe jaate hain. Uska kharcha usi hisaab se nikalta hai jo Chapter

6.8 mein tha.

Kuch bhi naya nahi hoga. Sirf ek aur parat, isi dher pe.

Aur agar Book 1 ne apna kaam kiya hai, toh Book 2 aapko ek nayi duniya nahi lagegi.
Woh ek aur chapter lagega.

Glossary

Har shabd, aur woh chapter jahan aapne use kamaaya.

abstraction (0.4) — neech wali cheez ko chhupa dena, taaki uske upar wala usse soche bina kaam kar sake. Poori computer ki duniya isi ka dher hai.

adder (1.5) — switchon se bana woh hissa jo do numbers jodta hai. Usse ghatana, guna aur bhaag sab nikalte hain.

AND / OR / NOT (1.3) — teen shartein. AND: dono chalu hon. OR: koi ek. NOT: ulta. In teeno se har woh cheez ban sakti hai jo koi computer karta hai.

API (6.4) — tay-shuda tareeka jisse ek program doosre se baat karta hai. Ek waada, ek darwaza, aur ek taala, teeno.

assembly (3.1) — machine code ki woh shakal jismein har nirdesh ko ek chhota naam mil jaata hai.

backup (5.7) — kisi purane waqt ki copy, alag rakhi hui. Woh aapki apni galti se bachata hai. Jaanchi na ho toh backup hai hi nahi.

bandwidth (4.3) — ek second mein kitna data aa sakta hai. Sadak kitni chaudi hai. Badhayi ja sakti hai.

BGP (4.7) — woh tareeka jisse padosi machinein ek doosre ko batati hain ki kahan se kya pahuncha ja sakta hai.

binary (1.4) — sirf do nishaanon wali ginti, 0 aur 1.

binary search (5.2) — har kadam pe aadha hissa chhod dena. Das crore mein se ek cheez lagbhag sattaais kadam mein.

bit (1.4) — ek switch ki haalat. Ek haan-na. Sabse chhoti cheez.

bug (7.1) — galti. Woh isliye rehti hai kyunki haalatein anginat hain aur poora jaanchna namumkin hai.

byte (2.1) — aath bit. Aam taur pe ek angrezi akshar.

cache (5.5) — paas rakhi hui copy. Woh tez isliye hai kyunki woh purani ho sakti hai. Woh do alag baatein nahi hain.

cascading failure (6.6) — bojh khud ko badhata jaata hai. Log intezaar chhod kar dobara koshish karte hain, jisse bojh aur badhta hai.

CDN (5.5) — duniya bhar mein rakhi copies, taaki cheez user ke paas se aaye. Speed ke liye bhi, aur paise ke liye bhi.

certificate authority (4.6) — woh teesra jo batata hai ki koi sach mein wahi hai jo keh raha hai. Uski list aapke device mein pehle se hoti hai.

CI (7.2) — woh machine jo har badlav pe saari jaanch apne aap chalati hai.

client (4.1 / 6.3) — woh cheez jo sawal poochti hai. User ke device pe chalti hai, isliye uspe kabhi bharosa nahi kiya jaata.

cloud (6.5) — kisi aur ki machinein, kirye pe. Per machine mehnga, aur phir bhi aksar sahi.

compiler (3.1, 3.2) — poore program ko pehle numbers mein badal deta hai, phir woh chalta hai.

compression (2.1) — dohrav hatao, ya sirf farak rakho. Woh dohrav pe chalta hai, isliye jahan dohrav nahi wahan bekaar hai.

congestion control (4.3) — dheere se badhao, kuch khone lage toh peeche hato. Isse poori sadak bina kisi manager ke baat jaati hai.

container (7.4) — program aur uski saari cheezein ek dabbe mein, taaki woh har machine pe ek jaisa chale.

CPU (2.3) — woh hissa jahan nirdesh chalte hain. Uthao, samjho, karo, baar baar.

database (5.1) — woh program jo chaar problem sambhalti hai: dhoondhna, ek saath likhna, beech mein rukna, dohrav.

DNS (4.5) — naam ko pate mein badalne wala jaal. Ek pedh, jismein har shakha apna hissa sambhalti hai.

encoding (1.4) — woh tay-shuda list jo batati hai ki kaunsa number kis cheez ka matlab hai. Matlab andar nahi hota, isi list se aata hai.

front / back (6.3) — dikhne wala hissa user ke device pe, na dikhne wala server pe. Sach hamesha peeche tay hota hai.

git (7.3) — har badlav ka record. Uska asli faayda purani copy nahi, himmat hai.

HTTP (4.4) — web ka protocol. Kya maanga jaa sakta hai aur jawab kaisa dikhega.

index (5.2) — peeche wali list. Padhna tez karta hai, likhna dheema, aur jagah leta hai.

interpreter (3.2) — chalte chalte translate karta hai, line by line.

IP address (4.2) — machine ka pata. Woh network mein jagah batata hai, zameen pe jagah nahi.

latency (4.3, 6.7) — ek chakkar mein kitna waqt. Sadak kitni lambi hai. Roshni ki seema hai, isliye badhayi nahi ja sakti.

load balancer (6.1) — saamne khadi cheez jo users ko machinon pe baantti hai.

lock-in (6.5) — jitni kisi ki khaas cheezein istemaal ki, utna wahan se nikalna mushkil.

logic gate (1.3) — bijli ko shart ke hisaab se rokne ya jaane dene wala hissa.

machine code (3.1) — seedhe numbers, jinhe chip samajhti hai.

N+1 (6.7) — ek sawal se sau cheezein aayin, aur phir har cheez ke liye ek alag sawal. Speed ki sabse aam galti.

NAT (4.2) — ek public pate ke peeche kai andar wale pate. Isiliye aapka phone bahar se pahuncha nahi ja sakta.

network effect (4.1) — har naya judne wala pehle se jude hue sabki value badha deta hai.

open source (3.6) — jiska code sabke liye khula ho. Uski wajah aksar daan nahi, hisaab hoti hai.

operating system (3.4) — woh manager jo baari, jagah aur saaman baantta hai, aur sabko ek doosre se alag rakhta hai.

overflow (1.5) — jagah khatam ho gayi. Machine rukti nahi, woh chup-chaap galat jawab de deti hai.

packet (4.3) — bade sandesh ka ek chhota tukda, jisme pata aur number likha hota hai.

process (3.5) — chalta hua program. File ek nuskha hai, process us nuskhe se bana khaana.

program (1.6) — nirdesh, jo khud bhi numbers hain. Isiliye machine badle bina kaam badal jaata hai.

protocol (4.4) — pehle se tay niyam. Woh bheja nahi ja sakta, isliye dono taraf pehle se hona chahiye.

public key / private key (4.6) — ek chaabi jo band karti hai aur sabko di ja sakti hai, doosri jo kholti hai aur kisi ko nahi.

queue (5.6) — kaam ki kataar. Woh kaam kam nahi karti, woh intezaar hata deti hai.

RAM (2.4) — tez memory jo bijli jaate hi khaali ho jaati hai.

registers (2.3) — CPU ke ekdam paas ke chhote khaane. Sabse tez, sabse kam.

replica (5.7) — turant banti nakal. Machine kharaab hone se bachati hai, aapki galti se bilkul nahi.

routing (4.7) — har machine sirf agla kadam jaanti hai. Koi poora raasta nahi jaanta.

scale up / scale out (6.1) — ek machine ko bada karo, ya kai chhoti lagao.

server (4.1) — woh machine jo ek jagah rehti hai aur jo bhi aaye uska jawab deti hai.

SQL / NoSQL / graph (5.3) — data ki teen shakalein. Shakal data se nahi, us sawal se aati hai jo aap sabse zyada poochoge.

stateless (6.1) — machine ke andar aisi koi jaankari na ho jo sirf usi ke paas ho.

TCP / UDP (4.3) — har tukda pahunchna zaroori, ya pahunche toh theek na pahunche toh chalo aage.

TLS / HTTPS (4.6) — woh taala. Woh sirf itna kehta hai ki raaste mein koi padh nahi raha aur naam wahi hai jo dikh raha hai.

transaction (5.4) — ya toh sab kaam ho, ya kuch bhi na ho. Aadha kabhi nahi.

transistor (1.1) — chip ke andar ka chhota switch. Use doosra switch daba sakta hai, aur wahi sab kuch mumkin banata hai.

Unicode (1.4) — duniya ki lagbhag har likhavat ke liye ek hi list.

uptime (6.2) — kitna waqt chalu raha. Naukon mein ginte hain, aur har nau daam ko lagbhag guna kar deta hai.

version control (7.3) — dekho git.

XOR (1.5) — “dono alag hain” wali shart. Adder ka aadha hissa.

Kahan dhoondhein

Chhe mahine baad jab koi sawal aaye, poora dobara mat padho. Yeh table dekho.

ROZ KE SAWAL

"Mera phone dheema hai / app band ho rahe hain"

- 2.4 RAM aur disk ka farak
- 3.5 process, aur "khula" hona matlab "chal raha" nahi

"Internet dheema hai" ka asli matlab kya hai

- 4.3 bandwidth aur latency alag cheezein hain
- 6.7 waqt ke paanch hisse

"Yeh site surakshit hai?" taala kya kehta hai

- 4.6 taala sirf do baatein kehta hai, teesri nahi

"Maine badla lekin purana dikh raha hai"

- 5.5 cache, aur woh raaste mein paanch jagah hota hai
- 4.5 DNS ka jawab bhi kuch ghante rakha jaata hai

"Mera data cloud pe hai, surakshit hai?"

- 5.7 replica aur backup do alag cheezein hain

"Yeh app muft kaise hai?"

- 3.6 free cheezein exist kyun karti hain
- 6.8 kharcha kaun utha raha hai

KUCH BANANE KE SAWAL

"Isse banane mein kitna waqt lagega?"

- 8.2 chaar pehchaanein, aur mota andaza
- 7.5 anumaan gyaan ka naap hai

"Kisi ko kya batana chahiye ki kya banana hai?"

- 8.3 spec ke paanch hisse

"Mujhe database chahiye ya file kaafi hai?"

- 5.1 chaar problem, aur unmein se kaunsi aapko hai

"SQL ya NoSQL?"

- 5.3 shakal us sawal se aati hai jo aap poochoge

"Kaunsi programming language seekhun?"

- 3.3 pehli mushkil hai, doosri ek hafte ki

"Server khareedun ya kirye pe loon?"

- 6.5 ubhaar-chadhaav wala bojh = kirya

"Kya main yeh khud banaun ya koi bana banaya lun?"

- 8.2 khareedo ya banao wala hissa

JAB KUCH TOOTA HO

"Site das lakh users pe baith gayi"

- 5.2 kahin poori list padhi ja rahi hai
- 6.6 ya seema paar ho gayi aur shrinkhla chal padi

"Ek machine se sab kuch ruk gaya"

- 6.2 koi ek jagah aisi na ho jo sab rok de

"Do log ek saath badalte hain aur data galat ho jaata hai"

- 5.4 transaction aur taala

"Mere laptop pe chalta hai, server pe nahi"

- 7.4 container

"Sab jaanch paas hai aur phir bhi toot gaya"

- 7.1 galti aksar hisson ke BEECH mein hoti hai
- 8.4 asli ghatnaayein

KHABAR SAMAJHNE KE SAWAL

"Yeh number bada hai ya nahi?"

- 2.2 shoonya ginno, ank nahi
- pratishat = khabar, guna = badlav

"Yeh nayi cheez sach mein badlegi kya?"

- 8.5 jeet ki paanch wajahein
- aur unmein se koi bhi "achha hona" nahi hai

"Yeh technology kis kaam ki hai?"

- 8.1 teen sawal: kaunsi parat, kaunsa dard, kaunsa sauda

"Internet band kaise kiya jaata hai jab uska koi maalik nahi?"

- 4.7 aur 4.8 chaar jagahein jahan taakat hai

PAISE AUR CAREER KE SAWAL

"Mujhe kitna milega, yeh kya tay karta hai?"

- 0.2 size x scarcity x leverage, aur ghante usmein nahi hain

"Kaunsi field chunun?"

- 0.2 har field mein dono column hain

"AI mein jaane se paisa milega?"

- 0.3 level se kuch nahi milta, position se milta hai

"Leverage kya hai, software mein?"

- 6.8 ek aur user aane se aapka kharcha kitna badhta hai

"Paisa kahan rukta hai?"

- 6.9 aata neeche se hai, rukta scarcity pe hai

SIRF SPINE PADHNA HO TOH

37 chapters. Yeh poora naksha de dete hain, mechanism ke bina.

0.1	0.2	0.3	0.4	0.5				
1.1	1.2	1.3	1.4	1.6				
2.1	2.2	2.4						
3.1	3.2	3.3	3.4	3.6				
4.1	4.2	4.4	4.5	4.6	4.7			
5.1	5.2	5.3	5.5	5.6	5.7			
6.1	6.2	6.3	6.4	6.5	6.7	6.8	6.9	
7.1	7.2	7.3	7.5					
8.1	8.2	8.3	8.5	8.6				

Baaki DEPTH chapters tab padho jab jaanna ho ki andar se kaam kaise karta hai.

Poora naksha

Ek page. Sab kuch. Isse deewar pe laga sakte ho.

LEVEL 1	INSAAN KI ZAROORAT	khaana, paani, chhat, sehat, suraksha, gyaan, judav, matlab ← paisa SIRF yahan se ghusta hai
LEVEL 2	ORGANISATION	ajnabiyon ko scale pe serve karna
LEVEL 3	TECHNOLOGY	← YEH KITAAB, poora ka poora
LEVEL 4	AI	← BOOK 2

pay = problem ka size x kitne kam log x kitni baar wapas aata
(ghante isme kahin nahi hain)

LEVEL 3 KE ANDAR, NEECHE SE UPAR

8	JODNA	spec, kya aasan kya mushkil, jeet ki wajahein
7	BANANA	bug, jaanch, itihaas, container, anumaan
6	SERVE KARNA	kai machinein, uptime, front/back, API, cloud, shrinkhla, latency, kharcha
5	YAAD RAKHNA	database, index, shakal, transaction, cache, queue, replica, backup
4	JUDNA	pata, packet, protocol, DNS, taala, internet
3	BATANA	language, translator, OS, process, open source
2	NUMBERS	byte, compression, paimana, CPU, RAM/disk
1	NEEV	switch → shartein → binary → program

DAS BAATEIN JO POORI KITAAB MEIN DOHRAYI GAYIN
--

1. Andar sirf numbers hain. Matlab hamesha bahar se aata hai.
2. Har parat neeche wali ko chhupati hai.
Isiliye achhi technology samajh nahi aati.
3. Machine badalti nahi. Sirf numbers badalte hain.
4. Hisaab sasta hai. Data laana mehnga hai.
Jab kuch dheema ho, pehle poocho ki data kahan se aa raha hai.
5. Har cheez ka ek sauda hai.
Agar nahi dikh raha, toh aapne dhoondha nahi.
index → padhna tez, likhna dheema
cache → tez, aur purana ho sakta hai
taala → sahi, aur dheema
container → ek jaisa, aur alagav poora nahi
6. Bojh hat ta nahi. Woh khiskta hai.
Ek jagah baanto, agli jagah pe pahunch jaata hai.
7. Systems dheere-dheere kharaab nahi hote.
Woh ek seema tak theek chalte hain, phir jhatke mein girte hain.
8. Jo jaancha nahi gaya, woh hai hi nahi.
Backup, doosri machine, wapas jaane ka raasta, sab.
9. Har cheez kisi dard ka jawab hai.
Dard nahi, toh cheez nahi chahiye.
10. Jeet cheez ki quality se nahi aati.
Woh pehle wahan hone se, logon ke wahan hone se,
aas-paas ke saaman se, aur aasani se aati hai.

EK SHAKAL JO CHAAR JAGAH DIKHTI HAI

Jab ek cheez ho aur maangne wale kai, tab koi beech mein baithkar baantta hai, aur use yaad rakhna padta hai kaun kahan hai.

- | | |
|-------------------|----------------------------------|
| 3.4 OS | programs ko ek machine pe |
| 6.1 load balancer | users ko machinon pe |
| 7.4 Kubernetes | dabbon ko machinon ke jhund pe |
| 4.2 NAT ka router | andar walon ko ek public pate pe |

Har baar wahi sawal: manager mar jaaye toh? Do atak jaayein toh?
Ek sab kuch le le toh?

KISI BHI NAYI CHEEZ KO 60 SECOND MEIN RAKHNE KA TAREEKA

- | | |
|-----------------------------------|--------------------------------|
| 1. Yeh kaunsi parat pe hai? | (upar wala dher dekho) |
| 2. Yeh kis dard ke liye bani thi? | (na bata sako toh shak karo) |
| 3. Iska sauda kya hai? | (kuch deti hai, kuch leti hai) |